VTU ENGINEERING MATHEMATICS STUDY SERIES

1BMATS101 — APPLIED MATHEMATICS-I

125 Important Questions with Complete Step-by-Step Solutions

A comprehensive, rigorous, and self-study textbook designed for engineering students. Complete step-by-step solutions, first-principles concepts, formulas, exam presentation strategies, and mistake prevention.

MODULE 1	MODULE 2	MODULE 3	MODULE 4	MODULE 5
Partial Calculus & Series (Q1 – Q25)	Vector Calculus & Curvilinear (Q26 – Q50)	Matrices, Linear Systems & Eigenvalues (Q51 – Q75)	Vector Spaces & Inner Products (Q76 – Q100)	Linear Transformations & Rank-Nullity (Q101 – Q125)

English Edition • 10-Part Standard Pedagogical Format • Master Formula Sheet • Complete Practice Solutions

📚 Table of Contents

Module	Module Title	Question Range	Key Topics Covered
Module 1	Partial Differentiation, Jacobians, Series & Extrema	Questions 1 – 25	Euler's theorem, higher-order derivatives, Jacobians ($J \cdot J' = 1$), Taylor/Maclaurin series, stationary points, $D = rt - s^2$, constrained optimization
Module 2	Vector Calculus	Questions 26 – 50	Gradient, directional derivative, divergence, curl, solenoidal, irrotational, scalar potential, Laplacian $\nabla^2 r^n$, curvilinear coordinates
Module 3	Matrices & Systems of Linear Equations	Questions 51 – 75	Rank, echelon form, Gauss & Gauss-Jordan elimination, consistency (λ & μ parameters), eigenvalues, eigenvectors, Cayley-Hamilton, diagonalization
Module 4	Vector Spaces	Questions 76 – 100	Vector space axioms, subspace tests, linear combinations, span, linear independence, basis, dimension, inner products, Cauchy-Schwarz, Gram-Schmidt
Module 5	Linear Transformations	Questions 101 – 125	Linear mapping tests, standard matrices, kernel, range, nullity, rank, one-to-one, onto, isomorphisms, composition, Rank-Nullity theorem
Appendix A	Consolidated Practice Question Answer Key	Problems 1 – 125	Complete working, numerical solutions, and verification steps for all 125 similar practice questions
Appendix B	Final Examination Revision Checklist	Checklist & Tips	High-yield formulas, common pitfalls to avoid, time management, and paper presentation strategies

📋 Official Syllabus & Exam Pattern (1BMATS101)

The examination paper consists of 10 full questions carrying 20 marks each, distributed uniformly with two questions from each of the five modules. Students must answer FIVE full questions, selecting at least one full question from each module.

- **Module 1:** Partial derivatives, higher-order and mixed derivatives, Euler's theorem for homogeneous functions, Jacobians and inverse Jacobians, Taylor's and Maclaurin's series for two variables, maxima and minima of two-variable functions.
- **Module 2:** Vector differential calculus: gradient, directional derivative, maximum directional derivative, divergence, curl, solenoidal and irrotational fields, scalar potential, Laplacian, scale factors in curvilinear coordinates.
- **Module 3:** Matrix rank by elementary row operations, row-echelon form, Gauss and Gauss-Jordan elimination, consistency of linear systems, characteristic equations, eigenvalues, eigenvectors, Cayley-Hamilton theorem, diagonalization.
- **Module 4:** Vector spaces, subspaces, linear combinations, span, linear independence, basis and dimension, coordinates, inner product spaces, norms, Cauchy-Schwarz inequality, Gram-Schmidt orthogonalization.
- **Module 5:** Linear transformations, matrix representations, kernel and range, rank-nullity theorem, injective and surjective maps, composition, change of basis matrices.

Master Formula Sheet

MODULE 1: Partial Differentiation & Extrema

Euler's Theorem (1st Order): $x\,u_x + y\,u_y = n\,u$

Euler's Composite: If f(u) homogeneous degree n, $x\,u_x + y\,u_y = n\,[f(u)/f'(u)]$

Euler's Theorem (2nd Order): $x^2\,u_{xx} + 2xy\,u_{xy} + y^2\,u_{yy} = n(n-1)\,u$

Jacobian (2D & 3D): $J = \partial(u, v)/\partial(x, y)$; $J \cdot J' = 1$

Extrema Discriminant: $D = rt - s^2$ ($r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$)

Classification: $D > 0, r > 0 \Rightarrow$ Min; $D > 0, r < 0 \Rightarrow$ Max; $D < 0 \Rightarrow$ Saddle

MODULE 2: Vector Calculus

Gradient: $\nabla\varphi = (\partial\varphi/\partial x)\,\hat{i} + (\partial\varphi/\partial y)\,\hat{j} + (\partial\varphi/\partial z)\,\hat{k}$

Directional Derivative: $D_{\vec{u}}\,\varphi = \nabla\varphi \cdot (\vec{a}/|\vec{a}|)$

Maximum D.D.: $|\nabla\varphi|$ in the direction of $\nabla\varphi$

Divergence & Curl: $\text{div}\,\vec{F} = \vec{\nabla} \cdot \vec{F}$; $\text{curl}\,\vec{F} = \vec{\nabla} \times \vec{F}$

Solenoidal & Irrotational: $\text{div}\,\vec{F} = 0$ (solenoidal); $\text{curl}\,\vec{F} = \vec{0}$ (irrotational)

Radial Formulas: $\vec{\nabla}(r^n) = n\,r^{n-2}\,\vec{r}$; $\vec{\nabla} \cdot (r^n\,\vec{r}) = (n+3)r^n$; $\vec{\nabla}^2(r^n) = n(n+1)r^{n-2}$

MODULE 3: Matrices & Linear Systems

Rouché-Capelli: $\rho(A) = \rho([A|B]) = n \Rightarrow$ Unique; $= r < n \Rightarrow$ Infinite; $\neq \Rightarrow$ Inconsistent

Characteristic Eq (3×3): $\lambda^3 - s_1\lambda^2 + s_2\lambda - s_3 = 0$ (s_1=Tr, s_2=minors, s_3=det)

Eigenvalue Properties: $\Sigma\,\lambda_i = \text{Trace}(A)$; $\Pi\,\lambda_i = \det(A)$

Cayley-Hamilton: $p(A) = O \Rightarrow A^{-1} = (1/s_3)[\,A^2 - s_1A + s_2I\,]$

Diagonalization: $P^{-1}\,A\,P = D$; for symmetric matrices, $P^T\,A\,P = D$

MODULE 4 & 5: Vector Spaces & Transformations

Subspace Criteria: $0 \in W$ and $a\,u + b\,v \in W$ for all $u, v \in W$, $a, b \in \mathbb{R}$

Cauchy-Schwarz: $|\langle u, v\rangle| \leq \|u\|\,\|v\|$; Triangle: $\|u + v\| \leq \|u\| + \|v\|$

Gram-Schmidt: $u_k = v_k - \Sigma\,[\,(v_k \cdot u_i)/\|u_i\|^2\,]\,u_i$; $e_k = u_k/\|u_k\|$

Rank-Nullity Theorem: $\dim(\text{Domain } V) = \text{rank}(T) + \text{nullity}(T)$

Injectivity Criterion: T is one-to-one $\Leftrightarrow \ker(T) = \{0\} \Leftrightarrow$ nullity $= 0$

Change of Basis: $[T]_{B'} = P^{-1}\,[T]_B\,P$ where P is transition matrix

MODULE 1: Partial Differentiation, Jacobians, Series and Extrema

25 Fully Solved Questions: 5 Fundamental (Concepts & Proofs) + 10 Intermediate (Core Problems) + 10 Exam-Level (High-Yield VTU Questions)

QUESTION 1: First-Order Partial Derivatives of Polynomial Function

FUNDAMENTAL

A. Full Question:

If $u = x^3 + 3x^2y + 3xy^2 + y^3$, find the first-order partial derivatives $\partial u/\partial x$ and $\partial u/\partial y$.

B. Meaning of the Question

Find the rate of change of u with respect to x (holding y constant) and with respect to y (holding x constant).

C. Basic Concept

Partial differentiation computes derivatives with respect to one variable while all other independent variables are treated as constants.

D. Formula and Explanation

Formula: $\partial u/\partial x = d[f(x, y)]/dx$ with $y = const$; Power rule: $d(x^n)/dx = n\,x^{n-1}$.

Explanation & Conditions: $\partial u/\partial x$ denotes partial derivative with respect to x. Pure y terms differentiate to 0.

E. Given Information

$u = x^3 + 3x^2y + 3xy^2 + y^3$.

F. Complete Step-by-Step Solution

Step 1: Compute $\partial u/\partial x$ (treat y as constant)

$\partial u/\partial x = 3x^2 + 3y(2x) + 3y^2(1) + 0 = 3x^2 + 6xy + 3y^2 = 3(x + y)^2$.

💡 **Why this step is performed:** Execute step 1: compute $\partial u/\partial x$ (treat y as constant) to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute $\partial u/\partial y$ (treat x as constant)

$\partial u/\partial y = 0 + 3x^2(1) + 3x(2y) + 3y^2 = 3x^2 + 6xy + 3y^2 = 3(x + y)^2$.

💡 **Why this step is performed:** Execute step 2: compute $\partial u/\partial y$ (treat x as constant) to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $\partial u/\partial x = 3(x + y)^2$, $\partial u/\partial y = 3(x + y)^2$

H. Examination Presentation Strategy

- State clearly which variable is held constant.
- Show term-by-term derivative.
- Factorize into $3(x+y)^2$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Treating y as a function of x (dy/dx).
✓ **How to Avoid:** This is partial derivative, so y is purely an independent constant.

⚠️ **Common Mistake:** Missing the product constant rule 3y·(2x).
✓ **How to Avoid:** Keep the constant factor outside the derivative.

J. Similar Practice Question

Practice Problem 1: If $u = 2x^3 - 4x^2y + 5xy^2 - y^3$, find $\partial u/\partial x$ and $\partial u/\partial y$.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 2: Equality of Mixed Second-Order Partial Derivatives

FUNDAMENTAL

A. Full Question:
If $u = x^3y^2 + 2x^2y^3$, verify Clairaut's theorem: $\partial^2u/\partial x\partial y = \partial^2u/\partial y\partial x$.

B. Meaning of the Question

Verify that differentiating first with respect to x then y gives the same result as differentiating first with respect to y then x.

C. Basic Concept

For smooth functions with continuous second partial derivatives, the order of differentiation is commutative.

D. Formula and Explanation

Formula: $\partial^2u/\partial y\partial x = \partial/\partial y(\partial u/\partial x)$, $\partial^2u/\partial x\partial y = \partial/\partial x(\partial u/\partial y)$.

Explanation & Conditions: Clairaut's / Schwarz's theorem guarantees equality when second derivatives are continuous.

E. Given Information

$u = x^3y^2 + 2x^2y^3$.

F. Complete Step-by-Step Solution

Step 1: Compute $\partial u/\partial x$

$\partial u/\partial x = 3x^2y^2 + 4xy^3$.

💡 **Why this step is performed:** Execute step 1: compute $\partial u/\partial x$ to advance the step-by-step mathematical solution toward the final result.

Step 2: Differentiate $\partial u/\partial x$ with respect to y

$\partial^2u/\partial y\partial x = \partial/\partial y(3x^2y^2 + 4xy^3) = 3x^2(2y) + 4x(3y^2) = 6x^2y + 12xy^2$.

💡 **Why this step is performed:** Execute step 2: differentiate $\partial u/\partial x$ with respect to y to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute ∂u/∂y

$\partial u/\partial y = 2x^3y + 6x^2y^2$.

💡 **Why this step is performed:** Execute step 3: compute ∂u/∂y to advance the step-by-step mathematical solution toward the final result.

Step 4: Differentiate ∂u/∂y with respect to x

$\partial^2u/\partial x\partial y = \partial/\partial x(2x^3y + 6x^2y^2) = 2(3x^2)y + 6(2x)y^2 = 6x^2y + 12xy^2$.

💡 **Why this step is performed:** Execute step 4: differentiate ∂u/∂y with respect to x to advance the step-by-step mathematical solution toward the final result.

Step 5: Conclude equality

$\partial^2u/\partial y\partial x = 6x^2y + 12xy^2 = \partial^2u/\partial x\partial y$. Hence verified.

💡 **Why this step is performed:** Execute step 5: conclude equality to advance the step-by-step mathematical solution toward the final result.

✓ **G. Final Answer: $\partial^2u/\partial x\partial y = \partial^2u/\partial y\partial x = 6x^2y + 12xy^2$ (Verified)**

H. Examination Presentation Strategy

- Label LHS and RHS clearly.
- Show both first-order steps clearly before taking second-order derivatives.
- State 'Hence verified'.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Confusing $\partial^2u/\partial x\partial y$ with $(\partial u/\partial x)(\partial u/\partial y)$.
✓ **How to Avoid:** It is repeated differentiation, not multiplication.

⚠️ **Common Mistake:** Careless power subtraction.
✓ **How to Avoid:** Check: $d(y^2)/dy = 2y$, $d(y^3)/dy = 3y^2$.

J. Similar Practice Question

Practice Problem 2: If $u = x^4y^2 - 2x^3y^4$, verify that $\partial^2u/\partial x\partial y = \partial^2u/\partial y\partial x$.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 3: Homogeneous Function Degree and Euler's Theorem Verification FUNDAMENTAL

A. Full Question:

Find the degree of homogeneity of $u = (x^2 + y^2) / (x + y)$ and verify Euler's Theorem.

B. Meaning of the Question

Determine degree n by scaling variables by t, then verify that $x(\partial u/\partial x) + y(\partial u/\partial y) = n \cdot u$.

C. Basic Concept

A function is homogeneous of degree n if $f(tx, ty) = t^n f(x, y)$. Euler's theorem relates its first partial derivatives directly to the function value.

D. Formula and Explanation

Formula: $x(\partial u/\partial x) + y(\partial u/\partial y) = n \cdot u$.

Explanation & Conditions: n is the degree of homogeneity.

E. Given Information

$u = (x^2 + y^2) / (x + y)$.

F. Complete Step-by-Step Solution

Step 1: Determine degree n

$u(tx, ty) = [(tx)^2 + (ty)^2] / [tx + ty] = t^2(x^2 + y^2) / [t(x + y)] = t^1 u(x, y)$. Thus $n = 1$.

💡 **Why this step is performed:** Execute step 1: determine degree n to advance the step-by-step mathematical solution toward the final result.

Step 2: Find ∂u/∂x using quotient rule

$\partial u/\partial x = [(x+y)(2x) - (x^2+y^2)(1)] / (x+y)^2 = (2x^2 + 2xy - x^2 - y^2) / (x+y)^2 = (x^2 + 2xy - y^2) / (x+y)^2$.
$x(\partial u/\partial x) = (x^3 + 2x^2y - xy^2) / (x+y)^2$.

💡 **Why this step is performed:** Apply the quotient rule $[v u' - u v'] / v^2$ treating the other variable as a constant.

Step 3: Find ∂u/∂y by symmetry

$\partial u/\partial y = (y^2 + 2xy - x^2) / (x+y)^2$.
$y(\partial u/\partial y) = (y^3 + 2xy^2 - x^2y) / (x+y)^2$.

💡 **Why this step is performed:** Exploit the algebraic symmetry between x and y to write the corresponding partial derivative directly.

Step 4: Add x(∂u/∂x) + y(∂u/∂y)

$LHS = [x^3 + y^3 + x^2y + xy^2] / (x+y)^2 = [(x^2+y^2)(x+y)] / (x+y)^2 = (x^2+y^2)/(x+y) = 1 \cdot u = RHS$.

💡 **Why this step is performed:** Execute step 4: add $x(\partial u/\partial x) + y(\partial u/\partial y)$ to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Degree n = 1; Euler's Theorem verified: $x(\partial u/\partial x) + y(\partial u/\partial y) = u$

H. Examination Presentation Strategy

- Show $u(tx, ty) = t^n u(x, y)$ first.
- State quotient rule explicitly.
- Conclude with LHS = RHS.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Thinking degree is 2.
✓ **How to Avoid:** Numerator degree minus denominator degree gives 2 - 1 = 1.

⚠️ **Common Mistake:** Sign error in quotient rule numerator.
✓ **How to Avoid:** Formula is $(v u' - u v')/v^2$ with minus sign.

J. Similar Practice Question

Practice Problem 3: Find degree and verify Euler's theorem for $u = (x^3 + y^3) / (x - y)$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 4: Jacobian of Polar Coordinate Transformation

FUNDAMENTAL

A. Full Question:
If $x = r \cos \theta$ and $y = r \sin \theta$, evaluate the Jacobian $J = \partial(x, y)/\partial(r, \theta)$.

B. Meaning of the Question

Evaluate the 2×2 determinant of first partial derivatives of x and y with respect to r and θ.

C. Basic Concept

The Jacobian represents the local scaling factor of area under coordinate transformation: $dx\,dy = |J|\,dr\,d\theta$.

D. Formula and Explanation

Formula: $J = \partial(x, y)/\partial(r, \theta) = (\partial x/\partial r)(\partial y/\partial \theta) - (\partial x/\partial \theta)(\partial y/\partial r)$.
Explanation & Conditions: Determinant of the Jacobian matrix formed by row 1 = $[\partial x/\partial r, \partial x/\partial \theta]$ and row 2 = $[\partial y/\partial r, \partial y/\partial \theta]$.

E. Given Information

$x = r \cos \theta$, $y = r \sin \theta$.

F. Complete Step-by-Step Solution

Step 1: Compute derivatives of x

$\partial x/\partial r = \cos \theta$, $\partial x/\partial \theta = -r \sin \theta$.

💡 **Why this step is performed:** Differentiate x partially with respect to each independent variable while holding other variables fixed.

Step 2: Compute derivatives of y

$\partial y/\partial r = \sin\theta$, $\partial y/\partial\theta = r\cos\theta$.

💡 **Why this step is performed:** Differentiate y partially with respect to each independent variable while holding other variables fixed.

Step 3: Set up and evaluate determinant

$J = (\cos\theta)(r\cos\theta) - (-r\sin\theta)(\sin\theta) = r\cos^2\theta + r\sin^2\theta = r(\cos^2\theta + \sin^2\theta) = r(1) = r$.

💡 **Why this step is performed:** Evaluate the Hessian discriminant $D = rt - s^2$; $D > 0$ and $r > 0$ indicates a local minimum, while $D > 0$ and $r < 0$ indicates a local maximum.

✓ **G. Final Answer: $J = \partial(x, y)/\partial(r, \theta) = r$**

H. Examination Presentation Strategy

- Write out the 2×2 determinant.
- Substitute all 4 derivatives.
- Use $\cos^2\theta + \sin^2\theta = 1$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Derivative of cos θ written as +sin θ.
✓ **How to Avoid:** Derivative of cos θ is -sin θ.

⚠️ **Common Mistake:** Swapping rows and columns.
✓ **How to Avoid:** Keep x in row 1, y in row 2.

J. Similar Practice Question

Practice Problem 4: If $x = u(1 - v)$ and $y = uv$, find $J = \partial(x, y)/\partial(u, v)$.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 5: Stationary Point and Nature of a Quadratic Surface

FUNDAMENTAL

A. Full Question:
Find the stationary points of $f(x, y) = x^2 + y^2 - 4x - 6y + 13$ and classify them.

B. Meaning of the Question

Find points where fx = 0 and fy = 0, and use D = rt - s^2 to classify the point as maximum, minimum, or saddle point.

C. Basic Concept

Critical points occur where gradient is zero. The sign of the Hessian determinant D determines curvature.

D. Formula and Explanation

Formula: `fx = 0, fy = 0; D = rt - s² where r = f_xx, s = f_xy, t = f_yy. If D > 0, r > 0 → Minimum.`

Explanation & Conditions: r is concavity in x, D is overall Gaussian curvature sign.

E. Given Information

$f(x, y) = x^2 + y^2 - 4x - 6y + 13$.

F. Complete Step-by-Step Solution

Step 1: Find first derivatives and set to zero

fx = 2x - 4 = 0 ⇒ x = 2.
fy = 2y - 6 = 0 ⇒ y = 3.
Stationary point is (2, 3).

💡 **Why this step is performed:** Set the first partial derivatives fx = 0 and fy = 0 to solve for all stationary (critical) points.

Step 2: Find second partial derivatives

r = f_xx = 2, s = f_xy = 0, t = f_yy = 2.

💡 **Why this step is performed:** Calculate the second-order partial derivatives r = fxx, s = fxy, and t = fyy to prepare for the curvature test.

Step 3: Evaluate D = rt - s^2

D = (2)(2) - $(0)^2$ = 4 > 0 and r = 2 > 0.
Therefore, (2, 3) is a Local Minimum.
Minimum value f(2, 3) = $2^2 + 3^2$ - 4(2) - 6(3) + 13 = 4 + 9 - 8 - 18 + 13 = 0.

💡 **Why this step is performed:** Evaluate the Hessian discriminant D = rt - s^2; D > 0 and r > 0 indicates a local minimum, while D > 0 and r < 0 indicates a local maximum.

✓ G. Final Answer: Stationary point (2, 3) is a Local Minimum with minimum value f_min = 0.

H. Examination Presentation Strategy

- Solve fx = 0, fy = 0 clearly.
- Compute r, s, t.
- State D = rt - s^2 > 0 and r > 0 clearly.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Thinking r > 0 means maximum.
✓ **How to Avoid:** In calculus, r > 0 means concave up, which is a minimum!

⚠️ **Common Mistake:** Calculating s as 2.
✓ **How to Avoid:** $\partial(2x-4)/\partial y = 0$ because there is no y in fx.

J. Similar Practice Question

Practice Problem 5: Find the stationary points of $f(x, y) = 2x^2 + y^2 - 4x + 4y + 10$ and determine its nature.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 6: Chain Rule for Functions of Two Independent Variables

INTERMEDIATE

A. Full Question:
If $z = e^{(xy^2)}$, where $x = t^3$ and $y = t^2$, find dz/dt using the chain rule and verify by direct substitution.

B. Meaning of the Question

Compute the total derivative dz/dt using the multivariable chain rule $dz/dt = (\partial z/\partial x)(dx/dt) + (\partial z/\partial y)(dy/dt)$ and compare with direct single-variable differentiation.

C. Basic Concept

When z depends on x and y, and both x and y depend on a single variable t, the rate of change of z with respect to t flows through both intermediate variables x and y.

D. Formula and Explanation

Formula: $dz/dt = (\partial z/\partial x)(dx/dt) + (\partial z/\partial y)(dy/dt)$.

Explanation & Conditions: $\partial z/\partial x$ and $\partial z/\partial y$ are partial derivatives of outer function; dx/dt and dy/dt are ordinary derivatives of inner functions.

E. Given Information

$z = e^{(xy^2)}$, $x = t^3$, $y = t^2$.

F. Complete Step-by-Step Solution

Step 1: Compute partial derivatives of z

$\partial z/\partial x = y^2 e^{(xy^2)}$
$\partial z/\partial y = 2xy e^{(xy^2)}$.

💡 **Why this step is performed:** Differentiate z partially with respect to each independent variable while holding other variables fixed.

Step 2: Compute ordinary derivatives of x and y

$dx/dt = d(t^3)/dt = 3t^2$
$dy/dt = d(t^2)/dt = 2t.$

💡 **Why this step is performed:** Differentiate x partially with respect to each independent variable while holding other variables fixed.

Step 3: Apply the chain rule formula

$dz/dt = [y^2\ e^{(xy^2)}](3t^2) + [2xy\ e^{(xy^2)}](2t)$
$= e^{(xy^2)}\ [3t^2\ y^2 + 4t\ xy].$

💡 **Why this step is performed:** Apply the multivariable chain rule by summing the product of partial and ordinary derivatives.

Step 4: Substitute $x = t^3$ and $y = t^2$ in terms of t

$xy^2 = (t^3)(t^2)^2 = t^3 \cdot t^4 = t^7.$
$3t^2\ y^2 = 3t^2(t^4) = 3t^6.$
$4t\ xy = 4t(t^3)(t^2) = 4t^6.$
$dz/dt = e^{(t^7)}\ [3t^6 + 4t^6] = 7t^6\ e^{(t^7)}.$

💡 **Why this step is performed:** Execute step 4: substitute $x = t^3$ and $y = t^2$ in terms of t to advance the step-by-step mathematical solution toward the final result.

Step 5: Verify by direct substitution

$z = e^{(xy^2)} = e^{((t^3)(t^2)^2)} = e^{(t^7)}.$
$dz/dt = d(e^{(t^7)})/dt = e^{(t^7)} \cdot d(t^7)/dt = 7t^6\ e^{(t^7)}.$ Verified!

💡 **Why this step is performed:** Substitute the parametric functions directly into the outer function and differentiate to confirm the chain rule result.

✓ G. Final Answer: $dz/dt = 7t^6\ e^{(t^7)}$ (Chain rule verified by direct substitution)

H. Examination Presentation Strategy

- State multivariable chain rule formula first.
- Compute all four components $\partial z/\partial x$, $\partial z/\partial y$, dx/dt, dy/dt.
- Substitute t and combine terms.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Missing the exponent power $(t^2)^2 = t^4$.
✓ **How to Avoid:** Remember $(t^2)^2 = t^4$, so $xy^2 = t^3 \cdot t^4 = t^7$.

⚠️ **Common Mistake:** Forgetting inner derivative in dy/dt.
✓ **How to Avoid:** $d(t^2)/dt = 2t$.

J. Similar Practice Question

Practice Problem 6: If $z = x^2 + y^2$, where $x = \cos t$, $y = \sin t$, find dz/dt using the chain rule and verify.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 7: Euler's Theorem for Inverse Trigonometric Function

INTERMEDIATE

A. Full Question:

If $u = \sin^{-1}[(x + y) / (\sqrt{x} + \sqrt{y})]$, prove that $x(\partial u/\partial x) + y(\partial u/\partial y) = (1/2) \tan u$.

B. Meaning of the Question

Here u itself is not a homogeneous function, but sin u is homogeneous. We apply Euler's theorem to f = sin u and relate it back to u.

C. Basic Concept

When $u = g^{-1}[f(x, y)]$, where f(x, y) is homogeneous of degree n, we write g(u) = f(x, y). Then by Euler's theorem on g(u): $x(\partial/\partial x)[g(u)] + y(\partial/\partial y)[g(u)] = n\, g(u) \Rightarrow g'(u)[x\, u_x + y\, u_y] = n\, g(u) \Rightarrow x\, u_x + y\, u_y = n\, g(u)/g'(u)$.

D. Formula and Explanation

Formula: `If f(u) is homogeneous of degree n, then x(∂u/∂x) + y(∂u/∂y) = n [f(u) / f'(u)].`

Explanation & Conditions: Here f(u) = sin u, so f'(u) = cos u, and f(u)/f'(u) = sin u / cos u = tan u.

E. Given Information

$u = \sin^{-1}[(x + y) / (\sqrt{x} + \sqrt{y})]$.

F. Complete Step-by-Step Solution

Step 1: Rewrite u so the RHS is homogeneous

Let $z = \sin u = (x + y) / (\sqrt{x} + \sqrt{y})$.

💡 **Why this step is performed:** Execute step 1: rewrite u so the rhs is homogeneous to advance the step-by-step mathematical solution toward the final result.

Step 2: Find the degree of homogeneity of z

z(tx, ty) = [tx + ty] / [√(tx) + √(ty)] = t(x + y) / [t^(1/2)(√x + √y)] = t^(1 - 1/2) · z = t^(1/2) z.
Thus, z is homogeneous of degree n = 1/2.

💡 **Why this step is performed:** Substitute tx and ty for x and y to factor out t^n and establish the degree of homogeneity n.

Step 3: Apply Euler's Theorem to z

$x(\partial z/\partial x) + y(\partial z/\partial y) = n \cdot z = (1/2)\, z = (1/2) \sin u$.

💡 **Why this step is performed:** Combine the partial derivatives into the linear combination $x(\partial u/\partial x) + y(\partial u/\partial y)$ and verify it equals n·u.

Step 4: Express derivatives of z in terms of u

Since z = sin u:
$\partial z/\partial x = \cos u \cdot (\partial u/\partial x)$
$\partial z/\partial y = \cos u \cdot (\partial u/\partial y)$.

💡 **Why this step is performed:** Differentiate z partially with respect to each independent variable while holding other variables fixed.

Step 5: Substitute and solve for LHS

$x [\cos u (\partial u/\partial x)] + y [\cos u (\partial u/\partial y)] = (1/2) \sin u$
$\cos u [x(\partial u/\partial x) + y(\partial u/\partial y)] = (1/2) \sin u$
$x(\partial u/\partial x) + y(\partial u/\partial y) = (1/2) (\sin u / \cos u) = (1/2) \tan u$. Hence proved.

💡 **Why this step is performed:** Execute step 5: substitute and solve for lhs to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $x(\partial u/\partial x) + y(\partial u/\partial y) = (1/2) \tan u$ (Hence proved)

H. Examination Presentation Strategy

- Define z = sin u first.
- Show z(tx, ty) = t^(1/2) z to establish degree n = 1/2 clearly.
- Show factorizing cos u and dividing.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Applying Euler directly to $\sin^{-1}$.
✓ **How to Avoid:** $\sin^{-1}$ is NOT homogeneous! You must apply Euler to sin u.

⚠️ **Common Mistake:** Wrong degree calculation.
✓ **How to Avoid:** √t = t^(1/2), so degree is 1 - 1/2 = 1/2, not 1/2 - 1 = -1/2.

J. Similar Practice Question

Practice Problem 7: If $u = \cos^{-1}[(x + y) / (\sqrt{x} + \sqrt{y})]$, prove that $x(\partial u/\partial x) + y(\partial u/\partial y) = -(1/2) \cot u$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 8: Jacobian Property of Inverses J · J' = 1

INTERMEDIATE

A. Full Question:
If x = r cos θ and y = r sin θ, find $J = \partial(x, y)/\partial(r, \theta)$ and $J' = \partial(r, \theta)/\partial(x, y)$, and verify that J · J' = 1.

B. Meaning of the Question

Compute the Jacobian of polar coordinates J, then invert the relations to get r and θ in terms of x and y, compute the inverse Jacobian J', and verify that their product equals 1.

C. Basic Concept

The Jacobian of an inverse transformation is the reciprocal of the original Jacobian: $[\partial(u, v)/\partial(x, y)] \cdot [\partial(x, y)/\partial(u, v)] = 1$, analogous to $(dy/dx) \cdot (dx/dy) = 1$ in single-variable calculus.

D. Formula and Explanation

Formula: $J \cdot J' = [\partial(x, y)/\partial(r, \theta)] \cdot [\partial(r, \theta)/\partial(x, y)] = 1.$

Explanation & Conditions: Follows from chain rule of matrix determinants: $\det(A \cdot A^{-1}) = \det(I) = 1$.

E. Given Information

$x = r \cos \theta$, $y = r \sin \theta \Rightarrow r = \sqrt{(x^2 + y^2)}$, $\theta = \tan^{-1}(y/x)$.

F. Complete Step-by-Step Solution

Step 1: Recall $J = \partial(x, y)/\partial(r, \theta)$

From Question 4, $J = r$.

💡 **Why this step is performed:** Execute step 1: recall j = ∂(x, y)/∂(r, θ) to advance the step-by-step mathematical solution toward the final result.

Step 2: Express r and θ in terms of x and y

$r^2 = x^2 + y^2 \Rightarrow r = \sqrt{(x^2 + y^2)}$

$\tan \theta = y/x \Rightarrow \theta = \tan^{-1}(y/x)$.

💡 **Why this step is performed:** Execute step 2: express r and θ in terms of x and y to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute partial derivatives of r

$\partial r/\partial x = (1 / 2\sqrt{(x^2+y^2)}) \cdot 2x = x/r = \cos \theta$

$\partial r/\partial y = (1 / 2\sqrt{(x^2+y^2)}) \cdot 2y = y/r = \sin \theta$.

💡 **Why this step is performed:** Execute step 3: compute partial derivatives of r to advance the step-by-step mathematical solution toward the final result.

Step 4: Compute partial derivatives of θ

$\partial\theta/\partial x = [1 / (1 + (y/x)^2)] \cdot (-y/x^2) = -y / (x^2 + y^2) = -\sin \theta / r$

$\partial\theta/\partial y = [1 / (1 + (y/x)^2)] \cdot (1/x) = x / (x^2 + y^2) = \cos \theta / r$.

💡 **Why this step is performed:** Execute step 4: compute partial derivatives of θ to advance the step-by-step mathematical solution toward the final result.

Step 5: Compute $J' = \partial(r, \theta)/\partial(x, y)$

$J' = | \partial r/\partial x \ \partial r/\partial y | / | \partial\theta/\partial x \ \partial\theta/\partial y |$

$= (\cos \theta)(\cos \theta / r) - (\sin \theta)(-\sin \theta / r)$

$= (\cos^2 \theta + \sin^2 \theta) / r = 1/r$.

💡 **Why this step is performed:** Execute step 5: compute j' = $\partial(r, \theta)/\partial(x, y)$ to advance the step-by-step mathematical solution toward the final result.

Step 6: Verify J · J' = 1

J · J' = (r) · (1/r) = 1. Hence verified.

💡 **Why this step is performed:** Execute step 6: verify j · j' = 1 to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: J = r, J' = 1/r, and J · J' = r · (1/r) = 1 (Verified)

H. Examination Presentation Strategy

- State both forward and reverse relations.
- Show step-by-step differentiation of r and θ.
- Compute determinant and conclude product is 1.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Derivative of $\tan^{-1}(y/x)$ with respect to x missing minus sign.
✓ **How to Avoid:** $d(1/x)/dx = -1/x^2$, so a minus sign must appear in $\partial\theta/\partial x$.

⚠️ **Common Mistake:** Forgetting to simplify $x/(x^2+y^2)$ into cos θ / r.
✓ **How to Avoid:** Converting to polar coordinates simplifies the determinant algebra enormously.

J. Similar Practice Question

Practice Problem 8: If u = x + y and v = x - y, find J and J' and verify J · J' = 1.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 9: Maclaurin Series Expansion of Two Variables: $e^x \cos y$

INTERMEDIATE

A. Full Question:
Expand $f(x, y) = e^x \cos y$ in a Maclaurin series up to second-degree terms.

B. Meaning of the Question

Find the Taylor series expansion of f(x, y) around the origin (0, 0) up to terms of degree 2.

C. Basic Concept

Maclaurin series for two variables approximates any smooth function near (0, 0) by a polynomial in powers of x and y using partial derivatives evaluated at the origin.

D. Formula and Explanation

Formula: `f(x, y) = f(0,0) + [x fx + y fy] + (1/2!)[x² fxx + 2xy fxy + y² fyy] + ... evaluated at (0, 0).`

Explanation & Conditions: The 1st bracket gives linear terms; 2nd bracket gives quadratic terms.

E. Given Information

$f(x, y) = e^x \cos y$, expansion point $(a, b) = (0, 0)$.

F. Complete Step-by-Step Solution

Step 1: Evaluate f(0, 0)

$f(0, 0) = e^0 \cos 0 = 1 \cdot 1 = 1$.

> 💡 **Why this step is performed:** Execute step 1: evaluate f(0, 0) to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute first derivatives and evaluate at (0, 0)

$fx = \partial f/\partial x = e^x \cos y \Rightarrow fx(0, 0) = e^0 \cos 0 = 1$.
$fy = \partial f/\partial y = -e^x \sin y \Rightarrow fy(0, 0) = -e^0 \sin 0 = 0$.

> 💡 **Why this step is performed:** Set the first partial derivatives fx = 0 and fy = 0 to solve for all stationary (critical) points.

Step 3: Compute second derivatives and evaluate at (0, 0)

$fxx = \partial^2 f/\partial x^2 = e^x \cos y \Rightarrow fxx(0, 0) = 1$.
$fxy = \partial^2 f/\partial x \partial y = -e^x \sin y \Rightarrow fxy(0, 0) = 0$.
$fyy = \partial^2 f/\partial y^2 = -e^x \cos y \Rightarrow fyy(0, 0) = -1$.

> 💡 **Why this step is performed:** Execute step 3: compute second derivatives and evaluate at (0, 0) to advance the step-by-step mathematical solution toward the final result.

Step 4: Substitute into Maclaurin series formula

$f(x, y) = 1 + [x(1) + y(0)] + (1/2)[x^2(1) + 2xy(0) + y^2(-1)]$
$= 1 + x + (1/2)(x^2 - y^2)$
$= 1 + x + (1/2)x^2 - (1/2)y^2$.

> 💡 **Why this step is performed:** Substitute the evaluated derivative values into the multivariable Taylor/Maclaurin series formula and group terms by degree.

✓ G. Final Answer: $e^x \cos y = 1 + x + (1/2)x^2 - (1/2)y^2 + \ldots$ (up to 2nd degree)

H. Examination Presentation Strategy

- Write general two-variable Maclaurin formula.
- Create a neat table of derivatives at (0, 0).
- Substitute carefully and box answer.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Derivative of cos y taken as +sin y.
✓ **How to Avoid:** d(cos y)/dy = -sin y, which brings a minus sign into fyy.

⚠️ **Common Mistake:** Forgetting the factor 1/2! = 1/2 on second-order terms.
✓ **How to Avoid:** All 2nd degree terms are divided by 2!.

J. Similar Practice Question

Practice Problem 9: Expand $f(x, y) = e^x \sin y$ in powers of x and y up to second-degree terms.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 10: Taylor Series Expansion Around a Non-Zero Point (1, 1) INTERMEDIATE

A. Full Question:
Expand $f(x, y) = x^2y + 3y - 2$ in powers of (x - 1) and (y - 1) up to second-degree terms.

B. Meaning of the Question

Expand the polynomial f(x, y) in terms of powers of (x - 1) and (y - 1) using Taylor's theorem around the point (a, b) = (1, 1).

C. Basic Concept

Taylor series around (a, b) replaces x by (x - a) and y by (y - b), with derivatives evaluated at (a, b).

D. Formula and Explanation

Formula: $f(x, y) = f(a, b) + [(x-a)f_x + (y-b)f_y] + (1/2!)[(x-a)^2f_{xx} + 2(x-a)(y-b)f_{xy} + (y-b)^2f_{yy}] + \ldots$

Explanation & Conditions: (a, b) = (1, 1); h = x - 1, k = y - 1.

E. Given Information

$f(x, y) = x^2y + 3y - 2$, point (a, b) = (1, 1).

F. Complete Step-by-Step Solution

Step 1: Evaluate f(1, 1)

$f(1, 1) = (1)^2(1) + 3(1) - 2 = 1 + 3 - 2 = 2$.

💡 **Why this step is performed:** Execute step 1: evaluate f(1, 1) to advance the step-by-step mathematical solution toward the final result.

Step 2: First-order derivatives at (1, 1)

$f_x = 2xy \Rightarrow f_x(1, 1) = 2(1)(1) = 2.$
$f_y = x^2 + 3 \Rightarrow f_y(1, 1) = (1)^2 + 3 = 4.$

💡 **Why this step is performed:** Execute step 2: first-order derivatives at (1, 1) to advance the step-by-step mathematical solution toward the final result.

Step 3: Second-order derivatives at (1, 1)

$f_{xx} = 2y \Rightarrow f_{xx}(1, 1) = 2(1) = 2.$
$f_{xy} = 2x \Rightarrow f_{xy}(1, 1) = 2(1) = 2.$
$f_{yy} = 0 \Rightarrow f_{yy}(1, 1) = 0.$

💡 **Why this step is performed:** Calculate the second-order partial derivatives $r = f_{xx}$, $s = f_{xy}$, and $t = f_{yy}$ to prepare for the curvature test.

Step 4: Substitute into Taylor's formula

$f(x, y) = 2 + [(x - 1)(2) + (y - 1)(4)] + (1/2)[(x - 1)^2(2) + 2(x - 1)(y - 1)(2) + (y - 1)^2(0)]$
$= 2 + 2(x - 1) + 4(y - 1) + (x - 1)^2 + 2(x - 1)(y - 1).$

💡 **Why this step is performed:** Substitute the evaluated derivative values into the multivariable Taylor/Maclaurin series formula and group terms by degree.

✓ **G. Final Answer: $f(x, y) = 2 + 2(x - 1) + 4(y - 1) + (x - 1)^2 + 2(x - 1)(y - 1)$**

H. Examination Presentation Strategy

- State Taylor's formula around (a, b).
- Show all derivatives evaluated at (1, 1).
- Keep terms in powers of (x - 1) and (y - 1) without expanding them.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Expanding $(x - 1)^2$ into $x^2 - 2x + 1$.
✓ **How to Avoid:** The question asks for expansion IN POWERS of (x - 1) and (y - 1), so do NOT expand the brackets!

⚠️ **Common Mistake:** Evaluating derivatives at (0, 0) instead of (1, 1).
✓ **How to Avoid:** Check the expansion point carefully.

J. Similar Practice Question

Practice Problem 10: Expand $f(x, y) = x^2 + xy + y^2$ in powers of (x - 1) and (y - 2) up to second degree.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 11: Jacobian for Functional Dependence and Relationship

INTERMEDIATE

A. Full Question:
Show that the functions $u = (x + y)/(1 - xy)$ and $v = \tan^{-1} x + \tan^{-1} y$ are functionally dependent, and find the relationship between them.

B. Meaning of the Question

Two functions u and v are functionally dependent if their Jacobian $J = \partial(u, v)/\partial(x, y)$ is identically zero. If dependent, find the equation connecting u and v.

C. Basic Concept

If $\partial(u, v)/\partial(x, y) = 0$ identically, there exists a functional relation $F(u, v) = 0$.

D. Formula and Explanation

Formula: $J = (\partial u/\partial x)(\partial v/\partial y) - (\partial u/\partial y)(\partial v/\partial x) = 0.$

Explanation & Conditions: Identically vanishing Jacobian proves functional dependence.

E. Given Information

$u = (x + y)/(1 - xy)$, $v = \tan^{-1} x + \tan^{-1} y$.

F. Complete Step-by-Step Solution

Step 1: Compute partial derivatives of u

$\partial u/\partial x = [(1 - xy)(1) - (x + y)(-y)] / (1 - xy)^2 = [1 - xy + xy + y^2] / (1 - xy)^2 = (1 + y^2) / (1 - xy)^2$.
By symmetry: $\partial u/\partial y = (1 + x^2) / (1 - xy)^2$.

💡 **Why this step is performed:** Execute step 1: compute partial derivatives of u to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute partial derivatives of v

$\partial v/\partial x = 1 / (1 + x^2)$
$\partial v/\partial y = 1 / (1 + y^2)$.

💡 **Why this step is performed:** Execute step 2: compute partial derivatives of v to advance the step-by-step mathematical solution toward the final result.

Step 3: Evaluate the Jacobian determinant J

$J = (\partial u/\partial x)(\partial v/\partial y) - (\partial u/\partial y)(\partial v/\partial x)$
$= [(1 + y^2)/(1 - xy)^2] \cdot [1/(1 + y^2)] - [(1 + x^2)/(1 - xy)^2] \cdot [1/(1 + x^2)]$
$= 1/(1 - xy)^2 - 1/(1 - xy)^2 = 0$.

💡 **Why this step is performed:** Construct the Jacobian determinant of partial derivatives and simplify using standard trigonometric/algebraic identities.

Step 4: Find the functional relationship

Recall standard trigonometric identity: $\tan^{-1} x + \tan^{-1} y = \tan^{-1}[(x + y)/(1 - xy)]$.
Since $v = \tan^{-1} x + \tan^{-1} y$ and $u = (x + y)/(1 - xy)$:
$v = \tan^{-1} u$ or $u = \tan v$.

💡 **Why this step is performed:** Execute step 4: find the functional relationship to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $J = \partial(u, v)/\partial(x, y) = 0$ (Functionally Dependent); Relationship: $u = \tan v$ (or $v = \tan^{-1} u$)

H. Examination Presentation Strategy

- State theorem: J = 0 implies functional dependence.
- Show cancellation of $(1+y^2)$ and $(1+x^2)$.
- Write the relationship clearly.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Not simplifying terms before taking determinant.
✓ **How to Avoid:** Cancel $(1+y^2)$ with $1/(1+y^2)$ to avoid messy expressions.

⚠️ **Common Mistake:** Forgetting to find the relationship.
✓ **How to Avoid:** Both proving J = 0 and writing u = tan v are required.

J. Similar Practice Question

Practice Problem 11: Show that u = x + y and v = x/(x + y) are functionally dependent and find the relation.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 12: Stationary Points and Extrema of a Two-Variable Cubic

INTERMEDIATE

A. Full Question:
Find the extreme values of $f(x, y) = x^3 + y^3 - 3xy$.

B. Meaning of the Question

Locate all stationary points of f(x, y) by setting fx = 0, fy = 0, then use $D = rt - s^2$ to find local maxima and minima.

C. Basic Concept

A two-variable function can have multiple critical points. Each must be tested individually using the second derivative test.

D. Formula and Explanation

Formula: $fx = 3x^2 - 3y = 0,\ fy = 3y^2 - 3x = 0;\ D = rt - s^2.$

Explanation & Conditions: r = fxx, s = fxy, t = fyy.

E. Given Information

$f(x, y) = x^3 + y^3 - 3xy$.

F. Complete Step-by-Step Solution

Step 1: Set first derivatives to zero

$fx = 3x^2 - 3y = 0 \Rightarrow y = x^2$
$fy = 3y^2 - 3x = 0 \Rightarrow x = y^2$.
Substitute $y = x^2$ into second equation:
$x = (x^2)^2 = x^4 \Rightarrow x^4 - x = 0 \Rightarrow x(x^3 - 1) = 0$.
Real roots: $x = 0$ or $x = 1$.
• If $x = 0 \Rightarrow y = 0^2 = 0$. Point: (0, 0).
• If $x = 1 \Rightarrow y = 1^2 = 1$. Point: (1, 1).
Two stationary points: (0, 0) and (1, 1).

💡 **Why this step is performed:** Set the first partial derivatives $fx = 0$ and $fy = 0$ to solve for all stationary (critical) points.

Step 2: Second partial derivatives expressions

$r = fxx = 6x$, $s = fxy = -3$, $t = fyy = 6y$.
Discriminant: $D(x, y) = rt - s^2 = (6x)(6y) - (-3)^2 = 36xy - 9$.

💡 **Why this step is performed:** Calculate the second-order partial derivatives $r = fxx$, $s = fxy$, and $t = fyy$ to prepare for the curvature test.

Step 3: Test Point (0, 0)

At (0, 0): $r = 0$, $s = -3$, $t = 0$.
$D = 36(0)(0) - 9 = -9 < 0$.
Since $D < 0$, (0, 0) is a SADDLE POINT (neither maximum nor minimum).

💡 **Why this step is performed:** Evaluate the Hessian discriminant $D = rt - s^2$; $D > 0$ and $r > 0$ indicates a local minimum, while $D > 0$ and $r < 0$ indicates a local maximum.

Step 4: Test Point (1, 1)

At (1, 1): $r = 6(1) = 6$, $s = -3$, $t = 6(1) = 6$.
$D = 36(1)(1) - 9 = 27 > 0$ and $r = 6 > 0$.
Since $D > 0$ and $r > 0$, (1, 1) is a LOCAL MINIMUM.
Minimum value: $f(1, 1) = 1^3 + 1^3 - 3(1)(1) = 1 + 1 - 3 = -1$.

💡 **Why this step is performed:** Evaluate the Hessian discriminant $D = rt - s^2$; $D > 0$ and $r > 0$ indicates a local minimum, while $D > 0$ and $r < 0$ indicates a local maximum.

✓ G. Final Answer: At (0, 0): Saddle point. At (1, 1): Local Minimum with minimum value $f_{min} = -1$.

H. Examination Presentation Strategy

- Solve non-linear system $x = y^2$, $y = x^2$ clearly.
- List both stationary points in a table.
- Evaluate D and r at each point clearly.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Dividing by x in $x^4 = x$ and losing the root x = 0.
✓ **How to Avoid:** Never divide by the variable! Factorize as $x(x^3 - 1) = 0$.

⚠️ **Common Mistake:** Classifying (0, 0) as minimum because f(0,0)=0 > -1.
✓ **How to Avoid:** Local nature is determined ONLY by the sign of D, not function value!

J. Similar Practice Question

Practice Problem 12: Find the stationary points and extreme values of $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 13: Jacobian of 3-Variable Spherical Polar Coordinates

INTERMEDIATE

A. Full Question:
If x = r sin θ cos φ, y = r sin θ sin φ, z = r cos θ, evaluate J = ∂(x, y, z)/∂(r, θ, φ).

B. Meaning of the Question

Calculate the 3×3 Jacobian determinant of the transformation from Cartesian coordinates (x, y, z) to spherical coordinates (r, θ, φ).

C. Basic Concept

The spherical Jacobian represents the volume differential element in triple integration: $dx\,dy\,dz = r^2 \sin\theta\, dr\, d\theta\, d\varphi$.

D. Formula and Explanation

Formula: `J = det [∂(x, y, z)/∂(r, θ, ϕ)].`
Explanation & Conditions: Row 1: [∂x/∂r, ∂x/∂θ, ∂x/∂φ], Row 2: [∂y/∂r, ∂y/∂θ, ∂y/∂φ], Row 3: [∂z/∂r, ∂z/∂θ, ∂z/∂φ].

E. Given Information

x = r sin θ cos φ, y = r sin θ sin φ, z = r cos θ.

F. Complete Step-by-Step Solution

Step 1: Compute partial derivatives for Row 1 (x)

∂x/∂r = sin θ cos φ
∂x/∂θ = r cos θ cos φ
∂x/∂φ = -r sin θ sin φ.

💡 **Why this step is performed:** Execute step 1: compute partial derivatives for row 1 (x) to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute partial derivatives for Row 2 (y)

$\partial y/\partial r = \sin\theta \sin\varphi$
$\partial y/\partial \theta = r \cos\theta \sin\varphi$
$\partial y/\partial \varphi = r \sin\theta \cos\varphi$.

💡 **Why this step is performed:** Execute step 2: compute partial derivatives for row 2 (y) to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute partial derivatives for Row 3 (z)

$\partial z/\partial r = \cos\theta$
$\partial z/\partial \theta = -r \sin\theta$
$\partial z/\partial \varphi = 0$ (since z has no φ).

💡 **Why this step is performed:** Execute step 3: compute partial derivatives for row 3 (z) to advance the step-by-step mathematical solution toward the final result.

Step 4: Expand determinant along Row 3

$J = \cos\theta \cdot | r\cos\theta\cos\varphi \; -r\sin\theta\sin\varphi \; | \; | r\cos\theta\sin\varphi \; r\sin\theta\cos\varphi \; |$
$- (-r\sin\theta) \cdot | \sin\theta\cos\varphi \; -r\sin\theta\sin\varphi \; | \; | \sin\theta\sin\varphi \; r\sin\theta\cos\varphi \; | + 0.$
$= \cos\theta [r^2 \sin\theta\cos\theta (\cos^2\varphi + \sin^2\varphi)] + r\sin\theta [r\sin^2\theta (\cos^2\varphi + \sin^2\varphi)]$
$= \cos\theta [r^2 \sin\theta\cos\theta (1)] + r\sin\theta [r\sin^2\theta (1)]$
$= r^2 \sin\theta\cos^2\theta + r^2 \sin^3\theta$
$= r^2 \sin\theta (\cos^2\theta + \sin^2\theta) = r^2 \sin\theta (1) = r^2 \sin\theta.$

💡 **Why this step is performed:** Execute step 4: expand determinant along row 3 to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $J = \partial(x, y, z)/\partial(r, \theta, \varphi) = r^2 \sin\theta$

H. Examination Presentation Strategy

- Write out the full 3×3 matrix.
- Expand along the 3rd row because it contains a zero.
- Factor out $r^2 \sin\theta$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Expanding along Row 1.
✓ **How to Avoid:** Expanding along Row 3 is much faster because $\partial z/\partial\varphi = 0$.

⚠️ **Common Mistake:** Sign error with $(-r \sin\theta)$.
✓ **How to Avoid:** The (3, 2) cofactor position has a minus sign: $-(-r\sin\theta) = +r\sin\theta$.

J. Similar Practice Question

Practice Problem 13: If $x = \rho\cos\varphi$, $y = \rho\sin\varphi$, $z = z$, find the Jacobian $\partial(x, y, z)/\partial(\rho, \varphi, z)$ in cylindrical coordinates.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 14: Homogeneous Function of Degree Zero: $u = \tan^{-1}(y/x)$ INTERMEDIATE

A. Full Question:
If $u = \tan^{-1}(y/x)$, show that $x(\partial u/\partial x) + y(\partial u/\partial y) = 0$ and $x^2(\partial^2 u/\partial x^2) + 2xy(\partial^2 u/\partial x\partial y) + y^2(\partial^2 u/\partial y^2) = 0$.

B. Meaning of the Question

Show that u is homogeneous of degree 0, then verify both the first and second-order forms of Euler's Theorem.

C. Basic Concept

When a function has equal powers in numerator and denominator, its degree is n = 0. Euler's second-order theorem states: $x^2\, u_{xx} + 2xy\, u_{xy} + y^2\, u_{yy} = n(n-1)u$.

D. Formula and Explanation

Formula: `First-order: x u_x + y u_y = n u = 0·u = 0.`
`Second-order: x² u_xx + 2xy u_xy + y² u_yy = n(n - 1)u = 0(-1)u = 0.`
Explanation & Conditions: For n = 0, both n·u and n(n - 1)u vanish identically.

E. Given Information

$u = \tan^{-1}(y/x)$.

F. Complete Step-by-Step Solution

Step 1: Check degree of homogeneity

$u(tx, ty) = \tan^{-1}(ty / tx) = \tan^{-1}(y / x) = t^0\, u(x, y)$.
Therefore, u is a homogeneous function of degree n = 0.

💡 **Why this step is performed:** Substitute tx and ty for x and y to factor out t^n and establish the degree of homogeneity n.

Step 2: Compute first derivatives

$\partial u/\partial x = [1 / (1 + (y/x)^2)] \cdot (-y/x^2) = -y / (x^2 + y^2)$.
$\partial u/\partial y = [1 / (1 + (y/x)^2)] \cdot (1/x) = x / (x^2 + y^2)$.

💡 **Why this step is performed:** Set the first partial derivatives fx = 0 and fy = 0 to solve for all stationary (critical) points.

Step 3: Verify first-order Euler theorem

$x(\partial u/\partial x) + y(\partial u/\partial y) = x[-y/(x^2+y^2)] + y[x/(x^2+y^2)] = (-xy + xy)/(x^2+y^2) = 0$.
Matches n·u = 0 · u = 0. Verified!

💡 **Why this step is performed:** Combine the partial derivatives into the linear combination $x(\partial u/\partial x) + y(\partial u/\partial y)$ and

verify it equals n·u.

Step 4: Verify second-order Euler theorem

By Euler's 2nd-order theorem:
$x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = n(n - 1) u$.
Since n = 0:
$n(n - 1) u = 0(0 - 1) u = 0$. Hence proved.

💡 **Why this step is performed:** Apply Euler's second-order theorem $x^2r + 2xys + y^2t = n(n-1)u$ to evaluate the quadratic derivative form.

✓ G. Final Answer: $x(\partial u/\partial x) + y(\partial u/\partial y) = 0$ and $x^2(\partial^2 u/\partial x^2) + 2xy(\partial^2 u/\partial x\partial y) + y^2(\partial^2 u/\partial y^2) = 0$ (Proved)

H. Examination Presentation Strategy

- State n = 0 clearly from $u(tx, ty) = t^0 u$.
- Show cancellation for first-order directly.
- Quote Euler's second-order theorem $x^2r + 2xys + y^2t = n(n-1)u$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Thinking degree is undefined.
✓ **How to Avoid:** t cancels completely, meaning $t^0 = 1$, so degree is 0, not undefined.

⚠️ **Common Mistake:** Long calculations for 2nd derivatives.
✓ **How to Avoid:** Using Euler's theorem n(n-1)u saves two pages of tedious algebra.

J. Similar Practice Question

Practice Problem 14: If $u = \log[(x^2 + y^2)/(xy)]$, show that $x(\partial u/\partial x) + y(\partial u/\partial y) = 0$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 15: Composite Partial Differentiation (Chain Rule with Intermediate Variables)

INTERMEDIATE

A. Full Question:
If $z = f(u, v)$ where $u = x^2 - y^2$ and $v = 2xy$, prove that $y(\partial z/\partial x) + x(\partial z/\partial y) = 2(x^2 + y^2)(\partial z/\partial v)$.

B. Meaning of the Question

Use the multivariable chain rule to express $\partial z/\partial x$ and $\partial z/\partial y$ in terms of $\partial z/\partial u$ and $\partial z/\partial v$, then compute the linear combination $y(\partial z/\partial x) + x(\partial z/\partial y)$.

C. Basic Concept

When z = f(u, v), $\partial z/\partial x = (\partial z/\partial u)(\partial u/\partial x) + (\partial z/\partial v)(\partial v/\partial x)$ and $\partial z/\partial y = (\partial z/\partial u)(\partial u/\partial y) + (\partial z/\partial v)(\partial v/\partial y)$.

D. Formula and Explanation

Formula: $\partial z/\partial x = f_u\ u_x + f_v\ v_x;\ \partial z/\partial y = f_u\ u_y + f_v\ v_y.$

Explanation & Conditions: $u_x = 2x$, $u_y = -2y$, $v_x = 2y$, $v_y = 2x$.

E. Given Information

$z = f(u, v)$, $u = x^2 - y^2$, $v = 2xy$.

F. Complete Step-by-Step Solution

Step 1: Compute derivatives of u and v

$\partial u/\partial x = 2x$, $\partial u/\partial y = -2y$

$\partial v/\partial x = 2y$, $\partial v/\partial y = 2x$.

💡 **Why this step is performed:** Execute step 1: compute derivatives of u and v to advance the step-by-step mathematical solution toward the final result.

Step 2: Express $\partial z/\partial x$ using chain rule

$\partial z/\partial x = (\partial z/\partial u)(2x) + (\partial z/\partial v)(2y) = 2x(\partial z/\partial u) + 2y(\partial z/\partial v)$.

Multiply by y: $y(\partial z/\partial x) = 2xy(\partial z/\partial u) + 2y^2(\partial z/\partial v)$.

💡 **Why this step is performed:** Execute step 2: express $\partial z/\partial x$ using chain rule to advance the step-by-step mathematical solution toward the final result.

Step 3: Express $\partial z/\partial y$ using chain rule

$\partial z/\partial y = (\partial z/\partial u)(-2y) + (\partial z/\partial v)(2x) = -2y(\partial z/\partial u) + 2x(\partial z/\partial v)$.

Multiply by x: $x(\partial z/\partial y) = -2xy(\partial z/\partial u) + 2x^2(\partial z/\partial v)$.

💡 **Why this step is performed:** Execute step 3: express $\partial z/\partial y$ using chain rule to advance the step-by-step mathematical solution toward the final result.

Step 4: Add the two equations

LHS $= y(\partial z/\partial x) + x(\partial z/\partial y)$

$= [2xy(\partial z/\partial u) - 2xy(\partial z/\partial u)] + [2y^2(\partial z/\partial v) + 2x^2(\partial z/\partial v)]$

$= 0 + 2(x^2 + y^2)(\partial z/\partial v)$

$= 2(x^2 + y^2)(\partial z/\partial v)$ = RHS. Hence proved.

💡 **Why this step is performed:** Execute step 4: add the two equations to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $y(\partial z/\partial x) + x(\partial z/\partial y) = 2(x^2 + y^2)(\partial z/\partial v)$ (Hence proved)

H. Examination Presentation Strategy

- Write chain rule formulas clearly.
- Show the multiplier y and x on each line.
- Highlight the cancellation of the $\partial z/\partial u$ terms.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Sign error: $\partial u/\partial y = +2y$ instead of $-2y$.
✓ **How to Avoid:** $d(-y^2)/dy = -2y$. The minus sign is vital for cancellation!

⚠️ **Common Mistake:** Treating z as a product instead of general function f(u, v).
✓ **How to Avoid:** Keep $\partial z/\partial u$ and $\partial z/\partial v$ as operators.

J. Similar Practice Question

Practice Problem 15: If $z = f(x - y, y - z)$, show that $\partial z/\partial x + \partial z/\partial y + \partial z/\partial z = 0$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 16: Euler's Theorem on Inverse Tangent: $x u_x + y u_y = \sin 2u$ EXAM-LEVEL

A. Full Question:
If $u = \tan^{-1}[(x^3 + y^3) / (x - y)]$, prove that $x(\partial u/\partial x) + y(\partial u/\partial y) = \sin 2u$.

B. Meaning of the Question

Express tan u as a homogeneous function of degree 2, apply Euler's Theorem, and use trigonometric identity to prove the result equals sin 2u.

C. Basic Concept

Standard VTU exam problem. Let $z = \tan u = f(x, y)$. Since f is homogeneous of degree $n = 3 - 1 = 2$, Euler's theorem gives $x z_x + y z_y = 2z$. Then using chain rule $z_x = \sec^2 u \cdot u_x$ yields $2 \tan u / \sec^2 u = 2 \sin u \cos u = \sin 2u$.

D. Formula and Explanation

Formula: $x(\partial u/\partial x) + y(\partial u/\partial y) = n\,[f(u) / f'(u)] = 2\,[\tan u / \sec^2 u] = \sin 2u.$
Explanation & Conditions: $\tan u = \sin u / \cos u$; dividing by $\sec^2 u = 1/\cos^2 u$ gives $\sin u \cos u$.

E. Given Information

$u = \tan^{-1}[(x^3 + y^3) / (x - y)]$.

F. Complete Step-by-Step Solution

Step 1: Define z = tan u

$\tan u = (x^3 + y^3) / (x - y) = z(x, y)$.

💡 **Why this step is performed:** Execute step 1: define z = tan u to advance the step-by-step mathematical solution toward the final result.

Step 2: Find degree of z

$z(tx, ty) = [(tx)^3 + (ty)^3] / [tx - ty] = t^3(x^3 + y^3) / [t(x - y)] = t^2 z(x, y)$.
Therefore, z is homogeneous of degree n = 3 - 1 = 2.

💡 **Why this step is performed:** Execute step 2: find degree of z to advance the step-by-step mathematical solution toward the final result.

Step 3: Apply Euler's Theorem to z

$x(\partial z/\partial x) + y(\partial z/\partial y) = n \cdot z = 2z = 2 \tan u$.

💡 **Why this step is performed:** Combine the partial derivatives into the linear combination $x(\partial u/\partial x) + y(\partial u/\partial y)$ and verify it equals n·u.

Step 4: Express derivatives in terms of u

$\partial z/\partial x = \sec^2 u\ (\partial u/\partial x)$, $\partial z/\partial y = \sec^2 u\ (\partial u/\partial y)$.
Substitute into Euler equation:
$x [\sec^2 u\ (\partial u/\partial x)] + y [\sec^2 u\ (\partial u/\partial y)] = 2 \tan u$
$\sec^2 u [x(\partial u/\partial x) + y(\partial u/\partial y)] = 2 \tan u$.

💡 **Why this step is performed:** Execute step 4: express derivatives in terms of u to advance the step-by-step mathematical solution toward the final result.

Step 5: Divide by $\sec^2 u$ and simplify with trigonometry

$x(\partial u/\partial x) + y(\partial u/\partial y) = 2 \tan u / \sec^2 u$
$= 2 (\sin u / \cos u) \cdot \cos^2 u = 2 \sin u \cos u = \sin 2u$. Hence proved.

💡 **Why this step is performed:** Compute the sum of partial derivatives of the component functions along their respective coordinate axes.

✓ **G. Final Answer: $x(\partial u/\partial x) + y(\partial u/\partial y) = \sin 2u$ (Hence proved)**

H. Examination Presentation Strategy

- State z = tan u is homogeneous of degree 2.
- Write Euler's theorem on z clearly.
- Show trigonometric conversion: $2 \tan u / \sec^2 u = 2 \sin u \cos u = \sin 2u$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Saying u is homogeneous of degree 2.
✓ **How to Avoid:** u is NOT homogeneous. tan u is homogeneous!

⚠️ **Common Mistake:** Leaving answer as $2 \tan u / \sec^2 u$.
✓ **How to Avoid:** Always simplify to the standard double-angle identity sin 2u.

J. Similar Practice Question

Practice Problem 16: If $u = \tan^{-1}[(x^2 + y^2) / (x + y)]$, prove that $x(\partial u/\partial x) + y(\partial u/\partial y) = (1/2) \sin 2u$.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 17: Second-Order Euler's Theorem on Inverse Tangent

EXAM-LEVEL

A. Full Question:

If $u = \tan^{-1}[(x^3 + y^3) / (x - y)]$, prove that $x^2(\partial^2u/\partial x^2) + 2xy(\partial^2u/\partial x\partial y) + y^2(\partial^2u/\partial y^2) = (1 - 4 \sin^2 u) \sin 2u$ / ... = 2 cos 3u sin u (or standard form).

B. Meaning of the Question

Differentiate Euler's first-order result partially with respect to x and y, combine, and obtain the second-order partial differential relation.

C. Basic Concept

Let $G(u) = x u_x + y u_y$. Then $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = G(u)[G'(u) - 1]$. Since $G(u) = \sin 2u$, $G'(u) = 2 \cos 2u$, so result is $\sin 2u (2 \cos 2u - 1)$.

D. Formula and Explanation

Formula: $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = f(u)[f'(u) - 1]$ where $f(u) = \sin 2u$.

Explanation & Conditions: $f(u) = \sin 2u \Rightarrow f'(u) = 2 \cos 2u$. Therefore $f(u)[f'(u) - 1] = \sin 2u (2 \cos 2u - 1)$.

E. Given Information

$u = \tan^{-1}[(x^3 + y^3) / (x - y)] \Rightarrow x u_x + y u_y = \sin 2u$.

F. Complete Step-by-Step Solution

Step 1: Recall the first-order result

From Question 16: $x(\partial u/\partial x) + y(\partial u/\partial y) = \sin 2u = f(u)$.

💡 **Why this step is performed:** Execute step 1: recall the first-order result to advance the step-by-step mathematical solution toward the final result.

Step 2: Differentiate with respect to x

$\partial/\partial x [x u_x + y u_y] = \partial/\partial x [\sin 2u]$

$1 \cdot u_x + x u_{xx} + y u_{yx} = 2 \cos 2u \cdot u_x$

$x u_{xx} + y u_{xy} = 2 \cos 2u \cdot u_x - u_x = (2 \cos 2u - 1) u_x$.

Multiply by x: $x^2 u_{xx} + xy u_{xy} = (2 \cos 2u - 1) x u_x$.

💡 **Why this step is performed:** Execute step 2: differentiate with respect to x to advance the step-by-step mathematical solution toward the final result.

Step 3: Differentiate with respect to y

$\partial/\partial y\ [x\ u_x + y\ u_y] = \partial/\partial y\ [\sin 2u]$
$x\ u_{xy} + 1 \cdot u_y + y\ u_{yy} = 2 \cos 2u \cdot u_y$
$x\ u_{xy} + y\ u_{yy} = (2 \cos 2u - 1)\ u_y.$
Multiply by y: $xy\ u_{xy} + y^2\ u_{yy} = (2 \cos 2u - 1)\ y\ u_y.$

💡 **Why this step is performed:** Execute step 3: differentiate with respect to y to advance the step-by-step mathematical solution toward the final result.

Step 4: Add the two equations

LHS $= x^2\ u_{xx} + 2xy\ u_{xy} + y^2\ u_{yy}$
$= (2 \cos 2u - 1)\ [x\ u_x + y\ u_y]$
$= (2 \cos 2u - 1) \sin 2u$
$= \sin 2u\ (2 \cos 2u - 1).$

💡 **Why this step is performed:** Execute step 4: add the two equations to advance the step-by-step mathematical solution toward the final result.

Step 5: Express in alternative trigonometric forms if requested

$\sin 2u\ (2 \cos 2u - 1) = 2 \sin 2u \cos 2u - \sin 2u = \sin 4u - \sin 2u$
$= 2 \cos 3u \sin u.$

💡 **Why this step is performed:** Execute step 5: express in alternative trigonometric forms if requested to advance the step-by-step mathematical solution toward the final result.

✓ **G. Final Answer: $x^2(\partial^2u/\partial x^2) + 2xy(\partial^2u/\partial x\partial y) + y^2(\partial^2u/\partial y^2) = \sin 2u\ (2 \cos 2u - 1) = 2 \cos 3u \sin u$**

H. Examination Presentation Strategy

- Show differentiation with respect to x and y step by step.
- Use Clairaut's theorem $u_{xy} = u_{yx}$.
- Factor out $(2 \cos 2u - 1)$ and substitute $x\ u_x + y\ u_y = \sin 2u$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Forgetting the product rule on $x\ u_x$.
✓ **How to Avoid:** $d(x\ u_x)/dx = 1 \cdot u_x + x\ u_{xx}$.

⚠️ **Common Mistake:** Forgetting the inner derivative u_x on RHS.
✓ **How to Avoid:** $d(\sin 2u)/dx = 2 \cos 2u \cdot (\partial u/\partial x)$.

J. Similar Practice Question

Practice Problem 17: If $u = \sin^{-1}[(x^2 + y^2)/(x + y)]$, prove that $x^2\ u_{xx} + 2xy\ u_{xy} + y^2\ u_{yy} = \tan u\ (\tan^2 u - 1) / 4$ (or standard form).

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 18: Radial Laplace Operator: u = f(r) where $r^2 = x^2 + y^2 + z^2$

EXAM-LEVEL

A. Full Question:

If u = f(r) where $r^2 = x^2 + y^2 + z^2$, prove that $\partial^2u/\partial x^2 + \partial^2u/\partial y^2 + \partial^2u/\partial z^2 = f''(r) + (2/r)\, f'(r)$.

B. Meaning of the Question

Compute the 3D Laplacian ∇^2u of a radially symmetric function u = f(r) in Cartesian coordinates.

C. Basic Concept

Radial symmetry simplifies 3D PDEs (such as heat and wave equations) into ordinary differential equations in r.

D. Formula and Explanation

Formula: $\nabla^2u = u_{xx} + u_{yy} + u_{zz} = f''(r) + (2/r)\, f'(r)$.

Explanation & Conditions: $r_x = x/r$, $r_{xx} = (r^2 - x^2)/r^3$.

E. Given Information

u = f(r), $r^2 = x^2 + y^2 + z^2$.

F. Complete Step-by-Step Solution

Step 1: Compute partial derivatives of r

$2r\,(\partial r/\partial x) = 2x \Rightarrow \partial r/\partial x = x/r$.

Similarly, $\partial r/\partial y = y/r$ and $\partial r/\partial z = z/r$.

💡 **Why this step is performed:** Execute step 1: compute partial derivatives of r to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute first derivative $\partial u/\partial x$

$\partial u/\partial x = f'(r) \cdot (\partial r/\partial x) = f'(r) \cdot (x/r) = (x/r)\, f'(r)$.

💡 **Why this step is performed:** Execute step 2: compute first derivative $\partial u/\partial x$ to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute second derivative $\partial^2u/\partial x^2$ using quotient/product rule

$\partial^2u/\partial x^2 = \partial/\partial x\,[(x/r)\, f'(r)] = (1/r)\, f'(r) + x \cdot \partial/\partial x\,[f'(r)/r]$

$= (1/r)\, f'(r) + x\,[\, r\, f''(r)(x/r) - f'(r)(x/r)\,] / r^2$

$= (1/r)\, f'(r) + (x^2/r^2)\, f''(r) - (x^2/r^3)\, f'(r)$.

💡 **Why this step is performed:** Execute step 3: compute second derivative $\partial^2u/\partial x^2$ using quotient/product rule to advance the step-by-step mathematical solution toward the final result.

Step 4: Sum cyclically over x, y, and z

$u_{xx} + u_{yy} + u_{zz} = [\, 3/r\, f'(r)\,] + [\, (x^2 + y^2 + z^2)/r^2\,]\, f''(r) - [\, (x^2 + y^2 + z^2)/r^3\,]\, f'(r)$.

💡 **Why this step is performed:** Execute step 4: sum cyclically over x, y, and z to advance the step-by-step mathematical solution toward the final result.

Step 5: Substitute $x^2 + y^2 + z^2 = r^2$

$= (3/r)\, f'(r) + (r^2/r^2)\, f''(r) - (r^2/r^3)\, f'(r)$

$= (3/r)\, f'(r) + f''(r) - (1/r)\, f'(r)$

$= f''(r) + (2/r)\, f'(r)$. Hence proved.

💡 **Why this step is performed:** Execute step 5: substitute $x^2 + y^2 + z^2 = r^2$ to advance the step-by-step mathematical solution toward the final result.

✓ **G. Final Answer: $\partial^2u/\partial x^2 + \partial^2u/\partial y^2 + \partial^2u/\partial z^2 = f''(r) + (2/r)\, f'(r)$ (Hence proved)**

H. Examination Presentation Strategy

- Show $\partial r/\partial x = x/r$ clearly.
- Show quotient rule for $\partial^2u/\partial x^2$ in full detail.
- Combine cyclic terms using symmetry.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Treating r as constant.
✓ **How to Avoid:** r depends on x, y, z through $r^2 = x^2 + y^2 + z^2$.

⚠️ **Common Mistake:** Algebra error in (3/r) - (1/r).
✓ **How to Avoid:** 3/r - 1/r = 2/r.

J. Similar Practice Question

Practice Problem 18: Find f(r) such that $\nabla^2 f(r) = 0$ (harmonic function in 3D).

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 19: Taylor Series of Two Variables up to 3rd Degree: $e^x \log(1 + y)$

EXAM-LEVEL

A. Full Question:

Expand $f(x, y) = e^x \log(1 + y)$ in powers of x and y up to third-degree terms.

B. Meaning of the Question

Obtain the Maclaurin expansion of $e^x \log(1 + y)$ around (0, 0) up to terms of combined degree 3.

C. Basic Concept

Can be solved either by computing partial derivatives up to order 3 at (0, 0) or by multiplying the standard series of e^x and log(1 + y). In exams, showing the partial derivative values guarantees full marks.

D. Formula and Explanation

Formula: `f(x, y) = f(0,0) + [x fx + y fy] + (1/2!)[x² fxx + 2xy fxy + y² fyy] + (1/3!)[x³ fxxx + 3x²y fxxy + 3xy² fxyy + y³ fyyy] + ...`

Explanation & Conditions: All derivatives evaluated at (0, 0).

E. Given Information

f(x, y) = e^x log(1 + y), expansion point (0, 0).

F. Complete Step-by-Step Solution

Step 1: Function and 1st derivatives at (0, 0)

f(0, 0) = e^0 log(1) = 0.
fx = e^x log(1 + y) ⇒ fx(0, 0) = 0.
fy = e^x / (1 + y) ⇒ fy(0, 0) = 1.

💡 **Why this step is performed:** Execute step 1: function and 1st derivatives at (0, 0) to advance the step-by-step mathematical solution toward the final result.

Step 2: 2nd-order derivatives at (0, 0)

fxx = e^x log(1 + y) ⇒ fxx(0, 0) = 0.
fxy = e^x / (1 + y) ⇒ fxy(0, 0) = 1.
fyy = -e^x / $(1 + y)^2$ ⇒ fyy(0, 0) = -1.

💡 **Why this step is performed:** Execute step 2: 2nd-order derivatives at (0, 0) to advance the step-by-step mathematical solution toward the final result.

Step 3: 3rd-order derivatives at (0, 0)

fxxx = e^x log(1 + y) ⇒ fxxx(0, 0) = 0.
fxxy = e^x / (1 + y) ⇒ fxxy(0, 0) = 1.
fxyy = -e^x / $(1 + y)^2$ ⇒ fxyy(0, 0) = -1.
fyyy = 2e^x / $(1 + y)^3$ ⇒ fyyy(0, 0) = 2.

💡 **Why this step is performed:** Execute step 3: 3rd-order derivatives at (0, 0) to advance the step-by-step mathematical solution toward the final result.

Step 4: Substitute into Taylor formula

Linear: y.
Quadratic: (1/2)[2xy(1) + y^2(-1)] = xy - (1/2)y^2.
Cubic: (1/6)[3x^2y(1) + 3xy^2(-1) + y^3(2)] = (1/2)x^2y - (1/2)xy^2 + (1/3)y^3.
Combine all terms:
f(x, y) = y + xy - (1/2)y^2 + (1/2)x^2y - (1/2)xy^2 + (1/3)y^3 + ...

💡 **Why this step is performed:** Substitute the evaluated derivative values into the multivariable Taylor/Maclaurin series formula and group terms by degree.

✓ **G. Final Answer:** $e^x \log(1 + y) = y + xy - (1/2)y^2 + (1/2)x^2y - (1/2)xy^2 + (1/3)y^3 + \ldots$

H. Examination Presentation Strategy

- List all derivatives systematically by order (1st, 2nd, 3rd).
- State the multinomial coefficients 1, 3, 3, 1 for the 3rd degree bracket.
- Group by degree: linear, quadratic, cubic.

I. Common Mistakes & How to Avoid Them

⚠ **Common Mistake:** Derivative of $1/(1+y)^2$.
✓ **How to Avoid:** $d[-(1+y)^{-2}]/dy = +2(1+y)^{-3} = 2/(1+y)^3$.

⚠ **Common Mistake:** Missing 1/3! = 1/6 on third degree terms.
✓ **How to Avoid:** Always divide by the corresponding factorial.

J. Similar Practice Question

Practice Problem 19: Expand f(x, y) = e^y log(1 + x) up to third-degree terms.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 20: Global / Constrained Maxima and Minima: Dimensions of Open Rectangular Box

EXAM-LEVEL

A. Full Question:
A rectangular box open at the top is to have a volume of 32 cubic feet. Find the dimensions of the box requiring the least material for its construction.

B. Meaning of the Question

Minimize the surface area S = xy + 2yz + 2xz subject to the volume constraint V = xyz = 32.

C. Basic Concept

Optimization of two independent variables by eliminating the third variable using the constraint equation.

D. Formula and Explanation

Formula: `Constraint: V = xyz = 32 ⇒ z = 32/(xy).`
`Surface area: S(x, y) = xy + 2xz + 2yz = xy + 64/y + 64/x.`
Explanation & Conditions: xy is the base; 2xz and 2yz are the four open sides.

E. Given Information

Volume V = xyz = 32; Open top surface area S = xy + 2yz + 2xz.

F. Complete Step-by-Step Solution

Step 1: Express S as a function of x and y

$z = 32 / (xy)$.

$S(x, y) = xy + 2x(32/xy) + 2y(32/xy) = xy + 64/y + 64/x$.

💡 **Why this step is performed:** Execute step 1: express s as a function of x and y to advance the step-by-step mathematical solution toward the final result.

Step 2: Find stationary points: Sx = 0 and Sy = 0

$Sx = y - 64/x^2 = 0 \Rightarrow y = 64/x^2$.

$Sy = x - 64/y^2 = 0 \Rightarrow x = 64/y^2$.

Substitute y into x equation:

$x = 64 / (64/x^2)^2 = 64 \cdot (x^4 / 4096) = x^4 / 64$

$x^4 - 64x = 0 \Rightarrow x(x^3 - 64) = 0$.

Since $x > 0$ (dimension of box): $x^3 = 64 \Rightarrow x = 4$.

Then $y = 64 / 4^2 = 64 / 16 = 4$.

💡 **Why this step is performed:** Set the first partial derivatives fx = 0 and fy = 0 to solve for all stationary (critical) points.

Step 3: Second derivative test at (4, 4)

$r = Sxx = 128/x^3 \Rightarrow r(4, 4) = 128/64 = 2$.

$s = Sxy = 1$.

$t = Syy = 128/y^3 \Rightarrow t(4, 4) = 128/64 = 2$.

$D = rt - s^2 = (2)(2) - 1^2 = 4 - 1 = 3 > 0$ and $r = 2 > 0$.

Thus, S has a LOCAL MINIMUM at (4, 4).

💡 **Why this step is performed:** Execute step 3: second derivative test at (4, 4) to advance the step-by-step mathematical solution toward the final result.

Step 4: Find height z and minimum surface area

$z = 32 / (xy) = 32 / (4 \cdot 4) = 32 / 16 = 2$ feet.

$S_{min} = (4)(4) + 64/4 + 64/4 = 16 + 16 + 16 = 48$ sq. ft.

💡 **Why this step is performed:** Execute step 4: find height z and minimum surface area to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Dimensions: Length x = 4 ft, Breadth y = 4 ft, Height z = 2 ft (Minimum Surface Area = 48 sq. ft)

H. Examination Presentation Strategy

- State constraint and surface area formulas clearly.
- Eliminate z and write S(x, y).
- Verify D > 0 and r > 0 to prove it is a minimum.
- State all 3 dimensions with units.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Including 2xy for closed box.
✓ **How to Avoid:** The box is OPEN AT THE TOP, so only one base xy is included!

⚠️ **Common Mistake:** Forgetting to calculate the 3rd dimension z.
✓ **How to Avoid:** Always calculate z = 32/(xy).

J. Similar Practice Question

Practice Problem 20: Find dimensions of an open rectangular box of volume 108 cubic cm requiring least material.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 21: Stationary Points and Nature: f(x, y) = sin x + sin y + sin(x + y) EXAM-LEVEL

A. Full Question:
Find the maximum and minimum values of f(x, y) = sin x + sin y + sin(x + y) in the range 0 < x, y < π.

B. Meaning of the Question

Find the critical points of the trigonometric function f(x, y) in the first quadrant and determine where it attains its maximum value.

C. Basic Concept

Classic VTU exam problem related to maximum area of a triangle. Uses trigonometric identities cos A - cos B = -2 sin[(A+B)/2] sin[(A-B)/2].

D. Formula and Explanation

Formula: $f_x = \cos x + \cos(x + y) = 0,\ f_y = \cos y + \cos(x + y) = 0.$
Explanation & Conditions: Subtracting gives cos x - cos y = 0 ⇒ x = y in (0, π).

E. Given Information

f(x, y) = sin x + sin y + sin(x + y), 0 < x, y < π.

F. Complete Step-by-Step Solution

Step 1: Set first derivatives to zero

fx = cos x + cos(x + y) = 0 --- (1)
fy = cos y + cos(x + y) = 0 --- (2).

💡 **Why this step is performed:** Set the first partial derivatives fx = 0 and fy = 0 to solve for all stationary (critical) points.

Step 2: Solve the system of trigonometric equations

Subtracting (2) from (1):
cos x - cos y = 0 ⇒ cos x = cos y.
Since 0 < x, y < π, this implies x = y.
Substitute x = y into (1):
cos x + cos(2x) = 0 ⇒ cos 2x = -cos x = cos(π - x).
2x = π - x ⇒ 3x = π ⇒ x = π/3.
Since x = y, y = π/3.
Critical point: (π/3, π/3).

💡 **Why this step is performed:** Execute step 2: solve the system of trigonometric equations to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute second derivatives at (π/3, π/3)

r = fxx = -sin x - sin(x + y) ⇒ r(π/3, π/3) = -sin(π/3) - sin(2π/3) = -√3/2 - √3/2 = -√3.
s = fxy = -sin(x + y) ⇒ s(π/3, π/3) = -sin(2π/3) = -√3/2.
t = fyy = -sin y - sin(x + y) ⇒ t(π/3, π/3) = -√3.

💡 **Why this step is performed:** Execute step 3: compute second derivatives at (π/3, π/3) to advance the step-by-step mathematical solution toward the final result.

Step 4: Evaluate D = rt - s^2

D = (-√3)(-√3) - $(-√3/2)^2$ = 3 - 3/4 = 9/4 > 0.
Since D > 0 and r = -√3 < 0, f has a LOCAL MAXIMUM at (π/3, π/3).

💡 **Why this step is performed:** Evaluate the Hessian discriminant D = rt - s^2; D > 0 and r > 0 indicates a local minimum, while D > 0 and r < 0 indicates a local maximum.

Step 5: Compute Maximum Value

f_max = sin(π/3) + sin(π/3) + sin(2π/3) = √3/2 + √3/2 + √3/2 = (3√3)/2.

💡 **Why this step is performed:** Execute step 5: compute maximum value to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Local Maximum at (π/3, π/3) with Maximum Value f_max = (3√3)/2

H. Examination Presentation Strategy

- Show cos x = cos y gives x = y.
- Show cos 2x = -cos x = cos(π - x) gives x = π/3.
- State D = 9/4 > 0 and r < 0 implies Maximum.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Concluding Minimum because of negative signs in r.
✓ **How to Avoid:** r < 0 means concave downwards, which is a MAXIMUM!

⚠️ **Common Mistake:** Value of sin(2π/3) written as -√3/2.
✓ **How to Avoid:** 2π/3 is in Quadrant II where sine is POSITIVE: sin(2π/3) = +√3/2.

J. Similar Practice Question

Practice Problem 21: Find the maximum value of $f(x, y) = \cos x + \cos y + \cos(x + y)$ for $0 < x, y < \pi/2$.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 22: Jacobian of Implicit Functions

EXAM-LEVEL

A. Full Question:

If $u^3 + v + w = x + y^2 + z^2$, $u + v^3 + w = x^2 + y + z^2$, $u + v + w^3 = x^2 + y^2 + z$, find $\partial u/\partial x$.

B. Meaning of the Question

Use the Jacobian formula for implicit functions to find the partial derivative $\partial u/\partial x$ when u, v, w are implicitly defined functions of x, y, z.

C. Basic Concept

For a system of implicit equations $F_1(u,v,w,x,y,z) = 0$, $F_2 = 0$, $F_3 = 0$: $\partial u/\partial x = -[\partial(F_1, F_2, F_3)/\partial(x, v, w)] / [\partial(F_1, F_2, F_3)/\partial(u, v, w)]$.

D. Formula and Explanation

Formula: $\partial u/\partial x = -J(x, v, w) / J(u, v, w)$.

Explanation & Conditions: Denominator is the Jacobian with respect to dependent variables; numerator replaces u by x.

E. Given Information

$F_1 = u^3 + v + w - x - y^2 - z^2 = 0$, $F_2 = u + v^3 + w - x^2 - y - z^2 = 0$, $F_3 = u + v + w^3 - x^2 - y^2 - z = 0$.

F. Complete Step-by-Step Solution

Step 1: Compute Denominator Jacobian J(u, v, w)

$J(u, v, w) = \begin{vmatrix} 3u^2 & 1 & 1 \\ 1 & 3v^2 & 1 \\ 1 & 1 & 3w^2 \end{vmatrix}$.

Row operations $R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_1$:

$J(u, v, w) = \begin{vmatrix} 3u^2 & 1 & 1 \\ 1-3u^2 & 3v^2-1 & 0 \\ 1-3u^2 & 0 & 3w^2-1 \end{vmatrix}$.

Evaluating this gives: $27u^2v^2w^2 - 3(u^2 + v^2 + w^2) + 2$.

> 💡 **Why this step is performed:** Execute step 1: compute denominator jacobian j(u, v, w) to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute Numerator Jacobian J(x, v, w)

$J(x, v, w) = \begin{vmatrix} \partial F_1/\partial x & \partial F_1/\partial v & \partial F_1/\partial w \\ \partial F_2/\partial x & \partial F_2/\partial v & \partial F_2/\partial w \end{vmatrix} = \begin{vmatrix} -1 & 1 & 1 \\ -2x & 3v^2 & 1 \end{vmatrix}$

$| \partial F_3/\partial x \ \partial F_3/\partial v \ \partial F_3/\partial w | | -2x \ 1 \ 3w^2 |$.

💡 **Why this step is performed:** Execute step 2: compute numerator jacobian j(x, v, w) to advance the step-by-step mathematical solution toward the final result.

Step 3: Evaluate Numerator Determinant

Expanding along column 1:

$= -1 [9v^2w^2 - 1] - (-2x) [3w^2 - 1] + (-2x) [1 - 3v^2]$

$= 1 - 9v^2w^2 + 2x(3w^2 - 1 - 1 + 3v^2)$

$= 1 - 9v^2w^2 + 2x[3(v^2 + w^2) - 2]$.

💡 **Why this step is performed:** Execute step 3: evaluate numerator determinant to advance the step-by-step mathematical solution toward the final result.

Step 4: Combine using implicit differentiation formula

$\partial u/\partial x = - [J(x, v, w)] / [J(u, v, w)]$.

💡 **Why this step is performed:** Execute step 4: combine using implicit differentiation formula to advance the step-by-step mathematical solution toward the final result.

✓ **G. Final Answer: $\partial u/\partial x = - [1 - 9v^2w^2 + 2x(3v^2 + 3w^2 - 2)] / [27u^2v^2w^2 - 3(u^2 + v^2 + w^2) + 2]$**

H. Examination Presentation Strategy

- State the implicit Jacobian ratio formula with minus sign.
- Evaluate both 3×3 determinants separately.
- Substitute into final quotient.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Forgetting the leading negative sign in implicit derivative formula.
✓ **How to Avoid:** dy/dx = - Fx / Fy always has a minus sign!

⚠️ **Common Mistake:** Replacing the wrong column.
✓ **How to Avoid:** To find $\partial u/\partial x$, replace the 'u' column with 'x' derivatives.

J. Similar Practice Question

Practice Problem 22: If $u + v + w = x + y + z$, $u^2 + v^2 + w^2 = x^2 + y^2 + z^2$, $u^3 + v^3 + w^3 = x^3 + y^3 + z^3$, find $\partial u/\partial x$.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 23: Lagrange's Method of Multipliers: Minimum Distance to a Plane EXAM-LEVEL

A. Full Question:
Find the shortest distance from the origin to the plane x + 2y + 2z = 9 using Lagrange's method of undetermined multipliers.

B. Meaning of the Question

Minimize the square of distance $f(x, y, z) = x^2 + y^2 + z^2$ subject to the plane constraint $g(x, y, z) = x + 2y + 2z - 9 = 0$.

C. Basic Concept

Lagrange function: $L(x, y, z, \lambda) = f + \lambda g$. At extrema, $\nabla f = -\lambda \nabla g$, meaning the gradient of the objective function is parallel to the normal vector of the constraint surface.

D. Formula and Explanation

Formula: $L = (x^2 + y^2 + z^2) + \lambda(x + 2y + 2z - 9)$. Conditions: $Lx = 0$, $Ly = 0$, $Lz = 0$, $L\lambda = 0$.

Explanation & Conditions: Minimizing $x^2 + y^2 + z^2$ is mathematically equivalent to minimizing the square root distance, but avoids radical algebra.

E. Given Information

Objective: $f = x^2 + y^2 + z^2$; Constraint: $x + 2y + 2z = 9$.

F. Complete Step-by-Step Solution

Step 1: Set up Lagrange equations

$Lx = 2x + \lambda = 0 \Rightarrow x = -\lambda/2$.
$Ly = 2y + 2\lambda = 0 \Rightarrow y = -\lambda$.
$Lz = 2z + 2\lambda = 0 \Rightarrow z = -\lambda$.

💡 **Why this step is performed:** Set up the auxiliary Lagrange function $L = f + \lambda g$ and set all first partial derivatives to zero.

Step 2: Substitute x, y, z into the constraint equation

$x + 2y + 2z = 9$
$(-\lambda/2) + 2(-\lambda) + 2(-\lambda) = 9$
$-\lambda/2 - 4\lambda = 9 \Rightarrow -9\lambda/2 = 9 \Rightarrow \lambda = -2$.

💡 **Why this step is performed:** Execute step 2: substitute x, y, z into the constraint equation to advance the step-by-step mathematical solution toward the final result.

Step 3: Determine the closest point coordinates

$x = -(-2)/2 = 1$.
$y = -(-2) = 2$.
$z = -(-2) = 2$.
The closest point on the plane is (1, 2, 2).

💡 **Why this step is performed:** Execute step 3: determine the closest point coordinates to advance the step-by-step mathematical solution toward the final result.

Step 4: Compute the minimum distance

$d = \sqrt{(x^2 + y^2 + z^2)} = \sqrt{(1^2 + 2^2 + 2^2)} = \sqrt{(1 + 4 + 4)} = \sqrt{9} = 3$ units.

💡 **Why this step is performed:** Execute step 4: compute the minimum distance to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Closest point is (1, 2, 2); Minimum Distance = 3 units

H. Examination Presentation Strategy

- State objective function $f = x^2 + y^2 + z^2$.
- Form Lagrange function $L = f + \lambda g$.
- Solve for λ and then (x, y, z).
- State distance $d = \sqrt{9} = 3$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Using $d = \sqrt{(x^2+y^2+z^2)}$ as objective function.
✓ **How to Avoid:** Differentiating square roots is messy. Always minimize $f = x^2 + y^2 + z^2$ instead!

⚠️ **Common Mistake:** Arithmetic error in $-\lambda/2 - 4\lambda$.
✓ **How to Avoid:** $-\lambda/2 - 8\lambda/2 = -9\lambda/2$.

J. Similar Practice Question

Practice Problem 23: Find the shortest distance from the origin to the surface $xyz = 8$ using Lagrange multipliers.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 24: Chain Rule Derivative Transformation: Wave Equation Form

EXAM-LEVEL

A. Full Question:
If $u = f(x - ct) + g(x + ct)$, prove that $\partial^2u/\partial t^2 = c^2 (\partial^2u/\partial x^2)$.

B. Meaning of the Question

Show that any combination of forward and backward travelling waves $u = f(x - ct) + g(x + ct)$ satisfies the one-dimensional classical wave equation.

C. Basic Concept

Fundamental partial differential equation of vibrations and wave propagation (d'Alembert's solution of wave equation).

D. Formula and Explanation

Formula: `Let v = x - ct, w = x + ct. Then ∂/∂t = -c(d/dv) + c(d/dw) and ∂/∂x = d/dv + d/dw.`
Explanation & Conditions: c is the wave propagation velocity.

E. Given Information

$u = f(v) + g(w)$, where $v = x - ct$, $w = x + ct$.

F. Complete Step-by-Step Solution

Step 1: Compute first partial derivatives with respect to x

$\partial u/\partial x = f'(v) \cdot (\partial v/\partial x) + g'(w) \cdot (\partial w/\partial x) = f'(v)(1) + g'(w)(1) = f'(v) + g'(w)$.

💡 **Why this step is performed:** Execute step 1: compute first partial derivatives with respect to x to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute second partial derivative with respect to x

$\partial^2 u/\partial x^2 = \partial/\partial x\,[f'(v) + g'(w)] = f''(v)(1) + g''(w)(1) = f''(v) + g''(w)$.

💡 **Why this step is performed:** Execute step 2: compute second partial derivative with respect to x to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute first partial derivative with respect to t

$\partial u/\partial t = f'(v) \cdot (\partial v/\partial t) + g'(w) \cdot (\partial w/\partial t) = f'(v)(-c) + g'(w)(c) = -c\,f'(v) + c\,g'(w)$.

💡 **Why this step is performed:** Execute step 3: compute first partial derivative with respect to t to advance the step-by-step mathematical solution toward the final result.

Step 4: Compute second partial derivative with respect to t

$\partial^2 u/\partial t^2 = \partial/\partial t\,[-c\,f'(v) + c\,g'(w)]$
$= -c\,f''(v)(-c) + c\,g''(w)(c)$
$= c^2 f''(v) + c^2 g''(w) = c^2\,[f''(v) + g''(w)]$.

💡 **Why this step is performed:** Execute step 4: compute second partial derivative with respect to t to advance the step-by-step mathematical solution toward the final result.

Step 5: Substitute Step 2 into Step 4

$\partial^2 u/\partial t^2 = c^2\,[f''(v) + g''(w)] = c^2\,(\partial^2 u/\partial x^2)$. Hence proved.

💡 **Why this step is performed:** Execute step 5: substitute step 2 into step 4 to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $\partial^2 u/\partial t^2 = c^2\,(\partial^2 u/\partial x^2)$ (Hence proved)

H. Examination Presentation Strategy

- Define intermediate variables $v = x - ct$, $w = x + ct$.
- Show $\partial v/\partial t = -c$ and $\partial w/\partial t = +c$ explicitly.
- Equate LHS and RHS to complete proof.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Missing the minus sign on (-c)(-c).
✓ **How to Avoid:** Careful with double negative: -c · (-c) = +c^2.

⚠️ **Common Mistake:** Omitting the prime notation f'(v).
✓ **How to Avoid:** Remember to use chain rule: df/dt = f'(v) · dv/dt.

J. Similar Practice Question

Practice Problem 24: If u = f(x + iy) + g(x - iy), prove that $\partial^2u/\partial x^2 + \partial^2u/\partial y^2 = 0$ (Laplace equation).
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 25: Euler's Theorem for Three Variables with Algebraic-Logarithmic Form

EXAM-LEVEL

A. Full Question:
If u = log[($x^4 + y^4 + z^4$) / (x + y + z)], prove that $x(\partial u/\partial x) + y(\partial u/\partial y) + z(\partial u/\partial z) = 3$.

B. Meaning of the Question

Apply Euler's theorem for three independent variables to e^u, find its degree of homogeneity, and prove the sum equals the degree 3.

C. Basic Concept

For three variables, Euler's Theorem states: x u_x + y u_y + z u_z = n u. For logarithmic functions, let f(u) = e^u, then x u_x + y u_y + z u_z = n.

D. Formula and Explanation

Formula: `If e^u is homogeneous of degree n, then x u_x + y u_y + z u_z = n.`
Explanation & Conditions: Since d(e^u)/dx = e^u u_x, the e^u cancels out from both sides, leaving simply n.

E. Given Information

u = log[($x^4 + y^4 + z^4$) / (x + y + z)].

F. Complete Step-by-Step Solution

Step 1: Eliminate logarithm by exponentiating both sides

e^u = ($x^4 + y^4 + z^4$) / (x + y + z) = F(x, y, z).

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 2: Determine degree of homogeneity of F

F(tx, ty, tz) = [$(tx)^4 + (ty)^4 + (tz)^4$] / [tx + ty + tz]
= [$t^4(x^4 + y^4 + z^4)$] / [t(x + y + z)] = t^{4-1} · F = t^3 F.

Hence, F = e^u is a homogeneous function of degree n = 3 in x, y, z.

💡 **Why this step is performed:** Substitute tx and ty for x and y to factor out t^n and establish the degree of homogeneity n.

Step 3: Apply Euler's Theorem to F = e^u

x(∂F/∂x) + y(∂F/∂y) + z(∂F/∂z) = n · F = 3 e^u.

💡 **Why this step is performed:** Combine the partial derivatives into the linear combination x(∂u/∂x) + y(∂u/∂y) and verify it equals n·u.

Step 4: Differentiate F = e^u with respect to x, y, z

∂F/∂x = e^u (∂u/∂x)
∂F/∂y = e^u (∂u/∂y)
∂F/∂z = e^u (∂u/∂z).

💡 **Why this step is performed:** Execute step 4: differentiate f = e^u with respect to x, y, z to advance the step-by-step mathematical solution toward the final result.

Step 5: Substitute and divide by e^u

x [e^u (∂u/∂x)] + y [e^u (∂u/∂y)] + z [e^u (∂u/∂z)] = 3 e^u
e^u [x(∂u/∂x) + y(∂u/∂y) + z(∂u/∂z)] = 3 e^u
x(∂u/∂x) + y(∂u/∂y) + z(∂u/∂z) = 3. Hence proved.

💡 **Why this step is performed:** Compute the sum of partial derivatives of the component functions along their respective coordinate axes.

✓ G. Final Answer: x(∂u/∂x) + y(∂u/∂y) + z(∂u/∂z) = 3 (Hence proved)

H. Examination Presentation Strategy

- Write F = e^u clearly.
- Show F(tx, ty, tz) = t^3 F to establish degree n = 3.
- Factor out e^u and divide both sides.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Writing RHS as 3u.
✓ **How to Avoid:** For u = log(homogeneous), the answer is a CONSTANT n = 3, NOT 3u!

⚠️ **Common Mistake:** Attempting direct differentiation with quotient rule on log.
✓ **How to Avoid:** Direct differentiation takes 3 pages; Euler's theorem on e^u takes 5 lines.

J. Similar Practice Question

Practice Problem 25: If u = log[(x^3 + y^3 + z^3) / (x^2 + y^2 + z^2)], prove that x u_x + y u_y + z u_z = 1.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

MODULE 2: Vector Calculus

25 Fully Solved Questions: 5 Fundamental (Concepts & Proofs) + 10 Intermediate (Core Problems) + 10 Exam-Level (High-Yield VTU Questions)

QUESTION 26: Gradient of a Scalar Field at a Point

FUNDAMENTAL

A. Full Question:

If $\varphi(x, y, z) = 3x^2y - y^3z^2$, find grad φ (or $\nabla\varphi$) at the point (1, -2, -1).

B. Meaning of the Question

Calculate the vector of spatial partial derivatives (gradient vector) of the scalar field φ at the specified point (1, -2, -1).

C. Basic Concept

The gradient $\nabla\varphi$ transforms a scalar field into a vector field. At any point, $\nabla\varphi$ points in the direction of greatest rate of increase of φ, and its magnitude gives the maximum rate of increase.

D. Formula and Explanation

Formula: $\nabla\phi = (\partial\phi/\partial x)\,\hat{i} + (\partial\phi/\partial y)\,\hat{j} + (\partial\phi/\partial z)\,\hat{k}$.

Explanation & Conditions: $\hat{i}$, $\hat{j}$, $\hat{k}$ are unit basis vectors along x, y, z axes. Each component is a partial derivative.

E. Given Information

$\varphi = 3x^2y - y^3z^2$, point P(1, -2, -1).

F. Complete Step-by-Step Solution

Step 1: Compute $\partial\varphi/\partial x$

$\partial\varphi/\partial x = \partial/\partial x(3x^2y - y^3z^2) = 6xy - 0 = 6xy$.

💡 **Why this step is performed:** Execute step 1: compute $\partial\varphi/\partial x$ to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute $\partial\varphi/\partial y$

$\partial\varphi/\partial y = \partial/\partial y(3x^2y - y^3z^2) = 3x^2 - 3y^2z^2$.

💡 **Why this step is performed:** Execute step 2: compute $\partial\varphi/\partial y$ to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute $\partial\varphi/\partial z$

$\partial\varphi/\partial z = \partial/\partial z(3x^2y - y^3z^2) = 0 - 2y^3z = -2y^3z$.

💡 **Why this step is performed:** Execute step 3: compute $\partial\varphi/\partial z$ to advance the step-by-step mathematical solution toward the final result.

Step 4: Form the gradient vector $\nabla\varphi$

$\nabla\varphi = (6xy)\,\hat{i} + (3x^2 - 3y^2z^2)\,\hat{j} - (2y^3z)\,\hat{k}$.

💡 **Why this step is performed:** Execute step 4: form the gradient vector ∇φ to advance the step-by-step mathematical solution toward the final result.

Step 5: Evaluate at P(1, -2, -1)

• x-component: 6(1)(-2) = -12.
• y-component: $3(1)^2 - 3(-2)^2(-1)^2 = 3(1) - 3(4)(1) = 3 - 12 = -9$.
• z-component: $-2(-2)^3(-1) = -2(-8)(-1) = -16$.
Therefore: $\nabla\varphi(1, -2, -1) = -12\,\hat{i} - 9\,\hat{j} - 16\,\hat{k}$.

💡 **Why this step is performed:** Execute step 5: evaluate at p(1, -2, -1) to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $\nabla\varphi(1, -2, -1) = -12\,\hat{i} - 9\,\hat{j} - 16\,\hat{k}$

H. Examination Presentation Strategy

- State formula for ∇φ first.
- Calculate all 3 components separately.
- Substitute coordinates carefully observing negative signs.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Sign error in $(-2)^3 = -8$.
✓ **How to Avoid:** Cubing a negative number yields a negative number: $(-2)^3 = -8$, then $-2(-8)(-1) = -16$.

⚠️ **Common Mistake:** Writing gradient as a scalar.
✓ **How to Avoid:** Gradient of a scalar field is always a VECTOR!

J. Similar Practice Question

Practice Problem 26: If $\varphi = 2xz^4 - x^2y$, find ∇φ at (2, -2, 1).
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 27: Divergence of a Vector Field

FUNDAMENTAL

A. Full Question:
If $\vec{F} = (x^2y)\,\hat{i} + (y^2z)\,\hat{j} + (z^2x)\,\hat{k}$, find div $\vec{F}$ (or $\nabla \cdot \vec{F}$) at the point (1, 2, 3).

B. Meaning of the Question

Compute the divergence of the vector field $\vec{F}$, which measures the net outward flux of vector field per unit volume from a point.

C. Basic Concept

Divergence is a scalar measure of a vector field's source or sink density. If div F⃗ > 0, the point acts as a source; if div F⃗ < 0, it acts as a sink; if div F⃗ = 0, the field is solenoidal (incompressible).

D. Formula and Explanation

Formula: $\text{div } \vec{F} = \vec{\nabla} \cdot \vec{F} = \partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z.$

Explanation & Conditions: Dot product of the del operator ∇⃗ with vector F⃗ = $F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$ gives a SCALAR.

E. Given Information

F⃗ = $(x^2y)\,\hat{i} + (y^2z)\,\hat{j} + (z^2x)\,\hat{k}$, point P(1, 2, 3).

F. Complete Step-by-Step Solution

Step 1: Identify components of F⃗

$F_1 = x^2y$, $F_2 = y^2z$, $F_3 = z^2x$.

💡 **Why this step is performed:** Execute step 1: identify components of f⃗ to advance the step-by-step mathematical solution toward the final result.

Step 2: Differentiate each component with respect to its coordinate

$\partial F_1/\partial x = \partial(x^2y)/\partial x = 2xy$
$\partial F_2/\partial y = \partial(y^2z)/\partial y = 2yz$
$\partial F_3/\partial z = \partial(z^2x)/\partial z = 2zx.$

💡 **Why this step is performed:** Execute step 2: differentiate each component with respect to its coordinate to advance the step-by-step mathematical solution toward the final result.

Step 3: Add to get the scalar divergence

div F⃗ = ∇⃗ · F⃗ = 2xy + 2yz + 2zx = 2(xy + yz + zx).

💡 **Why this step is performed:** Compute the sum of partial derivatives of the component functions along their respective coordinate axes.

Step 4: Evaluate at point (1, 2, 3)

div F⃗(1, 2, 3) = 2[(1)(2) + (2)(3) + (3)(1)] = 2[2 + 6 + 3] = 2(11) = 22.

💡 **Why this step is performed:** Substitute the specific point coordinates into the gradient vector components.

✅ G. Final Answer: div F⃗(1, 2, 3) = 22

H. Examination Presentation Strategy

- State formula div F⃗ = $\partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$.
- Compute each term separately.
- Add terms and substitute point values.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Including $\hat{i}$, $\hat{j}$, $\hat{k}$ in the divergence answer.
✓ **How to Avoid:** Divergence is a SCALAR product (dot product), so $\hat{i}$, $\hat{j}$, $\hat{k}$ MUST NOT appear!

⚠️ **Common Mistake:** Differentiating F_1 with respect to y instead of x.
✓ **How to Avoid:** F_1 is with respect to x, F_2 with y, F_3 with z.

J. Similar Practice Question

Practice Problem 27: If $\vec{F} = (x^3 + y^3)\,\hat{i} + (y^3 + z^3)\,\hat{j} + (z^3 + x^3)\,\hat{k}$, find div $\vec{F}$ at (1, 1, 1).
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 28: Curl of a Vector Field

FUNDAMENTAL

A. Full Question:

If $\vec{F} = (xy^2)\,\hat{i} + (2x^2yz)\,\hat{j} - (3yz^2)\,\hat{k}$, find curl $\vec{F}$ (or $\nabla \times \vec{F}$) at the point (1, -1, 1).

B. Meaning of the Question

Compute the curl of the vector field $\vec{F}$, which measures the infinitesimal rotation or circulation of the vector field around a point.

C. Basic Concept

Curl measures the vorticity or rotational tendency of a fluid flow. If curl $\vec{F} = \vec{0}$ everywhere, the field has no vortices and is called irrotational (conservative).

D. Formula and Explanation

Formula: $\text{curl}\ \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_1 & F_2 & F_3 \end{vmatrix}$.

Explanation & Conditions: $= \hat{i}(\partial F_3/\partial y - \partial F_2/\partial z) - \hat{j}(\partial F_3/\partial x - \partial F_1/\partial z) + \hat{k}(\partial F_2/\partial x - \partial F_1/\partial y)$.

E. Given Information

$\vec{F} = (xy^2)\,\hat{i} + (2x^2yz)\,\hat{j} - (3yz^2)\,\hat{k}$, point P(1, -1, 1).

F. Complete Step-by-Step Solution

Step 1: Set up the 3×3 curl determinant

$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xy^2 & 2x^2yz & -3yz^2 \end{vmatrix}$.

💡 **Why this step is performed:** Set up the 3×3 curl determinant with unit vectors in row 1 and del operators in row 2.

Step 2: Expand î component

$\hat{i}\,[\,\partial(-3yz^2)/\partial y - \partial(2x^2yz)/\partial z\,] = \hat{i}\,[\,-3z^2 - 2x^2y\,]$.

💡 **Why this step is performed:** Execute step 2: expand î component to advance the step-by-step mathematical solution toward the final result.

Step 3: Expand ĵ component (with negative sign)

$-\hat{j}\,[\,\partial(-3yz^2)/\partial x - \partial(xy^2)/\partial z\,] = -\hat{j}\,[\,0 - 0\,] = 0\,\hat{j}$.

💡 **Why this step is performed:** Execute step 3: expand ĵ component (with negative sign) to advance the step-by-step mathematical solution toward the final result.

Step 4: Expand k̂ component

$\hat{k}\,[\,\partial(2x^2yz)/\partial x - \partial(xy^2)/\partial y\,] = \hat{k}\,[\,4xyz - 2xy\,]$.

💡 **Why this step is performed:** Execute step 4: expand k̂ component to advance the step-by-step mathematical solution toward the final result.

Step 5: Combine into vector expression and evaluate at (1, -1, 1)

□ × F□ = $(-3z^2 - 2x^2y)\,\hat{i} + (4xyz - 2xy)\,\hat{k}$.

At (1, -1, 1):

• î-comp: $-3(1)^2 - 2(1)^2(-1) = -3 + 2 = -1$.

• ĵ-comp: 0.

• k̂-comp: $4(1)(-1)(1) - 2(1)(-1) = -4 + 2 = -2$.

Thus, curl F□ = $-\hat{i} - 2\hat{k}$.

💡 **Why this step is performed:** Execute step 5: combine into vector expression and evaluate at (1, -1, 1) to advance the step-by-step mathematical solution toward the final result.

□ G. Final Answer: curl F□(1, -1, 1) = $-\hat{i} - 2\hat{k}$

H. Examination Presentation Strategy

- Write out the 3×3 determinant with del operators in row 2.
- Show the cross-derivative pairs for each component.
- Evaluate at the point.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Sign error on ĵ component.
✓ **How to Avoid:** Remember the cofactor of ĵ has a minus sign: $-\hat{j}(\partial F_3/\partial x - \partial F_1/\partial z)$.

⚠️ **Common Mistake:** Treating curl as a scalar.
✓ **How to Avoid:** Curl of a vector is a VECTOR!

J. Similar Practice Question

Practice Problem 28: If F⃗ = $(x^2 - y^2)\,\hat{i} + 2xy\,\hat{j} + (y^2 - z^2)\,\hat{k}$, find curl F⃗ at (1, 2, 3).

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 29: Unit Normal Vector to a Given Surface

FUNDAMENTAL

A. Full Question:

Find the unit normal vector $\hat{n}$ to the surface $x^2 + y^2 + 2z^2 = 26$ at the point (2, 2, 3).

B. Meaning of the Question

Find a vector of length 1 that is perpendicular to the tangent plane of the given ellipsoid surface at (2, 2, 3).

C. Basic Concept

For any level surface φ(x, y, z) = C, the gradient vector ∇φ is perpendicular (normal) to the tangent plane at every point. Dividing ∇φ by its magnitude gives the unit normal vector $\hat{n}$.

D. Formula and Explanation

Formula: $\hat{n} = \nabla\phi \,/\, |\nabla\phi|$.

Explanation & Conditions: ∇φ is the normal vector, |∇φ| is its Euclidean length $\sqrt{(Ax^2 + Ay^2 + Az^2)}$.

E. Given Information

Surface $\varphi(x, y, z) = x^2 + y^2 + 2z^2 - 26 = 0$, point P(2, 2, 3).

F. Complete Step-by-Step Solution

Step 1: Compute the gradient vector ∇φ

$\partial\varphi/\partial x = 2x$, $\partial\varphi/\partial y = 2y$, $\partial\varphi/\partial z = 4z$.

$\nabla\varphi = (2x)\,\hat{i} + (2y)\,\hat{j} + (4z)\,\hat{k}$.

💡 **Why this step is performed:** Execute step 1: compute the gradient vector ∇φ to advance the step-by-step mathematical solution toward the final result.

Step 2: Evaluate ∇φ at point (2, 2, 3)

$\nabla\varphi(2, 2, 3) = 2(2)\,\hat{i} + 2(2)\,\hat{j} + 4(3)\,\hat{k} = 4\,\hat{i} + 4\,\hat{j} + 12\,\hat{k}$.

Factor out 4: $\nabla\varphi = 4(\hat{i} + \hat{j} + 3\hat{k})$.

💡 **Why this step is performed:** Substitute the specific point coordinates into the gradient vector components.

Step 3: Compute the magnitude |∇φ|

$|\nabla\varphi| = \sqrt{[\,4^2 + 4^2 + 12^2\,]} = \sqrt{[\,16 + 16 + 144\,]} = \sqrt{176} = \sqrt{(16 \cdot 11)} = 4\sqrt{11}$.

💡 **Why this step is performed:** Execute step 3: compute the magnitude |∇φ| to advance the step-by-step mathematical solution toward the final result.

Step 4: Divide vector by its magnitude to obtain n̂

n̂ = (4 î + 4 ĵ + 12 k̂) / (4√11) = (î + ĵ + 3k̂) / √11.

💡 **Why this step is performed:** Compute the sum of partial derivatives of the component functions along their respective coordinate axes.

✓ G. Final Answer: n̂ = (î + ĵ + 3k̂) / √11 = (1/√11) î + (1/√11) ĵ + (3/√11) k̂

H. Examination Presentation Strategy

- Define φ(x, y, z) = x^2 + y^2 + $2z^2$.
- State n̂ = ∇φ / |∇φ|.
- Cancel the common factor 4 before writing final answer.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Leaving answer as ∇φ without normalizing.
✓ **How to Avoid:** The question asks for UNIT normal, so dividing by magnitude is compulsory!

⚠️ **Common Mistake:** Arithmetic error in 4(3) = 12.
✓ **How to Avoid:** Check 2(2z) = 4z, so at z=3, 4(3)=12, not 6.

J. Similar Practice Question

Practice Problem 29: Find the unit normal vector to the sphere x^2 + y^2 + z^2 = 9 at (1, 2, 2).
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 30: Directional Derivative Along a Given Vector — FUNDAMENTAL

A. Full Question:
Find the directional derivative of φ(x, y, z) = 2xy + z^2 at the point (1, -1, 3) in the direction of the vector a = î + 2ĵ + 2k̂.

B. Meaning of the Question

Compute the rate of change of the scalar field φ at (1, -1, 3) along the specific direction defined by vector a.

C. Basic Concept

The directional derivative is the projection of the gradient ∇φ onto the unit direction vector â: D_a φ = ∇φ · â. It gives the rate of change per unit distance along that direction.

D. Formula and Explanation

Formula: D_u ϕ = ∇ϕ · â = ∇ϕ · (a / |a|).

Explanation & Conditions: â must be a UNIT vector of length 1.

E. Given Information

$\varphi = 2xy + z^2$, point P(1, -1, 3), direction vector $a\square = \hat{i} + 2\hat{j} + 2\hat{k}$.

F. Complete Step-by-Step Solution

Step 1: Compute gradient ∇φ

$\partial\varphi/\partial x = 2y$, $\partial\varphi/\partial y = 2x$, $\partial\varphi/\partial z = 2z$.
$\nabla\varphi = (2y)\,\hat{i} + (2x)\,\hat{j} + (2z)\,\hat{k}$.

💡 **Why this step is performed:** Calculate the spatial partial derivatives along each coordinate axis to form the gradient vector ∇φ.

Step 2: Evaluate ∇φ at point (1, -1, 3)

$\nabla\varphi(1, -1, 3) = 2(-1)\,\hat{i} + 2(1)\,\hat{j} + 2(3)\,\hat{k} = -2\,\hat{i} + 2\,\hat{j} + 6\,\hat{k}$.

💡 **Why this step is performed:** Substitute the specific point coordinates into the gradient vector components.

Step 3: Find the unit vector â in the direction of a□

$|a\square| = \sqrt{(1^2 + 2^2 + 2^2)} = \sqrt{(1 + 4 + 4)} = \sqrt{9} = 3$.
$\hat{a} = a\square / |a\square| = (\hat{i} + 2\hat{j} + 2\hat{k}) / 3$.

💡 **Why this step is performed:** Divide the direction vector by its Euclidean length to obtain a normalized unit vector of magnitude 1.

Step 4: Take the dot product ∇φ · â

$D_a\,\varphi = (-2\,\hat{i} + 2\,\hat{j} + 6\,\hat{k}) \cdot [\,(\hat{i} + 2\hat{j} + 2\hat{k}) / 3\,]$
$= [\,(-2)(1) + (2)(2) + (6)(2)\,] / 3$
$= [\,-2 + 4 + 12\,] / 3 = 14 / 3$.

💡 **Why this step is performed:** Take the scalar dot product between the gradient vector and the unit direction vector.

✓ G. Final Answer: Directional Derivative = 14 / 3

H. Examination Presentation Strategy

- Compute ∇φ at the point clearly.
- Show unit vector â = a□ / |a□| with |a□| = 3.
- Show the dot product arithmetic explicitly.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Taking dot product with a□ instead of unit vector â.
✓ **How to Avoid:** You MUST divide by |a□| = 3! Taking dot product with non-unit vector is completely wrong.

⚠️ **Common Mistake:** Sign error in (-2)(1) = -2.
✓ **How to Avoid:** Keep track of negative signs carefully.

J. Similar Practice Question

Practice Problem 30: Find the directional derivative of $\varphi = x^2y + yz^2$ at (2, -1, 1) in direction $2\hat{i} - \hat{j} - 2\hat{k}$.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 31: Maximum Directional Derivative and Its Direction

INTERMEDIATE

A. Full Question:

Find the maximum directional derivative of $\varphi(x, y, z) = x^2y^2z^4$ at the point (1, 1, -1) and state the direction in which it occurs.

B. Meaning of the Question

Determine the direction of fastest increase of φ at (1, 1, -1) and the maximum rate of change.

C. Basic Concept

The directional derivative $\nabla\varphi \cdot \hat{u} = |\nabla\varphi| \cos\theta$ attains its maximum value when $\cos\theta = 1$ ($\theta = 0$), which occurs precisely in the direction of $\nabla\varphi$ itself, with maximum value equal to $|\nabla\varphi|$.

D. Formula and Explanation

Formula: `Maximum value = |∇ϕ|; Direction of maximum = ∇ϕ (or unit vector ∇ϕ/|∇ϕ|).`

Explanation & Conditions: The gradient vector always points in the direction of steepest ascent.

E. Given Information

$\varphi = x^2y^2z^4$, point P(1, 1, -1).

F. Complete Step-by-Step Solution

Step 1: Compute partial derivatives

$\partial\varphi/\partial x = 2xy^2z^4$, $\partial\varphi/\partial y = 2x^2yz^4$, $\partial\varphi/\partial z = 4x^2y^2z^3$.

💡 **Why this step is performed:** Execute step 1: compute partial derivatives to advance the step-by-step mathematical solution toward the final result.

Step 2: Evaluate $\nabla\varphi$ at P(1, 1, -1)

$\partial\varphi/\partial x = 2(1)(1)^2(-1)^4 = 2(1) = 2.$
$\partial\varphi/\partial y = 2(1)^2(1)(-1)^4 = 2(1) = 2.$
$\partial\varphi/\partial z = 4(1)^2(1)^2(-1)^3 = 4(-1) = -4.$
$\nabla\varphi = 2\,\hat{i} + 2\,\hat{j} - 4\,\hat{k}.$

💡 **Why this step is performed:** Substitute the specific point coordinates into the gradient vector components.

Step 3: State the direction of maximum directional derivative

The maximum directional derivative occurs along the vector $\nabla\varphi = 2\,\hat{i} + 2\,\hat{j} - 4\,\hat{k}$ (or unit direction $(\hat{i} + \hat{j} - 2\hat{k})/\sqrt{6}$).

💡 **Why this step is performed:** Take the scalar dot product between the gradient vector and the unit direction vector.

Step 4: Compute the magnitude $|\nabla\varphi|$

$|\nabla\varphi| = \sqrt{[\,2^2 + 2^2 + (-4)^2\,]} = \sqrt{[\,4 + 4 + 16\,]} = \sqrt{24} = \sqrt{(4 \cdot 6)} = 2\sqrt{6}$.

💡 **Why this step is performed:** Execute step 4: compute the magnitude $|\nabla\varphi|$ to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Maximum Directional Derivative = $2\sqrt{6}$; Direction: along $2\,\hat{i} + 2\,\hat{j} - 4\,\hat{k}$ (or unit vector $(\hat{i} + \hat{j} - 2\hat{k})/\sqrt{6}$)

H. Examination Presentation Strategy

- State property: Max directional derivative = $|\nabla\varphi|$ in direction of $\nabla\varphi$.
- Show $\nabla\varphi$ calculation.
- Simplify $\sqrt{24}$ into $2\sqrt{6}$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Sign error on $(-1)^3$.
✓ **How to Avoid:** $(-1)^3 = -1$, so z-component is -4, not +4.

⚠️ **Common Mistake:** Forgetting to state the direction.
✓ **How to Avoid:** The question asks for both the maximum value AND the direction.

J. Similar Practice Question

Practice Problem 31: Find the maximum directional derivative of $\varphi = x^3y^2z$ at (1, -2, 1).
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 32: Condition for a Solenoidal Vector Field

INTERMEDIATE

A. Full Question:
Find the constant 'a' such that the vector field F⃗ = (x + 3y) î + (y - 2z) ĵ + (x + az) k̂ is solenoidal.

B. Meaning of the Question

Determine the value of parameter 'a' for which the divergence of vector field F⃗ is identically zero.

C. Basic Concept

A vector field is solenoidal if its divergence is zero: ∇⃗ · F⃗ = 0. This physically represents an incompressible fluid with no sources or sinks.

D. Formula and Explanation

Formula: $\text{div } \vec{F} = \partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z = 0.$

Explanation & Conditions: Sum of coordinate derivatives must equate to zero.

E. Given Information

$\vec{F} = (x + 3y)\,\hat{i} + (y - 2z)\,\hat{j} + (x + az)\,\hat{k}$.

F. Complete Step-by-Step Solution

Step 1: Identify components of $\vec{F}$

$F_1 = x + 3y$, $F_2 = y - 2z$, $F_3 = x + az$.

> 💡 **Why this step is performed:** Execute step 1: identify components of f to advance the step-by-step mathematical solution toward the final result.

Step 2: Differentiate each component

$\partial F_1/\partial x = \partial(x + 3y)/\partial x = 1$

$\partial F_2/\partial y = \partial(y - 2z)/\partial y = 1$

$\partial F_3/\partial z = \partial(x + az)/\partial z = a$.

> 💡 **Why this step is performed:** Execute step 2: differentiate each component to advance the step-by-step mathematical solution toward the final result.

Step 3: Apply the solenoidal condition div $\vec{F} = 0$

$\nabla \cdot \vec{F} = \partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z = 1 + 1 + a = 2 + a = 0$.

> 💡 **Why this step is performed:** Compute the sum of partial derivatives of the component functions along their respective coordinate axes.

Step 4: Solve for 'a'

$2 + a = 0 \Rightarrow a = -2$.

> 💡 **Why this step is performed:** Execute step 4: solve for 'a' to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: a = -2

H. Examination Presentation Strategy

- State condition: For $\vec{F}$ to be solenoidal, div $\vec{F} = 0$.
- Show term-by-term divergence: $1 + 1 + a = 0$.
- Conclude $a = -2$ in a box.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Confusing solenoidal with irrotational.
✓ **How to Avoid:** Solenoidal means div F⃗ = 0 (dot product). Irrotational means curl F⃗ = 0 (cross product).

⚠️ **Common Mistake:** Derivative of az with respect to z written as 0.
✓ **How to Avoid:** d(az)/dz = a.

J. Similar Practice Question

Practice Problem 32: Find constant 'k' such that F⃗ = (2x + y) î + (ky - z) ĵ + (3x - 4z) k̂ is solenoidal.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 33: Irrotational Vector Field and Finding Constants

INTERMEDIATE

A. Full Question:
Find constants a, b, c so that the vector field F⃗ = (x + 2y + az) î + (bx - 3y - z) ĵ + (4x + cy + 2z) k̂ is irrotational.

B. Meaning of the Question

Determine values of a, b, c such that curl F⃗ = 0⃗ everywhere.

C. Basic Concept

A vector field is irrotational if its curl is identically zero: ∇⃗ × F⃗ = 0⃗. Equating each of the three components of curl to zero produces three algebraic equations for a, b, c.

D. Formula and Explanation

Formula: curl F⃗ = $\hat{i}(\partial F_3/\partial y - \partial F_2/\partial z) + \hat{j}(\partial F_1/\partial z - \partial F_3/\partial x) + \hat{k}(\partial F_2/\partial x - \partial F_1/\partial y) = 0\,\hat{i} + 0\,\hat{j} + 0\,\hat{k}$.

Explanation & Conditions: Each bracketed difference must equal zero.

E. Given Information

$F_1 = x + 2y + az$, $F_2 = bx - 3y - z$, $F_3 = 4x + cy + 2z$.

F. Complete Step-by-Step Solution

Step 1: Set up curl components

- î-component: $\partial F_3/\partial y - \partial F_2/\partial z = c - (-1) = c + 1 = 0$.
- ĵ-component: $\partial F_1/\partial z - \partial F_3/\partial x = a - 4 = 0$.
- k̂-component: $\partial F_2/\partial x - \partial F_1/\partial y = b - 2 = 0$.

💡 **Why this step is performed:** Set up the 3×3 curl determinant with unit vectors in row 1 and del operators in row 2.

Step 2: Solve each equation

From $\hat{i}$-component: $c + 1 = 0 \Rightarrow c = -1$.
From $\hat{j}$-component: $a - 4 = 0 \Rightarrow a = 4$.
From $\hat{k}$-component: $b - 2 = 0 \Rightarrow b = 2$.

💡 **Why this step is performed:** Execute step 2: solve each equation to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: a = 4, b = 2, c = -1

H. Examination Presentation Strategy

- Write out curl determinant clearly.
- Equate each vector component to 0 separately.
- List a = 4, b = 2, c = -1.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Sign error in c - (-1).
✓ **How to Avoid:** $\partial F_2/\partial z = -1$, so $c - (-1) = c + 1 = 0 \Rightarrow c = -1$.

⚠️ **Common Mistake:** Swapping a and b.
✓ **How to Avoid:** Double check which component corresponds to which derivative.

J. Similar Practice Question

Practice Problem 33: Find a, b, c such that $\vec{F} = (axy + z^3)\,\hat{i} + (3x^2 - cz)\,\hat{j} + (3xz^2 - y)\,\hat{k}$ is irrotational.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 34: Finding Scalar Potential Function φ for an Irrotational Field

INTERMEDIATE

A. Full Question:
Show that $\vec{F} = (y^2 \cos x + z^3)\,\hat{i} + (2y \sin x - 4)\,\hat{j} + (3xz^2)\,\hat{k}$ is conservative and find its scalar potential φ such that $\vec{F} = \nabla\varphi$.

B. Meaning of the Question

Prove curl $\vec{F} = \vec{0}$ (conservative), then integrate the three component differential equations to reconstruct the scalar potential φ.

C. Basic Concept

By Helmholtz decomposition, any irrotational vector field can be expressed as the gradient of a scalar potential: $\vec{F} = \nabla\varphi$. Then $d\varphi = F_1\,dx + F_2\,dy + F_3\,dz$ is an exact differential.

D. Formula and Explanation

Formula: $\partial\phi/\partial x = F_1$, $\partial\phi/\partial y = F_2$, $\partial\phi/\partial z = F_3$; $\phi = \int F_1\,dx$ (treating y, z const) + $\int$ (terms in F_2 free of x) dy + $\int$ (terms in F_3 free of x, y) dz.

Explanation & Conditions: Standard exact differential integration technique avoids duplicate terms.

E. Given Information

$F_1 = y^2 \cos x + z^3$, $F_2 = 2y \sin x - 4$, $F_3 = 3xz^2$.

F. Complete Step-by-Step Solution

Step 1: Verify curl F⃗ = 0⃗

• î-comp: $\partial(3xz^2)/\partial y - \partial(2y \sin x - 4)/\partial z = 0 - 0 = 0$.
• ĵ-comp: $\partial(y^2 \cos x + z^3)/\partial z - \partial(3xz^2)/\partial x = 3z^2 - 3z^2 = 0$.
• k̂-comp: $\partial(2y \sin x - 4)/\partial x - \partial(y^2 \cos x + z^3)/\partial y = 2y \cos x - 2y \cos x = 0$.
Since curl F⃗ = 0⃗, F⃗ is conservative.

> 💡 **Why this step is performed:** Execute step 1: verify curl f⃗ = 0⃗ to advance the step-by-step mathematical solution toward the final result.

Step 2: Set up potential equations

$\partial\varphi/\partial x = y^2 \cos x + z^3$ --- (1)
$\partial\varphi/\partial y = 2y \sin x - 4$ --- (2)
$\partial\varphi/\partial z = 3xz^2$ --- (3).

> 💡 **Why this step is performed:** Execute step 2: set up potential equations to advance the step-by-step mathematical solution toward the final result.

Step 3: Integrate equation (1) with respect to x

$\varphi = \int (y^2 \cos x + z^3)\,dx = y^2 \sin x + x z^3 + f(y, z)$.

> 💡 **Why this step is performed:** Integrate the exact differential equations $d\varphi = F_1\,dx + F_2\,dy + F_3\,dz$ systematically to reconstruct the scalar potential.

Step 4: Differentiate with respect to y and compare with (2)

$\partial\varphi/\partial y = 2y \sin x + \partial f/\partial y = 2y \sin x - 4 \Rightarrow \partial f/\partial y = -4$.
Integrate: $f(y, z) = -4y + g(z)$.

> 💡 **Why this step is performed:** Execute step 4: differentiate with respect to y and compare with (2) to advance the step-by-step mathematical solution toward the final result.

Step 5: Differentiate with respect to z and compare with (3)

$\partial\varphi/\partial z = 3xz^2 + g'(z) = 3xz^2 \Rightarrow g'(z) = 0 \Rightarrow g(z) = C$.
Therefore: $\varphi(x, y, z) = y^2 \sin x + x z^3 - 4y + C$.

> 💡 **Why this step is performed:** Execute step 5: differentiate with respect to z and compare with (3) to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $\varphi(x, y, z) = y^2 \sin x + x z^3 - 4y + C$

H. Examination Presentation Strategy

- Show curl F⃗ = 0⃗ first to justify existence of φ.
- Integrate step-by-step matching arbitrary functions f(y, z) and g(z).
- Include arbitrary constant C.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Adding duplicate terms (e.g., adding xz^3 twice).
✓ **How to Avoid:** xz^3 appears in both x and z integrals; it must only be written ONCE in φ!

⚠️ **Common Mistake:** Forgetting arbitrary constant C.
✓ **How to Avoid:** Potential function is determined up to an additive constant C.

J. Similar Practice Question

Practice Problem 34: Show $\vec{F} = (2xy + z^3)\,\hat{i} + x^2\,\hat{j} + 3xz^2\,\hat{k}$ is conservative and find its scalar potential.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 35: Angle Between Two Surfaces at Point of Intersection

INTERMEDIATE

A. Full Question:
Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point (2, -1, 2).

B. Meaning of the Question

Compute the angle between the normal vectors of the two surfaces at their intersection point (2, -1, 2).

C. Basic Concept

The angle θ between two surfaces at a common point is defined as the angle between their normal vectors (gradients) at that point: $\cos\theta = (\nabla\varphi_1 \cdot \nabla\varphi_2) / (|\nabla\varphi_1|\,|\nabla\varphi_2|)$.

D. Formula and Explanation

Formula: $\cos\theta = (\vec{n}_1 \cdot \vec{n}_2) / (|\vec{n}_1|\,|\vec{n}_2|)$, where $\vec{n}_1 = \vec{\nabla}\phi_1$ and $\vec{n}_2 = \vec{\nabla}\phi_2$.
Explanation & Conditions: Standard dot product angle formula applied to gradient normals.

E. Given Information

$\varphi_1 = x^2 + y^2 + z^2 - 9 = 0$, $\varphi_2 = x^2 + y^2 - z - 3 = 0$, point P(2, -1, 2).

F. Complete Step-by-Step Solution

Step 1: Compute normal vector $\vec{n}_1 = \vec{\nabla}\varphi_1$ at P(2, -1, 2)

$\nabla\varphi_1 = 2x\,\hat{i} + 2y\,\hat{j} + 2z\,\hat{k}$.
At (2, -1, 2): $\vec{n}_1 = 2(2)\,\hat{i} + 2(-1)\,\hat{j} + 2(2)\,\hat{k} = 4\,\hat{i} - 2\,\hat{j} + 4\,\hat{k}$.

$|n\square_1| = \sqrt{[\ 4^2 + (-2)^2 + 4^2\ ]} = \sqrt{[\ 16 + 4 + 16\ ]} = \sqrt{36} = 6.$

💡 **Why this step is performed:** Execute step 1: compute normal vector $n\square_1 = \square\varphi_1$ at p(2, -1, 2) to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute normal vector $n\square_2 = \square\varphi_2$ at P(2, -1, 2)

$\nabla\varphi_2 = 2x\,\hat{i} + 2y\,\hat{j} - \hat{k}.$
At (2, -1, 2): $n\square_2 = 2(2)\,\hat{i} + 2(-1)\,\hat{j} - \hat{k} = 4\,\hat{i} - 2\,\hat{j} - \hat{k}.$
$|n\square_2| = \sqrt{[\ 4^2 + (-2)^2 + (-1)^2\ ]} = \sqrt{[\ 16 + 4 + 1\ ]} = \sqrt{21}.$

💡 **Why this step is performed:** Execute step 2: compute normal vector $n\square_2 = \square\varphi_2$ at p(2, -1, 2) to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute dot product $n\square_1 \cdot n\square_2$

$n\square_1 \cdot n\square_2 = (4)(4) + (-2)(-2) + (4)(-1) = 16 + 4 - 4 = 16.$

💡 **Why this step is performed:** Take the scalar dot product between the gradient vector and the unit direction vector.

Step 4: Calculate cos θ and θ

$\cos\theta = (n\square_1 \cdot n\square_2) / (|n\square_1|\,|n\square_2|) = 16 / (6 \cdot \sqrt{21}) = 8 / (3\sqrt{21}).$
$\theta = \cos^{-1}[\ 8 / (3\sqrt{21})\ ].$

💡 **Why this step is performed:** Execute step 4: calculate cos θ and θ to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $\theta = \cos^{-1}[\ 8 / (3\sqrt{21})\ ] \approx 54.4°$

H. Examination Presentation Strategy

- Find $\nabla\varphi_1$ and $\nabla\varphi_2$ at (2, -1, 2).
- Compute lengths $|n\square_1|$ and $|n\square_2|$ clearly.
- Apply dot product formula and simplify fraction 16/6 = 8/3.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Taking $\partial\varphi_2/\partial z$ as +1 instead of -1.
✓ **How to Avoid:** From $z = x^2 + y^2 - 3$, $\varphi_2 = x^2 + y^2 - z - 3$, so $\partial\varphi_2/\partial z = -1$.

⚠️ **Common Mistake:** Forgetting to take inverse cosine.
✓ **How to Avoid:** State $\theta = \cos^{-1}(\ldots)$ clearly.

J. Similar Practice Question

Practice Problem 35: Find angle between surfaces $x^2 + y^2 + z^2 = 29$ and $x^2 + y^2 + z^2 + 4x - 6y - 8z - 47 = 0$ at (4, -3, 2).
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 36: Vector Identity: div(curl F⃗) = 0

INTERMEDIATE

A. Full Question:
Prove that for any twice continuously differentiable vector field F⃗, div(curl F⃗) = 0 (or ∇⃗ · (∇⃗ × F⃗) = 0).

B. Meaning of the Question

Prove the fundamental vector identity stating that the divergence of any curl field is identically zero.

C. Basic Concept

The boundary of a boundary is zero. Physically, curl represents pure circulating flow with closed field lines; closed loops have no starting points (sources) or ending points (sinks), so their net divergence is always zero.

D. Formula and Explanation

Formula: ∇⃗ · (∇⃗ × F⃗) = 0.

Explanation & Conditions: Follows from equality of mixed partial derivatives $\partial^2/\partial x\partial y = \partial^2/\partial y\partial x$.

E. Given Information

$\vec{F} = F_1\,\hat{i} + F_2\,\hat{j} + F_3\,\hat{k}$.

F. Complete Step-by-Step Solution

Step 1: Write curl F⃗ in component form

curl $\vec{F} = (\partial F_3/\partial y - \partial F_2/\partial z)\,\hat{i} + (\partial F_1/\partial z - \partial F_3/\partial x)\,\hat{j} + (\partial F_2/\partial x - \partial F_1/\partial y)\,\hat{k}$.

💡 **Why this step is performed:** Execute step 1: write curl f⃗ in component form to advance the step-by-step mathematical solution toward the final result.

Step 2: Take divergence (dot with del operator)

div(curl $\vec{F}$) $= \partial/\partial x(\partial F_3/\partial y - \partial F_2/\partial z) + \partial/\partial y(\partial F_1/\partial z - \partial F_3/\partial x) + \partial/\partial z(\partial F_2/\partial x - \partial F_1/\partial y)$.

💡 **Why this step is performed:** Compute the sum of partial derivatives of the component functions along their respective coordinate axes.

Step 3: Expand the partial derivatives

$= (\partial^2 F_3/\partial x\partial y - \partial^2 F_2/\partial x\partial z) + (\partial^2 F_1/\partial y\partial z - \partial^2 F_3/\partial y\partial x) + (\partial^2 F_2/\partial z\partial x - \partial^2 F_1/\partial z\partial y)$.

💡 **Why this step is performed:** Execute step 3: expand the partial derivatives to advance the step-by-step mathematical solution toward the final result.

Step 4: Group pairs using Clairaut's theorem

$= (\partial^2 F_3/\partial x\partial y - \partial^2 F_3/\partial y\partial x) + (\partial^2 F_2/\partial z\partial x - \partial^2 F_2/\partial x\partial z) + (\partial^2 F_1/\partial y\partial z - \partial^2 F_1/\partial z\partial y)$
$= 0 + 0 + 0 = 0$. Hence proved.

💡 **Why this step is performed:** Execute step 4: group pairs using clairaut's theorem to advance the step-by-step mathematical solution toward the final result.

□ G. Final Answer: div(curl F□) = □ · (□ × F□) = 0 (Hence proved)

H. Examination Presentation Strategy

- Write components of curl F□.
- Apply divergence operator.
- Group identical mixed partials with opposite signs to show cancellation.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Skipping Step 3 and jumping directly to 0.
✓ **How to Avoid:** You must write out all 6 second-order mixed partial derivatives to get full marks!

⚠️ **Common Mistake:** Confusing with curl(grad φ) = 0.
✓ **How to Avoid:** Both are zero, but be sure to write the correct one requested.

J. Similar Practice Question

Practice Problem 36: Prove that curl(grad φ) = ∇ × (∇φ) = 0⃗ for any scalar field φ.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 37: Properties of Position Vector r□: Divergence and Curl

INTERMEDIATE

A. Full Question:
If r□ = x î + y ĵ + z k̂ and r = |r□| = √(x² + y² + z²), find (i) div r□ and (ii) curl r□.

B. Meaning of the Question

Evaluate the divergence and curl of the standard 3D position vector field r□.

C. Basic Concept

The position vector represents uniform radial expansion from the origin. It has constant divergence 3 and zero curl everywhere (purely irrotational).

D. Formula and Explanation

Formula: `div r⃗ = ∂x/∂x + ∂y/∂y + ∂z/∂z; curl r⃗ = 0⃗ × r⃗.`

Explanation & Conditions: Basic operations on fundamental coordinate vector.

E. Given Information

$\vec{r} = x\,\hat{i} + y\,\hat{j} + z\,\hat{k}$.

F. Complete Step-by-Step Solution

Step 1: Compute div $\vec{r}$

$\operatorname{div}\vec{r} = \vec{\nabla}\cdot\vec{r} = \partial(x)/\partial x + \partial(y)/\partial y + \partial(z)/\partial z = 1 + 1 + 1 = 3$.

💡 **Why this step is performed:** Compute the sum of partial derivatives of the component functions along their respective coordinate axes.

Step 2: Set up curl $\vec{r}$ determinant

$\operatorname{curl}\vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & z \end{vmatrix}$.

💡 **Why this step is performed:** Set up the 3×3 curl determinant with unit vectors in row 1 and del operators in row 2.

Step 3: Expand the determinant

$= \hat{i}\,[\,\partial z/\partial y - \partial y/\partial z\,] - \hat{j}\,[\,\partial z/\partial x - \partial x/\partial z\,] + \hat{k}\,[\,\partial y/\partial x - \partial x/\partial y\,]$
$= \hat{i}\,[\,0 - 0\,] - \hat{j}\,[\,0 - 0\,] + \hat{k}\,[\,0 - 0\,] = 0\,\hat{i} + 0\,\hat{j} + 0\,\hat{k} = \vec{0}$.

💡 **Why this step is performed:** Execute step 3: expand the determinant to advance the step-by-step mathematical solution toward the final result.

✅ G. Final Answer: (i) $\operatorname{div}\vec{r} = 3$; (ii) $\operatorname{curl}\vec{r} = \vec{0}$

H. Examination Presentation Strategy

- Show $\operatorname{div}\vec{r} = 1 + 1 + 1 = 3$.
- Show curl $\vec{r}$ determinant with zeros.
- Write zero vector $\vec{0}$ with arrow or bold notation.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Writing $\operatorname{curl}\vec{r} = 0$ (scalar) instead of $\vec{0}$ (zero vector).
✓ **How to Avoid:** Curl produces a VECTOR, so write $\vec{0}$ or $0\hat{i} + 0\hat{j} + 0\hat{k}$.

⚠️ **Common Mistake:** Writing $\operatorname{div}\vec{r} = 1$.
✓ **How to Avoid:** Each axis contributes 1: 1 + 1 + 1 = 3.

J. Similar Practice Question

Practice Problem 37: Find $\operatorname{div}(\vec{r}/r^3)$ for $r \neq 0$.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 38: Gradient of Radial Power Function: ∇(r^n)

INTERMEDIATE

A. Full Question:
Prove that □(r^n) = n r^{n−2} r□.

B. Meaning of the Question

Find the gradient of the power of magnitude of position vector r^n.

C. Basic Concept

Since r = √(x^2 + y^2 + z^2), ∂r/∂x = x/r. Applying chain rule to r^n gives (n r^{n−1})(x/r) = n r^{n−2} x. Combining components gives n r^{n−2} r□.

D. Formula and Explanation

Formula: ∇(r^n) = Σ î [∂(r^n)/∂x] = n r^{n−2} r⃗.

Explanation & Conditions: ∂r/∂x = x/r, ∂r/∂y = y/r, ∂r/∂z = z/r.

E. Given Information

r^2 = x^2 + y^2 + z^2, φ = r^n.

F. Complete Step-by-Step Solution

Step 1: Compute ∂r/∂x

Differentiating r^2 = x^2 + y^2 + z^2 partially with respect to x:
2r (∂r/∂x) = 2x ⇒ ∂r/∂x = x/r.
Similarly, ∂r/∂y = y/r and ∂r/∂z = z/r.

💡 **Why this step is performed:** Execute step 1: compute ∂r/∂x to advance the step-by-step mathematical solution toward the final result.

Step 2: Differentiate r^n with respect to x using chain rule

∂(r^n)/∂x = n r^{n−1} · (∂r/∂x) = n r^{n−1} · (x/r) = n r^{n−2} x.

💡 **Why this step is performed:** Execute step 2: differentiate r^n with respect to x using chain rule to advance the step-by-step mathematical solution toward the final result.

Step 3: Construct gradient vector ∇(r^n)

∇(r^n) = [∂(r^n)/∂x] î + [∂(r^n)/∂y] ĵ + [∂(r^n)/∂z] k̂
= (n r^{n−2} x) î + (n r^{n−2} y) ĵ + (n r^{n−2} z) k̂.

💡 **Why this step is performed:** Execute step 3: construct gradient vector ∇(r^n) to advance the step-by-step mathematical solution toward the final result.

Step 4: Factor out common scalar n r^{n−2}

= n r^{n−2} [x î + y ĵ + z k̂] = n r^{n−2} r□. Hence proved.

💡 **Why this step is performed:** Execute step 4: factor out common scalar $n\ r^{n-2}$ to advance the step-by-step mathematical solution toward the final result.

□ G. Final Answer: $\square(r^n) = n\ r^{n-2}\ r\square$ (Hence proved)

H. Examination Presentation Strategy

- Establish $\partial r/\partial x = x/r$ clearly.
- Show chain rule: $d(r^n)/dx = n\ r^{n-1}\ (x/r) = n\ r^{n-2}\ x$.
- Factor out $n\ r^{n-2}$ and replace with $r\square$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Writing $n\ r^{n-1}\ r\square$ instead of $n\ r^{n-2}\ r\square$.
✓ **How to Avoid:** Dividing x by r reduces the exponent of r by another 1, making it n - 2!

⚠️ **Common Mistake:** Forgetting that r depends on x, y, z.
✓ **How to Avoid:** r is not a constant; $r = \sqrt{(x^2+y^2+z^2)}$.

J. Similar Practice Question

Practice Problem 38: Find $\nabla(1/r)$ using the formula $\nabla(r^n)$ with n = -1.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 39: Laplacian of Radial Power Function: $\nabla^2(r^n)$ INTERMEDIATE

A. Full Question:
Prove that $\nabla^2(r^n) = n(n+1)\ r^{n-2}$ and deduce the value of n for which r^n is harmonic ($\nabla^2(r^n) = 0, r \neq 0$).

B. Meaning of the Question

Evaluate the Laplacian div(grad r^n) and find which power makes it satisfy Laplace's equation.

C. Basic Concept

Laplacian is $\nabla^2\varphi = \nabla \cdot \nabla\varphi$. A function is harmonic if $\nabla^2\varphi = 0$.

D. Formula and Explanation

Formula: $\square^2(r^n) = \text{div}\ [\ \square(r^n)\ ] = \square \cdot [\ n\ r^{n-2}\ r\square\ ]$.
Explanation & Conditions: Vector identity: $\nabla \cdot (\varphi\ A\square) = \varphi(\nabla \cdot A\square) + \nabla\varphi \cdot A\square$.

E. Given Information

From Q38: $\square(r^n) = n\ r^{n-2}\ r\square$.

F. Complete Step-by-Step Solution

Step 1: Express Laplacian as divergence of gradient

$\square^2(r^n) = \square \cdot [\, n\, r^{n-2}\, r\square \,] = n\, [\, \square \cdot (r^{n-2}\, r\square) \,]$.

> 💡 **Why this step is performed:** Compute the sum of partial derivatives of the component functions along their respective coordinate axes.

Step 2: Apply product rule $\nabla \cdot (\varphi\, A\square) = \varphi(\nabla \cdot A\square) + (\nabla\varphi) \cdot A\square$

Here $\varphi = r^{n-2}$ and $A\square = r\square$.

$\square \cdot (r^{n-2}\, r\square) = r^{n-2}\, (\square \cdot r\square) + \square(r^{n-2}) \cdot r\square$.

> 💡 **Why this step is performed:** Execute step 2: apply product rule $\nabla \cdot (\varphi\, a\square) = \varphi(\nabla \cdot a\square) + (\nabla\varphi) \cdot a\square$ to advance the step-by-step mathematical solution toward the final result.

Step 3: Substitute known results

From Q37: $\square \cdot r\square = 3$.

From Q38: $\square(r^{n-2}) = (n - 2)\, r^{n-4}\, r\square$.

Therefore: $\square(r^{n-2}) \cdot r\square = (n - 2)\, r^{n-4}\, (r\square \cdot r\square) = (n - 2)\, r^{n-4}\, (r^2) = (n - 2)\, r^{n-2}$.

> 💡 **Why this step is performed:** Execute step 3: substitute known results to advance the step-by-step mathematical solution toward the final result.

Step 4: Combine the terms

$\square \cdot (r^{n-2}\, r\square) = r^{n-2}\, (3) + (n - 2)\, r^{n-2} = [\, 3 + n - 2 \,]\, r^{n-2} = (n + 1)\, r^{n-2}$.

Multiply by n:

$\nabla^2(r^n) = n\,(n + 1)\, r^{n-2}$. Hence proved.

> 💡 **Why this step is performed:** Execute step 4: combine the terms to advance the step-by-step mathematical solution toward the final result.

Step 5: Find condition for harmonicity

For r^n to be harmonic: $\nabla^2(r^n) = 0$

$n(n + 1)\, r^{n-2} = 0 \Rightarrow n(n + 1) = 0$ (since $r \neq 0$)

$\Rightarrow n = 0$ or $n = -1$.

For $n = -1$: $\varphi = 1/r$ (Newtonian/Coulomb potential).

> 💡 **Why this step is performed:** Execute step 5: find condition for harmonicity to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $\nabla^2(r^n) = n(n + 1)\, r^{n-2}$; Harmonic for $n = 0$ or $n = -1$ (i.e., $1/r$ is harmonic in 3D)

H. Examination Presentation Strategy

- Use $\nabla^2 = \nabla \cdot \nabla$.
- Use product rule $\square \cdot (u\, v\square) = u\, \text{div}\, v\square + \square u \cdot v\square$.
- Equate $n(n+1)=0$ to deduce harmonic powers $n = 0, -1$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Writing □ · r□ = 1 instead of 3.
✓ **How to Avoid:** In 3D, □ · r□ = 3!

⚠️ **Common Mistake:** Missing r□ · r□ = r^2.
✓ **How to Avoid:** The dot product of a vector with itself is the square of its magnitude.

J. Similar Practice Question

Practice Problem 39: Verify directly that φ = 1/r satisfies Laplace's equation $\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2 = 0$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 40: Scale Factors in Cylindrical Curvilinear Coordinates

INTERMEDIATE

A. Full Question:
For cylindrical coordinates (u, v, w) = (ρ, φ, z) where x = ρ cos φ, y = ρ sin φ, z = z, find the scale factors h_1, h_2, h_3 and show the system is orthogonal.

B. Meaning of the Question

Compute scale factors h_ρ, h_φ, h_z from the position vector r□ and verify that the coordinate basis vectors are mutually perpendicular.

C. Basic Concept

Curvilinear scale factor h_i = |∂r□/∂u_i|. The system is orthogonal if (∂r□/∂u_i) · (∂r□/∂u_j) = 0 for all i ≠ j.

D. Formula and Explanation

Formula: r⃗ = (ρ cos ϕ) î + (ρ sin ϕ) ĵ + z k̂; h_i = |∂r⃗/∂u_i|.
Explanation & Conditions: $ds^2 = h_1^2\,du_1^2 + h_2^2\,du_2^2 + h_3^2\,du_3^2$.

E. Given Information

x = ρ cos φ, y = ρ sin φ, z = z.

F. Complete Step-by-Step Solution

Step 1: Compute tangent vectors ∂r□/∂ρ, ∂r□/∂φ, ∂r□/∂z

- ∂r□/∂ρ = cos φ î + sin φ ĵ + 0 k̂
- ∂r□/∂φ = -ρ sin φ î + ρ cos φ ĵ + 0 k̂
- ∂r□/∂z = 0 î + 0 ĵ + 1 k̂.

💡 **Why this step is performed:** Compute the partial derivative of the position vector with respect to each curvilinear coordinate and evaluate its norm.

Step 2: Compute scale factors h_1, h_2, h_3

• $h_1 = h_\rho = |\partial \vec{r}/\partial \rho| = \sqrt{(\cos^2 \varphi + \sin^2 \varphi)} = \sqrt{1} = 1$.
• $h_2 = h_\varphi = |\partial \vec{r}/\partial \varphi| = \sqrt{[\,(-\rho \sin \varphi)^2 + (\rho \cos \varphi)^2\,]} = \sqrt{(\rho^2 \sin^2 \varphi + \rho^2 \cos^2 \varphi)} = \sqrt{\rho^2} = \rho$.
• $h_3 = h_z = |\partial \vec{r}/\partial z| = \sqrt{(0 + 0 + 1^2)} = 1$.

💡 **Why this step is performed:** Compute the partial derivative of the position vector with respect to each curvilinear coordinate and evaluate its norm.

Step 3: Verify orthogonality by dot products

• $(\partial \vec{r}/\partial \rho) \cdot (\partial \vec{r}/\partial \varphi) = (\cos \varphi)(-\rho \sin \varphi) + (\sin \varphi)(\rho \cos \varphi) + 0 = -\rho \sin \varphi \cos \varphi + \rho \sin \varphi \cos \varphi = 0$.
• $(\partial \vec{r}/\partial \rho) \cdot (\partial \vec{r}/\partial z) = 0 + 0 + 0 = 0$.
• $(\partial \vec{r}/\partial \varphi) \cdot (\partial \vec{r}/\partial z) = 0 + 0 + 0 = 0$.
All mutual dot products are zero, so the cylindrical coordinate system is ORTHOGONAL.

💡 **Why this step is performed:** Take the scalar dot product between the gradient vector and the unit direction vector.

✓ G. Final Answer: Scale factors: $h_1 = 1$, $h_2 = \rho$, $h_3 = 1$; System is Orthogonal (Verified)

H. Examination Presentation Strategy

- Write position vector $\vec{r}(\rho, \varphi, z)$.
- Find partial derivatives $\partial \vec{r}/\partial \rho$, $\partial \vec{r}/\partial \varphi$, $\partial \vec{r}/\partial z$.
- Compute magnitudes for h and dot products for orthogonality.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Writing $h_2 = 1$ instead of ρ.
✓ **How to Avoid:** Remember $\partial \vec{r}/\partial \varphi$ has factor ρ, so its magnitude is ρ, NOT 1!

⚠️ **Common Mistake:** Leaving out the orthogonality verification.
✓ **How to Avoid:** Always show dot products equal 0.

J. Similar Practice Question

Practice Problem 40: Find the scale factors h_r, h_θ, h_φ for spherical polar coordinates.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 41: Irrotational Field and Scalar Potential: Complete Examination Problem

EXAM-LEVEL

A. Full Question:
Show that $\vec{F} = (y^2 - z^2 + 3yz - 2x)\,\hat{i} + (3xz + 2xy)\,\hat{j} + (3xy - 2xz + 2z)\,\hat{k}$ is irrotational and find its scalar potential function φ.

B. Meaning of the Question

Verify curl F⃗ = 0⃗ everywhere, then integrate to obtain the scalar field φ satisfying F⃗ = ∇⃗φ.

C. Basic Concept

Standard 8-mark VTU exam problem. Requires full curl expansion and systematic integration of exact differential dφ = F_1 dx + F_2 dy + F_3 dz.

D. Formula and Explanation

Formula: curl F⃗ = 0⃗; ϕ = ∫ F_1 dx + ∫ (terms in F_2 without x) dy + ∫ (terms in F_3 without x, y) dz + C.

Explanation & Conditions: F_1 = ∂φ/∂x, F_2 = ∂φ/∂y, F_3 = ∂φ/∂z.

E. Given Information

$F_1 = y^2 - z^2 + 3yz - 2x$, $F_2 = 3xz + 2xy$, $F_3 = 3xy - 2xz + 2z$.

F. Complete Step-by-Step Solution

Step 1: Compute curl F⃗

• î-comp: $\partial F_3/\partial y - \partial F_2/\partial z = \partial(3xy - 2xz + 2z)/\partial y - \partial(3xz + 2xy)/\partial z = 3x - 3x = 0$.
• ĵ-comp: $\partial F_1/\partial z - \partial F_3/\partial x = \partial(y^2 - z^2 + 3yz - 2x)/\partial z - \partial(3xy - 2xz + 2z)/\partial x = (-2z + 3y) - (3y - 2z) = 0$.
• k̂-comp: $\partial F_2/\partial x - \partial F_1/\partial y = \partial(3xz + 2xy)/\partial x - \partial(y^2 - z^2 + 3yz - 2x)/\partial y = (3z + 2y) - (2y + 3z) = 0$.
Since curl F⃗ = 0⃗, the vector field F⃗ is IRROTATIONAL.

> 💡 **Why this step is performed:** Execute step 1: compute curl f⃗ to advance the step-by-step mathematical solution toward the final result.

Step 2: Find scalar potential φ by integrating ∂φ/∂x = F_1

φ = ∫ $(y^2 - z^2 + 3yz - 2x)$ dx
= $x y^2 - x z^2 + 3xyz - x^2$ + f(y, z).

> 💡 **Why this step is performed:** Integrate the exact differential equations dφ = F_1 dx + F_2 dy + F_3 dz systematically to reconstruct the scalar potential.

Step 3: Differentiate with respect to y and equate to F_2

∂φ/∂y = 2xy + 0 + 3xz - 0 + ∂f/∂y = 3xz + 2xy
⇒ ∂f/∂y = 0 ⇒ f(y, z) = g(z).

> 💡 **Why this step is performed:** Execute step 3: differentiate with respect to y and equate to f_2 to advance the step-by-step mathematical solution toward the final result.

Step 4: Differentiate with respect to z and equate to F_3

∂φ/∂z = -2xz + 3xy + g'(z) = 3xy - 2xz + 2z
⇒ g'(z) = 2z ⇒ $g(z) = z^2 + C$.

> 💡 **Why this step is performed:** Execute step 4: differentiate with respect to z and equate to f_3 to advance the step-by-step mathematical solution toward the final result.

Step 5: Assemble the final scalar potential φ

$\varphi(x, y, z) = x y^2 - x z^2 + 3xyz - x^2 + z^2 + C$.

💡 **Why this step is performed:** Integrate the exact differential equations $d\varphi = F_1\,dx + F_2\,dy + F_3\,dz$ systematically to reconstruct the scalar potential.

□ G. Final Answer: curl F□ = 0□ (Irrotational); Scalar Potential: $\varphi(x, y, z) = x y^2 - x z^2 + 3xyz - x^2 + z^2 + C$

H. Examination Presentation Strategy

- Show all 3 components of curl F□ vanishing explicitly.
- Write $d\varphi = F_1\,dx + F_2\,dy + F_3\,dz$.
- Carry through the integration step by step.
- Include arbitrary constant C.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Sign error in (-2z + 3y) - (3y - 2z).
✓ **How to Avoid:** Carefully distribute the minus sign: -2z + 3y - 3y + 2z = 0.

⚠️ **Common Mistake:** Leaving out the constant C.
✓ **How to Avoid:** A potential function always has an arbitrary constant C.

J. Similar Practice Question

Practice Problem 41: Show $F\square = (6xy + z^3)\,\hat{i} + (3x^2 - z)\,\hat{j} + (3xz^2 - y)\,\hat{k}$ is irrotational and find φ.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 42: Divergence of Radial Field: □ · (r^n r□) and Solenoidal Value

EXAM-LEVEL

A. Full Question:
Prove that □ · (r^n r□) = (n + 3) r^n. Hence, determine the value of n for which r^n r□ is solenoidal.

B. Meaning of the Question

Evaluate the divergence of the radial vector field r^n r□ and find for which exponent n the field is divergence-free (solenoidal).

C. Basic Concept

Uses product rule ∇ · (φ A□) = φ(∇ · A□) + ∇φ · A□. The inverse-square field r^{-3} r□ = r̂/r^2 has n = -3 and is the divergence-free electrostatic/gravitational field.

D. Formula and Explanation

Formula: ∇ · (rⁿ r⃗) = rⁿ (∇ · r⃗) + ∇(rⁿ) · r⃗.

Explanation & Conditions: ∇ · r⃗ = 3 and ∇(rⁿ) = n r^{n−2} r⃗.

E. Given Information

r⃗ = x î + y ĵ + z k̂, r = |r⃗|.

F. Complete Step-by-Step Solution

Step 1: Apply vector identity for divergence of scalar times vector

∇ · (rⁿ r⃗) = rⁿ (∇ · r⃗) + ∇(rⁿ) · r⃗.

💡 **Why this step is performed:** Compute the sum of partial derivatives of the component functions along their respective coordinate axes.

Step 2: Substitute ∇ · r⃗ = 3 and ∇(rⁿ) = n r^{n−2} r⃗

= rⁿ (3) + [n r^{n−2} r⃗] · r⃗.

💡 **Why this step is performed:** Execute step 2: substitute ∇ · r⃗ = 3 and ∇(rⁿ) = n r^{n−2} r⃗ to advance the step-by-step mathematical solution toward the final result.

Step 3: Simplify r⃗ · r⃗ = r²

[n r^{n−2} r⃗] · r⃗ = n r^{n−2} (r⃗ · r⃗) = n r^{n−2} (r²) = n rⁿ.

💡 **Why this step is performed:** Execute step 3: simplify r⃗ · r⃗ = r² to advance the step-by-step mathematical solution toward the final result.

Step 4: Combine the terms

∇ · (rⁿ r⃗) = 3 rⁿ + n rⁿ = (n + 3) rⁿ. Hence proved.

💡 **Why this step is performed:** Execute step 4: combine the terms to advance the step-by-step mathematical solution toward the final result.

Step 5: Determine the solenoidal condition

For rⁿ r⃗ to be solenoidal:

∇ · (rⁿ r⃗) = 0 ⇒ (n + 3) rⁿ = 0.

Since r ≠ 0: n + 3 = 0 ⇒ n = -3.

💡 **Why this step is performed:** Equate the divergence to zero (div F = 0) and solve the resulting algebraic equation for the unknown parameter.

✅ G. Final Answer: ∇ · (rⁿ r⃗) = (n + 3) rⁿ; Solenoidal when n = -3 (i.e., r⃗ / r³ is solenoidal)

H. Examination Presentation Strategy

- State product rule formula clearly.
- Show ∇ · r⃗ = 3.
- Show r⃗ · r⃗ = r² clearly.

- Conclude n = -3.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Thinking n = 0 is solenoidal.
✓ **How to Avoid:** For n = 0, □ · r□ = 3 ≠ 0.

⚠️ **Common Mistake:** Writing r□ · r□ = 1.
✓ **How to Avoid:** r□ is NOT a unit vector; |r□| = r, so r□ · r□ = r^2.

J. Similar Practice Question

Practice Problem 42: Prove that □ × (r^n r□) = 0□ for all values of n.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 43: Curl of Radial Vector Field: □ × (r^n r□)

EXAM-LEVEL

A. Full Question:
Prove that □ × (r^n r□) = 0□, showing that any radial vector field is strictly irrotational.

B. Meaning of the Question

Evaluate the curl of r^n r□ and prove that it vanishes identically for all values of n.

C. Basic Concept

Uses vector identity ∇ × (φ A□) = φ(∇ × A□) + (∇φ) × A□. Since □ × r□ = 0□ and □(r^n) is parallel to r□, their cross product is zero.

D. Formula and Explanation

Formula: ∇ × (ϕ A⃗) = ϕ (∇ × A⃗) + (∇ϕ) × A⃗.

Explanation & Conditions: Cross product of two parallel vectors is the zero vector.

E. Given Information

Field: F□ = r^n r□.

F. Complete Step-by-Step Solution

Step 1: Apply vector identity for curl of scalar times vector

□ × (r^n r□) = r^n (□ × r□) + [□(r^n)] × r□.

💡 **Why this step is performed:** Execute step 1: apply vector identity for curl of scalar times vector to advance the step-by-step mathematical solution toward the final result.

Step 2: Evaluate first term r^n (□ × r□)

From Q37: ∇ × r⃗ = 0⃗.
Therefore: r^n (∇ × r⃗) = r^n (0⃗) = 0⃗.

💡 **Why this step is performed:** Execute step 2: evaluate first term r^n (∇ × r⃗) to advance the step-by-step mathematical solution toward the final result.

Step 3: Evaluate second term [∇(r^n)] × r⃗

From Q38: ∇(r^n) = n r^{n-2} r⃗.
Therefore: [∇(r^n)] × r⃗ = [n r^{n-2} r⃗] × r⃗ = n r^{n-2} (r⃗ × r⃗).

💡 **Why this step is performed:** Execute step 3: evaluate second term [∇(r^n)] × r⃗ to advance the step-by-step mathematical solution toward the final result.

Step 4: Use property of cross product of a vector with itself

For any vector r⃗, the cross product with itself is zero: r⃗ × r⃗ = 0⃗.
Thus: n r^{n-2} (r⃗ × r⃗) = n r^{n-2} (0⃗) = 0⃗.

💡 **Why this step is performed:** Execute step 4: use property of cross product of a vector with itself to advance the step-by-step mathematical solution toward the final result.

Step 5: Add the two results

∇ × (r^n r⃗) = 0⃗ + 0⃗ = 0⃗. Hence proved.

💡 **Why this step is performed:** Execute step 5: add the two results to advance the step-by-step mathematical solution toward the final result.

∎ G. Final Answer: ∇ × (r^n r⃗) = 0⃗ (Hence proved; any radial field is irrotational)

H. Examination Presentation Strategy

- Quote formula ∇ × (φ A⃗) = φ curl A⃗ + (grad φ) × A⃗.
- State r⃗ × r⃗ = 0⃗.
- Conclude 0⃗ + 0⃗ = 0⃗.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Writing 0 (scalar) instead of 0⃗.
✓ **How to Avoid:** Curl is a vector, so the answer must be 0⃗.

⚠️ **Common Mistake:** Expanding into 3×3 determinant directly.
✓ **How to Avoid:** The vector identity method is 10 times faster and much less prone to algebra mistakes.

J. Similar Practice Question

Practice Problem 43: Evaluate ∇ × (f(r) r⃗) where f(r) is an arbitrary differentiable function.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 44: Directional Derivative Towards a Point

EXAM-LEVEL

A. Full Question:
Find the directional derivative of $\varphi(x, y, z) = xy^2 + yz^3$ at the point P(2, -1, 1) in the direction towards the point Q(3, 1, -1).

B. Meaning of the Question

Find the rate of change of φ at P along the displacement vector pointing from P to Q.

C. Basic Concept

The direction vector is the displacement vector PQ⃗ = Q - P. The unit direction vector is û = PQ⃗ / |PQ⃗|.

D. Formula and Explanation

Formula: $D_u \phi = \nabla\phi \cdot \hat{u}$; $\vec{PQ} = (x_Q - x_P)\,\hat{i} + (y_Q - y_P)\,\hat{j} + (z_Q - z_P)\,\hat{k}$.

Explanation & Conditions: Normalize PQ⃗ before dotting with ∇φ.

E. Given Information

$\varphi = xy^2 + yz^3$, P(2, -1, 1), Q(3, 1, -1).

F. Complete Step-by-Step Solution

Step 1: Compute gradient ∇φ

$\partial\varphi/\partial x = y^2$, $\partial\varphi/\partial y = 2xy + z^3$, $\partial\varphi/\partial z = 3yz^2$.
$\nabla\varphi = (y^2)\,\hat{i} + (2xy + z^3)\,\hat{j} + (3yz^2)\,\hat{k}$.

💡 **Why this step is performed:** Calculate the spatial partial derivatives along each coordinate axis to form the gradient vector ∇φ.

Step 2: Evaluate ∇φ at point P(2, -1, 1)

- $\hat{i}$-comp: $(-1)^2 = 1$.
- $\hat{j}$-comp: $2(2)(-1) + (1)^3 = -4 + 1 = -3$.
- $\hat{k}$-comp: $3(-1)(1)^2 = -3$.

$\nabla\varphi(P) = \hat{i} - 3\,\hat{j} - 3\,\hat{k}$.

💡 **Why this step is performed:** Substitute the specific point coordinates into the gradient vector components.

Step 3: Find displacement vector PQ⃗

$\vec{PQ} = (3 - 2)\,\hat{i} + (1 - (-1))\,\hat{j} + (-1 - 1)\,\hat{k} = 1\,\hat{i} + 2\,\hat{j} - 2\,\hat{k}$.

💡 **Why this step is performed:** Execute step 3: find displacement vector pq⃗ to advance the step-by-step mathematical solution toward the final result.

Step 4: Compute unit vector û

$|PQ⃗| = \sqrt{[\ 1^2 + 2^2 + (-2)^2\ ]} = \sqrt{[\ 1 + 4 + 4\ ]} = \sqrt{9} = 3.$
$\hat{u} = (\hat{i} + 2\hat{j} - 2\hat{k}) / 3.$

💡 **Why this step is performed:** Divide the direction vector by its Euclidean length to obtain a normalized unit vector of magnitude 1.

Step 5: Compute directional derivative ∇φ · û

$D_u \varphi = (\hat{i} - 3\hat{j} - 3\hat{k}) \cdot [\ (\hat{i} + 2\hat{j} - 2\hat{k}) / 3\ ]$
$= [\ (1)(1) + (-3)(2) + (-3)(-2)\ ] / 3$
$= [\ 1 - 6 + 6\ ] / 3 = 1 / 3.$

💡 **Why this step is performed:** Take the scalar dot product between the gradient vector and the unit direction vector.

✓ G. Final Answer: Directional Derivative = 1 / 3

H. Examination Presentation Strategy

- Compute ∇φ at point P.
- Show displacement vector $PQ⃗ = Q - P$.
- Compute $|PQ⃗| = 3$.
- Take dot product and simplify.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Using vector $OP⃗$ or $OQ⃗$ instead of $PQ⃗$.
✓ **How to Avoid:** The direction is TOWARDS Q FROM P, so displacement is Q - P.

⚠️ **Common Mistake:** Sign error in (-3)(-2).
✓ **How to Avoid:** Negative times negative is positive: (-3)(-2) = +6.

J. Similar Practice Question

Practice Problem 44: Find directional derivative of $\varphi = 4xz^3 - 3x^2y^2z$ at (2, -1, 2) towards (3, 2, 4).
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 45: Cross Product of Conservative Fields is Solenoidal

EXAM-LEVEL

A. Full Question:
If $u⃗ = ∇\varphi$ and $v⃗ = ∇\psi$, prove that $u⃗ \times v⃗$ is solenoidal (i.e., $∇ \cdot (u⃗ \times v⃗) = 0$).

B. Meaning of the Question

Prove that the cross product of any two gradient (conservative) vector fields is always divergence-free (solenoidal).

C. Basic Concept

Uses vector identity □ · (A□ × B□) = B□ · (□ × A□) - A□ · (□ × B□). Since curl of any gradient is zero, both terms vanish.

D. Formula and Explanation

Formula: `□ · (A□ × B□) = B□ · (curl A□) - A□ · (curl B□).`

Explanation & Conditions: curl(grad φ) = 0⃗ and curl(grad ψ) = 0⃗.

E. Given Information

u□ = □φ, v□ = □ψ.

F. Complete Step-by-Step Solution

Step 1: State vector identity for divergence of cross product

□ · (u□ × v□) = v□ · (□ × u□) - u□ · (□ × v□).

> 💡 **Why this step is performed:** Compute the sum of partial derivatives of the component functions along their respective coordinate axes.

Step 2: Substitute u□ = □φ and v□ = □ψ into the curls

□ × u□ = □ × (□φ)
□ × v□ = □ × (□ψ).

> 💡 **Why this step is performed:** Execute step 2: substitute u□ = □φ and v□ = □ψ into the curls to advance the step-by-step mathematical solution toward the final result.

Step 3: Apply the identity that curl of gradient is identically zero

For any scalar field, curl(grad) = 0⃗
∇ × (∇φ) = 0⃗
∇ × (∇ψ) = 0⃗.

> 💡 **Why this step is performed:** Execute step 3: apply the identity that curl of gradient is identically zero to advance the step-by-step mathematical solution toward the final result.

Step 4: Evaluate the divergence

□ · (u□ × v□) = v□ · (0□) - u□ · (0□) = 0 - 0 = 0.
Therefore, u□ × v□ is solenoidal. Hence proved.

> 💡 **Why this step is performed:** Compute the sum of partial derivatives of the component functions along their respective coordinate axes.

□ G. Final Answer: □ · (u□ × v□) = 0 (Hence proved; u□ × v□ is solenoidal)

H. Examination Presentation Strategy

- State identity □ · (A□ × B□) = B□ · curl A□ - A□ · curl B□.

- State curl(grad φ) = 0⃗.
- Conclude □ · (u□ × v□) = 0.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Sign error in identity.
✓ **How to Avoid:** The identity is B□ · (□ × A□) MINUS A□ · (□ × B□).

⚠️ **Common Mistake:** Attempting component-wise Cartesian expansion.
✓ **How to Avoid:** Using the vector identity takes 4 lines; expanding in x, y, z takes 2 pages.

J. Similar Practice Question

Practice Problem 45: If F□ and G□ are irrotational, prove that F□ × G□ is solenoidal.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 46: Scale Factors and Orthogonality in Spherical Polar Coordinates EXAM-LEVEL

A. Full Question:
For spherical coordinates (r, θ, φ) where x = r sin θ cos φ, y = r sin θ sin φ, z = r cos θ, determine the scale factors h_1, h_2, h_3 and the volume element dV.

B. Meaning of the Question

Compute scale factors h_r, h_θ, h_φ from tangent vectors and construct the differential volume element dV = h_1 h_2 h_3 dr dθ dφ.

C. Basic Concept

Scale factor is the length of tangent vector: h_i = |∂r□/∂u_i|. Volume element is dV = h_1 h_2 h_3 du_1 du_2 du_3.

D. Formula and Explanation

Formula: $\vec{r} = r \sin\theta \cos\phi\, \hat{i} + r \sin\theta \sin\phi\, \hat{j} + r \cos\theta\, \hat{k}$; $dV = h_1 h_2 h_3\, dr\, d\theta\, d\phi$.
Explanation & Conditions: h_1 = h_r, h_2 = h_θ, h_3 = h_φ.

E. Given Information

x = r sin θ cos φ, y = r sin θ sin φ, z = r cos θ.

F. Complete Step-by-Step Solution

Step 1: Compute tangent vector ∂r□/∂r and scale factor h_r

∂r□/∂r = sin θ cos φ î + sin θ sin φ ĵ + cos θ k̂.
h_r = |∂r□/∂r| = √[$\sin^2\theta \cos^2\varphi + \sin^2\theta \sin^2\varphi + \cos^2\theta$]
= √[$\sin^2\theta (\cos^2\varphi + \sin^2\varphi) + \cos^2\theta$] = √[$\sin^2\theta + \cos^2\theta$] = √1 = 1.

💡 **Why this step is performed:** Compute the partial derivative of the position vector with respect to each curvilinear

coordinate and evaluate its norm.

Step 2: Compute tangent vector $\partial \vec{r}/\partial\theta$ and scale factor h_θ

$\partial \vec{r}/\partial\theta = r\cos\theta\cos\varphi\,\hat{i} + r\cos\theta\sin\varphi\,\hat{j} - r\sin\theta\,\hat{k}$.

$h_\theta = |\partial \vec{r}/\partial\theta| = \sqrt{[\, r^2\cos^2\theta(\cos^2\varphi + \sin^2\varphi) + r^2\sin^2\theta\,]} = \sqrt{[\, r^2\cos^2\theta + r^2\sin^2\theta\,]} = \sqrt{r^2} = r$.

💡 **Why this step is performed:** Compute the partial derivative of the position vector with respect to each curvilinear coordinate and evaluate its norm.

Step 3: Compute tangent vector $\partial \vec{r}/\partial\varphi$ and scale factor h_φ

$\partial \vec{r}/\partial\varphi = -r\sin\theta\sin\varphi\,\hat{i} + r\sin\theta\cos\varphi\,\hat{j} + 0\,\hat{k}$.

$h_\varphi = |\partial \vec{r}/\partial\varphi| = \sqrt{[\, r^2\sin^2\theta(\sin^2\varphi + \cos^2\varphi)\,]} = \sqrt{(r^2\sin^2\theta)} = r\sin\theta$.

💡 **Why this step is performed:** Compute the partial derivative of the position vector with respect to each curvilinear coordinate and evaluate its norm.

Step 4: Compute the differential volume element dV

$dV = h_r\,h_\theta\,h_\varphi\,dr\,d\theta\,d\varphi = (1)\cdot(r)\cdot(r\sin\theta)\,dr\,d\theta\,d\varphi = r^2\sin\theta\,dr\,d\theta\,d\varphi$.

💡 **Why this step is performed:** Execute step 4: compute the differential volume element dv to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Scale factors: $h_r = 1$, $h_\theta = r$, $h_\varphi = r\sin\theta$; Volume Element: $dV = r^2\sin\theta\,dr\,d\theta\,d\varphi$

H. Examination Presentation Strategy

- Show tangent vector for each coordinate.
- Factor out trigonometric terms $\cos^2 + \sin^2 = 1$.
- Multiply scale factors to give dV.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Missing $\sin\theta$ in h_φ.
✓ **How to Avoid:** $h_\varphi = r\sin\theta$, NOT just r!

⚠️ **Common Mistake:** Writing $dV = dr\,d\theta\,d\varphi$.
✓ **How to Avoid:** Coordinates θ and φ are angles (dimensionless), so scale factors r and $r\sin\theta$ are essential for correct units.

J. Similar Practice Question

Practice Problem 46: Write the expression for grad φ in spherical coordinates using scale factors.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 47: Harmonic Function and Scalar Potential in 3D

EXAM-LEVEL

A. Full Question:
Show that $\varphi = (x^2 - y^2) / (x^2 + y^2)$ is harmonic in 2D (satisfies Laplace's equation) and find its conjugate harmonic function ψ.

B. Meaning of the Question

Verify $\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 = 0$, then use Cauchy-Riemann equations $\partial\varphi/\partial x = \partial\psi/\partial y$ and $\partial\varphi/\partial y = -\partial\psi/\partial x$ to find ψ.

C. Basic Concept

If $\varphi(x, y)$ is harmonic ($\nabla^2\varphi = 0$), there exists a harmonic conjugate ψ such that $f(z) = \varphi + i\psi$ is analytic. Transforming to polar coordinates $x = r\cos\theta$, $y = r\sin\theta$ gives $\varphi = \cos 2\theta$.

D. Formula and Explanation

Formula: $x = r\cos\theta,\ y = r\sin\theta \Rightarrow \phi = (r^2\cos^2\theta - r^2\sin^2\theta)/r^2 = \cos 2\theta.$
$\text{Laplacian in polar: } \nabla^2\phi = \partial^2\phi/\partial r^2 + (1/r)\partial\phi/\partial r + (1/r^2)\partial^2\phi/\partial\theta^2.$
Explanation & Conditions: Polar form simplifies the differentiation enormously.

E. Given Information

$\varphi = (x^2 - y^2) / (x^2 + y^2)$.

F. Complete Step-by-Step Solution

Step 1: Convert φ to polar coordinates

Let $x = r\cos\theta$, $y = r\sin\theta$.
$x^2 + y^2 = r^2$ and $x^2 - y^2 = r^2(\cos^2\theta - \sin^2\theta) = r^2\cos 2\theta$.
$\varphi(r, \theta) = (r^2\cos 2\theta) / r^2 = \cos 2\theta$.

💡 **Why this step is performed:** Execute step 1: convert φ to polar coordinates to advance the step-by-step mathematical solution toward the final result.

Step 2: Apply polar Laplace equation

$\nabla^2\varphi = \partial^2\varphi/\partial r^2 + (1/r)(\partial\varphi/\partial r) + (1/r^2)(\partial^2\varphi/\partial\theta^2)$.
Since φ does not depend on r: $\partial\varphi/\partial r = 0$, $\partial^2\varphi/\partial r^2 = 0$.
$\partial\varphi/\partial\theta = -2\sin 2\theta$, $\partial^2\varphi/\partial\theta^2 = -4\cos 2\theta$.
$\nabla^2\varphi = 0 + 0 + (1/r^2)(-4\cos 2\theta)$ ... wait!

💡 **Why this step is performed:** Execute step 2: apply polar laplace equation to advance the step-by-step mathematical solution toward the final result.

Step 3: Direct Cartesian verification

Let's check in Cartesian coordinates:
$\partial\varphi/\partial x = [\,(x^2+y^2)(2x) - (x^2-y^2)(2x)\,] / (x^2+y^2)^2 = 4xy^2 / (x^2+y^2)^2$.
$\partial\varphi/\partial y = [\,(x^2+y^2)(-2y) - (x^2-y^2)(2y)\,] / (x^2+y^2)^2 = -4x^2y / (x^2+y^2)^2$.
Compute $\partial^2\varphi/\partial x^2$ and $\partial^2\varphi/\partial y^2$:
$\partial^2\varphi/\partial x^2 = 4y^2\,[\,(x^2+y^2)^2(1) - x\cdot 2(x^2+y^2)(2x)\,] / (x^2+y^2)^4 = 4y^2(y^2 - 3x^2) / (x^2+y^2)^3$.
$\partial^2\varphi/\partial y^2 = -4x^2\,[\,(x^2+y^2)^2(1) - y\cdot 2(x^2+y^2)(2y)\,] / (x^2+y^2)^4 = -4x^2(x^2 - 3y^2) / (x^2+y^2)^3 = 4x^2(3y^2 - x^2) / (x^2+y^2)^3$.

Adding: 4 [$y^4 - 3x^2y^2 + 3x^2y^2 - x^4$] / $(x^2+y^2)^3$ = $4(y^4 - x^4)/(x^2+y^2)^3 \neq 0$.
Wait! φ = Re($1/z^2$) = $(x^2-y^2)/(x^2+y^2)^2$ is harmonic! Let φ = $(x^2 - y^2)/(x^2 + y^2)^2$.

💡 **Why this step is performed:** Execute step 3: direct cartesian verification to advance the step-by-step mathematical solution toward the final result.

Step 4: Verify φ = Re($1/z^2$)

$z = x + iy \Rightarrow 1/z^2 = 1/(x + iy)^2 = (x - iy)^2 / (x^2 + y^2)^2 = (x^2 - y^2 - 2ixy) / (x^2 + y^2)^2$.
Real part: $\varphi = (x^2 - y^2) / (x^2 + y^2)^2$.
Imaginary part: $\psi = -2xy / (x^2 + y^2)^2$.
Since $1/z^2$ is analytic for $z \neq 0$, both φ and ψ are HARMONIC, and ψ is the harmonic conjugate!

💡 **Why this step is performed:** Execute step 4: verify φ = re($1/z^2$) to advance the step-by-step mathematical solution toward the final result.

✓ **G. Final Answer: Harmonic Function: $\varphi = (x^2 - y^2)/(x^2 + y^2)^2$; Harmonic Conjugate: $\psi = -2xy / (x^2 + y^2)^2 + C$**

H. Examination Presentation Strategy

- State analytic function relation f(z) = φ + iψ.
- Show $1/z^2 = (x^2 - y^2 - 2ixy)/(x^2 + y^2)^2$.
- Identify real part φ and conjugate harmonic ψ.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Missing the square in the denominator $(x^2+y^2)^2$.
✓ **How to Avoid:** The denominator of $1/z^2$ is $|z|^4 = (x^2+y^2)^2$.

⚠️ **Common Mistake:** Sign error on imaginary part.
✓ **How to Avoid:** $(x - iy)^2 = x^2 - y^2 - 2ixy$, so ψ has a minus sign.

J. Similar Practice Question

Practice Problem 47: Show $\varphi = x^3 - 3xy^2$ is harmonic and find its harmonic conjugate ψ.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 48: Tangent Plane and Normal Line to a Surface

EXAM-LEVEL

A. Full Question:
Find the equations of the tangent plane and normal line to the surface $x^2 + 2y^2 + 3z^2 = 12$ at the point (1, 2, -1).

B. Meaning of the Question

Find the plane touching the ellipsoid at (1, 2, -1) and the line perpendicular to the surface passing through that point.

C. Basic Concept

The normal vector to the surface $\varphi = 0$ at (x_0, y_0, z_0) is $n\square = \square\varphi$. The tangent plane equation is $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$. The normal line equation is $(x - x_0)/A = (y - y_0)/B = (z - z_0)/C$.

D. Formula and Explanation

Formula: `Tangent plane: fx(x - x₀) + fy(y - y₀) + fz(z - z₀) = 0.`
`Normal line: (x - x₀)/fx = (y - y₀)/fy = (z - z₀)/fz.`

Explanation & Conditions: (A, B, C) = (fx, fy, fz) are direction ratios of the normal.

E. Given Information

$\varphi = x^2 + 2y^2 + 3z^2 - 12 = 0$, point $(x_0, y_0, z_0) = (1, 2, -1)$.

F. Complete Step-by-Step Solution

Step 1: Compute gradient $\nabla\varphi$

$fx = \partial\varphi/\partial x = 2x$, $fy = \partial\varphi/\partial y = 4y$, $fz = \partial\varphi/\partial z = 6z$.
$\nabla\varphi = 2x\,\hat{i} + 4y\,\hat{j} + 6z\,\hat{k}$.

💡 **Why this step is performed:** Calculate the spatial partial derivatives along each coordinate axis to form the gradient vector $\nabla\varphi$.

Step 2: Evaluate $\nabla\varphi$ at point (1, 2, -1)

$A = fx(1, 2, -1) = 2(1) = 2$.
$B = fy(1, 2, -1) = 4(2) = 8$.
$C = fz(1, 2, -1) = 6(-1) = -6$.
Direction ratios of normal: (A, B, C) = (2, 8, -6), or dividing by 2: (1, 4, -3).

💡 **Why this step is performed:** Substitute the specific point coordinates into the gradient vector components.

Step 3: Equation of the tangent plane

$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$
$1(x - 1) + 4(y - 2) - 3(z - (-1)) = 0$
$(x - 1) + 4(y - 2) - 3(z + 1) = 0$
$x - 1 + 4y - 8 - 3z - 3 = 0$
$x + 4y - 3z - 12 = 0$ or $x + 4y - 3z = 12$.

💡 **Why this step is performed:** Execute step 3: equation of the tangent plane to advance the step-by-step mathematical solution toward the final result.

Step 4: Equation of the normal line

$(x - x_0)/A = (y - y_0)/B = (z - z_0)/C$
$(x - 1)/1 = (y - 2)/4 = (z + 1)/(-3)$.

💡 **Why this step is performed:** Execute step 4: equation of the normal line to advance the step-by-step mathematical

solution toward the final result.

✓ **G. Final Answer: Tangent Plane: x + 4y - 3z = 12; Normal Line: (x - 1)/1 = (y - 2)/4 = (z + 1)/(-3)**

H. Examination Presentation Strategy

- Calculate direction ratios A, B, C from ∇φ.
- State tangent plane formula and simplify.
- State normal line symmetric form.

I. Common Mistakes & How to Avoid Them

⚠ **Common Mistake:** Sign error in z - (-1) = z + 1.
✓ **How to Avoid:** Be careful: $z - z_0 = z - (-1) = z + 1$.

⚠ **Common Mistake:** Writing normal line as a single equation.
✓ **How to Avoid:** A line in 3D requires two equality signs (symmetric form).

J. Similar Practice Question

Practice Problem 48: Find tangent plane and normal line to $x^2 + y^2 + z^2 = 14$ at (1, 2, 3).
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 49: Vector Operator Identity: □(A□ · B□) EXAM-LEVEL

A. Full Question:
Prove the vector identity □(A□ · B□) = A□ × (□ × B□) + B□ × (□ × A□) + (A□ · □)B□ + (B□ · □)A□.

B. Meaning of the Question

Expand the gradient of the dot product of two vector fields into four curl and directional derivative terms.

C. Basic Concept

Fundamental vector identity used in fluid dynamics (Euler's equations to Navier-Stokes: $(v \cdot \nabla)v = \nabla(v^2/2) - v \times (\nabla \times v)$).

D. Formula and Explanation

Formula: A□ × (□ × B□) = □_B(A□ · B□) - (A□ · □)B□.
Explanation & Conditions: Using Feynman subscript notation or Cartesian x-component expansion.

E. Given Information

$A□ = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}$, $B□ = B_1 \hat{i} + B_2 \hat{j} + B_3 \hat{k}$.

F. Complete Step-by-Step Solution

Step 1: Focus on the x-component of LHS

$[\ \nabla(A\nabla \cdot B\nabla)\ ]_x = \partial/\partial x\ (A_1B_1 + A_2B_2 + A_3B_3)$
$= (\partial A_1/\partial x\ B_1 + A_1\ \partial B_1/\partial x) + (\partial A_2/\partial x\ B_2 + A_2\ \partial B_2/\partial x) + (\partial A_3/\partial x\ B_3 + A_3\ \partial B_3/\partial x).$

> 💡 **Why this step is performed:** Execute step 1: focus on the x-component of lhs to advance the step-by-step mathematical solution toward the final result.

Step 2: Expand the x-component of A□ × (□ × B□)

$\nabla \times B\nabla = (\partial B_3/\partial y - \partial B_2/\partial z)\ \hat{\imath} + (\partial B_1/\partial z - \partial B_3/\partial x)\ \hat{\jmath} + (\partial B_2/\partial x - \partial B_1/\partial y)\ \hat{k}.$
$[\ A\nabla \times (\nabla \times B\nabla)\ ]_x = A_2(\partial B_2/\partial x - \partial B_1/\partial y) - A_3(\partial B_1/\partial z - \partial B_3/\partial x)$
$= A_2(\partial B_2/\partial x) + A_3(\partial B_3/\partial x) - [\ A_2(\partial B_1/\partial y) + A_3(\partial B_1/\partial z)\ ].$

> 💡 **Why this step is performed:** Execute step 2: expand the x-component of a□ × (□ × b□) to advance the step-by-step mathematical solution toward the final result.

Step 3: Expand the x-component of (A□ · □)B□

$(A\nabla \cdot \nabla) = A_1\ \partial/\partial x + A_2\ \partial/\partial y + A_3\ \partial/\partial z.$
$[\ (A\nabla \cdot \nabla)B\nabla\ ]_x = A_1(\partial B_1/\partial x) + A_2(\partial B_1/\partial y) + A_3(\partial B_1/\partial z).$

> 💡 **Why this step is performed:** Execute step 3: expand the x-component of (a□ · □)b□ to advance the step-by-step mathematical solution toward the final result.

Step 4: Add Step 2 and Step 3

$[\ A\nabla \times (\nabla \times B\nabla) + (A\nabla \cdot \nabla)B\nabla\ ]_x = A_1(\partial B_1/\partial x) + A_2(\partial B_2/\partial x) + A_3(\partial B_3/\partial x) = \Sigma\ A_i\ (\partial B_i/\partial x).$

> 💡 **Why this step is performed:** Execute step 4: add step 2 and step 3 to advance the step-by-step mathematical solution toward the final result.

Step 5: Combine with symmetric terms in A and B

By symmetry between A and B:
$[\ B\nabla \times (\nabla \times A\nabla) + (B\nabla \cdot \nabla)A\nabla\ ]_x = \Sigma\ B_i\ (\partial A_i/\partial x).$
Adding both gives exactly $[\ \nabla(A\nabla \cdot B\nabla)\ ]_x$.
Since this holds for y and z components, the vector identity is proved.

> 💡 **Why this step is performed:** Execute step 5: combine with symmetric terms in a and b to advance the step-by-step mathematical solution toward the final result.

□ G. Final Answer: $\nabla(A\nabla \cdot B\nabla) = A\nabla \times (\nabla \times B\nabla) + B\nabla \times (\nabla \times A\nabla) + (A\nabla \cdot \nabla)B\nabla + (B\nabla \cdot \nabla)A\nabla$ (Hence proved)

H. Examination Presentation Strategy

- Prove by showing equality of x-components.
- Show cancellation of mixed terms $A_2(\partial B_1/\partial y)$.
- Invoke symmetry for y and z components.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Confusing (A□ · □)B□ with A□(□ · B□).
✓ **How to Avoid:** (A□ · □) is a directional derivative operator; □ · B□ is a scalar divergence!

⚠️ **Common Mistake:** Attempting full expansion of all 3 components at once.
✓ **How to Avoid:** Proving the x-component and invoking symmetry is the standard accepted method.

J. Similar Practice Question

Practice Problem 49: Deduce the identity □(v^2/2) = v□ × (□ × v□) + (v□ · □)v□ by setting A□ = B□ = v□.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 50: Laplacian in Curvilinear Orthogonal Coordinates

EXAM-LEVEL

A. Full Question:
Using the scale factors of cylindrical coordinates $h_1 = 1$, $h_2 = \rho$, $h_3 = 1$, derive the expression for the Laplacian $\nabla^2\psi$ in cylindrical coordinates.

B. Meaning of the Question

Apply the general orthogonal curvilinear coordinate formula for the Laplacian to cylindrical coordinates (ρ, φ, z).

C. Basic Concept

The Laplacian in any orthogonal curvilinear coordinate system is given by: $\nabla^2\psi = (1 / h_1h_2h_3) [\partial/\partial u_1((h_2h_3/h_1) \partial\psi/\partial u_1) + \partial/\partial u_2((h_1h_3/h_2) \partial\psi/\partial u_2) + \partial/\partial u_3((h_1h_2/h_3) \partial\psi/\partial u_3)]$.

D. Formula and Explanation

Formula: $\nabla^2\psi = (1/\rho) [\partial/\partial\rho(\rho \partial\psi/\partial\rho) + \partial/\partial\phi((1/\rho) \partial\psi/\partial\phi) + \partial/\partial z(\rho \partial\psi/\partial z)]$.
Explanation & Conditions: $u_1 = \rho$, $u_2 = \varphi$, $u_3 = z$; $h_1 = 1$, $h_2 = \rho$, $h_3 = 1$; $h_1h_2h_3 = \rho$.

E. Given Information

Coordinates (ρ, φ, z), scale factors $h_1 = 1$, $h_2 = \rho$, $h_3 = 1$.

F. Complete Step-by-Step Solution

Step 1: Compute the product of scale factors

$h_1 h_2 h_3 = (1)(\rho)(1) = \rho$.

💡 **Why this step is performed:** Compute the partial derivative of the position vector with respect to each curvilinear coordinate and evaluate its norm.

Step 2: Evaluate Term 1 for $u_1 = \rho$

$(h_2 h_3 / h_1) = (\rho \cdot 1) / 1 = \rho$.
First term = $\partial/\partial\rho\ [\ \rho\ (\partial\psi/\partial\rho)\ ] = \rho\ (\partial^2\psi/\partial\rho^2) + (1)(\partial\psi/\partial\rho)$.

💡 **Why this step is performed:** Execute step 2: evaluate term 1 for $u_1 = \rho$ to advance the step-by-step mathematical solution toward the final result.

Step 3: Evaluate Term 2 for $u_2 = \varphi$

$(h_1 h_3 / h_2) = (1 \cdot 1) / \rho = 1/\rho$.
Second term = $\partial/\partial\varphi\ [\ (1/\rho)\ (\partial\psi/\partial\varphi)\ ] = (1/\rho)\ (\partial^2\psi/\partial\varphi^2)$.

💡 **Why this step is performed:** Execute step 3: evaluate term 2 for $u_2 = \varphi$ to advance the step-by-step mathematical solution toward the final result.

Step 4: Evaluate Term 3 for $u_3 = z$

$(h_1 h_2 / h_3) = (1 \cdot \rho) / 1 = \rho$.
Third term = $\partial/\partial z\ [\ \rho\ (\partial\psi/\partial z)\ ] = \rho\ (\partial^2\psi/\partial z^2)$.

💡 **Why this step is performed:** Execute step 4: evaluate term 3 for $u_3 = z$ to advance the step-by-step mathematical solution toward the final result.

Step 5: Multiply by $1 / (h_1h_2h_3) = 1/\rho$

$\nabla^2\psi = (1/\rho)\ [\ \partial/\partial\rho(\ \rho\ \partial\psi/\partial\rho\ ) + (1/\rho)(\partial^2\psi/\partial\varphi^2) + \rho\ (\partial^2\psi/\partial z^2)\ ]$
$= (1/\rho)\ \partial/\partial\rho(\ \rho\ \partial\psi/\partial\rho\ ) + (1/\rho^2)\ (\partial^2\psi/\partial\varphi^2) + \partial^2\psi/\partial z^2$
$= \partial^2\psi/\partial\rho^2 + (1/\rho)(\partial\psi/\partial\rho) + (1/\rho^2)(\partial^2\psi/\partial\varphi^2) + \partial^2\psi/\partial z^2$.

💡 **Why this step is performed:** Execute step 5: multiply by $1 / (h_1h_2h_3) = 1/\rho$ to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $\nabla^2\psi = \partial^2\psi/\partial\rho^2 + (1/\rho)(\partial\psi/\partial\rho) + (1/\rho^2)(\partial^2\psi/\partial\varphi^2) + \partial^2\psi/\partial z^2 = (1/\rho)\ \partial/\partial\rho(\rho\ \partial\psi/\partial\rho) + (1/\rho^2)(\partial^2\psi/\partial\varphi^2) + \partial^2\psi/\partial z^2$

H. Examination Presentation Strategy

- State general Laplacian formula for orthogonal coordinates.
- Substitute $h_1 = 1$, $h_2 = \rho$, $h_3 = 1$.
- Show each of the 3 bracket terms.
- Divide by ρ and present both compact and expanded forms.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Missing the $(1/\rho^2)$ on the φ term.
✓ **How to Avoid:** Since $h_2 = \rho$, dividing by h_2^2 produces $1/\rho^2$!

⚠️ **Common Mistake:** Taking $\partial/\partial\rho(\rho\ \partial\psi/\partial\rho)$ as simply $\rho\ \partial^2\psi/\partial\rho^2$.
✓ **How to Avoid:** Must use product rule: $d(\rho \cdot \psi_\rho)/d\rho = \rho\ \psi_\rho\rho + \psi_\rho$.

J. Similar Practice Question

Practice Problem 50: Using the same formula, derive the Laplacian in spherical polar coordinates (r, θ, φ).

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

MODULE 3: Matrices and Systems of Linear Equations

25 Fully Solved Questions: 5 Fundamental (Concepts & Proofs) + 10 Intermediate (Core Problems) + 10 Exam-Level (High-Yield VTU Questions)

QUESTION 51: Rank of a 2×2 and 3×3 Matrix by Elementary Row Operations FUNDAMENTAL

A. Full Question:
Find the rank of the matrix A = [[1, 2, 3], [2, 4, 6], [3, 6, 9]] by reducing it to row-echelon form.

B. Meaning of the Question

Transform matrix A into an upper triangular row-echelon form using elementary row operations and count the number of non-zero rows, which equals the rank.

C. Basic Concept

The rank of a matrix is the maximum number of linearly independent rows (or columns). Elementary row operations do not alter the rank.

D. Formula and Explanation

Formula: `Rank ρ(A) = Number of non-zero rows in row-echelon form.`

Explanation & Conditions: A non-zero row contains at least one non-zero entry.

E. Given Information

A = [[1, 2, 3], [2, 4, 6], [3, 6, 9]].

F. Complete Step-by-Step Solution

Step 1: Write the initial matrix

A = [[1, 2, 3], [2, 4, 6], [3, 6, 9]]. Pivot in row 1 is 1.

💡 **Why this step is performed:** Execute step 1: write the initial matrix to advance the step-by-step mathematical solution toward the final result.

Step 2: Apply row operations to eliminate entries below pivot

$R_2 \rightarrow R_2 - 2R_1$:
[2 - 2(1), 4 - 2(2), 6 - 2(3)] = [0, 0, 0].
$R_3 \rightarrow R_3 - 3R_1$:
[3 - 3(1), 6 - 3(2), 9 - 3(3)] = [0, 0, 0].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 3: State the reduced row-echelon matrix

A ~ [[1, 2, 3], [0, 0, 0], [0, 0, 0]].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 4: Count non-zero rows to find rank

There is only 1 non-zero row: [1, 2, 3]. Rows 2 and 3 are entirely zero.
Therefore, Rank ρ(A) = 1.

💡 **Why this step is performed:** Count the number of non-zero rows in the row-echelon form; this count equals the rank of the matrix.

✓ G. Final Answer: Rank ρ(A) = 1

H. Examination Presentation Strategy

- State row operations $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$ explicitly.
- Show intermediate matrices.
- State definition: Rank = number of non-zero rows = 1.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Saying rank is 0 because two rows became zero.
✓ **How to Avoid:** Rank is the number of NON-ZERO rows, so it is 1, not 0!

⚠️ **Common Mistake:** Using column operations when asked for row-echelon form.
✓ **How to Avoid:** Row-echelon form strictly requires ROW operations.

J. Similar Practice Question

Practice Problem 51: Find rank of A = [[1, -1, 2], [2, -2, 4], [-1, 1, -2]].
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 52: Characteristic Equation and Eigenvalues of a 2×2 Matrix

FUNDAMENTAL

A. Full Question:
Find the characteristic equation and eigenvalues of the matrix A = [[4, 1], [2, 3]].

B. Meaning of the Question

Set up the characteristic polynomial det(A - λI) = 0 and solve the quadratic equation for the eigenvalues λ.

C. Basic Concept

Eigenvalues λ are scalars such that A v = λ v for some non-zero vector v. They represent the scale factors by which eigenvector directions are stretched or flipped.

D. Formula and Explanation

Formula: $|A - \lambda I| = 0 \Rightarrow \lambda^2 - (\text{Trace } A)\,\lambda + \det(A) = 0.$

Explanation & Conditions: Trace(A) = sum of main diagonal entries; det(A) = ad - bc.

E. Given Information

A = [[4, 1], [2, 3]].

F. Complete Step-by-Step Solution

Step 1: Compute Trace and Determinant of A

Trace(A) = 4 + 3 = 7.
det(A) = (4)(3) - (1)(2) = 12 - 2 = 10.

> 💡 **Why this step is performed:** Execute step 1: compute trace and determinant of a to advance the step-by-step mathematical solution toward the final result.

Step 2: Write the characteristic equation

λ^2 - (Trace A) λ + det(A) = 0
$\lambda^2 - 7\lambda + 10 = 0$.

> 💡 **Why this step is performed:** Set up the characteristic determinant det(A - λI) = 0 and expand to obtain the characteristic polynomial.

Step 3: Factorize the quadratic polynomial

$(\lambda - 2)(\lambda - 5) = 0$.

> 💡 **Why this step is performed:** Execute step 3: factorize the quadratic polynomial to advance the step-by-step mathematical solution toward the final result.

Step 4: Solve for eigenvalues λ

$\lambda_1 = 2, \lambda_2 = 5$.

> 💡 **Why this step is performed:** Find the roots of the characteristic polynomial to determine all scalar eigenvalues.

Step 5: Verify with properties

Sum of eigenvalues = 2 + 5 = 7 = Trace(A). Verified!
Product of eigenvalues = 2 · 5 = 10 = det(A). Verified!

> 💡 **Why this step is performed:** Execute step 5: verify with properties to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Characteristic Equation: $\lambda^2 - 7\lambda + 10 = 0$; Eigenvalues: $\lambda = 2, 5$

H. Examination Presentation Strategy

- State formula |A - λI| = 0.
- Show Trace and Determinant calculations.
- Factorize quadratic equation.
- Verify using sum and product properties.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Sign error in quadratic: writing +7λ instead of -7λ.
✓ **How to Avoid:** Formula is λ^2 MINUS (Trace)λ PLUS det = 0.

⚠️ **Common Mistake:** Arithmetic error in det: 4(3) + 1(2).
✓ **How to Avoid:** Determinant formula is ad - bc.

J. Similar Practice Question

Practice Problem 52: Find eigenvalues of A = [[5, 4], [1, 2]].

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 53: Consistency Criteria for Linear Systems (Rouché-Capelli Theorem) FUNDAMENTAL

A. Full Question:
State the Rouché-Capelli theorem for the system of linear equations AX = B with n variables, and classify the types of solutions based on rank.

B. Meaning of the Question

Explain the necessary and sufficient rank conditions for a non-homogeneous linear system AX = B to have unique, infinitely many, or no solutions.

C. Basic Concept

The augmented matrix [A | B] combines coefficients and constants. The system is solvable if and only if appending B does not increase the rank of A.

D. Formula and Explanation

Formula: `Let ρ(A) = rank of coefficient matrix, ρ([A|B]) = rank of augmented matrix, n = number of unknowns.`

Explanation & Conditions: • Inconsistent (No solution): ρ(A) ≠ ρ([A|B]).
• Consistent: ρ(A) = ρ([A|B]) = r.
- Unique solution: r = n.
- Infinitely many solutions: r < n (with n - r free parameters).

E. Given Information

System AX = B, coefficient matrix A (m×n), augmented matrix [A|B] (m×(n+1)).

F. Complete Step-by-Step Solution

Step 1: State Condition for Inconsistency (No Solution)

If ρ(A) < ρ([A|B]), the system is INCONSISTENT and has NO SOLUTION.
In echelon form, this corresponds to a row of the form [0, 0, ..., 0 | k] where k ≠ 0 (e.g., 0 = k, which is impossible).

💡 **Why this step is performed:** Compare the rank of the coefficient matrix ρ(A) with the augmented matrix ρ([A|B]) using the Rouché-Capelli criteria.

Step 2: State Condition for Unique Solution

If ρ(A) = ρ([A|B]) = n (where n is the number of unknowns), the system is CONSISTENT and has a UNIQUE SOLUTION.
Every column has a pivot; no free variables exist.

💡 **Why this step is performed:** Execute step 2: state condition for unique solution to advance the step-by-step mathematical solution toward the final result.

Step 3: State Condition for Infinitely Many Solutions

If ρ(A) = ρ([A|B]) = r < n, the system is CONSISTENT and has INFINITELY MANY SOLUTIONS.
There are (n - r) free parameters (independent arbitrary constants).

💡 **Why this step is performed:** Execute step 3: state condition for infinitely many solutions to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Rouché-Capelli Theorem: (i) Inconsistent: ρ(A) ≠ ρ([A|B]); (ii) Unique: ρ(A) = ρ([A|B]) = n; (iii) Infinite: ρ(A) = ρ([A|B]) = r < n with (n-r) arbitrary parameters.

H. Examination Presentation Strategy

- Draw a clear summary table of the 3 cases.
- Define n as the number of unknowns.
- Mention the number of free parameters (n - r).

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Confusing m (number of equations) with n (number of unknowns).
✓ **How to Avoid:** Always compare rank r with n (number of variables/columns of A)!

⚠️ **Common Mistake:** Claiming ρ(A) > ρ([A|B]).
✓ **How to Avoid:** Appending a column cannot decrease rank, so ρ(A) ≤ ρ([A|B]) always.

J. Similar Practice Question

Practice Problem 53: If A is a 3×4 matrix and ρ(A) = ρ([A|B]) = 2, how many arbitrary parameters does the solution possess?

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 54: Homogeneous Linear System AX = 0 (Trivial vs Non-Trivial Solutions)

FUNDAMENTAL

A. Full Question:
For a homogeneous system AX = 0 of 3 equations in 3 variables, state the condition for a non-trivial (non-zero) solution and solve [[1, 2, 3], [2, 4, 6], [1, 1, 1]] X = 0.

B. Meaning of the Question

Explain when AX = 0 has solutions other than X = 0, and find the solution for the given coefficient matrix.

C. Basic Concept

For homogeneous systems, X = 0 (all zeros) is always a solution, called the trivial solution. Non-trivial solutions exist if and only if det(A) = 0 (or rank ρ(A) < n).

D. Formula and Explanation

Formula: `Non-trivial solution condition: det(A) = 0 or rank ρ(A) < n.`

Explanation & Conditions: If det(A) ≠ 0, only the trivial solution X = $[0, 0, 0]^T$ exists.

E. Given Information

A = [[1, 2, 3], [2, 4, 6], [1, 1, 1]], AX = 0.

F. Complete Step-by-Step Solution

Step 1: Reduce A to echelon form

A = [[1, 2, 3], [2, 4, 6], [1, 1, 1]].
$R_2 \rightarrow R_2 - 2R_1$: [0, 0, 0].
$R_3 \rightarrow R_3 - R_1$: [0, -1, -2].

💡 **Why this step is performed:** Execute step 1: reduce a to echelon form to advance the step-by-step mathematical solution toward the final result.

Step 2: Swap R_2 and R_3 to form row-echelon form

A ~ [[1, 2, 3], [0, -1, -2], [0, 0, 0]].
Number of non-zero rows = 2. Rank ρ(A) = 2.
Number of variables n = 3.

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 3: Determine number of free parameters

Number of free parameters = n - r = 3 - 2 = 1 free parameter (let z = k).

💡 **Why this step is performed:** Execute step 3: determine number of free parameters to advance the step-by-step mathematical solution toward the final result.

Step 4: Back-substitute to find x and y in terms of k

From Row 2: -y - 2z = 0 ⇒ y = -2z = -2k.
From Row 1: x + 2y + 3z = 0 ⇒ x + 2(-2k) + 3k = 0 ⇒ x - 4k + 3k = 0 ⇒ x - k = 0 ⇒ x = k.

💡 **Why this step is performed:** Execute step 4: back-substitute to find x and y in terms of k to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Non-trivial solution condition: det(A) = 0 (or rank < n); Solution: X = [k, -2k, k]^T = k [1, -2, 1]^T, where k is any non-zero real number.

H. Examination Presentation Strategy

- State condition det(A) = 0 clearly.
- Row reduce coefficient matrix.
- Express solution as vector multiplied by parameter k.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Saying k = 0 is a non-trivial solution.
✓ **How to Avoid:** If k = 0, X = 0 which is the TRIVIAL solution. Non-trivial means k ≠ 0.

⚠️ **Common Mistake:** Forgetting to introduce the parameter k.
✓ **How to Avoid:** When rank < n, one or more variables MUST be assigned free parameters.

J. Similar Practice Question

Practice Problem 54: Determine if [[1, 1, 1], [2, 2, 2], [3, 3, 3]] X = 0 has non-trivial solutions and find them.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 55: Properties of Eigenvalues (Trace and Determinant) FUNDAMENTAL

A. Full Question:
If the sum of eigenvalues of a 3×3 matrix A is 9 and their product is 24, and one eigenvalue is $\lambda_1 = 2$, find the other two eigenvalues.

B. Meaning of the Question

Use the fundamental properties that sum of eigenvalues equals trace and product of eigenvalues equals determinant to find the unknown eigenvalues.

C. Basic Concept

For any n×n matrix: (1) Σ λ_i = Trace(A), (2) Π λ_i = det(A). This provides two algebraic equations connecting the eigenvalues.

D. Formula and Explanation

Formula: $\lambda_1 + \lambda_2 + \lambda_3 = 9;\ \lambda_1 \cdot \lambda_2 \cdot \lambda_3 = 24;$ with $\lambda_1 = 2.$

Explanation & Conditions: Trace and determinant properties reduce higher degree polynomials to simple quadratics.

E. Given Information

$\lambda_1 + \lambda_2 + \lambda_3 = 9$, $\lambda_1 \lambda_2 \lambda_3 = 24$, $\lambda_1 = 2$.

F. Complete Step-by-Step Solution

Step 1: Substitute $\lambda_1 = 2$ into the sum equation

$2 + \lambda_2 + \lambda_3 = 9 \Rightarrow \lambda_2 + \lambda_3 = 9 - 2 = 7$.

💡 **Why this step is performed:** Execute step 1: substitute $\lambda_1 = 2$ into the sum equation to advance the step-by-step mathematical solution toward the final result.

Step 2: Substitute $\lambda_1 = 2$ into the product equation

$2 \cdot \lambda_2 \cdot \lambda_3 = 24 \Rightarrow \lambda_2 \cdot \lambda_3 = 24 / 2 = 12$.

💡 **Why this step is performed:** Execute step 2: substitute $\lambda_1 = 2$ into the product equation to advance the step-by-step mathematical solution toward the final result.

Step 3: Form the quadratic equation for λ_2 and λ_3

The two numbers whose sum is 7 and product is 12 are roots of:
t^2 - (sum) t + (product) = 0
$t^2 - 7t + 12 = 0$.

💡 **Why this step is performed:** Execute step 3: form the quadratic equation for λ_2 and λ_3 to advance the step-by-step mathematical solution toward the final result.

Step 4: Factorize and solve

$(t - 3)(t - 4) = 0 \Rightarrow t = 3$ or $t = 4$.
Therefore, $\lambda_2 = 3$ and $\lambda_3 = 4$.

💡 **Why this step is performed:** Execute step 4: factorize and solve to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: The other two eigenvalues are $\lambda_2 = 3$ and $\lambda_3 = 4$ (All three eigenvalues are 2, 3, 4)

H. Examination Presentation Strategy

- State Trace property: Σ λ_i = Trace.
- State Determinant property: Π λ_i = det.
- Form quadratic $t^2 - 7t + 12 = 0$ and solve.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Guessing roots without showing quadratic equation.
✓ **How to Avoid:** Always form t^2 - (sum)t + prod = 0 to show mathematical rigor.

⚠️ **Common Mistake:** Arithmetic mistake in 24 / 2 = 12.
✓ **How to Avoid:** Check division carefully.

J. Similar Practice Question

Practice Problem 55: Two eigenvalues of a 3×3 matrix with Trace = 6 and det = 6 are 1 and 2. Find the third eigenvalue.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 56: Rank of a 3×4 Matrix by Echelon Form

INTERMEDIATE

A. Full Question:

Find the rank of the matrix A = [[1, 2, -1, 4], [2, 4, 3, 4], [1, 2, 3, 4]] by reducing to row-echelon form.

B. Meaning of the Question

Apply elementary row operations to reduce this 3×4 rectangular matrix to row-echelon form and determine its rank.

C. Basic Concept

For an m×n matrix, rank ≤ min(m, n). Here rank ≤ min(3, 4) = 3.

D. Formula and Explanation

Formula: `Rank = number of non-zero rows in row-echelon form.`

Explanation & Conditions: Pivot elements must step to the right in each successive row.

E. Given Information

A = [[1, 2, -1, 4], [2, 4, 3, 4], [1, 2, 3, 4]].

F. Complete Step-by-Step Solution

Step 1: Eliminate entries below pivot $a_{11} = 1$

$R_2 \to R_2 - 2R_1$:

[2 - 2(1), 4 - 2(2), 3 - 2(-1), 4 - 2(4)] = [0, 0, 5, -4].

$R_3 \to R_3 - R_1$:

[1 - 1, 2 - 2, 3 - (-1), 4 - 4] = [0, 0, 4, 0].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 2: Matrix after first sweep

A ~ [[1, 2, -1, 4], [0, 0, 5, -4], [0, 0, 4, 0]].

💡 **Why this step is performed:** Execute step 2: matrix after first sweep to advance the step-by-step mathematical solution toward the final result.

Step 3: Eliminate entry below pivot in column 3

$R_3 \to 5R_3 - 4R_2$:
[0, 0, 5(4) - 4(5), 5(0) - 4(-4)] = [0, 0, 0, 16].

> 💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 4: Examine row-echelon form

A ~ [[1, 2, -1, 4], [0, 0, 5, -4], [0, 0, 0, 16]].
All 3 rows contain non-zero elements (Row 1 has 1, Row 2 has 5, Row 3 has 16).
Number of non-zero rows = 3.
Therefore, Rank ρ(A) = 3.

> 💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

✓ G. Final Answer: Rank ρ(A) = 3

H. Examination Presentation Strategy

- Show each row operation with intermediate arithmetic.
- State the final echelon matrix clearly.
- Count non-zero rows = 3.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Arithmetic error in 4 - 2(-1) = 6 instead of 3 - (-2) = 5.
✓ **How to Avoid:** Double minus is positive: 3 - 2(-1) = 3 + 2 = 5.

⚠️ **Common Mistake:** Trying to make column 2 pivot.
✓ **How to Avoid:** Column 2 has all zeros below row 1, so the pivot moves to column 3.

J. Similar Practice Question

Practice Problem 56: Find rank of A = [[1, 1, 1, 1], [1, 2, 3, -4], [2, 3, 5, -5]].
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 57: Solving Linear System by Gauss Elimination (Unique Solution) INTERMEDIATE

A. Full Question:
Solve the system of equations using Gauss Elimination method:
2x + y + z = 10,
3x + 2y + 3z = 18,
x + 4y + 9z = 16.

B. Meaning of the Question

Convert the augmented matrix [A|B] to upper triangular row-echelon form using elementary row operations, then solve for x, y, z by back-substitution.

C. Basic Concept

Gauss elimination reduces the augmented matrix [A|B] to row-echelon form, transforming a coupled system into a triangular system that is solved from bottom to top.

D. Formula and Explanation

Formula: `[A | B] ~ [U | B'] where U is upper triangular.`

Explanation & Conditions: Back-substitution: solve z first, then y, then x.

E. Given Information

System of 3 equations in 3 variables.

F. Complete Step-by-Step Solution

Step 1: Set up Augmented Matrix and swap R_1 and R_3 for pivot 1

Augmented matrix [A|B]:
[[2, 1, 1 | 10], [3, 2, 3 | 18], [1, 4, 9 | 16]].
Swap $R_1 \leftrightarrow R_3$ to get pivot 1 in (1, 1):
[[1, 4, 9 | 16], [3, 2, 3 | 18], [2, 1, 1 | 10]].

💡 **Why this step is performed:** Execute step 1: set up augmented matrix and swap r_1 and r_3 for pivot 1 to advance the step-by-step mathematical solution toward the final result.

Step 2: Eliminate x from Row 2 and Row 3

$R_2 \rightarrow R_2 - 3R_1$:
[3-3, 2-12, 3-27 | 18-48] = [0, -10, -24 | -30] ⇒ divide by -2: [0, 5, 12 | 15].
$R_3 \rightarrow R_3 - 2R_1$:
[2-2, 1-8, 1-18 | 10-32] = [0, -7, -17 | -22].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 3: Eliminate y from Row 3

Current matrix:
[[1, 4, 9 | 16], [0, 5, 12 | 15], [0, -7, -17 | -22]].
$R_3 \rightarrow 5R_3 + 7R_2$:
[0, -35 + 35, 5(-17) + 7(12) | 5(-22) + 7(15)]
= [0, 0, -85 + 84 | -110 + 105] = [0, 0, -1 | -5] ⇒ [0, 0, 1 | 5].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 4: Back-substitution

From Row 3: z = 5.
From Row 2: 5y + 12z = 15 ⇒ 5y + 12(5) = 15 ⇒ 5y + 60 = 15 ⇒ 5y = -45 ⇒ y = -9.
From Row 1: x + 4y + 9z = 16 ⇒ x + 4(-9) + 9(5) = 16 ⇒ x - 36 + 45 = 16 ⇒ x + 9 = 16 ⇒ x = 7.

💡 **Why this step is performed:** Substitute known variable values upward from the bottom row equation to determine each variable uniquely.

Step 5: Verify in original Equation 1

2x + y + z = 2(7) + (-9) + 5 = 14 - 9 + 5 = 10. Exactly matches RHS! Verified.

💡 **Why this step is performed:** Execute step 5: verify in original equation 1 to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: x = 7, y = -9, z = 5

H. Examination Presentation Strategy

- State augmented matrix clearly.
- Show row operations explicitly.
- Show back-substitution step by step.
- Verify in at least one original equation.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Arithmetic error in 5(-22) + 7(15) = -110 + 105 = -5.
✓ **How to Avoid:** Be very careful with mental arithmetic; write every multiplication out!

⚠️ **Common Mistake:** Forgetting to apply row operations to the RHS column B.
✓ **How to Avoid:** Any row operation applies to the ENTIRE row, including the constant term!

J. Similar Practice Question

Practice Problem 57: Solve by Gauss elimination: x + y + z = 6, 2x - y + z = 3, x + 2y - z = 2.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 58: Solving Linear System by Gauss-Jordan Method

INTERMEDIATE

A. Full Question:
Solve the system using Gauss-Jordan elimination method:
x + y + z = 9,
2x - 3y + 4z = 13,
3x + 4y + 5z = 40.

B. Meaning of the Question

Reduce the augmented matrix [A|B] all the way to diagonal form [I | X] using elementary row operations, reading off solutions directly without back-substitution.

C. Basic Concept

While Gauss elimination stops at upper triangular form, Gauss-Jordan continues by eliminating entries ABOVE each pivot as well, converting A directly into the identity matrix I.

D. Formula and Explanation

Formula: $[A \mid B] \sim [I \mid X] \Rightarrow x = b'_1,\ y = b'_2,\ z = b'_3.$

Explanation & Conditions: Elimination is performed both downwards and upwards.

E. Given Information

System of 3 equations in 3 variables.

F. Complete Step-by-Step Solution

Step 1: Set up Augmented Matrix

[A|B] = [[1, 1, 1 | 9], [2, -3, 4 | 13], [3, 4, 5 | 40]]. Pivot is $a_{11} = 1$.

💡 **Why this step is performed:** Execute step 1: set up augmented matrix to advance the step-by-step mathematical solution toward the final result.

Step 2: Eliminate column 1 entries below pivot

$R_2 \to R_2 - 2R_1$: [0, -5, 2 | -5].
$R_3 \to R_3 - 3R_1$: [0, 1, 2 | 13].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 3: Swap $R_2 \leftrightarrow R_3$ to get pivot 1 in row 2

[[1, 1, 1 | 9], [0, 1, 2 | 13], [0, -5, 2 | -5]].

💡 **Why this step is performed:** Execute step 3: swap $r_2 \leftrightarrow r_3$ to get pivot 1 in row 2 to advance the step-by-step mathematical solution toward the final result.

Step 4: Eliminate column 2 entries above and below pivot

$R_1 \to R_1 - R_2$:
[1 - 0, 1 - 1, 1 - 2 | 9 - 13] = [1, 0, -1 | -4].
$R_3 \to R_3 + 5R_2$:
[0, -5 + 5, 2 + 10 | -5 + 65] = [0, 0, 12 | 60] ⇒ divide by 12: [0, 0, 1 | 5].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 5: Eliminate column 3 entries above pivot $a_{33} = 1$

$R_1 \to R_1 + R_3$:
[1, 0, -1 + 1 | -4 + 5] = [1, 0, 0 | 1].
$R_2 \to R_2 - 2R_3$:
[0, 1, 2 - 2(1) | 13 - 2(5)] = [0, 1, 0 | 3].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 6: Read off solution from identity matrix

[[1, 0, 0 | 1], [0, 1, 0 | 3], [0, 0, 1 | 5]].
Therefore: x = 1, y = 3, z = 5.

💡 **Why this step is performed:** Execute step 6: read off solution from identity matrix to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: x = 1, y = 3, z = 5

H. Examination Presentation Strategy

- State augmented matrix.
- Show both downward and upward elimination steps.
- State final [I | X] form.
- Write x = 1, y = 3, z = 5 clearly.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Doing back-substitution instead of Gauss-Jordan.
✓ **How to Avoid:** In Gauss-Jordan, you MUST reduce A to the identity matrix [I | X]; do not use back-substitution!

⚠️ **Common Mistake:** Arithmetic error in 13 - 10 = 3.
✓ **How to Avoid:** Take time on each subtraction step.

J. Similar Practice Question

Practice Problem 58: Solve by Gauss-Jordan method: x + y + z = 3, 2x + 3y + z = 6, x - y + 2z = 2.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 59: Eigenvalues and Eigenvectors of a 2×2 Symmetric Matrix — INTERMEDIATE

A. Full Question:
Find the eigenvalues and eigenvectors of the symmetric matrix A = [[3, 2], [2, 3]].

B. Meaning of the Question

Determine the scalar eigenvalues λ and the corresponding non-zero direction vectors v satisfying A v = λ v.

C. Basic Concept

For each eigenvalue λ, the eigenvector is found by solving the homogeneous system (A - λI) v = 0. For real symmetric matrices, eigenvectors corresponding to distinct eigenvalues are always orthogonal.

D. Formula and Explanation

Formula: `|A - λI| = 0; (A - λI) v = 0.`

Explanation & Conditions: v = [x, y]ᵀ is a non-zero vector spanning the null space of (A - λI).

E. Given Information

A = [[3, 2], [2, 3]].

F. Complete Step-by-Step Solution

Step 1: Characteristic equation

Trace(A) = 3 + 3 = 6.
det(A) = (3)(3) - (2)(2) = 9 - 4 = 5.
$\lambda^2 - 6\lambda + 5 = 0 \Rightarrow (\lambda - 1)(\lambda - 5) = 0$.
Eigenvalues: $\lambda_1 = 1, \lambda_2 = 5$.

> 💡 **Why this step is performed:** Set up the characteristic determinant det(A - λI) = 0 and expand to obtain the characteristic polynomial.

Step 2: Eigenvector for $\lambda_1 = 1$

$(A - 1 \cdot I)\, v_1 = 0 \Rightarrow$ [[3-1, 2], [2, 3-1]] [x, y]ᵀ = [0, 0]ᵀ
[[2, 2], [2, 2]] [x, y]ᵀ = [0, 0]ᵀ ⇒ 2x + 2y = 0 ⇒ x + y = 0 ⇒ y = -x.
Let x = 1 ⇒ y = -1.
Eigenvector v_1 = [1, -1]ᵀ.

> 💡 **Why this step is performed:** Solve the homogeneous linear system (A - λI)v = 0 to find the non-zero basis vectors spanning the eigenspace.

Step 3: Eigenvector for $\lambda_2 = 5$

$(A - 5 \cdot I)\, v_2 = 0 \Rightarrow$ [[3-5, 2], [2, 3-5]] [x, y]ᵀ = [0, 0]ᵀ
[[-2, 2], [2, -2]] [x, y]ᵀ = [0, 0]ᵀ ⇒ -2x + 2y = 0 ⇒ x = y.
Let x = 1 ⇒ y = 1.
Eigenvector v_2 = [1, 1]ᵀ.

> 💡 **Why this step is performed:** Solve the homogeneous linear system (A - λI)v = 0 to find the non-zero basis vectors spanning the eigenspace.

Step 4: Check Orthogonality of Eigenvectors

$v_1 \cdot v_2 = (1)(1) + (-1)(1) = 1 - 1 = 0$.
The eigenvectors are mutually orthogonal, confirming the symmetric matrix theorem!

> 💡 **Why this step is performed:** Solve the homogeneous linear system (A - λI)v = 0 to find the non-zero basis vectors spanning the eigenspace.

✓ **G. Final Answer: Eigenvalues: $\lambda_1 = 1, \lambda_2 = 5$; Eigenvectors: $v_1 = [1, -1]^T, v_2 = [1, 1]^T$**

H. Examination Presentation Strategy

- Find characteristic equation and roots λ = 1, 5.
- Set up (A - λI)v = 0 for each eigenvalue.
- Solve for x and y and state eigenvectors.
- Note orthogonality.

I. Common Mistakes & How to Avoid Them

⚠ **Common Mistake:** Setting v = [0, 0]ᵀ.
✓ **How to Avoid:** An eigenvector must be NON-ZERO by definition!

⚠ **Common Mistake:** Writing $v_1 = [1, 1]^T$ for λ = 1.
✓ **How to Avoid:** Check: 2(1) + 2(1) = 4 ≠ 0. The correct relation is y = -x, so $v_1 = [1, -1]^T$.

J. Similar Practice Question

Practice Problem 59: Find eigenvalues and eigenvectors of A = [[4, 2], [2, 1]].
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 60: Verification of Cayley-Hamilton Theorem and Finding Inverse Matrix

INTERMEDIATE

A. Full Question:
Verify Cayley-Hamilton theorem for the matrix A = [[1, 2], [3, 4]] and hence find A^{-1}.

B. Meaning of the Question

Verify that matrix A satisfies its own characteristic equation A^2 - (Trace)A + det(A)I = 0, and multiply by A^{-1} to compute the inverse matrix.

C. Basic Concept

Cayley-Hamilton Theorem states that every square matrix satisfies its own characteristic equation: p(A) = 0. Multiplying p(A) = 0 by A^{-1} provides an expression for A^{-1} in powers of A without computing adjugate matrices.

D. Formula and Explanation

Formula: $A^2 - (\text{Trace } A)\, A + \det(A)\, I = 0 \Rightarrow A^{-1} = (1/\det A)\, [\, (\text{Trace } A)\, I - A\,].$

Explanation & Conditions: Trace(A) = 1 + 4 = 5; det(A) = (1)(4) - (2)(3) = 4 - 6 = -2.

E. Given Information

A = [[1, 2], [3, 4]].

F. Complete Step-by-Step Solution

Step 1: Characteristic equation of A

Trace(A) = 5, det(A) = -2.
Characteristic equation: $\lambda^2 - 5\lambda - 2 = 0$.
Cayley-Hamilton theorem claims: $A^2 - 5A - 2I = 0$.

> 💡 **Why this step is performed:** Set up the characteristic determinant det(A - λI) = 0 and expand to obtain the characteristic polynomial.

Step 2: Compute A^2

A^2 = [[1, 2], [3, 4]] [[1, 2], [3, 4]]
= [[1(1)+2(3), 1(2)+2(4)], [3(1)+4(3), 3(2)+4(4)]]
= [[1+6, 2+8], [3+12, 6+16]] = [[7, 10], [15, 22]].

> 💡 **Why this step is performed:** Execute step 2: compute a^2 to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute $A^2 - 5A - 2I$ and verify zero matrix

$A^2 - 5A - 2I$ = [[7, 10], [15, 22]] - 5 [[1, 2], [3, 4]] - 2 [[1, 0], [0, 1]]
= [[7 - 5 - 2, 10 - 10 - 0], [15 - 15 - 0, 22 - 20 - 2]]
= [[0, 0], [0, 0]] = O.
Hence Cayley-Hamilton theorem is verified!

> 💡 **Why this step is performed:** Execute step 3: compute a^2 - 5a - 2i and verify zero matrix to advance the step-by-step mathematical solution toward the final result.

Step 4: Find A^{-1} using the verified matrix equation

$A^2 - 5A - 2I = 0$.
Multiply both sides by A^{-1}:
$A - 5I - 2A^{-1} = 0 \Rightarrow 2A^{-1} = A - 5I \Rightarrow A^{-1} = (1/2)(A - 5I)$.
$A - 5I$ = [[1-5, 2-0], [3-0, 4-5]] = [[-4, 2], [3, -1]].
Therefore: A^{-1} = (1/2) [[-4, 2], [3, -1]] = [[-2, 1], [3/2, -1/2]].

> 💡 **Why this step is performed:** Execute step 4: find a^{-1} using the verified matrix equation to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Cayley-Hamilton Verified: $A^2 - 5A - 2I = O$; $A^{-1} = [[-2, 1], [3/2, -1/2]]$

H. Examination Presentation Strategy

- State characteristic equation.
- Compute A^2 by matrix multiplication.
- Show term-by-term subtraction to get zero matrix O.
- Multiply by A^{-1} to isolate A^{-1}.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Using adj(A)/|A| to find A^{-1} when question says 'HENCE find A^{-1}'.
✓ **How to Avoid:** When the problem specifies 'hence', you MUST use the Cayley-Hamilton equation; direct formula loses marks!

⚠️ **Common Mistake:** Writing 2 instead of 2I in matrix equation.
✓ **How to Avoid:** Scalars must be multiplied by identity matrix I in matrix expressions.

J. Similar Practice Question

Practice Problem 60: Verify Cayley-Hamilton theorem for A = [[2, 1], [3, 5]] and find A^{-1}.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 61: Consistency and Infinite Solutions with Free Parameter

INTERMEDIATE

A. Full Question:
Test for consistency and solve if consistent:
x + 2y - z = 3,
3x - y + 2z = 1,
2x - 3y + 3z = -2.

B. Meaning of the Question

Use rank criteria on augmented matrix [A|B] to test consistency and express the infinite solution set using a free parameter k.

C. Basic Concept

If ρ(A) = ρ([A|B]) = 2 < 3 (number of unknowns), the system is consistent with infinitely many solutions dependent on 3 - 2 = 1 parameter.

D. Formula and Explanation

Formula: $\rho(A) = \rho([A|B]) = r < n \Rightarrow$ infinitely many solutions.
Explanation & Conditions: Set z = k and back-substitute.

E. Given Information

System of 3 equations in x, y, z.

F. Complete Step-by-Step Solution

Step 1: Set up Augmented Matrix

[A|B] = [[1, 2, -1 | 3], [3, -1, 2 | 1], [2, -3, 3 | -2]].

💡 **Why this step is performed:** Execute step 1: set up augmented matrix to advance the step-by-step mathematical solution toward the final result.

Step 2: Eliminate x from rows 2 and 3

$R_2 \to R_2 - 3R_1$: [0, -7, 5 | -8].
$R_3 \to R_3 - 2R_1$: [0, -7, 5 | -8].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 3: Eliminate row 3

$R_3 \to R_3 - R_2$: [0, 0, 0 | 0].
Matrix in row-echelon form:
[[1, 2, -1 | 3], [0, -7, 5 | -8], [0, 0, 0 | 0]].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 4: Analyze ranks and consistency

$\rho(A) = 2$, $\rho([A|B]) = 2$.
Since $\rho(A) = \rho([A|B]) = 2 < n = 3$, the system is CONSISTENT and has INFINITELY MANY SOLUTIONS with (3 - 2) = 1 free parameter.

💡 **Why this step is performed:** Compare the rank of the coefficient matrix $\rho(A)$ with the augmented matrix $\rho([A|B])$ using the Rouché-Capelli criteria.

Step 5: Solve in terms of parameter k

Let $z = k$.
From Row 2: $-7y + 5z = -8 \Rightarrow 7y = 5k + 8 \Rightarrow y = (5k + 8) / 7$.
From Row 1: $x = 3 - 2y + z = 3 - 2(5k+8)/7 + k = [21 - 10k - 16 + 7k] / 7 = (5 - 3k) / 7$.

💡 **Why this step is performed:** Execute step 5: solve in terms of parameter k to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Consistent with infinitely many solutions: $x = (5 - 3k)/7$, $y = (5k + 8)/7$, $z = k$, where k is any real number.

H. Examination Presentation Strategy

- State both ranks $\rho(A) = 2$ and $\rho([A|B]) = 2$.
- Quote Rouché-Capelli consistency condition.
- Assign $z = k$ and show algebraic back-substitution.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Concluding no solution because a row became all zeros.
✓ **How to Avoid:** A row of all zeros [0, 0, 0 | 0] means CONSISTENT with infinite solutions! [0, 0, 0 | non-zero] means no solution.

⚠️ **Common Mistake:** Algebra error in (21 - 16)/7 = 5/7.
✓ **How to Avoid:** Double check fraction addition.

J. Similar Practice Question

Practice Problem 61: Test for consistency: x + y + z = 4, 2x + 5y - 2z = 3, x + 7y - 7z = 5.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 62: Inconsistent System (No Solution)

INTERMEDIATE

A. Full Question:

Show that the system of equations is inconsistent and hence has no solution:
x + 2y - 3z = -2,
3x - y - 2z = 1,
2x + 3y - 5z = k (where k ≠ -3, say k = 3).

B. Meaning of the Question

Reduce [A|B] to row-echelon form and show that ρ(A) ≠ ρ([A|B]), which proves the system is inconsistent.

C. Basic Concept

A linear system has no solution if and only if the rank of the augmented matrix is strictly greater than the rank of the coefficient matrix: ρ(A) < ρ([A|B]).

D. Formula and Explanation

Formula: `ρ(A) ≠ ρ([A|B]) ⇒ Inconsistent.`

Explanation & Conditions: Produces an impossible row 0x + 0y + 0z = c (where c ≠ 0).

E. Given Information

x + 2y - 3z = -2, 3x - y - 2z = 1, 2x + 3y - 5z = 3.

F. Complete Step-by-Step Solution

Step 1: Set up Augmented Matrix

[A|B] = [[1, 2, -3 | -2], [3, -1, -2 | 1], [2, 3, -5 | 3]].

💡 **Why this step is performed:** Execute step 1: set up augmented matrix to advance the step-by-step mathematical solution toward the final result.

Step 2: Row operations to eliminate column 1 below pivot

$R_2 \to R_2 - 3R_1$: [0, -7, 7 | 7] ⇒ divide by -7: [0, 1, -1 | -1].
$R_3 \to R_3 - 2R_1$: [0, -1, 1 | 7].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 3: Eliminate row 3 column 2

$R_3 \rightarrow R_3 + R_2$:
[0, -1 + 1, 1 - 1 | 7 + (-1)] = [0, 0, 0 | 6].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 4: Compare ranks of A and [A|B]

Matrix in echelon form:
[[1, 2, -3 | -2], [0, 1, -1 | -1], [0, 0, 0 | 6]].
• In matrix A, the 3rd row is [0, 0, 0] ⇒ ρ(A) = 2.
• In augmented matrix [A|B], the 3rd row is [0, 0, 0 | 6] ⇒ ρ([A|B]) = 3.
Since ρ(A) = 2 ≠ ρ([A|B]) = 3, the system is INCONSISTENT.

💡 **Why this step is performed:** Compare the rank of the coefficient matrix ρ(A) with the augmented matrix ρ([A|B]) using the Rouché-Capelli criteria.

Step 5: Physical interpretation

The third row represents the equation 0·x + 0·y + 0·z = 6, i.e., 0 = 6, which is an impossible contradiction. Hence, NO SOLUTION exists.

💡 **Why this step is performed:** Execute step 5: physical interpretation to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: ρ(A) = 2 ≠ ρ([A|B]) = 3; The system is INCONSISTENT and has NO SOLUTION.

H. Examination Presentation Strategy

- Show echelon reduction clearly.
- State ρ(A) = 2 and ρ([A|B]) = 3 explicitly.
- Highlight row [0, 0, 0 | 6] and statement 0 ≠ 6.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Saying rank of [A|B] is 2.
✓ **How to Avoid:** The entry 6 makes row 3 a non-zero row in [A|B], so rank is 3!

⚠️ **Common Mistake:** Attempting to solve for variables when inconsistent.
✓ **How to Avoid:** Once ρ(A) ≠ ρ([A|B]), stop immediately and conclude 'no solution'.

J. Similar Practice Question

Practice Problem 62: Show that the system x + y + z = 1, x + 2y + 3z = 4, x + 3y + 5z = 5 is inconsistent.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 63: Rank of Matrix Containing an Unknown Parameter λ

INTERMEDIATE

A. Full Question:

Find the values of λ for which the matrix A = [[1, 1, 1], [1, 2, 4], [1, 4, λ]] has rank (i) 3, (ii) less than 3.

B. Meaning of the Question

Evaluate the determinant of matrix A in terms of λ and determine the conditions under which det(A) = 0 (rank < 3) and det(A) ≠ 0 (rank = 3).

C. Basic Concept

For an n×n square matrix, rank = n if and only if det(A) ≠ 0. If det(A) = 0, rank < n.

D. Formula and Explanation

Formula: `det(A) ≠ 0 ⇒ rank = 3; det(A) = 0 ⇒ rank < 3.`

Explanation & Conditions: Row operations can also be used directly.

E. Given Information

A = [[1, 1, 1], [1, 2, 4], [1, 4, λ]].

F. Complete Step-by-Step Solution

Step 1: Reduce A using elementary row operations

A = [[1, 1, 1], [1, 2, 4], [1, 4, λ]].

$R_2 \rightarrow R_2 - R_1$: [0, 1, 3].

$R_3 \rightarrow R_3 - R_1$: [0, 3, λ - 1].

💡 **Why this step is performed:** Execute step 1: reduce a using elementary row operations to advance the step-by-step mathematical solution toward the final result.

Step 2: Eliminate column 2 in row 3

$R_3 \rightarrow R_3 - 3R_2$:

[0, 3 - 3(1), (λ - 1) - 3(3)] = [0, 0, λ - 10].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 3: State the echelon matrix

A ~ [[1, 1, 1], [0, 1, 3], [0, 0, λ - 10]].

💡 **Why this step is performed:** Execute step 3: state the echelon matrix to advance the step-by-step mathematical solution toward the final result.

Step 4: Case (i): Rank = 3

For rank to be 3, all 3 rows must be non-zero.

This requires the pivot λ - 10 ≠ 0 ⇒ λ ≠ 10.

💡 **Why this step is performed:** Execute step 4: case (i): rank = 3 to advance the step-by-step mathematical solution toward the final result.

Step 5: Case (ii): Rank < 3

If λ - 10 = 0, i.e., λ = 10, the 3rd row becomes [0, 0, 0].
Then there are only 2 non-zero rows, so Rank ρ(A) = 2 < 3.

💡 **Why this step is performed:** Execute step 5: case (ii): rank < 3 to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: (i) Rank = 3 when λ ≠ 10; (ii) Rank < 3 (Rank = 2) when λ = 10

H. Examination Presentation Strategy

- Show row operations leading to [0, 0, λ - 10].
- State clearly: pivot non-zero ⇒ rank 3.
- State clearly: pivot zero ⇒ rank 2.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Arithmetic error: (λ - 1) - 9 = λ - 10.
✓ **How to Avoid:** Be careful: -1 - 9 = -10, NOT -8!

⚠️ **Common Mistake:** Claiming rank becomes 1 when λ = 10.
✓ **How to Avoid:** Rows 1 and 2 are linearly independent, so rank is 2, not 1.

J. Similar Practice Question

Practice Problem 63: Find the value of k for which the matrix [[1, 2, 3], [2, k, 7], [3, 6, 10]] has rank 2.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 64: Eigenvalues of Special Matrices: Triangular, Orthogonal and Idempotent

INTERMEDIATE

A. Full Question:
Prove that: (i) The eigenvalues of a triangular matrix are its diagonal entries, and (ii) The eigenvalues of an idempotent matrix ($A^2 = A$) are either 0 or 1.

B. Meaning of the Question

Prove two fundamental theoretical properties regarding eigenvalues of structured matrices.

C. Basic Concept

Eigenvalue equations det(A - λI) = 0 and A v = λ v translate matrix algebraic properties into scalar polynomial equations in λ.

D. Formula and Explanation

Formula: (i) $\det(T - \lambda I) = (t_{11} - \lambda)(t_{22} - \lambda)...(t_{nn} - \lambda) = 0$.
(ii) $A v = \lambda v \Rightarrow A^2 v = \lambda^2 v$; but $A^2 = A \Rightarrow \lambda^2 = \lambda$.

Explanation & Conditions: $\lambda^2 - \lambda = 0$ has only two roots: 0 and 1.

E. Given Information

(i) Upper triangular matrix T; (ii) Idempotent matrix A satisfying $A^2 = A$.

F. Complete Step-by-Step Solution

Step 1: Proof for Triangular Matrix

Let T be an upper triangular matrix with diagonal entries $d_1, d_2, ..., d_n$.
Then (T - λI) is also an upper triangular matrix with diagonal entries $(d_1 - \lambda), (d_2 - \lambda), ..., (d_n - \lambda)$.
The determinant of any triangular matrix is simply the product of its diagonal entries:
$\det(T - \lambda I) = (d_1 - \lambda)(d_2 - \lambda)...(d_n - \lambda) = 0$.
The roots of this equation are precisely $\lambda = d_1, d_2, ..., d_n$.
Hence, the eigenvalues are the diagonal entries. Proved!

💡 **Why this step is performed:** Execute step 1: proof for triangular matrix to advance the step-by-step mathematical solution toward the final result.

Step 2: Proof for Idempotent Matrix ($A^2 = A$)

Let λ be an eigenvalue of A with corresponding non-zero eigenvector v:
A v = λ v --- (1).
Multiply both sides on the left by A:
$A(A v) = A(\lambda v) \Rightarrow A^2 v = \lambda (A v)$.
Substitute (1) into the RHS:
$A^2 v = \lambda (\lambda v) = \lambda^2 v$ --- (2).

💡 **Why this step is performed:** Execute step 2: proof for idempotent matrix ($a^2 = a$) to advance the step-by-step mathematical solution toward the final result.

Step 3: Equate using $A^2 = A$

Since A is idempotent, $A^2 = A$.
Therefore, the LHS of (2) is $A^2 v = A v = \lambda v$.
Equating with RHS of (2):
$\lambda v = \lambda^2 v \Rightarrow (\lambda^2 - \lambda) v = 0$.
Since v is an eigenvector, $v \neq 0$. Therefore:
$\lambda^2 - \lambda = 0 \Rightarrow \lambda(\lambda - 1) = 0 \Rightarrow \lambda = 0$ or $\lambda = 1$. Proved!

💡 **Why this step is performed:** Execute step 3: equate using $a^2 = a$ to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: (i) Eigenvalues of triangular matrix are its diagonal entries; (ii) Eigenvalues of idempotent matrix $A^2 = A$ are strictly 0 or 1. (Both proved)

H. Examination Presentation Strategy

- Present each proof with a clear heading.
- State determinant of triangular matrix property.
- State $v \neq 0$ to divide by v in $\lambda^2 - \lambda = 0$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Dividing by vector v directly like $(\lambda^2 - \lambda) = 0/v$.
✓ **How to Avoid:** You cannot divide by a vector! State 'Since v is a non-zero vector, the scalar coefficient $(\lambda^2 - \lambda)$ must be 0'.

⚠️ **Common Mistake:** Forgetting $\lambda = 0$ as a valid eigenvalue.
✓ **How to Avoid:** 0 is a completely valid eigenvalue.

J. Similar Practice Question

Practice Problem 64: Prove that the eigenvalues of an orthogonal matrix ($A^T A = I$) have absolute value $|\lambda| = 1$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 65: Diagonalization of a 2×2 Matrix

INTERMEDIATE

A. Full Question:
Diagonalize the matrix A = [[1, 4], [2, 3]] by finding the modal matrix P and computing $P^{-1}AP$.

B. Meaning of the Question

Find eigenvalues, form modal matrix P from eigenvectors, compute P^{-1}, and verify that $P^{-1}AP$ is the diagonal matrix D of eigenvalues.

C. Basic Concept

A matrix with distinct eigenvalues is diagonalizable: $P^{-1} A P = D = \text{diag}(\lambda_1, \lambda_2)$. The columns of P are the eigenvectors.

D. Formula and Explanation

Formula: $D = P^{-1} A P$ where $P = [v_1 \mid v_2]$ and $D = [[\lambda_1, 0], [0, \lambda_2]]$.
Explanation & Conditions: P is called the modal matrix; D is the spectral diagonal matrix.

E. Given Information

A = [[1, 4], [2, 3]].

F. Complete Step-by-Step Solution

Step 1: Find eigenvalues of A

Trace(A) = 1 + 3 = 4, det(A) = (1)(3) - (4)(2) = 3 - 8 = -5.
$\lambda^2 - 4\lambda - 5 = 0 \Rightarrow (\lambda - 5)(\lambda + 1) = 0$.

Eigenvalues: $\lambda_1 = 5, \lambda_2 = -1$.

💡 **Why this step is performed:** Execute step 1: find eigenvalues of a to advance the step-by-step mathematical solution toward the final result.

Step 2: Find eigenvector v_1 for $\lambda_1 = 5$

$(A - 5I)\, v_1 = 0 \Rightarrow [\,[-4, 4], [2, -2]\,]\, [x, y]^T = [0, 0]^T \Rightarrow -4x + 4y = 0 \Rightarrow x = y$.
Let $y = 1 \Rightarrow x = 1$. $v_1 = [1, 1]^T$.

💡 **Why this step is performed:** Solve the homogeneous linear system $(A - \lambda I)v = 0$ to find the non-zero basis vectors spanning the eigenspace.

Step 3: Find eigenvector v_2 for $\lambda_2 = -1$

$(A + I)\, v_2 = 0 \Rightarrow [\,[2, 4], [2, 4]\,]\, [x, y]^T = [0, 0]^T \Rightarrow 2x + 4y = 0 \Rightarrow x = -2y$.
Let $y = 1 \Rightarrow x = -2$. $v_2 = [-2, 1]^T$.

💡 **Why this step is performed:** Solve the homogeneous linear system $(A - \lambda I)v = 0$ to find the non-zero basis vectors spanning the eigenspace.

Step 4: Construct modal matrix P and compute P^{-1}

$P = [\, v_1 \mid v_2 \,] = [\,[1, -2], [1, 1]\,]$.
$\det(P) = (1)(1) - (-2)(1) = 1 + 2 = 3$.
$P^{-1} = (1/3)\, [\,[1, 2], [-1, 1]\,]$.

💡 **Why this step is performed:** Execute step 4: construct modal matrix p and compute p^{-1} to advance the step-by-step mathematical solution toward the final result.

Step 5: Compute AP and $P^{-1}(AP)$

$AP = [\,[1, 4], [2, 3]\,]\, [\,[1, -2], [1, 1]\,] = [\,[5, 2], [5, -1]\,]$.
$P^{-1}\, AP = (1/3)\, [\,[1, 2], [-1, 1]\,]\, [\,[5, 2], [5, -1]\,]$
$= (1/3)\, [\,[5+10, 2-2], [-5+5, -2-1]\,] = (1/3)\, [\,[15, 0], [0, -3]\,] = [\,[5, 0], [0, -1]\,] = D$.
Matches eigenvalues 5 and -1! Successfully diagonalized.

💡 **Why this step is performed:** Execute step 5: compute ap and p^{-1}(ap) to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Modal Matrix P = [[1, -2], [1, 1]]; Diagonal Matrix D = $P^{-1}AP$ = [[5, 0], [0, -1]]

H. Examination Presentation Strategy

- Find eigenvalues $\lambda = 5, -1$.
- Find eigenvectors v_1, v_2.
- Write $P = [v_1 \mid v_2]$ and P^{-1}.
- Multiply $P^{-1}AP$ step by step to produce D.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Ordering mismatch between P and D.
✓ **How to Avoid:** If column 1 of P is v_1 (for $\lambda = 5$), then D_{11} MUST be 5!

⚠️ **Common Mistake:** Determinant error in P^{-1}.
✓ **How to Avoid:** det(P) = 1 - (-2) = 3.

J. Similar Practice Question

Practice Problem 65: Diagonalize A = [[3, 1], [1, 3]].
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 66: Investigation of System Consistency for Parameters λ and μ

EXAM-LEVEL

A. Full Question:
Investigate the values of λ and μ for which the system of equations:
x + y + z = 6,
x + 2y + 3z = 10,
x + 2y + λz = μ
has (i) no solution, (ii) a unique solution, and (iii) infinitely many solutions.

B. Meaning of the Question

Determine the conditions on parameters λ and μ by row reduction of [A|B] to classify the system into the three Rouché-Capelli solution cases.

C. Basic Concept

Classic VTU exam question appearing in nearly every examination. The pivot in the last row (λ - 3) and constant (μ - 10) control the rank of A and [A|B].

D. Formula and Explanation

Formula: `[A|B] ~ echelon form; compare ρ(A) and ρ([A|B]).`
Explanation & Conditions: Row 3 becomes [0, 0, λ - 3 | μ - 10].

E. Given Information

System with parameters λ and μ.

F. Complete Step-by-Step Solution

Step 1: Write the Augmented Matrix [A|B]

[A|B] = [[1, 1, 1 | 6], [1, 2, 3 | 10], [1, 2, λ | μ]].

💡 **Why this step is performed:** Execute step 1: write the augmented matrix [a|b] to advance the step-by-step mathematical solution toward the final result.

Step 2: Eliminate x from rows 2 and 3

$R_2 \to R_2 - R_1$: [0, 1, 2 | 4].
$R_3 \to R_3 - R_1$: [0, 1, λ - 1 | μ - 6].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 3: Eliminate y from row 3

$R_3 \to R_3 - R_2$:
[0, 1 - 1, (λ - 1) - 2 | (μ - 6) - 4] = [0, 0, λ - 3 | μ - 10].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 4: State the reduced echelon form

[A|B] ~ [[1, 1, 1 | 6], [0, 1, 2 | 4], [0, 0, λ - 3 | μ - 10]].

💡 **Why this step is performed:** Execute step 4: state the reduced echelon form to advance the step-by-step mathematical solution toward the final result.

Step 5: Case (i): No Solution (Inconsistent)

For no solution: ρ(A) ≠ ρ([A|B]).
This happens when λ - 3 = 0 and μ - 10 ≠ 0.
Therefore: λ = 3 and μ ≠ 10.
Then ρ(A) = 2 while ρ([A|B]) = 3. System is inconsistent.

💡 **Why this step is performed:** Execute step 5: case (i): no solution (inconsistent) to advance the step-by-step mathematical solution toward the final result.

Step 6: Case (ii): Unique Solution

For unique solution: ρ(A) = ρ([A|B]) = 3 (number of unknowns).
This happens when the pivot λ - 3 ≠ 0, regardless of μ.
Therefore: λ ≠ 3 and μ can have ANY real value.

💡 **Why this step is performed:** Execute step 6: case (ii): unique solution to advance the step-by-step mathematical solution toward the final result.

Step 7: Case (iii): Infinitely Many Solutions

For infinite solutions: ρ(A) = ρ([A|B]) = 2 < 3.
This requires the entire last row to become all zeros: [0, 0, 0 | 0].
Therefore: λ - 3 = 0 and μ - 10 = 0 ⇒ λ = 3 and μ = 10.

💡 **Why this step is performed:** Execute step 7: case (iii): infinitely many solutions to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: (i) No Solution: λ = 3 and μ ≠ 10; (ii) Unique Solution: λ ≠ 3, μ ∈ ℝ; (iii) Infinitely Many Solutions: λ = 3 and μ = 10

H. Examination Presentation Strategy

- Show the row reduction clearly to [0, 0, λ - 3 | μ - 10].
- State the 3 cases with separate headings.
- Quote rank values ρ(A) and ρ([A|B]) in every case.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Saying unique solution requires μ ≠ 10.
✓ **How to Avoid:** For a unique solution, μ can be ANYTHING! Only λ ≠ 3 is required.

⚠️ **Common Mistake:** Writing λ = 3, μ = 10 for no solution.
✓ **How to Avoid:** That is the condition for INFINITE solutions.

J. Similar Practice Question

Practice Problem 66: Investigate for what values of a and b the system x + y + z = 2, x + 2y + 3z = 5, x + 3y + az = b has unique, no, and infinite solutions.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 67: Complete Eigenvalues and Eigenvectors of a 3×3 Matrix

EXAM-LEVEL

A. Full Question:
Find all eigenvalues and eigenvectors of the matrix A = [[2, -1, 1], [-1, 2, -1], [1, -1, 2]].

B. Meaning of the Question

Expand the 3×3 characteristic equation, find the three eigenvalues, and solve (A - λI)v = 0 for each eigenvalue to find the corresponding eigenvectors.

C. Basic Concept

Symmetric 3×3 matrix with repeated eigenvalues. For a repeated eigenvalue of multiplicity m, we must find m linearly independent eigenvectors.

D. Formula and Explanation

Formula: $\det(A - \lambda I) = -\lambda^3 + (\text{Tr } A)\lambda^2 - (M_{11} + M_{22} + M_{33})\lambda + \det(A) = 0.$

Explanation & Conditions: Tr(A) = 2+2+2 = 6; $M_{11}+M_{22}+M_{33}$ = sum of principal minors = 3+3+3 = 9; det(A) = 4.

E. Given Information

A = [[2, -1, 1], [-1, 2, -1], [1, -1, 2]].

F. Complete Step-by-Step Solution

Step 1: Characteristic Equation using coefficients

• s_1 = Trace(A) = 2 + 2 + 2 = 6.
• $s_2 = M_{11} + M_{22} + M_{33} = |2\ -1; -1\ 2| + |2\ 1; 1\ 2| + |2\ -1; -1\ 2| = (4 - 1) + (4 - 1) + (4 - 1) = 3 + 3 + 3 = 9$.
• s_3 = det(A) = 2(4 - 1) - (-1)(-2 + 1) + 1(1 - 2) = 2(3) + 1(-1) + 1(-1) = 6 - 1 - 1 = 4.
Characteristic equation: $\lambda^3 - s_1\lambda^2 + s_2\lambda - s_3 = 0 \Rightarrow \lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$.

💡 **Why this step is performed:** Set up the characteristic determinant det(A - λI) = 0 and expand to obtain the characteristic polynomial.

Step 2: Solve the cubic polynomial for eigenvalues

By inspection, test $\lambda = 1$: $1^3 - 6(1)^2 + 9(1) - 4 = 1 - 6 + 9 - 4 = 0$. So $\lambda = 1$ is a root.
Synthetic division by $(\lambda - 1)$:
$(\lambda - 1)(\lambda^2 - 5\lambda + 4) = 0 \Rightarrow (\lambda - 1)(\lambda - 1)(\lambda - 4) = 0$.
Eigenvalues: $\lambda_1 = 4$, $\lambda_2 = 1$, $\lambda_3 = 1$ (repeated eigenvalue).

💡 **Why this step is performed:** Find the roots of the characteristic polynomial to determine all scalar eigenvalues.

Step 3: Eigenvector for λ = 4

$(A - 4I)\ v = 0 \Rightarrow [\ [-2, -1, 1], [-1, -2, -1], [1, -1, -2]\ ]\ [x, y, z]^T = 0$.
Taking first two equations:
-2x - y + z = 0
-x - 2y - z = 0.
Using cross-multiplication rule:
x / [(-1)(-1) - (1)(-2)] = y / [(1)(-1) - (-2)(-1)] = z / [(-2)(-2) - (-1)(-1)]
x / [1 + 2] = y / [-1 - 2] = z / [4 - 1]
x / 3 = y / (-3) = z / 3 ⇒ x / 1 = y / (-1) = z / 1.
Eigenvector $v_1 = [1, -1, 1]^T$.

💡 **Why this step is performed:** Solve the homogeneous linear system (A - λI)v = 0 to find the non-zero basis vectors spanning the eigenspace.

Step 4: Eigenvectors for repeated eigenvalue λ = 1

$(A - I)\ v = 0 \Rightarrow [\ [1, -1, 1], [-1, 1, -1], [1, -1, 1]\ ]\ [x, y, z]^T = 0$.
All three rows are scalar multiples of a single equation: x - y + z = 0 ⇒ y = x + z.
Since rank = 1, nullity = 3 - 1 = 2 (two linearly independent eigenvectors!):
• Choice 1: Let z = 0, x = 1 ⇒ y = 1 + 0 = 1. $v_2 = [1, 1, 0]^T$.
• Choice 2: Let x = 0, z = 1 ⇒ y = 0 + 1 = 1. $v_3 = [0, 1, 1]^T$.

💡 **Why this step is performed:** Solve the homogeneous linear system (A - λI)v = 0 to find the non-zero basis vectors spanning the eigenspace.

Step 5: Orthogonalize eigenvectors if needed

Notice $v_1 \cdot v_2 = 1(1) + (-1)(1) + 1(0) = 0$. Orthogonal!
$v_1 \cdot v_3 = 1(0) + (-1)(1) + 1(1) = 0$. Orthogonal!

💡 **Why this step is performed:** Solve the homogeneous linear system (A - λI)v = 0 to find the non-zero basis vectors spanning the eigenspace.

✓ **G. Final Answer: Eigenvalues: λ = 4, 1, 1; Eigenvectors: for λ = 4: $v_1 = [1, -1, 1]^T$; for λ = 1: $v_2 = [1, 1, 0]^T$ and $v_3 = [0, 1, 1]^T$**

H. Examination Presentation Strategy

- Compute s_1, s_2, s_3 clearly.
- Show synthetic division for cubic equation.
- Show cross-multiplication for λ = 4.
- Show nullity = 2 for λ = 1 and give two independent vectors.

I. Common Mistakes & How to Avoid Them

⚠ **Common Mistake:** Giving only one eigenvector for the repeated eigenvalue λ = 1.
✓ **How to Avoid:** λ = 1 has algebraic multiplicity 2; since rank(A-I) = 1, geometric multiplicity is 3 - 1 = 2, so you MUST find TWO independent eigenvectors!

⚠ **Common Mistake:** Arithmetic error in cross-multiplication signs.
✓ **How to Avoid:** Check: (-1)(-1) - (1)(-2) = 1 + 2 = 3.

J. Similar Practice Question

Practice Problem 67: Find eigenvalues and eigenvectors of A = [[1, 1, 3], [1, 5, 1], [3, 1, 1]].
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 68: Diagonalization of a 3×3 Symmetric Matrix

EXAM-LEVEL

A. Full Question:
Diagonalize the symmetric matrix A = [[3, -1, 1], [-1, 3, -1], [1, -1, 3]] by orthogonal transformation.

B. Meaning of the Question

Find eigenvalues and mutually orthogonal normalized eigenvectors, construct orthogonal matrix P (where $P^T = P^{-1}$), and compute $P^TAP = D$.

C. Basic Concept

Every real symmetric matrix can be orthogonally diagonalized: $P^T A P = D$, where P is an orthogonal matrix whose columns are normalized eigenvectors.

D. Formula and Explanation

Formula: $P^T A P = D$; $P = [u_1 \mid u_2 \mid u_3]$ where $u_i = v_i / |v_i|$.
Explanation & Conditions: For an orthogonal matrix, $P^{-1} = P^T$, avoiding matrix inversion.

E. Given Information

A = [[3, -1, 1], [-1, 3, -1], [1, -1, 3]].

F. Complete Step-by-Step Solution

Step 1: Compute eigenvalues of A

• s_1 = Trace = 3 + 3 + 3 = 9.
• s_2 = 3(9 - 1) = 3(8) = 24.
• s_3 = det(A) = 3(8) - (-1)(-3 + 1) + 1(1 - 3) = 24 - 2 - 2 = 20.
$\lambda^3 - 9\lambda^2 + 24\lambda - 20 = 0 \Rightarrow (\lambda - 5)(\lambda - 2)^2 = 0$.
Eigenvalues: $\lambda_1 = 5$, $\lambda_2 = 2$, $\lambda_3 = 2$.

💡 **Why this step is performed:** Execute step 1: compute eigenvalues of a to advance the step-by-step mathematical solution toward the final result.

Step 2: Eigenvector for λ = 5

(A - 5I) v = 0 ⇒ [[-2, -1, 1], [-1, -2, -1], [1, -1, -2]] v = 0.
As in Q67, cross-multiplication gives: $v_1 = [1, -1, 1]^T$.
Normalized unit eigenvector: $u_1 = (1/\sqrt{3})\ [1, -1, 1]^T$.

💡 **Why this step is performed:** Solve the homogeneous linear system (A - λI)v = 0 to find the non-zero basis vectors spanning the eigenspace.

Step 3: Two orthogonal eigenvectors for λ = 2

(A - 2I) v = 0 ⇒ [[1, -1, 1], [-1, 1, -1], [1, -1, 1]] v = 0 ⇒ x - y + z = 0.
Choose $v_2 = [1, 1, 0]^T$. Normalized: $u_2 = (1/\sqrt{2})\ [1, 1, 0]^T$.
Now choose $v_3 = [a, b, c]^T$ such that:
1) a - b + c = 0 (satisfies plane equation)
2) $v_3 \cdot v_2 = 0 \Rightarrow a(1) + b(1) + c(0) = a + b = 0 \Rightarrow b = -a$.
Substitute b = -a into (1): a - (-a) + c = 0 ⇒ 2a + c = 0 ⇒ c = -2a.
Let a = 1 ⇒ b = -1, c = -2. $v_3 = [1, -1, -2]^T$.
Normalized: $|v_3| = \sqrt{(1 + 1 + 4)} = \sqrt{6} \Rightarrow u_3 = (1/\sqrt{6})\ [1, -1, -2]^T$.

💡 **Why this step is performed:** Solve the homogeneous linear system (A - λI)v = 0 to find the non-zero basis vectors spanning the eigenspace.

Step 4: Form the Orthogonal Matrix P

P = [[1/√3, 1/√2, 1/√6], [-1/√3, 1/√2, -1/√6], [1/√3, 0, -2/√6]].

💡 **Why this step is performed:** Execute step 4: form the orthogonal matrix p to advance the step-by-step mathematical solution toward the final result.

Step 5: Verify Diagonal Form D = P^T A P

D = P^T A P = [[5, 0, 0], [0, 2, 0], [0, 0, 2]]. Diagonalized!

💡 **Why this step is performed:** Execute step 5: verify diagonal form d = p^T a p to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Orthogonal Matrix P = [[1/√3, 1/√2, 1/√6], [-1/√3, 1/√2, -1/√6], [1/√3, 0, -2/√6]]; Diagonal Matrix D = [[5, 0, 0], [0, 2, 0], [0, 0, 2]]

H. Examination Presentation Strategy

- State eigenvalues $\lambda = 5, 2, 2$.
- Normalize each eigenvector.
- Ensure eigenvectors for $\lambda = 2$ are mutually orthogonal.
- State P and D.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Not making the two eigenvectors for $\lambda = 2$ orthogonal to each other.
✓ **How to Avoid:** For orthogonal diagonalization, ALL columns of P must be mutually orthogonal!

⚠️ **Common Mistake:** Using P^{-1} instead of P^T.
✓ **How to Avoid:** For an orthogonal matrix, $P^{-1} = P^T$, which is much easier to write.

J. Similar Practice Question

Practice Problem 68: Orthogonally diagonalize A = [[2, 2], [2, 5]].
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 69: Homogeneous System with Non-Trivial Solutions Depending on Parameter

EXAM-LEVEL

A. Full Question:
Find the values of k for which the system of homogeneous equations has non-trivial solutions, and find the solutions:
2x - y + 2z = 0,
3x + y - z = 0,
4x + ky + 0z = 0 (i.e., 4x + ky = 0).

B. Meaning of the Question

Set det(A) = 0 to determine the parameter k, then find the one-parameter family of non-trivial solutions.

C. Basic Concept

A homogeneous linear system AX = 0 has non-trivial solutions if and only if the determinant of the coefficient matrix vanishes: det(A) = 0.

D. Formula and Explanation

Formula: `det(A) = 0.`
Explanation & Conditions: If det(A) ≠ 0, only $X = [0, 0, 0]^T$ exists.

E. Given Information

A = [[2, -1, 2], [3, 1, -1], [4, k, 0]].

F. Complete Step-by-Step Solution

Step 1: Set determinant det(A) = 0

det(A) = | 2 -1 2 |
| 3 1 -1 |
| 4 k 0 | = 0.

💡 **Why this step is performed:** Execute step 1: set determinant det(a) = 0 to advance the step-by-step mathematical solution toward the final result.

Step 2: Expand along Row 3

4 [(-1)(-1) - (2)(1)] - k [2(-1) - 2(3)] + 0 = 0
4 [1 - 2] - k [-2 - 6] = 0
4(-1) - k(-8) = 0
-4 + 8k = 0 ⇒ 8k = 4 ⇒ k = 1/2.

💡 **Why this step is performed:** Execute step 2: expand along row 3 to advance the step-by-step mathematical solution toward the final result.

Step 3: Substitute k = 1/2 into the system

Equations become:
2x - y + 2z = 0 --- (1)
3x + y - z = 0 --- (2)
4x + (1/2)y = 0 ⇒ 8x + y = 0 ⇒ y = -8x --- (3).

💡 **Why this step is performed:** Execute step 3: substitute k = 1/2 into the system to advance the step-by-step mathematical solution toward the final result.

Step 4: Solve for z in terms of x

From (2): z = 3x + y = 3x + (-8x) = -5x.
Check in (1): 2x - (-8x) + 2(-5x) = 2x + 8x - 10x = 0. Matches!

💡 **Why this step is performed:** Execute step 4: solve for z in terms of x to advance the step-by-step mathematical solution toward the final result.

Step 5: Express solution in parametric vector form

Let x = t (where t is any arbitrary real constant):
x = t, y = -8t, z = -5t.
X = t [1, -8, -5]^{T}.

💡 **Why this step is performed:** Execute step 5: express solution in parametric vector form to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: k = 1/2; Non-trivial solutions: X = t [1, -8, -5]^{T}, where t is any non-zero real scalar.

H. Examination Presentation Strategy

- State condition det(A) = 0.
- Show expansion along row 3.
- Solve 8k = 4 ⇒ k = 1/2.
- State solution vector with parameter t.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Sign error in -k(-2 - 6) = +8k.
✓ **How to Avoid:** Carefully evaluate cofactor signs: -k(-8) = +8k.

⚠️ **Common Mistake:** Leaving answer as k = 1/2 without finding the solutions.
✓ **How to Avoid:** The question explicitly says 'AND find the solutions'.

J. Similar Practice Question

Practice Problem 69: Find k such that (1-k)x + y - z = 0, 2x - ky - 2z = 0, x - y - (1+k)z = 0 has non-trivial solutions.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 70: Theorem: Eigenvalues of a Real Symmetric Matrix are Real

EXAM-LEVEL

A. Full Question:

Prove that all eigenvalues of a real symmetric matrix are strictly real numbers.

B. Meaning of the Question

Use complex conjugate transpose (Hermitian transpose) to prove that no eigenvalue of a real symmetric matrix can have a non-zero imaginary part.

C. Basic Concept

Fundamental theorem of linear algebra. Guarantees that quantum observables (Hermitian operators) always yield real measurement values.

D. Formula and Explanation

Formula: `A v = λ v ⇒ v* A v = λ v* v; take conjugate transpose.`

Explanation & Conditions: v* is the conjugate transpose ($\bar{v}^T$); since A is real symmetric, $A^* = A^T = A$.

E. Given Information

A is an n×n matrix with real entries, $A^T = A$.

F. Complete Step-by-Step Solution

Step 1: Set up the eigenvalue equation

Let λ be an eigenvalue of A with corresponding non-zero eigenvector v in $\mathbb{C}^n$:
A v = λ v --- (1).

💡 **Why this step is performed:** Execute step 1: set up the eigenvalue equation to advance the step-by-step mathematical solution toward the final result.

Step 2: Pre-multiply by v* (conjugate transpose)

$v^* (A v) = v^* (\lambda v)$

$v^* A v = \lambda (v^* v)$ --- (2).

💡 **Why this step is performed:** Execute step 2: pre-multiply by v* (conjugate transpose) to advance the step-by-step mathematical solution toward the final result.

Step 3: Take conjugate transpose of equation (2)

$(v^* A v)^* = [\lambda (v^* v)]^*$

$v^* A^* (v^*)^* = \bar{\lambda} (v^* v)^*$.

Since $(v^*)^* = v$, $(v^* v)^* = v^* v$, and A is real symmetric ($A^* = A^T = A$):

$v^* A v = \bar{\lambda} (v^* v)$ --- (3).

💡 **Why this step is performed:** Execute step 3: take conjugate transpose of equation (2) to advance the step-by-step mathematical solution toward the final result.

Step 4: Subtract equation (3) from equation (2)

LHS - LHS = $v^* A v - v^* A v = 0$.

RHS - RHS = $(\lambda - \bar{\lambda}) (v^* v) = 0$.

$(\lambda - \bar{\lambda}) (v^* v) = 0$.

💡 **Why this step is performed:** Execute step 4: subtract equation (3) from equation (2) to advance the step-by-step mathematical solution toward the final result.

Step 5: Conclude $\lambda = \bar{\lambda}$ (λ is real)

Since v is an eigenvector, $v \neq 0$.

The Euclidean norm squared $v^* v = |v_1|^2 + |v_2|^2 + \ldots + |v_n|^2 > 0$ (strictly positive).

Therefore, we can divide by $v^* v$:

$\lambda - \bar{\lambda} = 0 \Rightarrow \lambda = \bar{\lambda}$.

A complex number equals its complex conjugate if and only if its imaginary part is zero.

Therefore, λ is a REAL number. Hence proved.

💡 **Why this step is performed:** Execute step 5: conclude $\lambda = \bar{\lambda}$ (λ is real) to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: All eigenvalues of a real symmetric matrix are real numbers (Hence proved)

H. Examination Presentation Strategy

- Define conjugate transpose v^*.
- Use property $A^* = A$ for real symmetric matrix.
- State $v^* v > 0$ since $v \neq 0$.
- Conclude $\lambda = \bar{\lambda}$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Assuming v has real entries from the start.
✓ **How to Avoid:** You CANNOT assume v is real; you must prove λ is real allowing for complex entries in v!

⚠️ **Common Mistake:** Omitting the justification that $v^* v \neq 0$.
✓ **How to Avoid:** You must state $v^* v = \Sigma |v_i|^2 > 0$ because v is non-zero.

J. Similar Practice Question

Practice Problem 70: Prove that the eigenvalues of a skew-symmetric real matrix ($A^T = -A$) are either purely imaginary or zero.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 71: Theorem: Eigenvectors of Real Symmetric Matrix are Orthogonal EXAM-LEVEL

A. Full Question:
Prove that eigenvectors corresponding to distinct eigenvalues of a real symmetric matrix are orthogonal.

B. Meaning of the Question

Show that if $A = A^T$ and $A v_1 = \lambda_1 v_1$, $A v_2 = \lambda_2 v_2$ with $\lambda_1 \neq \lambda_2$, then $v_1 \cdot v_2 = 0$.

C. Basic Concept

Distinct eigenspaces of a symmetric matrix are mutually perpendicular.

D. Formula and Explanation

Formula: $v_1^T A v_2 = \lambda_2 (v_1^T v_2)$ and $(A v_1)^T v_2 = \lambda_1 (v_1^T v_2)$.

Explanation & Conditions: Since $(A v_1)^T = v_1^T A^T = v_1^T A$, both expressions equate, forcing $(\lambda_1 - \lambda_2)(v_1 \cdot v_2) = 0$.

E. Given Information

$A = A^T$ (real symmetric), $A v_1 = \lambda_1 v_1$, $A v_2 = \lambda_2 v_2$, with $\lambda_1 \neq \lambda_2$.

F. Complete Step-by-Step Solution

Step 1: Write the eigenvalue equations

$A v_1 = \lambda_1 v_1$ --- (1)
$A v_2 = \lambda_2 v_2$ --- (2)
Since A is real symmetric, all eigenvalues and eigenvectors can be chosen real.

💡 **Why this step is performed:** Execute step 1: write the eigenvalue equations to advance the step-by-step mathematical solution toward the final result.

Step 2: Pre-multiply (2) by v_1^T

$v_1^T (A v_2) = v_1^T (\lambda_2 v_2) = \lambda_2 (v_1^T v_2)$ --- (3).

💡 **Why this step is performed:** Execute step 2: pre-multiply (2) by v_1^T to advance the step-by-step mathematical solution toward the final result.

Step 3: Take transpose of (1) and post-multiply by v_2

$(A v_1)^T = (\lambda_1 v_1)^T \Rightarrow v_1^T A^T = \lambda_1 v_1^T$.
Since A is symmetric, $A^T = A$:
$v_1^T A = \lambda_1 v_1^T$.
Post-multiply by v_2:
$(v_1^T A) v_2 = \lambda_1 (v_1^T v_2)$ --- (4).

💡 **Why this step is performed:** Execute step 3: take transpose of (1) and post-multiply by v_2 to advance the step-by-step mathematical solution toward the final result.

Step 4: Equate LHS of (3) and (4)

Notice that the LHS of (3) is $v_1^T A v_2$, which is identical to the LHS of (4):
$\lambda_1 (v_1^T v_2) = \lambda_2 (v_1^T v_2)$
$(\lambda_1 - \lambda_2) (v_1^T v_2) = 0$.

💡 **Why this step is performed:** Execute step 4: equate lhs of (3) and (4) to advance the step-by-step mathematical solution toward the final result.

Step 5: Conclude orthogonality

We are given that the eigenvalues are distinct: $\lambda_1 \neq \lambda_2 \Rightarrow \lambda_1 - \lambda_2 \neq 0$.
Dividing by $(\lambda_1 - \lambda_2)$ gives:
$v_1^T v_2 = 0 \Rightarrow v_1 \cdot v_2 = 0$.
Therefore, the eigenvectors v_1 and v_2 are ORTHOGONAL. Hence proved.

💡 **Why this step is performed:** Execute step 5: conclude orthogonality to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $v_1 \cdot v_2 = 0$ (Eigenvectors of distinct eigenvalues are orthogonal; Hence proved)

H. Examination Presentation Strategy

- State eigenvalue equations.
- Show transpose step $v_1^T A^T = v_1^T A$.
- Equate scalars: $(\lambda_1 - \lambda_2)(v_1^T v_2) = 0$.
- Conclude $v_1 \cdot v_2 = 0$ since $\lambda_1 \neq \lambda_2$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Forgetting to state $\lambda_1 \neq \lambda_2$.
✓ **How to Avoid:** If $\lambda_1 = \lambda_2$, the vectors can be any linear combination and need not be orthogonal without Gram-Schmidt.

⚠️ **Common Mistake:** Transpose error: $(A v)^T = A^T v^T$.
✓ **How to Avoid:** Correct transpose rule for products is $(AB)^T = B^T A^T$!

J. Similar Practice Question

Practice Problem 71: Verify this theorem for matrix A = [[1, 2], [2, 1]] using its eigenvectors.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 72: Gauss Elimination for a 4-Variable System

EXAM-LEVEL

A. Full Question:

Solve the system of 4 linear equations using Gauss Elimination:

w + x + y + z = 4,

2w - x + y - z = 1,

w + 2x - y + z = 3,

3w + x + 2y - z = 5.

B. Meaning of the Question

Reduce the 4×5 augmented matrix to upper triangular echelon form and determine w, x, y, z by back-substitution.

C. Basic Concept

Systematic 4×4 row reduction. Shows mastery of multi-step pivot operations.

D. Formula and Explanation

Formula: `[A|B] (4×5) ~ [U|B'].`

Explanation & Conditions: Variables: (w, x, y, z).

E. Given Information

4 equations in 4 variables.

F. Complete Step-by-Step Solution

Step 1: Set up 4×5 Augmented Matrix

[[1, 1, 1, 1 | 4], [2, -1, 1, -1 | 1], [1, 2, -1, 1 | 3], [3, 1, 2, -1 | 5]]. Pivot is a_{11} = 1.

💡 **Why this step is performed:** Execute step 1: set up 4×5 augmented matrix to advance the step-by-step mathematical solution toward the final result.

Step 2: Eliminate column 1 entries below pivot

$R_2 \to R_2 - 2R_1$: [0, -3, -1, -3 | -7].

$R_3 \to R_3 - R_1$: [0, 1, -2, 0 | -1].

$R_4 \to R_4 - 3R_1$: [0, -2, -1, -4 | -7].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 3: Swap $R_2 \leftrightarrow R_3$ to get pivot 1 in row 2

[[1, 1, 1, 1 | 4], [0, 1, -2, 0 | -1], [0, -3, -1, -3 | -7], [0, -2, -1, -4 | -7]].

💡 **Why this step is performed:** Execute step 3: swap $r_2 \leftrightarrow r_3$ to get pivot 1 in row 2 to advance the step-by-step mathematical solution toward the final result.

Step 4: Eliminate column 2 entries below pivot

$R_3 \rightarrow R_3 + 3R_2$: [0, 0, -7, -3 | -10] ⇒ [0, 0, 7, 3 | 10].
$R_4 \rightarrow R_4 + 2R_2$: [0, 0, -5, -4 | -9] ⇒ [0, 0, 5, 4 | 9].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 5: Eliminate column 3 in row 4

$R_4 \rightarrow 7R_4 - 5R_3$:
[0, 0, 7(5) - 5(7), 7(4) - 5(3) | 7(9) - 5(10)]
= [0, 0, 0, 28 - 15 | 63 - 50] = [0, 0, 0, 13 | 13] ⇒ [0, 0, 0, 1 | 1].

💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 6: Back-substitution

From Row 4: z = 1.
From Row 3: 7y + 3z = 10 ⇒ 7y + 3(1) = 10 ⇒ 7y = 7 ⇒ y = 1.
From Row 2: x - 2y = -1 ⇒ x - 2(1) = -1 ⇒ x = 1.
From Row 1: w + x + y + z = 4 ⇒ w + 1 + 1 + 1 = 4 ⇒ w = 1.

💡 **Why this step is performed:** Substitute known variable values upward from the bottom row equation to determine each variable uniquely.

✓ G. Final Answer: w = 1, x = 1, y = 1, z = 1

H. Examination Presentation Strategy

- Show 4×5 matrix clearly.
- Document row swaps and elimination.
- Show back-substitution clearly for each variable.
- Verify in equation 2: 2(1) - 1 + 1 - 1 = 1. Matches!

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Arithmetic error in 7(4) - 5(3) = 28 - 15 = 13.
✓ **How to Avoid:** Double check arithmetic on integer combinations.

⚠️ **Common Mistake:** Dropping one equation.
✓ **How to Avoid:** All 4 equations must be processed.

J. Similar Practice Question

Practice Problem 72: Solve by Gauss elimination: w + 2x + y = 4, 2w + x - z = 2, x + y + 2z = 4, w + y + z = 3.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 73: Powers of Matrix Using Diagonalization ($A^n = P D^n P^{-1}$)

EXAM-LEVEL

A. Full Question:

If A = [[1, 0], [2, 3]], find A^{10} using the diagonalization theorem.

B. Meaning of the Question

Diagonalize $A = P D P^{-1}$ and compute the 10th power using the matrix identity $A^{10} = P D^{10} P^{-1}$.

C. Basic Concept

Computing A^{10} directly requires 9 tedious matrix multiplications. Since $A = P D P^{-1}$, $A^n = (P D P^{-1})(P D P^{-1})... = P D^n P^{-1}$. For a diagonal matrix, D^n is trivially obtained by raising diagonal entries to power n.

D. Formula and Explanation

Formula: $A^n = P D^n P^{-1}$, where $D^n = [[\lambda_1^n, 0], [0, \lambda_2^n]]$.

Explanation & Conditions: Diagonal matrices commute with powers.

E. Given Information

A = [[1, 0], [2, 3]], n = 10.

F. Complete Step-by-Step Solution

Step 1: Find eigenvalues of lower triangular matrix A

Since A is lower triangular, its eigenvalues are simply its diagonal entries:
$\lambda_1 = 1, \lambda_2 = 3$.

💡 **Why this step is performed:** Execute step 1: find eigenvalues of lower triangular matrix a to advance the step-by-step mathematical solution toward the final result.

Step 2: Find eigenvector for $\lambda_1 = 1$

$(A - I) v_1 = 0 \Rightarrow [[0, 0], [2, 2]] [x, y]^T = [0, 0]^T \Rightarrow 2x + 2y = 0 \Rightarrow x = -y$.
Let $y = -1 \Rightarrow x = 1$. $v_1 = [1, -1]^T$.

💡 **Why this step is performed:** Solve the homogeneous linear system $(A - \lambda I)v = 0$ to find the non-zero basis vectors spanning the eigenspace.

Step 3: Find eigenvector for $\lambda_2 = 3$

$(A - 3I)\ v_2 = 0 \Rightarrow [\ [-2, 0], [2, 0]\ ]\ [x, y]^T = [0, 0]^T \Rightarrow -2x = 0 \Rightarrow x = 0.$
y can be any value, choose $y = 1$. $v_2 = [0, 1]^T$.

> 💡 **Why this step is performed:** Solve the homogeneous linear system $(A - \lambda I)v = 0$ to find the non-zero basis vectors spanning the eigenspace.

Step 4: Construct modal matrix P and P^{-1}

$P = [\ [1, 0], [-1, 1]\ ]$.
$\det(P) = (1)(1) - (0)(-1) = 1$.
$P^{-1} = [\ [1, 0], [1, 1]\ ]$.

> 💡 **Why this step is performed:** Execute step 4: construct modal matrix p and p^{-1} to advance the step-by-step mathematical solution toward the final result.

Step 5: Compute D^{10} and $A^{10} = P\ D^{10}\ P^{-1}$

$D = [\ [1, 0], [0, 3]\ ] \Rightarrow D^{10} = [\ [1^{10}, 0], [0, 3^{10}]\ ] = [\ [1, 0], [0, 59049]\ ]$.
$P\ D^{10} = [\ [1, 0], [-1, 1]\ ]\ [\ [1, 0], [0, 3^{10}]\ ] = [\ [1, 0], [-1, 3^{10}]\ ]$.
$A^{10} = (P\ D^{10})\ P^{-1} = [\ [1, 0], [-1, 3^{10}]\ ]\ [\ [1, 0], [1, 1]\ ]$
$= [\ [1(1) + 0(1), 1(0) + 0(1)], [-1(1) + 3^{10}(1), -1(0) + 3^{10}(1)]\ ]$
$= [\ [1, 0], [3^{10} - 1, 3^{10}]\ ]$.
Since $3^{10} = 59049$: $3^{10} - 1 = 59048$.
$A^{10} = [\ [1, 0], [59048, 59049]\ ]$.

> 💡 **Why this step is performed:** Execute step 5: compute d^{10} and $a^{10} = p\ d^{10}\ p^{-1}$ to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $A^{10} = [[1, 0], [3^{10} - 1, 3^{10}]] = [[1, 0], [59048, 59049]]$

H. Examination Presentation Strategy

- State $A^n = P\ D^n\ P^{-1}$ formula.
- Compute eigenvalues and eigenvectors.
- Form P and P^{-1}.
- Show multiplication to reach final matrix.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Computing A^2 then A^4 then A^{10} directly.
✓ **How to Avoid:** The problem requires using DIAGONALIZATION, not brute force matrix multiplication!

⚠️ **Common Mistake:** Wrong power $3^{10} = 59049$.
✓ **How to Avoid:** Compute powers of 3 carefully: $3^2=9$, $3^4=81$, $3^8=6561$, $3^{10}=6561\times9=59049$.

J. Similar Practice Question

Practice Problem 73: Find A^8 where $A = [[3, 4], [1, 0]]$ using diagonalization.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 74: Rank of a 4×4 Matrix by Normal Form [I_r, 0] EXAM-LEVEL

A. Full Question:
Reduce the matrix A = [[0, 1, -3, -1], [1, 0, 1, 1], [3, 1, 0, 2], [1, 1, -2, 0]] to normal form [I_r 0; 0 0] using elementary row and column operations, and find its rank.

B. Meaning of the Question

Use both row and column operations to transform A into a canonical block matrix containing an identity matrix I_r surrounded by zeros, which reveals rank r.

C. Basic Concept

Every m×n matrix of rank r can be reduced to the normal form [I_r 0; 0 0] using both row and column elementary operations. The order of the identity block equals the rank.

D. Formula and Explanation

Formula: `PAQ = [ [I_r, 0], [0, 0] ] ⇒ Rank ρ(A) = r.`

Explanation & Conditions: Row operations clear column entries below pivots; column operations clear row entries to the right.

E. Given Information

4×4 matrix A.

F. Complete Step-by-Step Solution

Step 1: Get pivot 1 in (1, 1) by swapping $R_1 \leftrightarrow R_2$

[[1, 0, 1, 1], [0, 1, -3, -1], [3, 1, 0, 2], [1, 1, -2, 0]].

💡 **Why this step is performed:** Execute step 1: get pivot 1 in (1, 1) by swapping $r_1 \leftrightarrow r_2$ to advance the step-by-step mathematical solution toward the final result.

Step 2: Clear column 1 below pivot using row operations

$R_3 \rightarrow R_3 - 3R_1$: [0, 1, -3, -1].
$R_4 \rightarrow R_4 - R_1$: [0, 1, -3, -1].
Matrix becomes:
[[1, 0, 1, 1], [0, 1, -3, -1], [0, 1, -3, -1], [0, 1, -3, -1]].

💡 **Why this step is performed:** Execute step 2: clear column 1 below pivot using row operations to advance the step-by-step mathematical solution toward the final result.

Step 3: Clear row 1 to the right of pivot using column operations

$C_3 \rightarrow C_3 - C_1$: eliminates entry (1, 3).
$C_4 \rightarrow C_4 - C_1$: eliminates entry (1, 4).
Matrix becomes:
[[1, 0, 0, 0], [0, 1, -3, -1], [0, 1, -3, -1], [0, 1, -3, -1]].

💡 **Why this step is performed:** Execute step 3: clear row 1 to the right of pivot using column operations to advance the step-by-step mathematical solution toward the final result.

Step 4: Clear rows 3 and 4 using row operations

$R_3 \to R_3 - R_2$: [0, 0, 0, 0].
$R_4 \to R_4 - R_2$: [0, 0, 0, 0].
Matrix becomes:
[[1, 0, 0, 0], [0, 1, -3, -1], [0, 0, 0, 0], [0, 0, 0, 0]].

💡 **Why this step is performed:** Execute step 4: clear rows 3 and 4 using row operations to advance the step-by-step mathematical solution toward the final result.

Step 5: Clear row 2 to the right of pivot using column operations

$C_3 \to C_3 + 3C_2$: eliminates -3.
$C_4 \to C_4 + C_2$: eliminates -1.
Matrix becomes:
[[1, 0, 0, 0], [0, 1, 0, 0], [0, 0, 0, 0], [0, 0, 0, 0]] = [[I_2, O], [O, O]].

💡 **Why this step is performed:** Execute step 5: clear row 2 to the right of pivot using column operations to advance the step-by-step mathematical solution toward the final result.

Step 6: State rank from identity block size

The identity block is I_2 of size 2×2.
Therefore, Rank ρ(A) = 2.

💡 **Why this step is performed:** Execute step 6: state rank from identity block size to advance the step-by-step mathematical solution toward the final result.

✓ **G. Final Answer: Normal Form: [[I_2, O], [O, O]] where I_2 is a 2×2 identity matrix; Rank ρ(A) = 2**

H. Examination Presentation Strategy

- Show row operations on left and column operations on right.
- Write intermediate matrices clearly.
- State final normal block matrix.
- Conclude rank = 2.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Using only row operations.
✓ **How to Avoid:** Normal form REQUIRES both row AND column operations to reduce off-diagonal elements to zero!

⚠️ **Common Mistake:** Writing rank as 4.
✓ **How to Avoid:** Two rows and two columns are completely zero, so rank is 2.

J. Similar Practice Question

Practice Problem 74: Reduce A = [[1, 2, -1, 3], [4, 1, 2, 1], [3, -1, 1, 2], [1, 2, 0, 1]] to normal form and find rank.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 75: Cayley-Hamilton Theorem for 3×3 Matrix: Inverse and Higher Powers

EXAM-LEVEL

A. Full Question:

State and verify Cayley-Hamilton theorem for A = [[1, 0, -1], [1, 2, 1], [2, 2, 3]]. Hence compute A^{-1} and A^4.

B. Meaning of the Question

Verify $A^3 - s_1A^2 + s_2A - s_3I = 0$, then express A^{-1} and A^4 as linear combinations of A^2, A, and I.

C. Basic Concept

Cayley-Hamilton theorem allows expressing any higher power A^n and the inverse A^{-1} as a polynomial of degree at most (n - 1) in A.

D. Formula and Explanation

Formula: $A^3 - s_1A^2 + s_2A - s_3I = 0 \Rightarrow A^{-1} = (1/s_3)[\, A^2 - s_1A + s_2I \,]; \; A^4 = s_1A^3 - s_2A^2 + s_3A.$

Explanation & Conditions: s_1 = Trace, s_2 = sum of principal minors, s_3 = determinant.

E. Given Information

A = [[1, 0, -1], [1, 2, 1], [2, 2, 3]].

F. Complete Step-by-Step Solution

Step 1: Compute coefficients of characteristic equation

• s_1 = Trace = 1 + 2 + 3 = 6.
• s_2 = |2 1; 2 3| + |1 -1; 2 3| + |1 0; 1 2| = (6 - 2) + (3 + 2) + (2 - 0) = 4 + 5 + 2 = 11.
• s_3 = det(A) = 1(6 - 2) - 0 + (-1)(2 - 4) = 4 - 0 + 2 = 6.
Characteristic equation: $\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$.
Cayley-Hamilton equation: $A^3 - 6A^2 + 11A - 6I = O$.

💡 **Why this step is performed:** Set up the characteristic determinant det(A - λI) = 0 and expand to obtain the characteristic polynomial.

Step 2: Compute A^2

A^2 = [[1, 0, -1], [1, 2, 1], [2, 2, 3]] [[1, 0, -1], [1, 2, 1], [2, 2, 3]]
= [[-1, -2, -4], [5, 6, 4], [10, 10, 9]].

💡 **Why this step is performed:** Execute step 2: compute a^2 to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute A^3 and verify $A^3 - 6A^2 + 11A - 6I = O$

$A^3 = A^2 \cdot A$ = [[-11, -12, -13], [19, 20, 13], [38, 38, 27]].
Evaluating $A^3 - 6A^2 + 11A - 6I$ term by term yields [[0, 0, 0], [0, 0, 0], [0, 0, 0]] = O.
Cayley-Hamilton theorem is VERIFIED!

💡 **Why this step is performed:** Execute step 3: compute a^3 and verify $a^3 - 6a^2 + 11a - 6i$ = o to advance the step-by-step mathematical solution toward the final result.

Step 4: Compute A^{-1} from Cayley-Hamilton equation

Multiply by A^{-1}:
$A^2 - 6A + 11I - 6A^{-1} = 0 \Rightarrow 6A^{-1} = A^2 - 6A + 11I$
$A^{-1} = (1/6) [A^2 - 6A + 11I]$
= (1/6) [[4, -2, 2], [-1, 5, -2], [-2, -2, 2]].

💡 **Why this step is performed:** Execute step 4: compute a^{-1} from cayley-hamilton equation to advance the step-by-step mathematical solution toward the final result.

Step 5: Compute A^4 using Cayley-Hamilton equation

From $A^3 = 6A^2 - 11A + 6I$, multiply by A:
$A^4 = 6A^3 - 11A^2 + 6A = 6(6A^2 - 11A + 6I) - 11A^2 + 6A$
$= 36A^2 - 66A + 36I - 11A^2 + 6A$
$= 25A^2 - 60A + 36I$.
Substituting A^2 and A gives the required 3×3 matrix.

💡 **Why this step is performed:** Execute step 5: compute a^4 using cayley-hamilton equation to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Cayley-Hamilton Verified: $A^3 - 6A^2 + 11A - 6I = O$; $A^{-1} = (1/6)$ [[4, -2, 2], [-1, 5, -2], [-2, -2, 2]]; $A^4 = 25A^2 - 60A + 36I$

H. Examination Presentation Strategy

- Compute s_1, s_2, s_3 clearly.
- Show A^2 and A^3.
- Verify $A^3 - 6A^2 + 11A - 6I = O$ explicitly.
- Show algebraic derivation of A^{-1} and A^4.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Computing A^4 by multiplying A^2 by A^2.
✓ **How to Avoid:** The prompt asks to use Cayley-Hamilton; reduce using $A^3 = 6A^2 - 11A + 6I$!

⚠️ **Common Mistake:** Arithmetic error in determinant $s_3 = 6$.
✓ **How to Avoid:** 1(4) - (-1)(-2) = 4 + 2 = 6.

J. Similar Practice Question

Practice Problem 75: Verify Cayley-Hamilton theorem for A = [[2, -1, 1], [-1, 2, -1], [1, -1, 2]] and find A^{-1}.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

MODULE 4: Vector Spaces

25 Fully Solved Questions: 5 Fundamental (Concepts & Proofs) + 10 Intermediate (Core Problems) + 10 Exam-Level (High-Yield VTU Questions)

QUESTION 76: Definition of Vector Space over $\mathbb{R}$ and Its Fundamental Axioms FUNDAMENTAL

A. Full Question:
Define a real vector space V and state the ten fundamental axioms satisfied by vector addition and scalar multiplication.

B. Meaning of the Question

Provide the formal mathematical definition of a vector space over the field of real numbers $\mathbb{R}$, detailing the closure, abelian group, and scalar multiplication axioms.

C. Basic Concept

A vector space is an abstract algebraic structure consisting of a set V whose elements ('vectors') can be added together and multiplied ('scaled') by numbers from a field F, satisfying natural geometric rules.

D. Formula and Explanation

Formula: V with operations $+ : V \times V \to V$ and $\cdot : \mathbb{R} \times V \to V$.

Explanation & Conditions: 5 axioms for addition (+), 5 axioms for scalar multiplication (·).

E. Given Information

Set V, field of scalars $\mathbb{R}$.

F. Complete Step-by-Step Solution

Step 1: Axioms for Vector Addition (V is an Abelian group under +)

For all u, v, w ∈ V:
1. Closure under addition: u + v ∈ V.
2. Commutativity: u + v = v + u.
3. Associativity: (u + v) + w = u + (v + w).
4. Additive Identity: There exists a zero vector 0 ∈ V such that u + 0 = u.
5. Additive Inverse: For each u ∈ V, there exists -u ∈ V such that u + (-u) = 0.

💡 **Why this step is performed:** Execute step 1: axioms for vector addition (v is an abelian group under +) to advance the step-by-step mathematical solution toward the final result.

Step 2: Axioms for Scalar Multiplication

For all c, d ∈ $\mathbb{R}$ and u, v ∈ V:
6. Closure under scalar multiplication: c u ∈ V.
7. Distributivity over vector addition: c(u + v) = c u + c v.
8. Distributivity over scalar addition: (c + d)u = c u + d u.
9. Compatibility of multiplication: c(d u) = (c d)u.
10. Scalar identity: 1 · u = u for the real number 1.

💡 **Why this step is performed:** Execute step 2: axioms for scalar multiplication to advance the step-by-step mathematical solution toward the final result.

✓ **G. Final Answer: A real vector space V is a non-empty set with vector addition and scalar multiplication satisfying the 10 axioms (closure, commutativity, associativity, identity, inverse, and distributivities).**

H. Examination Presentation Strategy

- Group axioms neatly into Addition (1-5) and Scalar Multiplication (6-10).
- State universal quantifiers ∀ u, v, w ∈ V and c, d ∈ $\mathbb{R}$.

I. Common Mistakes & How to Avoid Them

⚠ **Common Mistake:** Omitting the scalar identity axiom 1·u = u.
✓ **How to Avoid:** Without 1·u = u, algebraic pathologies can occur!

⚠ **Common Mistake:** Writing c + u.
✓ **How to Avoid:** You cannot add a scalar to a vector! Only c·u or u + v is defined.

J. Similar Practice Question

Practice Problem 76: Show that the set of all 2×2 real matrices $M_{22}(\mathbb{R})$ forms a vector space over $\mathbb{R}$ under standard addition and scalar multiplication.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 77: Subspace Criteria (The Two-Step and One-Step Subspace Tests) FUNDAMENTAL

A. Full Question:
State the necessary and sufficient conditions for a non-empty subset W of a vector space V to be a subspace of V.

B. Meaning of the Question

State the conditions required to test whether a subset W inherits the vector space structure from the enclosing parent space V.

C. Basic Concept

Since W is inside V, it automatically inherits associativity, commutativity, and distributivity. We only need to check that W is closed under the operations and contains the origin.

D. Formula and Explanation

Formula: One-step test: W is a subspace if and only if for all u, v ∈ W and c ∈ $\mathbb{R}$, (u + c v) ∈ W (or a u + b v ∈ W).
Two-step test: (1) 0 ∈ W, (2) u + v ∈ W, (3) c u ∈ W.
Explanation & Conditions: Non-emptiness is most easily tested by checking if 0 ∈ W.

E. Given Information

Subset $W \subseteq V$.

F. Complete Step-by-Step Solution

Step 1: Two-Step Subspace Test

A subset W of vector space V is a subspace if and only if:
1. W is non-empty (typically verified by showing the zero vector $0 \in W$).
2. Closed under vector addition: $\forall\ u, v \in W \Rightarrow u + v \in W$.
3. Closed under scalar multiplication: $\forall\ c \in \mathbb{R}, u \in W \Rightarrow c\,u \in W$.

💡 **Why this step is performed:** Execute step 1: two-step subspace test to advance the step-by-step mathematical solution toward the final result.

Step 2: Equivalent One-Step Test (Linear Combination Test)

W is a subspace of V if and only if W is non-empty and for all scalars $a, b \in \mathbb{R}$ and all vectors $u, v \in W$:
$a\,u + b\,v \in W$.

💡 **Why this step is performed:** Execute step 2: equivalent one-step test (linear combination test) to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Subspace Criteria: (i) $0 \in W$, (ii) $u + v \in W$ for all $u, v \in W$, and (iii) $c\,u \in W$ for all $c \in \mathbb{R}$, $u \in W$ (Equivalently: $a\,u + b\,v \in W$).

H. Examination Presentation Strategy

- State both the 3-step test and the combined 1-step test.
- Highlight the quick check $0 \in W$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Forgetting to check if $0 \in W$.
✓ **How to Avoid:** If $0 \notin W$, W can NEVER be a subspace! This immediately disproves non-subspaces.

⚠️ **Common Mistake:** Testing all 10 vector space axioms.
✓ **How to Avoid:** Never test all 10 axioms for a subset; only test closure and $0 \in W$!

J. Similar Practice Question

Practice Problem 77: Use the subspace test to determine if $W = \{(x, y, 0) : x, y \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 78: Linear Combinations and Span of Vectors FUNDAMENTAL

A. Full Question:
Define linear combination and span of a set of vectors S = {v_1, v_2, ..., v_k} in a vector space V, and express v = (3, 7) as a linear combination of v_1 = (1, 1) and v_2 = (1, 2) in $\mathbb{R}^2$.

B. Meaning of the Question

Define what it means for a vector to be formed by scaling and adding set S, and find the specific scalar coefficients c_1, c_2 such that $v = c_1 v_1 + c_2 v_2$.

C. Basic Concept

A linear combination is any vector of the form $c_1 v_1 + c_2 v_2 + ... + c_k v_k$. The span of S, denoted span(S), is the set of ALL possible linear combinations of vectors in S, forming the smallest subspace containing S.

D. Formula and Explanation

Formula: $v = c_1 v_1 + c_2 v_2; \text{span}(S) = \{ \Sigma c_i v_i : c_i \in \mathbb{R} \}.$

Explanation & Conditions: Linear combination solves a system of linear equations for the unknown scalars c_i.

E. Given Information

v = (3, 7), v_1 = (1, 1), v_2 = (1, 2).

F. Complete Step-by-Step Solution

Step 1: Set up the vector equation

c_1 (1, 1) + c_2 (1, 2) = (3, 7)
(c_1 + c_2, c_1 + 2c_2) = (3, 7).

💡 **Why this step is performed:** Execute step 1: set up the vector equation to advance the step-by-step mathematical solution toward the final result.

Step 2: Equate components to form linear system

c_1 + c_2 = 3 --- (1)
c_1 + 2c_2 = 7 --- (2).

💡 **Why this step is performed:** Execute step 2: equate components to form linear system to advance the step-by-step mathematical solution toward the final result.

Step 3: Solve for c_1 and c_2

Subtract (1) from (2):
(c_1 + 2c_2) - (c_1 + c_2) = 7 - 3 ⇒ c_2 = 4.
Substitute c_2 = 4 into (1):
c_1 + 4 = 3 ⇒ c_1 = 3 - 4 = -1.

💡 **Why this step is performed:** Execute step 3: solve for c_1 and c_2 to advance the step-by-step mathematical solution toward the final result.

Step 4: Write the linear combination

v = -1 v_1 + 4 v_2 = -1(1, 1) + 4(1, 2).
Check: (-1 + 4, -1 + 8) = (3, 7). Exactly verified!

💡 **Why this step is performed:** Execute step 4: write the linear combination to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $v = -1\ v_1 + 4\ v_2$ (or $v = -(1, 1) + 4(1, 2)$)

H. Examination Presentation Strategy

- State definitions of linear combination and span clearly.
- Set up $c_1v_1 + c_2v_2 = v$.
- Solve the 2×2 system.
- Write final vector equation.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Arithmetic error in 3 - 4 = -1.
✓ **How to Avoid:** Check signs carefully.

⚠️ **Common Mistake:** Not verifying the result.
✓ **How to Avoid:** Always verify: -1(1, 1) + 4(1, 2) = (-1+4, -1+8) = (3, 7).

J. Similar Practice Question

Practice Problem 78: Express $v = (2, -5, 3)$ as a linear combination of $v_1 = (1, -3, 2)$, $v_2 = (2, -4, -1)$, and $v_3 = (1, -5, 7)$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 79: Testing Linear Independence of Vectors in $\mathbb{R}^3$ FUNDAMENTAL

A. Full Question:
Determine whether the set of vectors $S = \{(1, 2, 1), (2, 1, 0), (1, -1, -1)\}$ in $\mathbb{R}^3$ is linearly independent or linearly dependent.

B. Meaning of the Question

Test if $c_1\ v_1 + c_2\ v_2 + c_3\ v_3 = 0$ has only the trivial solution $c_1 = c_2 = c_3 = 0$ (independent) or has non-zero scalar solutions (dependent), by evaluating the determinant.

C. Basic Concept

Vectors are linearly independent if no vector in the set can be written as a linear combination of the others. For n vectors in $\mathbb{R}^n$, the set is linearly independent if and only if the matrix formed by them has non-zero determinant.

D. Formula and Explanation

Formula: `det [v₁ | v₂ | v₃] ≠ 0 ⇒ Linearly Independent; det = 0 ⇒ Linearly Dependent.`

Explanation & Conditions: $c_1v_1 + c_2v_2 + c_3v_3 = 0$ is equivalent to A c = 0.

E. Given Information

$v_1 = (1, 2, 1)$, $v_2 = (2, 1, 0)$, $v_3 = (1, -1, -1)$.

F. Complete Step-by-Step Solution

Step 1: Set up the matrix with vectors as rows or columns

A = | 1 2 1 |
| 2 1 0 |
| 1 -1 -1 |.

💡 **Why this step is performed:** Execute step 1: set up the matrix with vectors as rows or columns to advance the step-by-step mathematical solution toward the final result.

Step 2: Evaluate the determinant det(A)

Expand along Row 2 (because it has a zero):
det(A) = -2 [(2)(-1) - (1)(-1)] + 1 [(1)(-1) - (1)(1)] - 0
= -2 [-2 + 1] + 1 [-1 - 1]
= -2 [-1] + 1 [-2]
= 2 - 2 = 0.

💡 **Why this step is performed:** Execute step 2: evaluate the determinant det(a) to advance the step-by-step mathematical solution toward the final result.

Step 3: State the conclusion

Since det(A) = 0, the columns/rows are linearly dependent.
Therefore, the set S is LINEARLY DEPENDENT.
(Notice that $v_1 - v_2 + v_3 = (1-2+1, 2-1-1, 1-0-1) = (0, 0, 0)$ provides the non-trivial relation with $c_1=1$, $c_2=-1$, $c_3=1$).

💡 **Why this step is performed:** Execute step 3: state the conclusion to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: det(A) = 0; The set S is LINEARLY DEPENDENT.

H. Examination Presentation Strategy

- Form matrix A.
- Compute det(A) clearly.
- State: det = 0 implies Linearly Dependent (det ≠ 0 implies Independent).
- Provide non-trivial relation $v_1 - v_2 + v_3 = 0$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Concluding independent when det = 0.
✓ **How to Avoid:** det = 0 means DEPENDENT! det ≠ 0 means INDEPENDENT!

⚠️ **Common Mistake:** Sign error on cofactor of entry (2, 1).
✓ **How to Avoid:** Position (2, 1) has sign $(-1)^{2+1} = -1$.

J. Similar Practice Question

Practice Problem 79: Test whether S = {(1, 1, 1), (0, 1, 1), (0, 0, 1)} is linearly independent in $\mathbb{R}^3$.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 80: Basis and Dimension of a Vector Space

FUNDAMENTAL

A. Full Question:

Define a basis and dimension of a vector space V, and state the standard basis and dimension of $\mathbb{R}^n$, P_n(x) (polynomials of degree ≤ n), and M_{m×n}(ℝ).

B. Meaning of the Question

Provide standard definitions of basis (linearly independent spanning set) and dimension (number of elements in a basis), and list standard examples.

C. Basic Concept

A basis is a minimal spanning set and a maximal linearly independent set. Every vector in V can be uniquely represented as a linear combination of basis vectors. Dimension is the number of vectors in any basis.

D. Formula and Explanation

Formula: `B = {v₁, v₂, ..., v_n} is a basis if: (1) B is linearly independent, (2) span(B) = V. dim(V) = |B| = n.`

Explanation & Conditions: All bases of a finite-dimensional space have identical size.

E. Given Information

Vector spaces $\mathbb{R}^n$, P_n(x), M_{m×n}(ℝ).

F. Complete Step-by-Step Solution

Step 1: Definition of Basis and Dimension

A subset B of a vector space V is called a BASIS of V if:
1. B is linearly independent.
2. B spans V (span(B) = V).
The DIMENSION of V, denoted dim(V), is the number of vectors in any basis of V. If V = {0}, dim(V) = 0.

💡 **Why this step is performed:** Extract the linearly independent spanning vectors corresponding to the free parameters to form the basis.

Step 2: Standard Basis and Dimension of $\mathbb{R}^n$

Standard basis: B = {e_1, e_2, ..., e_n} where e_i has 1 in the i-th position and 0 elsewhere.
Dimension: dim($\mathbb{R}^n$) = n.

💡 **Why this step is performed:** Extract the linearly independent spanning vectors corresponding to the free parameters to form the basis.

Step 3: Standard Basis and Dimension of P_n(x)

$P_n(x)$ is the space of all polynomials of degree $\leq n$: $p(x) = a_0 + a_1x + \ldots + a_n x^n$.
Standard basis: $B = \{1, x, x^2, \ldots, x^n\}$.
Dimension: $\dim(P_n) = n + 1$ (since it includes the constant term 1).

💡 **Why this step is performed:** Extract the linearly independent spanning vectors corresponding to the free parameters to form the basis.

Step 4: Standard Basis and Dimension of $M_{m\times n}(\mathbb{R})$

Space of $m \times n$ matrices.
Standard basis: $B = \{E_{ij}\}$ where E_{ij} has 1 at entry (i, j) and 0 elsewhere.
Dimension: $\dim(M_{m\times n}) = m \cdot n$.

💡 **Why this step is performed:** Extract the linearly independent spanning vectors corresponding to the free parameters to form the basis.

✓ G. Final Answer: Basis: Linearly independent spanning set. Dimension: Number of basis vectors. $\dim(\mathbb{R}^n) = n$; $\dim(P_n) = n + 1$; $\dim(M_{m\times n}) = m \cdot n$.

H. Examination Presentation Strategy

- State both conditions for a basis.
- Define dimension.
- List the 3 standard spaces with their standard bases and dimensions.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Writing $\dim(P_n) = n$ instead of $n + 1$.
✓ **How to Avoid:** P_n includes degree 0 (constant 1), so $\{1, x, \ldots, x^n\}$ has $n + 1$ vectors!

⚠️ **Common Mistake:** Forgetting that a basis must satisfy BOTH independence AND spanning.
✓ **How to Avoid:** A set is not a basis if it fails either condition.

J. Similar Practice Question

Practice Problem 80: Find a basis and dimension for the subspace $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 81: Subspace Verification: Plane Passing Through the Origin INTERMEDIATE

A. Full Question:
Prove that the set W = {(x, y, z) ∈ $\mathbb{R}^3$: 2x - 3y + z = 0} is a subspace of $\mathbb{R}^3$, whereas U = {(x, y, z) ∈ $\mathbb{R}^3$: 2x - 3y + z = 5} is not a subspace.

B. Meaning of the Question

Apply the subspace test to show that a plane passing through the origin is closed under addition and scalar multiplication, while a plane not containing the origin fails the zero vector test.

C. Basic Concept

A flat plane in $\mathbb{R}^3$ is a subspace if and only if it passes through the origin (0, 0, 0). Any translation away from the origin destroys closure under scalar multiplication and eliminates the zero vector.

D. Formula and Explanation

Formula: `Test for W: (1) 0 ∈ W, (2) c u + v ∈ W. For U: Check (0, 0, 0) ∈ U.`

Explanation & Conditions: 2(0) - 3(0) + 0 = 0 satisfies W, but 0 = 5 is impossible for U.

E. Given Information

W: 2x - 3y + z = 0; U: 2x - 3y + z = 5.

F. Complete Step-by-Step Solution

Step 1: Check Zero Vector in W

For (0, 0, 0): 2(0) - 3(0) + 0 = 0. Thus (0, 0, 0) ∈ W. (Non-empty).

💡 **Why this step is performed:** Verify that the zero vector satisfies the defining equation, proving the subset is non-empty.

Step 2: Check Closure under Addition in W

Let u = (x_1, y_1, z_1) ∈ W and v = (x_2, y_2, z_2) ∈ W.
Then $2x_1 - 3y_1 + z_1 = 0$ and $2x_2 - 3y_2 + z_2 = 0$.
For u + v = ($x_1 + x_2$, $y_1 + y_2$, $z_1 + z_2$):
$2(x_1 + x_2) - 3(y_1 + y_2) + (z_1 + z_2) = (2x_1 - 3y_1 + z_1) + (2x_2 - 3y_2 + z_2) = 0 + 0 = 0$.
Therefore, u + v ∈ W.

💡 **Why this step is performed:** Take two arbitrary elements from the subset and prove their vector sum also satisfies the defining condition.

Step 3: Check Closure under Scalar Multiplication in W

For any scalar c ∈ $\mathbb{R}$ and u ∈ W:
c u = (c x_1, c y_1, c z_1).
$2(c\,x_1) - 3(c\,y_1) + (c\,z_1) = c\,(2x_1 - 3y_1 + z_1) = c\,(0) = 0$.
Therefore, c u ∈ W.
Since all conditions are satisfied, W is a SUBSPACE of $\mathbb{R}^3$.

💡 **Why this step is performed:** Multiply an arbitrary element by an arbitrary scalar and prove the resulting vector remains in the subset.

Step 4: Test subset U for Zero Vector

For the zero vector (0, 0, 0):
2(0) - 3(0) + 0 = 0 ≠ 5.
The zero vector does not satisfy the equation: (0, 0, 0) ∉ U.
Since every vector space must contain the zero vector, U is NOT a subspace of $\mathbb{R}^3$.

💡 **Why this step is performed:** Verify that the zero vector satisfies the defining equation, proving the subset is non-empty.

✓ G. Final Answer: W is a subspace of $\mathbb{R}^3$ (satisfies all closure axioms and 0 ∈ W); U is NOT a subspace because (0, 0, 0) ∉ U.

H. Examination Presentation Strategy

- Test W using (i) 0 ∈ W, (ii) addition closure, (iii) scalar closure.
- For U, directly evaluate (0, 0, 0) to show 0 ≠ 5.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Testing addition for U before checking the zero vector.
✓ **How to Avoid:** Always check the zero vector FIRST! If (0, 0, 0) ∉ U, it immediately disproves the subspace in 1 line.

⚠️ **Common Mistake:** Assuming all planes are subspaces.
✓ **How to Avoid:** Only planes PASSING THROUGH THE ORIGIN (constant term = 0) are subspaces.

J. Similar Practice Question

Practice Problem 81: Determine if W = {(x, y) ∈ $\mathbb{R}^2$: y = $3x^2$} is a subspace of $\mathbb{R}^2$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 82: Finding Basis and Dimension of a Subspace in $\mathbb{R}^4$ INTERMEDIATE

A. Full Question:
Find a basis and dimension for the subspace W of $\mathbb{R}^4$ defined by W = {(a, b, c, d) ∈ $\mathbb{R}^4$: a - 2b + c = 0 and 2a - 3b + d = 0}.

B. Meaning of the Question

Express the constrained 4-tuples in terms of free parameters, extract the generating vectors, and state the basis and dimension.

C. Basic Concept

For a subspace defined by k independent linear constraints in $\mathbb{R}^n$, dimension is dim(W) = n - k. Here n = 4 and k = 2, so dim(W) = 4 - 2 = 2.

D. Formula and Explanation

Formula: $a = 2b - c,\ d = 3b - 2a = 3b - 2(2b - c) = -b + 2c.$

Explanation & Conditions: b and c are chosen as the 2 free parameters.

E. Given Information

W = {(a, b, c, d) : a - 2b + c = 0, 2a - 3b + d = 0}.

F. Complete Step-by-Step Solution

Step 1: Express leading variables in terms of free variables

From equation 1: a = 2b - c.
From equation 2: d = 3b - 2a = 3b - 2(2b - c) = 3b - 4b + 2c = -b + 2c.
The free variables are b and c.

💡 **Why this step is performed:** Execute step 1: express leading variables in terms of free variables to advance the step-by-step mathematical solution toward the final result.

Step 2: Write general vector in W

v = (a, b, c, d) = (2b - c, b, c, -b + 2c).

💡 **Why this step is performed:** Execute step 2: write general vector in w to advance the step-by-step mathematical solution toward the final result.

Step 3: Decompose into independent parametric vectors

= (2b, b, 0, -b) + (-c, 0, c, 2c)
= b (2, 1, 0, -1) + c (-1, 0, 1, 2).

💡 **Why this step is performed:** Execute step 3: decompose into independent parametric vectors to advance the step-by-step mathematical solution toward the final result.

Step 4: Check Linear Independence of generating vectors

Let v_1 = (2, 1, 0, -1) and v_2 = (-1, 0, 1, 2).
Since neither vector is a scalar multiple of the other (v_1 has 0 in 3rd entry while v_2 has 1), they are linearly independent.
Since they span W and are linearly independent, B = {v_1, v_2} is a BASIS of W.

💡 **Why this step is performed:** Set a linear combination to zero and show that the only solution is the trivial solution where all scalar weights are zero.

Step 5: State the dimension

The basis contains 2 vectors.
Therefore, dim(W) = 2.

💡 **Why this step is performed:** Execute step 5: state the dimension to advance the step-by-step mathematical solution toward the final result.

✓ **G. Final Answer: Basis B = {(2, 1, 0, -1), (-1, 0, 1, 2)}; Dimension dim(W) = 2**

H. Examination Presentation Strategy

- Solve constraint equations for dependent variables.
- Write vector as b v_1 + c v_2.
- State that {v_1, v_2} are linearly independent and span W.
- Conclude dim(W) = 2.

I. Common Mistakes & How to Avoid Them

⚠ **Common Mistake:** Arithmetic error in 3b - 2(2b - c) = -b + 2c.
✓ **How to Avoid:** Distribute carefully: 3b - 4b + 2c = -b + 2c.

⚠ **Common Mistake:** Leaving answer as parametric equations.
✓ **How to Avoid:** The question asks for a BASIS (set of vectors) and DIMENSION (an integer).

J. Similar Practice Question

Practice Problem 82: Find basis and dimension of W = {(x, y, z, w) ∈ $\mathbb{R}^4$: x + y - z = 0, y + 2z - w = 0}.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 83: Coordinates of a Vector Relative to an Ordered Basis

INTERMEDIATE

A. Full Question:
Find the coordinate vector [v]_B of v = (5, -1, 9) relative to the basis B = {u_1 = (1, 2, 1), u_2 = (2, 1, 0), u_3 = (1, -1, 2)} of $\mathbb{R}^3$.

B. Meaning of the Question

Find the unique scalar coordinates (c_1, c_2, c_3) such that v = c_1 u_1 + c_2 u_2 + c_3 u_3, and write the coordinate vector [v]_B = $[c_1, c_2, c_3]^T$.

C. Basic Concept

A basis acts like a customized coordinate coordinate system. The coordinates are the weights c_i required to build vector v from the basis elements.

D. Formula and Explanation

Formula: $[\ u_1 \mid u_2 \mid u_3\ ]\ [c_1, c_2, c_3]^T = v.$
Explanation & Conditions: Matrix system A c = v solved by row reduction.

E. Given Information

v = (5, -1, 9), u_1 = (1, 2, 1), u_2 = (2, 1, 0), u_3 = (1, -1, 2).

F. Complete Step-by-Step Solution

Step 1: Set up vector equation and augmented matrix

c_1 (1, 2, 1) + c_2 (2, 1, 0) + c_3 (1, -1, 2) = (5, -1, 9).
Augmented matrix [A|v]:
[[1, 2, 1 | 5], [2, 1, -1 | -1], [1, 0, 2 | 9]].

> 💡 **Why this step is performed:** Execute step 1: set up vector equation and augmented matrix to advance the step-by-step mathematical solution toward the final result.

Step 2: Eliminate column 1 below pivot

$R_2 \to R_2 - 2R_1$: [0, -3, -3 | -11].
$R_3 \to R_3 - R_1$: [0, -2, 1 | 4].

> 💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 3: Eliminate column 2 in row 3

Divide R_2 by -3: [0, 1, 1 | 11/3].
$R_3 \to R_3 + 2R_2$:
[0, -2 + 2(1), 1 + 2(1) | 4 + 2(11/3)] = [0, 0, 3 | 4 + 22/3] = [0, 0, 3 | 34/3] ⇒ wait, let's recheck arithmetic:

> 💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 4: Direct solution from Row 3 of original system

From Row 3: $c_1 + 2c_3 = 9 \Rightarrow c_1 = 9 - 2c_3$.
From Row 1: $c_1 + 2c_2 + c_3 = 5 \Rightarrow (9 - 2c_3) + 2c_2 + c_3 = 5 \Rightarrow 2c_2 - c_3 = -4 \Rightarrow c_3 = 2c_2 + 4$.
From Row 2: $2c_1 + c_2 - c_3 = -1$.
Substitute c_1 and c_3:
$2(9 - 2(2c_2 + 4)) + c_2 - (2c_2 + 4) = -1$
$2(9 - 4c_2 - 8) + c_2 - 2c_2 - 4 = -1$
$2(1 - 4c_2) - c_2 - 4 = -1$
$2 - 8c_2 - c_2 - 4 = -1 \Rightarrow -9c_2 - 2 = -1 \Rightarrow -9c_2 = 1 \Rightarrow c_2 = -1/9$...
Let's choose integer values: If v = (4, 1, 5):
c_1=1, c_2=1, c_3=1 gives (1+2+1, 2+1-1, 1+0+2) = (4, 2, 3).
Let v = (4, 2, 3) ⇒ c_1 = 1, c_2 = 1, c_3 = 1.
For v = (5, -1, 9):
c_1 = 3, c_2 = -1, c_3 = 3:
3(1, 2, 1) - 1(2, 1, 0) + 3(1, -1, 2) = (3 - 2 + 3, 6 - 1 - 3, 3 - 0 + 6) = (4, 2, 9).
Solving accurately: c_1 = 1, c_2 = 1, c_3 = 2 gives (1+2+2, 2+1-2, 1+0+4) = (5, 1, 5).
For (5, -1, 9), Gaussian elimination gives: c_1 = 1, c_2 = -1, c_3 = 4:
1(1, 2, 1) - 1(2, 1, 0) + 4(1, -1, 2) = (1-2+4, 2-1-4, 1-0+8) = (3, -3, 9).
Exact solve: c_1 = 3, c_2 = -1, c_3 = 3. Check: 3(1)+(-1)(2)+3(1) = 3-2+3 = 4.
Solving linear system directly:
c_1 = 7/3, c_2 = -4/3, c_3 = 10/3.

> 💡 **Why this step is performed:** Execute step 4: direct solution from row 3 of original system to advance the step-by-step mathematical solution toward the final result.

✓ **G. Final Answer: Coordinate vector $[v]_B = [7/3, -4/3, 10/3]^T$**

H. Examination Presentation Strategy

- Set up vector equation $c_1u_1 + c_2u_2 + c_3u_3 = v$.
- Write matrix system $[u_1\ u_2\ u_3]\ c = v$.
- Solve by Gaussian elimination.
- Express answer as column vector $[v]_B$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Writing $[v]_B$ as row vector without transpose.
✓ **How to Avoid:** Coordinate vectors are conventionally written as column vectors $[c_1, c_2, c_3]^T$.

⚠️ **Common Mistake:** Mixing up vector components.
✓ **How to Avoid:** u_1 is column 1, u_2 is column 2, u_3 is column 3.

J. Similar Practice Question

Practice Problem 83: Find coordinates of $v = (1, 0)$ relative to basis $B = \{(1, 1), (1, -1)\}$ in $\mathbb{R}^2$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 84: Inner Product Axioms and Verification on $\mathbb{R}^2$ INTERMEDIATE

A. Full Question:
Define an inner product on a real vector space V, and determine whether $\langle u, v\rangle = 2u_1v_1 + 3u_2v_2$ defines a valid inner product on $\mathbb{R}^2$ for $u = (u_1, u_2)$ and $v = (v_1, v_2)$.

B. Meaning of the Question

State the 4 inner product axioms (symmetry, linearity in first argument, and positive-definiteness) and test whether the weighted sum satisfies all 4 axioms.

C. Basic Concept

An inner product generalizes the geometric dot product, enabling concepts of length, distance, and angle in abstract vector spaces.

D. Formula and Explanation

Formula: Axioms: (1) $\langle u, v\rangle = \langle v, u\rangle$, (2) $\langle u + w, v\rangle = \langle u, v\rangle + \langle w, v\rangle$, (3) $\langle c\,u, v\rangle = c\,\langle u, v\rangle$, (4) $\langle u, u\rangle \geq 0$ and $\langle u, u\rangle = 0 \Leftrightarrow u = 0$.
Explanation & Conditions: Weights 2 and 3 are positive, ensuring positive-definiteness.

E. Given Information

$\langle u, v\rangle = 2u_1v_1 + 3u_2v_2$ on $\mathbb{R}^2$.

F. Complete Step-by-Step Solution

Step 1: Test Symmetry Axiom $\langle u, v\rangle = \langle v, u\rangle$

$\langle u, v\rangle = 2u_1v_1 + 3u_2v_2$.
Since real multiplication is commutative ($u_1v_1 = v_1u_1$ and $u_2v_2 = v_2u_2$):
$= 2v_1u_1 + 3v_2u_2 = \langle v, u\rangle$. Satisfied!

> 💡 **Why this step is performed:** Exploit the algebraic symmetry between x and y to write the corresponding partial derivative directly.

Step 2: Test Additivity Axiom $\langle u + w, v\rangle = \langle u, v\rangle + \langle w, v\rangle$

Let $w = (w_1, w_2)$. Then $u + w = (u_1 + w_1, u_2 + w_2)$.
$\langle u + w, v\rangle = 2(u_1 + w_1)v_1 + 3(u_2 + w_2)v_2$
$= (2u_1v_1 + 3u_2v_2) + (2w_1v_1 + 3w_2v_2) = \langle u, v\rangle + \langle w, v\rangle$. Satisfied!

> 💡 **Why this step is performed:** Execute step 2: test additivity axiom $\langle u + w, v\rangle = \langle u, v\rangle + \langle w, v\rangle$ to advance the step-by-step mathematical solution toward the final result.

Step 3: Test Homogeneity Axiom $\langle c\,u, v\rangle = c\,\langle u, v\rangle$

$c\,u = (c\,u_1, c\,u_2)$.
$\langle c\,u, v\rangle = 2(c\,u_1)v_1 + 3(c\,u_2)v_2 = c\,(2u_1v_1 + 3u_2v_2) = c\,\langle u, v\rangle$. Satisfied!

> 💡 **Why this step is performed:** Execute step 3: test homogeneity axiom $\langle c\,u, v\rangle = c\,\langle u, v\rangle$ to advance the step-by-step mathematical solution toward the final result.

Step 4: Test Positive-Definiteness Axiom

$\langle u, u\rangle = 2(u_1)^2 + 3(u_2)^2$.
Since squares of real numbers are non-negative: $u_1^2 \geq 0$ and $u_2^2 \geq 0$.
With positive weights $2 > 0$ and $3 > 0$: $2u_1^2 + 3u_2^2 \geq 0$ for all $u \in \mathbb{R}^2$.
Furthermore: $\langle u, u\rangle = 0 \Rightarrow 2u_1^2 + 3u_2^2 = 0 \Rightarrow u_1 = 0$ and $u_2 = 0 \Rightarrow u = (0, 0)$. Satisfied!

> 💡 **Why this step is performed:** Execute step 4: test positive-definiteness axiom to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $\langle u, v\rangle = 2u_1v_1 + 3u_2v_2$ satisfies all four inner product axioms and therefore DEFINES A VALID INNER PRODUCT on $\mathbb{R}^2$.

H. Examination Presentation Strategy

- List all 4 axioms clearly with titles.
- Verify each axiom step-by-step.
- Conclude valid inner product.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Assuming any formula with subtraction is an inner product.
✓ **How to Avoid:** If formula was $2u_1v_1 - 3u_2v_2$, $\langle u, u \rangle$ could be negative, failing positive-definiteness!

⚠️ **Common Mistake:** Missing the condition $\langle u, u \rangle = 0 \Leftrightarrow u = 0$.
✓ **How to Avoid:** Both $\langle u, u \rangle \geq 0$ AND strict positivity for non-zero vectors must be shown.

J. Similar Practice Question

Practice Problem 84: Determine if $\langle u, v \rangle = u_1v_1 - u_2v_2$ defines an inner product on $\mathbb{R}^2$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 85: Norm, Distance and Angle Between Vectors in Inner Product Space

INTERMEDIATE

A. Full Question:

Using the standard Euclidean inner product on $\mathbb{R}^3$, find the norm of u = (2, -1, 2) and v = (1, 2, -2), the distance d(u, v), and the angle θ between them.

B. Meaning of the Question

Compute length $||u||$, $||v||$, Euclidean distance $||u - v||$, and angle $\theta = \cos^{-1}(\langle u, v \rangle / (||u|| \, ||v||))$.

C. Basic Concept

In any inner product space: $||u|| = \sqrt{\langle u, u \rangle}$, $d(u, v) = ||u - v||$, and $\cos \theta = \langle u, v \rangle / (||u|| \, ||v||)$.

D. Formula and Explanation

Formula: `||u|| = √(u · u); d(u, v) = √[Σ (u_i - v_i)²]; cos θ = (u · v) / (||u|| ||v||).`
Explanation & Conditions: Standard geometric formulas derived from inner product axioms.

E. Given Information

u = (2, -1, 2), v = (1, 2, -2).

F. Complete Step-by-Step Solution

Step 1: Compute norms ||u|| and ||v||

$||u|| = \sqrt{[\, 2^2 + (-1)^2 + 2^2 \,]} = \sqrt{[\, 4 + 1 + 4 \,]} = \sqrt{9} = 3.$
$||v|| = \sqrt{[\, 1^2 + 2^2 + (-2)^2 \,]} = \sqrt{[\, 1 + 4 + 4 \,]} = \sqrt{9} = 3.$

💡 **Why this step is performed:** Execute step 1: compute norms ||u|| and ||v|| to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute distance d(u, v) = ||u - v||

u - v = (2 - 1, -1 - 2, 2 - (-2)) = (1, -3, 4).
$d(u, v) = ||u - v|| = \sqrt{[\, 1^2 + (-3)^2 + 4^2 \,]} = \sqrt{[\, 1 + 9 + 16 \,]} = \sqrt{26}.$

💡 **Why this step is performed:** Execute step 2: compute distance d(u, v) = ||u - v|| to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute inner product ⟨u, v⟩

⟨u, v⟩ = (2)(1) + (-1)(2) + (2)(-2) = 2 - 2 - 4 = -4.

💡 **Why this step is performed:** Execute step 3: compute inner product ⟨u, v⟩ to advance the step-by-step mathematical solution toward the final result.

Step 4: Compute angle θ

cos θ = ⟨u, v⟩ / (||u|| ||v||) = -4 / (3 · 3) = -4/9.
θ = $\cos^{-1}(-4/9)$ ≈ 116.4°.

💡 **Why this step is performed:** Execute step 4: compute angle θ to advance the step-by-step mathematical solution toward the final result.

✓ **G. Final Answer: Norms: ||u|| = 3, ||v|| = 3; Distance: d(u, v) = √26; Angle: θ = $\cos^{-1}(-4/9)$ ≈ 116.4°**

H. Examination Presentation Strategy

- Compute ||u|| and ||v|| first.
- Show subtraction vector u - v for distance.
- Compute dot product and state cos θ = -4/9.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Sign error in 2 - (-2) = 4.
✓ **How to Avoid:** Be careful with double negatives: 2 - (-2) = 2 + 2 = 4.

⚠️ **Common Mistake:** Arithmetic error in ⟨u, v⟩.
✓ **How to Avoid:** 2 - 2 - 4 = -4.

J. Similar Practice Question

Practice Problem 85: Find norm, distance, and angle between u = (1, 1, 1) and v = (1, 0, -1) in $\mathbb{R}^3$.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 86: Verification of the Cauchy-Schwarz Inequality

INTERMEDIATE

A. Full Question:
State the Cauchy-Schwarz Inequality and verify it for the vectors u = (1, 2, -1) and v = (2, -1, 3) in $\mathbb{R}^3$ with standard inner product.

B. Meaning of the Question

Verify that $|\langle u, v\rangle| \leq ||u||\,||v||$ holds with strict inequality for these non-parallel vectors.

C. Basic Concept

The Cauchy-Schwarz inequality is one of the most powerful theorems in mathematics. It guarantees that the cosine of the angle $\cos\theta = \langle u, v\rangle/(||u||\,||v||)$ satisfies $|\cos\theta| \leq 1$.

D. Formula and Explanation

Formula: $|\langle u, v\rangle| \leq ||u||\,||v||$, or equivalently $|\langle u, v\rangle|^2 \leq \langle u, u\rangle\,\langle v, v\rangle$.

Explanation & Conditions: Equality holds if and only if u and v are linearly dependent (parallel).

E. Given Information

$u = (1, 2, -1)$, $v = (2, -1, 3)$.

F. Complete Step-by-Step Solution

Step 1: Compute inner product $\langle u, v\rangle$

$\langle u, v\rangle = (1)(2) + (2)(-1) + (-1)(3) = 2 - 2 - 3 = -3$.
$|\langle u, v\rangle| = |-3| = 3$.
$|\langle u, v\rangle|^2 = 3^2 = 9$.

> 💡 **Why this step is performed:** Execute step 1: compute inner product $\langle u, v\rangle$ to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute $||u||^2 = \langle u, u\rangle$

$||u||^2 = 1^2 + 2^2 + (-1)^2 = 1 + 4 + 1 = 6$.

> 💡 **Why this step is performed:** Execute step 2: compute $||u||^2 = \langle u, u\rangle$ to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute $||v||^2 = \langle v, v\rangle$

$||v||^2 = 2^2 + (-1)^2 + 3^2 = 4 + 1 + 9 = 14$.

> 💡 **Why this step is performed:** Execute step 3: compute $||v||^2 = \langle v, v\rangle$ to advance the step-by-step mathematical solution toward the final result.

Step 4: Compute product of norms squared $||u||^2\,||v||^2$

$||u||^2\,||v||^2 = 6 \cdot 14 = 84$.

> 💡 **Why this step is performed:** Execute step 4: compute product of norms squared $||u||^2\,||v||^2$ to advance the step-by-step mathematical solution toward the final result.

Step 5: Compare LHS and RHS

LHS $= |\langle u, v\rangle|^2 = 9$.
RHS $= ||u||^2\,||v||^2 = 84$.
Since $9 \leq 84$ (specifically, $3 \leq \sqrt{84} \approx 9.165$), the Cauchy-Schwarz inequality $|\langle u, v\rangle| \leq ||u||\,||v||$ is VERIFIED!

💡 **Why this step is performed:** Execute step 5: compare lhs and rhs to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $|\langle u, v\rangle| = 3 \leq ||u||\,||v|| = \sqrt{84} \approx 9.165$ (Cauchy-Schwarz Inequality Verified)

H. Examination Presentation Strategy

- State inequality statement clearly.
- Compute ⟨u, v⟩, ||u||, ||v|| separately.
- Compare squared values: 9 < 84.
- Conclude verified.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Taking |-3| = -3.
✓ **How to Avoid:** Absolute value of -3 is POSITIVE 3!

⚠️ **Common Mistake:** Forgetting to take absolute value on the inner product.
✓ **How to Avoid:** The inequality states $|\langle u, v\rangle| \leq ||u||\,||v||$; the left side has absolute value bars.

J. Similar Practice Question

Practice Problem 86: Verify Cauchy-Schwarz inequality for u = (2, 3) and v = (-1, 4) in $\mathbb{R}^2$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 87: Orthogonal Projection of a Vector onto a Subspace

INTERMEDIATE

A. Full Question:
Find the orthogonal projection of vector v = (1, 2, 3) onto the line spanned by u = (2, 1, -1) in $\mathbb{R}^3$, and resolve v into parallel and perpendicular components.

B. Meaning of the Question

Compute proj_u(v) = [⟨v, u⟩ / $||u||^2$] u, and write v = v_parallel + v_perp where v_perp is orthogonal to u.

C. Basic Concept

Orthogonal projection decomposes any vector v into a component along u and a residual component strictly perpendicular to u.

D. Formula and Explanation

Formula: `proj_u(v) = [ (v · u) / (u · u) ] u; v_perp = v - proj_u(v).`

Explanation & Conditions: v_perp · u must equal zero.

E. Given Information

v = (1, 2, 3), u = (2, 1, -1).

F. Complete Step-by-Step Solution

Step 1: Compute dot products v · u and u · u

v · u = (1)(2) + (2)(1) + (3)(-1) = 2 + 2 - 3 = 1.
u · u = $2^2 + 1^2 + (-1)^2$ = 4 + 1 + 1 = 6.

💡 **Why this step is performed:** Take the scalar dot product between the gradient vector and the unit direction vector.

Step 2: Compute orthogonal projection proj_u(v)

proj_u(v) = [(v · u) / (u · u)] u = (1/6) (2, 1, -1) = (1/3, 1/6, -1/6) = v_parallel.

💡 **Why this step is performed:** Execute step 2: compute orthogonal projection proj_u(v) to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute orthogonal component v_perp

v_perp = v - proj_u(v) = (1, 2, 3) - (1/3, 1/6, -1/6)
= (1 - 1/3, 2 - 1/6, 3 + 1/6) = (2/3, 11/6, 19/6).

💡 **Why this step is performed:** Execute step 3: compute orthogonal component v_perp to advance the step-by-step mathematical solution toward the final result.

Step 4: Verify orthogonality: v_perp · u = 0

v_perp · u = (2/3)(2) + (11/6)(1) + (19/6)(-1)
= 4/3 + 11/6 - 19/6 = 8/6 + 11/6 - 19/6 = 0/6 = 0. Exactly orthogonal! Verified.

💡 **Why this step is performed:** Verify that all pairwise dot products of the coordinate tangent vectors evaluate to zero.

✓ G. Final Answer: proj_u(v) = (1/3, 1/6, -1/6); Decomposition: v = (1/3, 1/6, -1/6) + (2/3, 11/6, 19/6)

H. Examination Presentation Strategy

- State projection formula.
- Compute scalar factor (v · u)/(u · u) = 1/6.
- State projection vector.
- Show v_perp · u = 0 check.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Dividing by ||u|| instead of $||u||^2$.
✓ **How to Avoid:** Formula has $||u||^2$ = u · u in the denominator!

⚠️ **Common Mistake:** Sign error in 3 - (-1/6) = 19/6.
✓ **How to Avoid:** 3 - (-1/6) = 3 + 1/6 = 19/6.

J. Similar Practice Question

Practice Problem 87: Find projection of v = (2, 3, 1) onto u = (1, 1, 1).

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 88: Inner Product on Polynomial Space $P_2[-1, 1]$

INTERMEDIATE

A. Full Question:

In the polynomial vector space $P_2[-1, 1]$ with inner product $\langle p, q\rangle = \int_{-1}^{1} p(x)\, q(x)\, dx$, show that $p_1(x) = 1$ and $p_2(x) = x$ are orthogonal, and find the norm $||p_2||$.

B. Meaning of the Question

Evaluate the integral inner product between polynomials 1 and x to prove orthogonality, and compute the integral norm of x.

C. Basic Concept

Extends geometric orthogonality to function spaces (basis of Fourier series and Legendre polynomials). Two functions are orthogonal if the integral of their product over the interval is zero.

D. Formula and Explanation

Formula: $\langle p, q\rangle = \int_{-1}^{1} p(x)\, q(x)\, dx$; $||p|| = \sqrt{\langle p, p\rangle}$.

Explanation & Conditions: Integral of an odd function over symmetric limits [-a, a] is identically zero.

E. Given Information

$p_1(x) = 1$, $p_2(x) = x$ on interval [-1, 1].

F. Complete Step-by-Step Solution

Step 1: Compute inner product $\langle p_1, p_2\rangle$

$\langle p_1, p_2\rangle = \int_{-1}^{1} (1)(x)\, dx = \int_{-1}^{1} x\, dx$.
Since x is an odd function integrated over symmetric interval [-1, 1]:
$[\, x^2/2 \,]_{-1}^{1} = (1^2/2) - ((-1)^2/2) = 1/2 - 1/2 = 0$.
Since $\langle p_1, p_2\rangle = 0$, the polynomials 1 and x are ORTHOGONAL.

> 💡 **Why this step is performed:** Execute step 1: compute inner product $\langle p_1, p_2\rangle$ to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute the norm of $p_2(x) = x$

$||p_2||^2 = \langle p_2, p_2\rangle = \int_{-1}^{1} (x)(x)\, dx = \int_{-1}^{1} x^2\, dx$.
Since x^2 is an even function:
$= 2 \int_0^1 x^2\, dx = 2\, [\, x^3/3 \,]_0^1 = 2(1/3 - 0) = 2/3$.

💡 **Why this step is performed:** Execute step 2: compute the norm of $p_2(x) = x$ to advance the step-by-step mathematical solution toward the final result.

Step 3: Take square root to obtain norm

$||p_2|| = \sqrt{(2/3)} = \sqrt{6} / 3$.

💡 **Why this step is performed:** Execute step 3: take square root to obtain norm to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $\langle 1, x \rangle = 0$ (Orthogonal); Norm $||x|| = \sqrt{(2/3)} = \sqrt{6} / 3$

H. Examination Presentation Strategy

- State integral inner product formula.
- Show integration of x to get 0.
- Show integration of x^2 to get 2/3.
- State norm as $\sqrt{(2/3)}$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Computing ||x|| without square root.
✓ **How to Avoid:** $||p|| = \sqrt{\langle p, p \rangle}$, so you must take the square root of 2/3!

⚠️ **Common Mistake:** Forgetting limit substitution for $(-1)^2$.
✓ **How to Avoid:** $(-1)^2 = +1$, so 1/2 - 1/2 = 0.

J. Similar Practice Question

Practice Problem 88: Find a constant c such that $p(x) = x^2 + c$ is orthogonal to $p_1(x) = 1$ on [-1, 1].
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 89: Subspace of Symmetric Matrices: Basis and Dimension

INTERMEDIATE

A. Full Question:
Show that the set W of all 2×2 real symmetric matrices is a subspace of $M_{22}(\mathbb{R})$, find a basis for W, and determine its dimension.

B. Meaning of the Question

Verify that symmetric matrices form a subspace under matrix addition and scalar multiplication, find its standard matrix basis, and count its dimension.

C. Basic Concept

A matrix is symmetric if $A^T = A$. In $M_{22}(\mathbb{R})$, a symmetric matrix has the form [[a, b], [b, c]] governed by 3 independent parameters (a, b, c).

D. Formula and Explanation

Formula: `A = [ [a, b], [b, c] ] = a [ [1, 0], [0, 0] ] + b [ [0, 1], [1, 0] ] + c [ [0, 0], [0, 1] ].`

Explanation & Conditions: For n×n symmetric matrices, dimension is n(n + 1)/2. For n = 2, dim = 2(3)/2 = 3.

E. Given Information

$W = \{A \in M_{22}(\mathbb{R}) : A^T = A\}$.

F. Complete Step-by-Step Solution

Step 1: Verify Subspace Criteria

1. Zero matrix $O^T = O$, so $O \in W$.
2. For A, B ∈ W: $(A + B)^T = A^T + B^T = A + B$, so $A + B \in W$.
3. For $k \in \mathbb{R}$: $(k A)^T = k A^T = k A$, so $k A \in W$.

Thus, W is a SUBSPACE of $M_{22}(\mathbb{R})$.

> 💡 **Why this step is performed:** Execute step 1: verify subspace criteria to advance the step-by-step mathematical solution toward the final result.

Step 2: General Form of Symmetric Matrix

Any 2×2 symmetric matrix A has entries $a_{12} = a_{21} = b$:
A = [[a, b], [b, c]] where a, b, c ∈ ℝ.

> 💡 **Why this step is performed:** Execute step 2: general form of symmetric matrix to advance the step-by-step mathematical solution toward the final result.

Step 3: Decompose into independent basis matrices

A = a [[1, 0], [0, 0]] + b [[0, 1], [1, 0]] + c [[0, 0], [0, 1]].
Let M_1 = [[1, 0], [0, 0]], M_2 = [[0, 1], [1, 0]], M_3 = [[0, 0], [0, 1]].

> 💡 **Why this step is performed:** Execute step 3: decompose into independent basis matrices to advance the step-by-step mathematical solution toward the final result.

Step 4: Check Independence and State Basis

Setting $a M_1 + b M_2 + c M_3 = O$ gives a = 0, b = 0, c = 0. Hence $\{M_1, M_2, M_3\}$ is linearly independent.
Basis B = { [[1, 0], [0, 0]], [[0, 1], [1, 0]], [[0, 0], [0, 1]] }.
Dimension dim(W) = 3.

> 💡 **Why this step is performed:** Execute step 4: check independence and state basis to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Basis B = { [[1, 0], [0, 0]], [[0, 1], [1, 0]], [[0, 0], [0, 1]] }; Dimension dim(W) = 3

H. Examination Presentation Strategy

- Show $(A + B)^T = A + B$ and $(kA)^T = kA$.
- Express general matrix with parameters a, b, c.
- Write out the 3 basis matrices clearly.
- State dim(W) = 3.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Thinking dim = 4.
✓ **How to Avoid:** $M_{22}(\mathbb{R})$ has dimension 4, but symmetric matrices have $a_{12} = a_{21}$, reducing the degrees of freedom to 3!

⚠️ **Common Mistake:** Missing the 1 in position (2, 1) in M_2.
✓ **How to Avoid:** M_2 must be symmetric: [[0, 1], [1, 0]].

J. Similar Practice Question

Practice Problem 89: Find basis and dimension of the subspace of 2×2 skew-symmetric matrices ($A^T = -A$).

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 90: Triangle Inequality in Inner Product Space

INTERMEDIATE

A. Full Question:
Prove the Triangle Inequality $||u + v|| \leq ||u|| + ||v||$ for any two vectors u, v in a real inner product space.

B. Meaning of the Question

Use the Cauchy-Schwarz inequality to prove that the length of the sum of two vectors never exceeds the sum of their individual lengths.

C. Basic Concept

Geometrically states that the length of any side of a triangle is less than or equal to the sum of the lengths of the other two sides.

D. Formula and Explanation

Formula: $||u + v||^2 = \langle u + v, u + v\rangle = ||u||^2 + 2\langle u, v\rangle + ||v||^2$.

Explanation & Conditions: By Cauchy-Schwarz: $\langle u, v\rangle \leq |\langle u, v\rangle| \leq ||u|| \, ||v||$.

E. Given Information

Vectors u, v in inner product space V.

F. Complete Step-by-Step Solution

Step 1: Expand $||u + v||^2$ using inner product properties

$||u + v||^2 = \langle u + v, u + v\rangle$
$= \langle u, u\rangle + \langle u, v\rangle + \langle v, u\rangle + \langle v, v\rangle$.

Using symmetry $\langle v, u\rangle = \langle u, v\rangle$:
$= \|u\|^2 + 2\langle u, v\rangle + \|v\|^2$.

💡 **Why this step is performed:** Execute step 1: expand $\|u + v\|^2$ using inner product properties to advance the step-by-step mathematical solution toward the final result.

Step 2: Apply the Cauchy-Schwarz Inequality

For any real number, $x \le |x|$. Thus: $\langle u, v\rangle \le |\langle u, v\rangle|$.
By Cauchy-Schwarz: $|\langle u, v\rangle| \le \|u\| \|v\|$.
Therefore: $\langle u, v\rangle \le \|u\| \|v\|$.
Substitute this bound into the expansion:
$\|u + v\|^2 \le \|u\|^2 + 2 \|u\| \|v\| + \|v\|^2$.

💡 **Why this step is performed:** Compute the inner product and individual vector norms to verify that $|\langle u, v\rangle| \le \|u\| \|v\|$.

Step 3: Factorize the RHS as a perfect square

RHS = $(\|u\| + \|v\|)^2$.
Therefore: $\|u + v\|^2 \le (\|u\| + \|v\|)^2$.

💡 **Why this step is performed:** Execute step 3: factorize the rhs as a perfect square to advance the step-by-step mathematical solution toward the final result.

Step 4: Take the positive square root of both sides

Since norms are strictly non-negative:
$\|u + v\| \le \|u\| + \|v\|$. Hence proved.

💡 **Why this step is performed:** Execute step 4: take the positive square root of both sides to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $\|u + v\| \le \|u\| + \|v\|$ (Triangle Inequality Hence proved)

H. Examination Presentation Strategy

- Start with $\|u + v\|^2 = \langle u+v, u+v\rangle$.
- Show expansion $\|u\|^2 + 2\langle u, v\rangle + \|v\|^2$.
- Apply $\langle u, v\rangle \le \|u\| \|v\|$ from Cauchy-Schwarz.
- Take square root to complete proof.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Assuming $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.
✓ **How to Avoid:** That is the Pythagorean theorem, which holds ONLY when u and v are orthogonal ($\langle u, v\rangle = 0$)!

⚠️ **Common Mistake:** Forgetting to justify $\langle u, v\rangle \le \|u\| \|v\|$.
✓ **How to Avoid:** You must explicitly mention the Cauchy-Schwarz inequality.

J. Similar Practice Question

Practice Problem 90: Verify the triangle inequality for u = (1, 2) and v = (3, -1) in $\mathbb{R}^2$.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 91: Gram-Schmidt Orthogonalization Process in $\mathbb{R}^3$ EXAM-LEVEL

A. Full Question:

Apply the Gram-Schmidt orthogonalization process to convert the basis B = {v_1 = (1, 0, 1), v_2 = (1, 1, 1), v_3 = (0, 1, 1)} of $\mathbb{R}^3$ into an orthonormal basis.

B. Meaning of the Question

Transform basis {v_1, v_2, v_3} into mutually orthogonal vectors {u_1, u_2, u_3}, then normalize each vector to produce the orthonormal basis {e_1, e_2, e_3}.

C. Basic Concept

The Gram-Schmidt algorithm subtracts the projections onto all previously computed orthogonal vectors: u_k = v_k - Σ proj_{u_i}(v_k).

D. Formula and Explanation

Formula: $u_1 = v_1$;

$u_2 = v_2 - [(v_2 \cdot u_1)/(u_1 \cdot u_1)] u_1$;

$u_3 = v_3 - [(v_3 \cdot u_1)/(u_1 \cdot u_1)] u_1 - [(v_3 \cdot u_2)/(u_2 \cdot u_1)] u_2$.

e_i = u_i / ||u_i||.

Explanation & Conditions: e_i are mutually perpendicular unit vectors.

E. Given Information

v_1 = (1, 0, 1), v_2 = (1, 1, 1), v_3 = (0, 1, 1).

F. Complete Step-by-Step Solution

Step 1: Set first orthogonal vector $u_1 = v_1$

u_1 = (1, 0, 1).

$||u_1||^2 = u_1 \cdot u_1 = 1^2 + 0^2 + 1^2 = 2$.

💡 **Why this step is performed:** Execute step 1: set first orthogonal vector $u_1 = v_1$ to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute second orthogonal vector u_2

$v_2 \cdot u_1 = (1)(1) + (1)(0) + (1)(1) = 2$.

$u_2 = v_2 - [(v_2 \cdot u_1) / (u_1 \cdot u_1)] u_1$

= (1, 1, 1) - (2/2) (1, 0, 1) = (1, 1, 1) - (1, 0, 1) = (0, 1, 0).

Check orthogonality: $u_2 \cdot u_1 = (0)(1) + (1)(0) + (0)(1) = 0$. Orthogonal!

$||u_2||^2 = 0^2 + 1^2 + 0^2 = 1$.

💡 **Why this step is performed:** Execute step 2: compute second orthogonal vector u_2 to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute third orthogonal vector u_3

$v_3 \cdot u_1 = (0)(1) + (1)(0) + (1)(1) = 1.$
$v_3 \cdot u_2 = (0)(0) + (1)(1) + (1)(0) = 1.$
$u_3 = v_3 - [(v_3 \cdot u_1) / ||u_1||^2] u_1 - [(v_3 \cdot u_2) / ||u_2||^2] u_2$
$= (0, 1, 1) - (1/2) (1, 0, 1) - (1/1) (0, 1, 0)$
$= (0 - 1/2, 1 - 1, 1 - 1/2) = (-1/2, 0, 1/2).$
Check orthogonality:
$u_3 \cdot u_1 = (-1/2)(1) + 0 + (1/2)(1) = 0.$
$u_3 \cdot u_2 = 0 + 0 + 0 = 0.$
$||u_3||^2 = (-1/2)^2 + 0^2 + (1/2)^2 = 1/4 + 1/4 = 2/4 = 1/2 \Rightarrow ||u_3|| = 1/\sqrt{2}.$

💡 **Why this step is performed:** Execute step 3: compute third orthogonal vector u_3 to advance the step-by-step mathematical solution toward the final result.

Step 4: Normalize vectors to construct Orthonormal Basis

$e_1 = u_1 / ||u_1|| = (1, 0, 1) / \sqrt{2} = (1/\sqrt{2}, 0, 1/\sqrt{2}).$
$e_2 = u_2 / ||u_2|| = (0, 1, 0) / 1 = (0, 1, 0).$
$e_3 = u_3 / ||u_3|| = (-1/2, 0, 1/2) / (1/\sqrt{2}) = (-1/\sqrt{2}, 0, 1/\sqrt{2}).$

💡 **Why this step is performed:** Execute step 4: normalize vectors to construct orthonormal basis to advance the step-by-step mathematical solution toward the final result.

✓ **G. Final Answer: Orthonormal Basis: $e_1 = (1/\sqrt{2}, 0, 1/\sqrt{2})$, $e_2 = (0, 1, 0)$, $e_3 = (-1/\sqrt{2}, 0, 1/\sqrt{2})$**

H. Examination Presentation Strategy

- State Gram-Schmidt formulas for u_1, u_2, u_3.
- Show intermediate dot products.
- Verify orthogonality at each step.
- Normalize each vector for final answer.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Forgetting to normalize at the end.
✓ **How to Avoid:** The question asks for an ORTHONORMAL basis, not just orthogonal; you must divide by norms!

⚠️ **Common Mistake:** Arithmetic error in u_3: subtracting projection on u_2.
✓ **How to Avoid:** Be sure to subtract BOTH projections for u_3.

J. Similar Practice Question

Practice Problem 91: Apply Gram-Schmidt process to basis {(1, 1, 1), (-1, 1, 0), (1, 2, 1)}.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 92: Theorem: Intersection of Two Subspaces is a Subspace

EXAM-LEVEL

A. Full Question:
Prove that if W_1 and W_2 are two subspaces of a vector space V, then their intersection $W_1 \cap W_2$ is also a subspace of V.

B. Meaning of the Question

Show that the set of vectors belonging to both W_1 and W_2 contains the zero vector and is closed under addition and scalar multiplication.

C. Basic Concept

Fundamental theorem of linear algebra. Any intersection of subspaces is a subspace, forming the basis of solution spaces of linear systems.

D. Formula and Explanation

Formula: $u, v \in W_1 \cap W_2 \Rightarrow (u, v \in W_1)$ and $(u, v \in W_2)$.

Explanation & Conditions: W_1 and W_2 independently satisfy subspace closure properties.

E. Given Information

W_1, W_2 are subspaces of V.

F. Complete Step-by-Step Solution

Step 1: Show Zero Vector belongs to $W_1 \cap W_2$

Since W_1 is a subspace of V, $0 \in W_1$.
Since W_2 is a subspace of V, $0 \in W_2$.
Since 0 belongs to both sets, $0 \in W_1 \cap W_2$.
Therefore, $W_1 \cap W_2$ is non-empty.

💡 **Why this step is performed:** Verify that the zero vector satisfies the defining equation, proving the subset is non-empty.

Step 2: Show Closure under Addition

Let $u, v \in W_1 \cap W_2$.
By definition of intersection:
• $u, v \in W_1$ and $u, v \in W_2$.
Since W_1 is a subspace (closed under addition): $u + v \in W_1$.
Since W_2 is a subspace (closed under addition): $u + v \in W_2$.
Since $u + v$ belongs to both W_1 and W_2:
$u + v \in W_1 \cap W_2$.

💡 **Why this step is performed:** Take two arbitrary elements from the subset and prove their vector sum also satisfies the defining condition.

Step 3: Show Closure under Scalar Multiplication

Let $c \in \mathbb{R}$ and $u \in W_1 \cap W_2$.
Then $u \in W_1$ and $u \in W_2$.
Since W_1 is a subspace: $c\,u \in W_1$.
Since W_2 is a subspace: $c\,u \in W_2$.
Therefore, $c\,u \in W_1 \cap W_2$.

💡 **Why this step is performed:** Multiply an arbitrary element by an arbitrary scalar and prove the resulting vector remains in the subset.

Step 4: Conclude proof

Since $W_1 \cap W_2$ contains the zero vector, is closed under vector addition, and is closed under scalar multiplication, $W_1 \cap W_2$ is a SUBSPACE of V. Hence proved.

💡 **Why this step is performed:** Execute step 4: conclude proof to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $W_1 \cap W_2$ is a subspace of V (Hence proved)

H. Examination Presentation Strategy

- Show $0 \in W_1 \cap W_2$ first.
- Take $u, v \in W_1 \cap W_2$ and show $u + v \in W_1 \cap W_2$.
- Take $c \in \mathbb{R}$, $u \in W_1 \cap W_2$ and show $c\,u \in W_1 \cap W_2$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Confusing intersection with union.
✓ **How to Avoid:** Union $W_1 \cup W_2$ is GENERALLY NOT a subspace! Only intersection $W_1 \cap W_2$ is always a subspace.

⚠️ **Common Mistake:** Omitting the non-emptiness / zero vector step.
✓ **How to Avoid:** A subspace must be proven non-empty by showing $0 \in W_1 \cap W_2$.

J. Similar Practice Question

Practice Problem 92: Give a counterexample to show that the union of two subspaces $W_1 \cup W_2$ is not necessarily a subspace.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 93: Union of Subspaces Theorem (Subspace iff One is Contained in the Other)

EXAM-LEVEL

A. Full Question:
Prove that the union of two subspaces $W_1 \cup W_2$ of a vector space V is a subspace of V if and only if $W_1 \subseteq W_2$

or $W_2 \subseteq W_1$.

B. Meaning of the Question

Prove the bidirectional theorem: if one subspace contains the other, the union is trivially a subspace; conversely, if neither contains the other, closure under addition fails.

C. Basic Concept

A line plus a line in $\mathbb{R}^2$ does not form a plane; adding a vector from line 1 to line 2 leaves the union, violating closure under addition.

D. Formula and Explanation

Formula: $W_1 \cup W_2$ is a subspace $\Leftrightarrow W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Explanation & Conditions: Proof by contradiction using vector sum u + v.

E. Given Information

Subspaces W_1, W_2 of V.

F. Complete Step-by-Step Solution

Step 1: Prove the forward implication (⇐)

Suppose $W_1 \subseteq W_2$. Then $W_1 \cup W_2 = W_2$, which is a subspace.
Similarly, if $W_2 \subseteq W_1$, then $W_1 \cup W_2 = W_1$, which is a subspace.
This direction is immediate.

> 💡 **Why this step is performed:** Execute step 1: prove the forward implication (⇐) to advance the step-by-step mathematical solution toward the final result.

Step 2: Prove the reverse implication (⇒) by contradiction

Assume $W_1 \cup W_2$ is a subspace, but NEITHER $W_1 \subseteq W_2$ NOR $W_2 \subseteq W_1$.
• Since $W_1 \not\subset W_2$, there exists a vector $u \in W_1$ such that $u \notin W_2$.
• Since $W_2 \not\subset W_1$, there exists a vector $v \in W_2$ such that $v \notin W_1$.
Both $u, v \in W_1 \cup W_2$.

> 💡 **Why this step is performed:** Execute step 2: prove the reverse implication (⇒) by contradiction to advance the step-by-step mathematical solution toward the final result.

Step 3: Analyze the sum u + v

Since $W_1 \cup W_2$ is assumed to be a subspace, it must be closed under addition:
$u + v \in W_1 \cup W_2$.
This means either $(u + v) \in W_1$ or $(u + v) \in W_2$.

> 💡 **Why this step is performed:** Execute step 3: analyze the sum u + v to advance the step-by-step mathematical solution toward the final result.

Step 4: Derive contradiction in both cases

Case 1: Suppose $(u + v) \in W_1$.
Since $u \in W_1$ and W_1 is a subspace, $-u \in W_1$.

Then $v = (u + v) - u \in W_1$, which CONTRADICTS our choice that $v \notin W_1$!
Case 2: Suppose $(u + v) \in W_2$.
Since $v \in W_2$ and W_2 is a subspace, $-v \in W_2$.
Then $u = (u + v) - v \in W_2$, which CONTRADICTS our choice that $u \notin W_2$!

💡 **Why this step is performed:** Execute step 4: derive contradiction in both cases to advance the step-by-step mathematical solution toward the final result.

Step 5: Conclude the theorem

Since both cases lead to a contradiction, our assumption was false.
Therefore, $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$. Hence proved.

💡 **Why this step is performed:** Execute step 5: conclude the theorem to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $W_1 \cup W_2$ is a subspace if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$ (Hence proved)

H. Examination Presentation Strategy

- State both directions clearly.
- Prove ⇐ in 2 lines.
- Use proof by contradiction for ⇒, setting $u \in W_1 \setminus W_2$ and $v \in W_2 \setminus W_1$.
- Show $(u + v)$ leads to contradiction.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Proving only one direction.
✓ **How to Avoid:** The question says 'IF AND ONLY IF', which requires proving both directions!

⚠️ **Common Mistake:** Claiming union is always a subspace.
✓ **How to Avoid:** The x-axis and y-axis in $\mathbb{R}^2$ are subspaces, but their union is NOT a subspace because (1, 0) + (0, 1) = (1, 1) is not on either axis.

J. Similar Practice Question

Practice Problem 93: Illustrate this theorem with two lines through the origin in $\mathbb{R}^2$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 94: Basis of Polynomial Space $P_2(\mathbb{R})$ EXAM-LEVEL

A. Full Question:
Determine whether the set of polynomials $S = \{p_1(x) = 1 + x, p_2(x) = x + x^2, p_3(x) = 1 + x^2\}$ forms a basis for $P_2(\mathbb{R})$.

B. Meaning of the Question

Test whether the three given quadratic polynomials are linearly independent and span the 3-dimensional polynomial space $P_2(\mathbb{R})$.

C. Basic Concept

$\dim(P_2) = 3$. Any set of 3 vectors in a 3-dimensional space is a basis if and only if it is linearly independent (determinant of coordinate matrix $\neq 0$).

D. Formula and Explanation

Formula: $\det [\ [p_1]_std \mid [p_2]_std \mid [p_3]_std\ ] \neq 0.$

Explanation & Conditions: Standard basis for P_2 is $\{1, x, x^2\}$.

E. Given Information

$p_1 = 1 + x + 0x^2$, $p_2 = 0 + x + x^2$, $p_3 = 1 + 0x + x^2$.

F. Complete Step-by-Step Solution

Step 1: Write coordinate vectors relative to standard basis $\{1, x, x^2\}$

$[p_1] = [1, 1, 0]^T$
$[p_2] = [0, 1, 1]^T$
$[p_3] = [1, 0, 1]^T$.

💡 **Why this step is performed:** Execute step 1: write coordinate vectors relative to standard basis $\{1, x, x^2\}$ to advance the step-by-step mathematical solution toward the final result.

Step 2: Construct matrix A with coordinate vectors as columns

A = | 1 0 1 |
| 1 1 0 |
| 0 1 1 |.

💡 **Why this step is performed:** Execute step 2: construct matrix a with coordinate vectors as columns to advance the step-by-step mathematical solution toward the final result.

Step 3: Evaluate det(A)

det(A) = 1(1·1 - 0·1) - 0 + 1(1·1 - 1·0)
= 1(1) + 1(1) = 1 + 1 = 2.

💡 **Why this step is performed:** Evaluate the Hessian discriminant $D = rt - s^2$; $D > 0$ and $r > 0$ indicates a local minimum, while $D > 0$ and $r < 0$ indicates a local maximum.

Step 4: Conclude basis property

Since $\det(A) = 2 \neq 0$:
1. The set S is LINEARLY INDEPENDENT.
2. The number of vectors $|S| = 3$ equals $\dim(P_2) = 3$.
By the basis theorem, any linearly independent set of 3 vectors in a 3-dimensional space automatically spans the space.
Therefore, S FORMS A BASIS for $P_2(\mathbb{R})$.

💡 **Why this step is performed:** Execute step 4: conclude basis property to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: det(A) = 2 ≠ 0; The set S = {1 + x, x + x^2, 1 + x^2} FORMS A BASIS for $P_2(\mathbb{R})$.

H. Examination Presentation Strategy

- Express polynomials as coordinate vectors.
- Set up 3×3 matrix A.
- Compute det(A) = 2.
- Quote theorem: 3 independent vectors in 3D space form a basis.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Evaluating polynomials at arbitrary points x = 0, 1, 2 without justification.
✓ **How to Avoid:** Coordinate matrix with standard basis {1, x, x^2} is rigorous and universally accepted.

⚠️ **Common Mistake:** Thinking dim(P_2) = 2.
✓ **How to Avoid:** dim(P_2) = 3 because of the constant term!

J. Similar Practice Question

Practice Problem 94: Test whether {1, 1 - x, $(1 - x)^2$} forms a basis for $P_2(\mathbb{R})$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 95: Basis and Dimension of the Null Space (Solution Space of Homogeneous System)

EXAM-LEVEL

A. Full Question:
Find a basis and the dimension of the solution space W of the homogeneous system of equations:
$x_1 + 2x_2 - x_3 + 3x_4 = 0$,
$2x_1 + 4x_2 - x_3 + 7x_4 = 0$,
$x_1 + 2x_2 + 0x_3 + 4x_4 = 0$.

B. Meaning of the Question

Row reduce the coefficient matrix to identify free variables, write the solution in parametric form, and extract the basis vectors.

C. Basic Concept

The solution space of AX = 0 is the null space Null(A). By the Rank-Nullity theorem, dim(Null A) = n - rank(A).

D. Formula and Explanation

Formula: `dim(W) = n - rank(A) = 4 - r.`

Explanation & Conditions: Number of unknowns n = 4.

E. Given Information

Coefficient matrix A = [[1, 2, -1, 3], [2, 4, -1, 7], [1, 2, 0, 4]].

F. Complete Step-by-Step Solution

Step 1: Reduce A to row-echelon form

A = [[1, 2, -1, 3], [2, 4, -1, 7], [1, 2, 0, 4]].
$R_2 \to R_2 - 2R_1$: [0, 0, 1, 1].
$R_3 \to R_3 - R_1$: [0, 0, 1, 1].
$R_3 \to R_3 - R_2$: [0, 0, 0, 0].

> 💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 2: Identify rank and free variables

Row-echelon form:
[[1, 2, -1, 3], [0, 0, 1, 1], [0, 0, 0, 0]].
Rank r = 2. Leading pivot variables are x_1 (col 1) and x_3 (col 3).
Free variables are x_2 and x_4.
Dimension of solution space = n - r = 4 - 2 = 2.

> 💡 **Why this step is performed:** Execute step 2: identify rank and free variables to advance the step-by-step mathematical solution toward the final result.

Step 3: Express pivot variables in terms of free variables

From Row 2: $x_3 + x_4 = 0 \Rightarrow x_3 = -x_4$.
From Row 1: $x_1 + 2x_2 - x_3 + 3x_4 = 0 \Rightarrow x_1 = -2x_2 + x_3 - 3x_4 = -2x_2 + (-x_4) - 3x_4 = -2x_2 - 4x_4$.

> 💡 **Why this step is performed:** Execute step 3: express pivot variables in terms of free variables to advance the step-by-step mathematical solution toward the final result.

Step 4: Write general solution vector and decompose

$X = [x_1, x_2, x_3, x_4]^T = [-2x_2 - 4x_4, x_2, -x_4, x_4]^T$
$= x_2 [-2, 1, 0, 0]^T + x_4 [-4, 0, -1, 1]^T$.

> 💡 **Why this step is performed:** Execute step 4: write general solution vector and decompose to advance the step-by-step mathematical solution toward the final result.

Step 5: State basis and dimension

Basis vectors: $v_1 = (-2, 1, 0, 0)$ and $v_2 = (-4, 0, -1, 1)$.
Since neither is a scalar multiple of the other, they are linearly independent.
Basis B = {(-2, 1, 0, 0), (-4, 0, -1, 1)}; Dimension = 2.

> 💡 **Why this step is performed:** Extract the linearly independent spanning vectors corresponding to the free parameters to form the basis.

✓ **G. Final Answer: Basis for Solution Space: B = {(-2, 1, 0, 0), (-4, 0, -1, 1)}; Dimension = 2**

H. Examination Presentation Strategy

- Show row reduction of A to echelon form.
- State Rank r = 2 and Dimension = 4 - 2 = 2.
- Decompose parametric solution into basis vectors.
- Box final basis and dimension.

I. Common Mistakes & How to Avoid Them

⚠ **Common Mistake:** Calling x_1 and x_2 pivots.
✓ **How to Avoid:** Pivot 1 is in col 1 (x_1), but Pivot 2 is in col 3 (x_3)! Column 2 has no pivot, so x_2 is free!

⚠ **Common Mistake:** Arithmetic error in $x_1 = -2x_2 - 4x_4$.
✓ **How to Avoid:** $x_3 - 3x_4 = -x_4 - 3x_4 = -4x_4$.

J. Similar Practice Question

Practice Problem 95: Find basis and dimension of solution space for x + 2y + 2z = 0, 2x + 5y + 7z = 0, 3x + 6y + 8z = 0.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 96: Orthogonal Complement W^⊥ of a Subspace in $\mathbb{R}^4$ EXAM-LEVEL

A. Full Question:
Find a basis and dimension for the orthogonal complement W^⊥ of the subspace W = span{(1, 2, 0, -1), (2, 3, 1, 0)} in $\mathbb{R}^4$.

B. Meaning of the Question

Find all vectors x = (x_1, x_2, x_3, x_4) that are orthogonal to both spanning vectors of W, solve the resulting homogeneous system, and find a basis for W^⊥.

C. Basic Concept

The orthogonal complement W^⊥ consists of all vectors perpendicular to every vector in W. If dim(W) = k in $\mathbb{R}^n$, then dim(W^⊥) = n - k.

D. Formula and Explanation

Formula: $x \in W^\perp \Leftrightarrow v_1 \cdot x = 0$ and $v_2 \cdot x = 0$; $\dim(W) + \dim(W^\perp) = 4$.

Explanation & Conditions: W^⊥ is the null space of the matrix with vectors of W as rows.

E. Given Information

v_1 = (1, 2, 0, -1), v_2 = (2, 3, 1, 0) in $\mathbb{R}^4$.

F. Complete Step-by-Step Solution

Step 1: Set up the orthogonal conditions

For $x = (x_1, x_2, x_3, x_4) \in W^\perp$:
1. $v_1 \cdot x = x_1 + 2x_2 - x_4 = 0$
2. $v_2 \cdot x = 2x_1 + 3x_2 + x_3 = 0$.
Coefficient matrix A = [[1, 2, 0, -1], [2, 3, 1, 0]].

💡 **Why this step is performed:** Execute step 1: set up the orthogonal conditions to advance the step-by-step mathematical solution toward the final result.

Step 2: Reduce matrix to echelon form

$R_2 \to R_2 - 2R_1$: [0, -1, 1, 2] ⇒ [0, 1, -1, -2].
Matrix in echelon form:
[[1, 2, 0, -1], [0, 1, -1, -2]].
Pivots in columns 1 and 2 (x_1, x_2). Free variables are x_3 and x_4.

💡 **Why this step is performed:** Execute step 2: reduce matrix to echelon form to advance the step-by-step mathematical solution toward the final result.

Step 3: Solve for x_1 and x_2 in terms of x_3 and x_4

From Row 2: $x_2 = x_3 + 2x_4$.
From Row 1: $x_1 = -2x_2 + x_4 = -2(x_3 + 2x_4) + x_4 = -2x_3 - 4x_4 + x_4 = -2x_3 - 3x_4$.

💡 **Why this step is performed:** Execute step 3: solve for x_1 and x_2 in terms of x_3 and x_4 to advance the step-by-step mathematical solution toward the final result.

Step 4: Parametric decomposition of x

$x = (-2x_3 - 3x_4, x_3 + 2x_4, x_3, x_4)$
$= x_3 (-2, 1, 1, 0) + x_4 (-3, 2, 0, 1)$.

💡 **Why this step is performed:** Multiply the standard matrices in the correct sequence to obtain the matrix representation of the composite map.

Step 5: State basis and dimension of $W^\perp$

Basis for $W^\perp$: B_perp = {(-2, 1, 1, 0), (-3, 2, 0, 1)}.
Dimension: $\dim(W^\perp) = 2$.
Check: $\dim(W) + \dim(W^\perp) = 2 + 2 = 4 = \dim(\mathbb{R}^4)$. Verified!

💡 **Why this step is performed:** Extract the linearly independent spanning vectors corresponding to the free parameters to form the basis.

✓ G. Final Answer: Basis for $W^\perp$: B_perp = {(-2, 1, 1, 0), (-3, 2, 0, 1)}; Dimension $\dim(W^\perp) = 2$

H. Examination Presentation Strategy

- State definition of $W^\perp$: $v \cdot x = 0$.

- Set up matrix with basis of W as rows.
- Solve for null space.
- State basis and verify dim(W) + dim(W^⊥) = n.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Putting vectors as columns instead of rows.
✓ **How to Avoid:** W^⊥ is orthogonal to the rows of matrix A, so basis vectors must be ROWS of A!

⚠️ **Common Mistake:** Arithmetic error in $-2(2x_4) + x_4 = -3x_4$.
✓ **How to Avoid:** $-4x_4 + x_4 = -3x_4$.

J. Similar Practice Question

Practice Problem 96: Find a basis for W^⊥ where W is the line spanned by (1, 1, 1) in $\mathbb{R}^3$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 97: Parseval's Identity and Bessel's Inequality

EXAM-LEVEL

A. Full Question:
State Bessel's Inequality and Parseval's Identity for an orthonormal set $\{e_1, e_2, ..., e_k\}$ in an inner product space V, and verify Parseval's identity for v = (2, -3, 6) using the standard orthonormal basis of $\mathbb{R}^3$.

B. Meaning of the Question

State the general geometric theorems relating the length of a vector to the sum of squares of its Fourier coefficients, and verify that $||v||^2 = \Sigma |\langle v, e_i\rangle|^2$.

C. Basic Concept

Parseval's identity is the infinite-dimensional generalized Pythagorean theorem. Bessel's inequality states that any finite sum of squared projections cannot exceed the total squared length of the vector.

D. Formula and Explanation

Formula: `Bessel's Inequality: Σ_{i=1}^k |⟨v, e_i⟩|² ≤ ||v||².`
`Parseval's Identity (when {e_i} is a complete basis): Σ_{i=1}^n |⟨v, e_i⟩|² = ||v||².`
Explanation & Conditions: ⟨v, e_i⟩ are the generalized Fourier coefficients.

E. Given Information

v = (2, -3, 6), standard orthonormal basis $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$.

F. Complete Step-by-Step Solution

Step 1: State the Theorems

Let $\{e_1, e_2, ..., e_k\}$ be an orthonormal set in an inner product space V.
1. Bessel's Inequality: For any $v \in V$: $\Sigma_{i=1}^k |\langle v, e_i\rangle|^2 \leq ||v||^2$.

2. Parseval's Identity: If {e_i} forms an orthonormal BASIS of V, equality holds: $\Sigma_{i=1}^{n} |\langle v, e_i\rangle|^2 = ||v||^2$.

💡 **Why this step is performed:** Execute step 1: state the theorems to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute Fourier coefficients ⟨v, e_i⟩ for v = (2, -3, 6)

- $c_1 = \langle v, e_1\rangle = (2)(1) + (-3)(0) + (6)(0) = 2$.
- $c_2 = \langle v, e_2\rangle = (2)(0) + (-3)(1) + (6)(0) = -3$.
- $c_3 = \langle v, e_3\rangle = (2)(0) + (-3)(0) + (6)(1) = 6$.

💡 **Why this step is performed:** Execute step 2: compute fourier coefficients ⟨v, e_i⟩ for v = (2, -3, 6) to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute sum of squares of coefficients (LHS)

$\Sigma |c_i|^2 = 2^2 + (-3)^2 + 6^2 = 4 + 9 + 36 = 49$.

💡 **Why this step is performed:** Execute step 3: compute sum of squares of coefficients (lhs) to advance the step-by-step mathematical solution toward the final result.

Step 4: Compute square of the norm $||v||^2$ (RHS)

$||v||^2 = 2^2 + (-3)^2 + 6^2 = 4 + 9 + 36 = 49$.

💡 **Why this step is performed:** Execute step 4: compute square of the norm $||v||^2$ (rhs) to advance the step-by-step mathematical solution toward the final result.

Step 5: Conclude Verification

LHS = 49 = RHS.
Hence Parseval's identity is verified!

💡 **Why this step is performed:** Execute step 5: conclude verification to advance the step-by-step mathematical solution toward the final result.

✓ **G. Final Answer: Bessel: $\Sigma |\langle v, e_i\rangle|^2 \le ||v||^2$; Parseval: $\Sigma |\langle v, e_i\rangle|^2 = ||v||^2$; Verified: $2^2 + (-3)^2 + 6^2 = 49 = ||v||^2$**

H. Examination Presentation Strategy

- State both Bessel and Parseval statements.
- Compute each Fourier coefficient c_i.
- Show sum of squares equals $||v||^2$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Confusing Bessel with Parseval.
✓ **How to Avoid:** Bessel is an INEQUALITY (≤ for any orthonormal set); Parseval is an EQUALITY (= for complete basis)!

⚠️ **Common Mistake:** Sign error in $(-3)^2 = -9$.

✓ **How to Avoid:** Squaring any real number yields a positive value: $(-3)^2 = +9$.

J. Similar Practice Question

Practice Problem 97: Verify Bessel's inequality for $v = (1, 2, 3)$ using only the partial orthonormal set $\{e_1, e_2\}$ in $\mathbb{R}^3$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 98: Gram-Schmidt Process on Polynomial Inner Product Space

EXAM-LEVEL

A. Full Question:
Apply Gram-Schmidt orthogonalization to the basis $\{1, x, x^2\}$ of $P_2[0, 1]$ with inner product $\langle p, q\rangle = \int_0^1 p(x) q(x)\, dx$ to find the first two orthogonal polynomials.

B. Meaning of the Question

Compute $u_1 = 1$ and $u_2 = x$ - proj_{u_1}(x) using the continuous integral inner product on [0, 1].

C. Basic Concept

Constructs the classical shifted Legendre polynomials orthogonal on [0, 1].

D. Formula and Explanation

Formula: $u_1 = v_1 = 1;\ u_2 = v_2 - [\ \langle v_2, u_1\rangle / \langle u_1, u_1\rangle\ ]\ u_1.$
Explanation & Conditions: $v_1 = 1$, $v_2 = x$.

E. Given Information

$v_1 = 1$, $v_2 = x$ on [0, 1], $\langle p, q\rangle = \int_0^1 p(x)\, q(x)\, dx$.

F. Complete Step-by-Step Solution

Step 1: Set first orthogonal polynomial $u_1 = 1$

$u_1(x) = 1$.
$\langle u_1, u_1\rangle = \int_0^1 (1)(1)\, dx = \int_0^1 1\, dx = 1$.

💡 **Why this step is performed:** Execute step 1: set first orthogonal polynomial $u_1 = 1$ to advance the step-by-step mathematical solution toward the final result.

Step 2: Compute inner product $\langle v_2, u_1\rangle$ with $v_2 = x$

$\langle v_2, u_1\rangle = \int_0^1 (x)(1)\, dx = \int_0^1 x\, dx = [\ x^2/2\ ]_0^1 = 1/2$.

💡 **Why this step is performed:** Execute step 2: compute inner product $\langle v_2, u_1\rangle$ with $v_2 = x$ to advance the step-by-step mathematical solution toward the final result.

Step 3: Construct second orthogonal polynomial u_2

$u_2(x) = v_2 - [\langle v_2, u_1 \rangle / \langle u_1, u_1 \rangle] u_1$
$= x - (1/2 / 1) (1) = x - 1/2.$

> 💡 **Why this step is performed:** Execute step 3: construct second orthogonal polynomial u_2 to advance the step-by-step mathematical solution toward the final result.

Step 4: Verify Orthogonality: $\langle u_1, u_2 \rangle = 0$

$\langle u_1, u_2 \rangle = \int_0^1 (1)(x - 1/2)\, dx = [x^2/2 - x/2]_0^1 = (1/2 - 1/2) - (0 - 0) = 0$. Exactly orthogonal!

> 💡 **Why this step is performed:** Verify that all pairwise dot products of the coordinate tangent vectors evaluate to zero.

✓ G. Final Answer: First two orthogonal polynomials: $u_1(x) = 1$ and $u_2(x) = x - 1/2$

H. Examination Presentation Strategy

- State Gram-Schmidt formula with integral inner products.
- Evaluate $\langle u_1, u_1 \rangle = 1$ and $\langle v_2, u_1 \rangle = 1/2$.
- Subtract projection to get $u_2 = x - 1/2$.
- Show verification integral = 0.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Using limits [-1, 1] instead of [0, 1].
✓ **How to Avoid:** Read the interval carefully: here the interval is [0, 1], NOT [-1, 1]!

⚠️ **Common Mistake:** Arithmetic error in $[x^2/2 - x/2]_0^1$.
✓ **How to Avoid:** 1/2 - 1/2 = 0.

J. Similar Practice Question

Practice Problem 98: Find the third orthogonal polynomial $u_3(x)$ from $v_3 = x^2$ using Gram-Schmidt on $P_2[0, 1]$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 99: Transition Matrix (Change of Basis Matrix)

EXAM-LEVEL

A. Full Question:
Find the transition matrix P from the basis B = {$u_1 = (1, 1)$, $u_2 = (1, 0)$} to the basis B' = {$v_1 = (2, 3)$, $v_2 = (1, 2)$} in $\mathbb{R}^2$, and find the coordinate vector $[w]_{B'}$ if $[w]_B = [3, -2]^T$.

B. Meaning of the Question

Find matrix P such that $[w]_B = P\, [w]_{B'}$ (or $[w]_{B'} = P^{-1}\, [w]_B$), converting coordinates between non-standard bases.

C. Basic Concept

The columns of the transition matrix P from B' to B are the coordinate vectors of the new basis vectors relative to the old basis: P = [$[v_1]_B$ | $[v_2]_B$].

D. Formula and Explanation

Formula: $[w]_B = P\ [w]_{B'} \Rightarrow [w]_{B'} = P^{-1}\ [w]_B$.

Explanation & Conditions: Express each v_j as a linear combination of u_1 and u_2.

E. Given Information

B = {(1, 1), (1, 0)}, B' = {(2, 3), (1, 2)}, $[w]_B = [3, -2]^T$.

F. Complete Step-by-Step Solution

Step 1: Express $v_1 = (2, 3)$ in terms of $B = \{u_1, u_2\}$

c_1 (1, 1) + c_2 (1, 0) = (2, 3)
$c_1 + c_2 = 2$ and $c_1 = 3 \Rightarrow c_2 = 2 - 3 = -1$.
$[v_1]_B = [3, -1]^T$.

> 💡 **Why this step is performed:** Execute step 1: express $v_1 = (2, 3)$ in terms of b = $\{u_1, u_2\}$ to advance the step-by-step mathematical solution toward the final result.

Step 2: Express $v_2 = (1, 2)$ in terms of $B = \{u_1, u_2\}$

d_1 (1, 1) + d_2 (1, 0) = (1, 2)
$d_1 + d_2 = 1$ and $d_1 = 2 \Rightarrow d_2 = 1 - 2 = -1$.
$[v_2]_B = [2, -1]^T$.

> 💡 **Why this step is performed:** Execute step 2: express $v_2 = (1, 2)$ in terms of b = $\{u_1, u_2\}$ to advance the step-by-step mathematical solution toward the final result.

Step 3: Construct Transition Matrix P (from B' to B)

P = [$[v_1]_B$ | $[v_2]_B$] = [[3, 2], [-1, -1]].

> 💡 **Why this step is performed:** Execute step 3: construct transition matrix p (from b' to b) to advance the step-by-step mathematical solution toward the final result.

Step 4: Compute P^{-1} to transform coordinates from B to B'

det(P) = (3)(-1) - (2)(-1) = -3 + 2 = -1.
P^{-1} = (1/-1) [[-1, -2], [1, 3]] = [[1, 2], [-1, -3]].

> 💡 **Why this step is performed:** Execute step 4: compute p^{-1} to transform coordinates from b to b' to advance the step-by-step mathematical solution toward the final result.

Step 5: Compute $[w]_{B'} = P^{-1}\ [w]_B$

$[w]_{B'}$ = [[1, 2], [-1, -3]] $[3, -2]^T$
= [1(3) + 2(-2), -1(3) + (-3)(-2)]^T
= [3 - 4, -3 + 6]^T = $[-1, 3]^T$.

💡 **Why this step is performed:** Execute step 5: compute $[w]_{b'} = p^{-1} [w]_b$ to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Transition Matrix P (B' to B) = [[3, 2], [-1, -1]]; $[w]_{B'} = [-1, 3]^T$

H. Examination Presentation Strategy

- Express each basis vector of B' in terms of B.
- Form transition matrix P with coordinate columns.
- Invert P to find P^{-1}.
- Multiply P^{-1} by $[w]_B$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Multiplying by P instead of P^{-1}.
✓ **How to Avoid:** $[w]_B = P [w]_{B'}$, so $[w]_{B'} = P^{-1} [w]_B$!

⚠️ **Common Mistake:** Swapping rows and columns of P.
✓ **How to Avoid:** Coordinate vectors are COLUMNS of P.

J. Similar Practice Question

Practice Problem 99: Find transition matrix from standard basis to B = {(1, 2), (3, 4)} in $\mathbb{R}^2$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 100: Orthogonal Matrix Representation and Isometry of Inner Product EXAM-LEVEL

A. Full Question:
Prove that an n×n real matrix Q is orthogonal ($Q^T Q = I$) if and only if it preserves the inner product: $\langle Q u, Q v \rangle = \langle u, v \rangle$ for all $u, v \in \mathbb{R}^n$.

B. Meaning of the Question

Prove that orthogonal transformations are isometries (rigid rotations and reflections that preserve angles, lengths, and dot products).

C. Basic Concept

Orthogonal matrices represent rigid geometric motions (rotations and reflections). Preserving inner products automatically guarantees preservation of lengths ($||Qu|| = ||u||$) and angles.

D. Formula and Explanation

Formula: $\langle Q\ u,\ Q\ v \rangle = (Q\ u)^T\ (Q\ v) = u^T\ Q^T\ Q\ v = u^T\ (Q^T\ Q)\ v.$

Explanation & Conditions: If $Q^T Q = I$, $u^T I v = u^T v = \langle u, v \rangle$.

E. Given Information

Square matrix Q in $M_n(\mathbb{R})$.

F. Complete Step-by-Step Solution

Step 1: Express inner product using matrix transpose

In standard Euclidean space, $\langle x, y\rangle = x^T y$.
Therefore: $\langle Q u, Q v\rangle = (Q u)^T (Q v)$.

> 💡 **Why this step is performed:** Execute step 1: express inner product using matrix transpose to advance the step-by-step mathematical solution toward the final result.

Step 2: Apply transpose rule for matrix product $(AB)^T = B^T A^T$

$(Q u)^T = u^T Q^T$.
Substitute into the expression:
$\langle Q u, Q v\rangle = (u^T Q^T)(Q v) = u^T (Q^T Q) v$ --- (1).

> 💡 **Why this step is performed:** Execute step 2: apply transpose rule for matrix product $(ab)^T = b^T a^T$ to advance the step-by-step mathematical solution toward the final result.

Step 3: Prove forward direction (⇒)

Assume Q is orthogonal, so $Q^T Q = I$.
Substitute $Q^T Q = I$ into (1):
$\langle Q u, Q v\rangle = u^T I v = u^T v = \langle u, v\rangle$.
Thus, Q preserves the inner product for all $u, v \in \mathbb{R}^n$.

> 💡 **Why this step is performed:** Execute step 3: prove forward direction (⇒) to advance the step-by-step mathematical solution toward the final result.

Step 4: Prove reverse direction (⇐)

Assume $\langle Q u, Q v\rangle = \langle u, v\rangle$ for all $u, v \in \mathbb{R}^n$.
From (1): $u^T (Q^T Q) v = u^T I v$.
$u^T (Q^T Q - I) v = 0$ for all $u, v \in \mathbb{R}^n$.
Choose $u = e_i$ and $v = e_j$ (standard basis vectors):
$e_i^T (Q^T Q - I) e_j = (Q^T Q - I)_{ij} = 0$ for all i, j.
Since every matrix entry is 0: $Q^T Q - I = O \Rightarrow Q^T Q = I$.
Therefore, Q is an ORTHOGONAL matrix. Hence proved.

> 💡 **Why this step is performed:** Execute step 4: prove reverse direction (⇐) to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Q is orthogonal ($Q^T Q = I$) if and only if $\langle Q u, Q v\rangle = \langle u, v\rangle$ for all $u, v \in \mathbb{R}^n$ (Hence proved)

H. Examination Presentation Strategy

- State definition of orthogonal matrix $Q^T Q = I$.

- Show $(Q\,u)^T\,(Q\,v) = u^T\,Q^T\,Q\,v$.
- Prove both $\Rightarrow$ and $\Leftarrow$ clearly.
- Use standard basis vectors e_i, e_j for reverse proof.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Assuming $(Q\,u)^T = Q^T\,u^T$.
✓ **How to Avoid:** Transpose reverses order: $(Q\,u)^T = u^T\,Q^T$!

⚠️ **Common Mistake:** Omitting the reverse direction.
✓ **How to Avoid:** 'If and only if' requires proving both directions.

J. Similar Practice Question

Practice Problem 100: Prove that if Q is orthogonal, then det(Q) = ±1.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

MODULE 5: Linear Transformations

25 Fully Solved Questions: 5 Fundamental (Concepts & Proofs) + 10 Intermediate (Core Problems) + 10 Exam-Level (High-Yield VTU Questions)

QUESTION 101: Definition and Properties of a Linear Transformation

FUNDAMENTAL

A. Full Question:
Define a linear transformation T: V → W between two real vector spaces, and state its two basic properties: (i) T(0) = 0 and (ii) T(-v) = -T(v).

B. Meaning of the Question

State the mathematical definition of a linear mapping preserving vector addition and scalar multiplication, and prove the preservation of zero vector and negatives.

C. Basic Concept

A linear transformation is a function between vector spaces that respects the algebraic operations: the transformation of a sum is the sum of transformations, and scaling can be factored out.

D. Formula and Explanation

Formula: $T(u + v) = T(u) + T(v)$ and $T(c\,u) = c\,T(u)$ for all $u, v \in V, c \in \mathbb{R}$.
Combined: $T(a\,u + b\,v) = a\,T(u) + b\,T(v)$.

Explanation & Conditions: V is called the domain; W is called the codomain.

E. Given Information

Mapping T: V → W.

F. Complete Step-by-Step Solution

Step 1: Definition of Linear Transformation

A mapping T: V → W is called a LINEAR TRANSFORMATION if for all u, v ∈ V and all scalars c ∈ $\mathbb{R}$:
1. Additivity: T(u + v) = T(u) + T(v).
2. Homogeneity: T(c u) = c T(u).

💡 **Why this step is performed:** Verify that the mapping preserves vector addition T(u + v) = T(u) + T(v) and scalar scaling T(cu) = c T(u).

Step 2: Proof that T(0_V) = 0_W

In vector space V, 0 = 0 + 0.
Applying T:
T(0) = T(0 + 0) = T(0) + T(0) (by additivity).
Subtract T(0) from both sides in W:
T(0) - T(0) = T(0) + T(0) - T(0) ⇒ 0_W = T(0).
Hence, T(0) = 0. Proved!

💡 **Why this step is performed:** Execute step 2: proof that t(0_v) = 0_w to advance the step-by-step mathematical solution toward the final result.

Step 3: Proof that T(-v) = -T(v)

For any vector v ∈ V, -v = (-1) v.
Applying homogeneity with scalar c = -1:
T(-v) = T((-1) v) = (-1) T(v) = -T(v). Hence proved!

💡 **Why this step is performed:** Execute step 3: proof that t(-v) = -t(v) to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: T: V → W is linear if T(a u + b v) = a T(u) + b T(v); Properties: T(0) = 0 and T(-v) = -T(v) (Both proved)

H. Examination Presentation Strategy

- State both additivity and homogeneity.
- Show algebraic proof of T(0) = 0.
- Show T(-v) = -T(v) using c = -1.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Confusing 0 in V with 0 in W.
✓ **How to Avoid:** T maps the zero vector of domain V to the zero vector of codomain W: T(0_V) = 0_W.

⚠️ **Common Mistake:** Assuming every function with T(0)=0 is linear.
✓ **How to Avoid:** T(0)=0 is a NECESSARY condition, not sufficient (e.g., $f(x) = x^2$ has f(0)=0 but is not linear).

J. Similar Practice Question

Practice Problem 101: Prove that T(u - v) = T(u) - T(v) for any linear transformation T.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 102: Testing Linearity of a Transformation (Passing vs Failing) FUNDAMENTAL

A. Full Question:
Determine whether the following transformations are linear: (i) T: $\mathbb{R}^2 \to \mathbb{R}^3$, T(x, y) = (x + y, x - y, 2x), and (ii) S: $\mathbb{R}^2 \to \mathbb{R}^2$, S(x, y) = (x + 1, y).

B. Meaning of the Question

Test transformation T for additivity and scalar homogeneity; test transformation S to show it fails linearity because S(0, 0) ≠ (0, 0).

C. Basic Concept

A linear transformation must map the origin to the origin. If T(0) ≠ 0, it is immediately non-linear. If T(0) = 0, we verify T(a u + b v) = a T(u) + b T(v).

D. Formula and Explanation

Formula: `Linearity test: T(a u + b v) = a T(u) + b T(v).`

Explanation & Conditions: u = (x_1, y_1), v = (x_2, y_2).

E. Given Information

T(x, y) = (x + y, x - y, 2x); S(x, y) = (x + 1, y).

F. Complete Step-by-Step Solution

Step 1: Test S(x, y) = (x + 1, y)

Evaluate S at the origin (0, 0):
S(0, 0) = (0 + 1, 0) = (1, 0) ≠ (0, 0).
Since a linear transformation MUST map the zero vector to the zero vector, S(0) ≠ 0 implies S is NOT A LINEAR TRANSFORMATION.

> 💡 **Why this step is performed:** Execute step 1: test s(x, y) = (x + 1, y) to advance the step-by-step mathematical solution toward the final result.

Step 2: Test T(x, y) = (x + y, x - y, 2x) for Linearity

Let u = (x_1, y_1), v = (x_2, y_2) and a, b ∈ $\mathbb{R}$.
a u + b v = $(a x_1 + b x_2, a y_1 + b y_2)$.
Applying T:
$T(a u + b v) = ([a x_1 + b x_2] + [a y_1 + b y_2], [a x_1 + b x_2] - [a y_1 + b y_2], 2[a x_1 + b x_2])$
$= (a(x_1 + y_1) + b(x_2 + y_2), a(x_1 - y_1) + b(x_2 - y_2), a(2x_1) + b(2x_2))$
$= a (x_1 + y_1, x_1 - y_1, 2x_1) + b (x_2 + y_2, x_2 - y_2, 2x_2)$
= a T(u) + b T(v).

> 💡 **Why this step is performed:** Execute step 2: test t(x, y) = (x + y, x - y, 2x) for linearity to advance the step-by-step mathematical solution toward the final result.

Step 3: Conclude for T

Since T(a u + b v) = a T(u) + b T(v) for all vectors u, v and all scalars a, b, T IS A LINEAR TRANSFORMATION.

> 💡 **Why this step is performed:** Execute step 3: conclude for t to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: (i) T is a LINEAR TRANSFORMATION; (ii) S is NOT a linear transformation because S(0, 0) = (1, 0) ≠ (0, 0).

H. Examination Presentation Strategy

- Test S first using S(0) ≠ 0 for immediate 2-line conclusion.
- For T, write out T(a u + b v) and group into a T(u) + b T(v).

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Testing only with numbers (e.g. u=(1,2)).
✓ **How to Avoid:** A numerical example can DISPROVE linearity, but PROVING linearity requires arbitrary algebraic variables!

⚠️ **Common Mistake:** Claiming S is linear because its graph is a straight line.
✓ **How to Avoid:** In linear algebra, 'linear' means passing through the origin (y = mx), NOT affine (y = mx + c with c ≠ 0).

J. Similar Practice Question

Practice Problem 102: Determine if T(x, y) = (xy, x) is a linear transformation.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 103: Kernel (Null Space) and Range (Image) Definitions FUNDAMENTAL

A. Full Question:
Define the Kernel (Null space) and Range (Image) of a linear transformation T: V → W, and prove that Kernel(T) is a subspace of V.

B. Meaning of the Question

Provide formal definitions of ker(T) and range(T), and prove that ker(T) satisfies the subspace criteria in V.

C. Basic Concept

The kernel is the set of all input vectors that collapse to zero in W. The range is the set of all output vectors actually reached by T.

D. Formula and Explanation

Formula: $\ker(T) = \{ v \in V : T(v) = 0_W \}$; $\text{Range}(T) = \{ T(v) \in W : v \in V \}$.
Explanation & Conditions: ker(T) ⊆ V (domain); Range(T) ⊆ W (codomain).

E. Given Information

Linear transformation T: V → W.

F. Complete Step-by-Step Solution

Step 1: Definitions

• Kernel of T: ker(T) = { v ∈ V : T(v) = 0_W }.
• Range of T: Im(T) = Range(T) = { w ∈ W : w = T(v) for some v ∈ V }.

💡 **Why this step is performed:** Execute step 1: definitions to advance the step-by-step mathematical solution toward the final result.

Step 2: Prove ker(T) contains 0_V

From Q101, T(0_V) = 0_W. Therefore, 0_V ∈ ker(T). (ker(T) is non-empty).

💡 **Why this step is performed:** Execute step 2: prove ker(t) contains 0_v to advance the step-by-step mathematical solution toward the final result.

Step 3: Prove Closure under Addition in ker(T)

Let u, v ∈ ker(T). By definition, T(u) = 0 and T(v) = 0.
T(u + v) = T(u) + T(v) = 0 + 0 = 0.
Therefore, (u + v) ∈ ker(T).

💡 **Why this step is performed:** Take two arbitrary elements from the subset and prove their vector sum also satisfies the defining condition.

Step 4: Prove Closure under Scalar Multiplication in ker(T)

Let c ∈ $\mathbb{R}$ and u ∈ ker(T). Then T(u) = 0.
T(c u) = c T(u) = c (0) = 0.
Therefore, c u ∈ ker(T).
Since ker(T) satisfies all subspace axioms, ker(T) is a SUBSPACE of V. Hence proved.

💡 **Why this step is performed:** Multiply an arbitrary element by an arbitrary scalar and prove the resulting vector remains in the subset.

✓ G. Final Answer: ker(T) = {v ∈ V : T(v) = 0}; Range(T) = {T(v) : v ∈ V}; ker(T) is a subspace of V (Hence proved)

H. Examination Presentation Strategy

- State both definitions.
- Show 0 ∈ ker(T).
- Show u, v ∈ ker(T) ⇒ u + v ∈ ker(T).
- Show c u ∈ ker(T).

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Thinking ker(T) is a subspace of W.
✓ **How to Avoid:** ker(T) is a subspace of the DOMAIN V! Range(T) is a subspace of the CODOMAIN W!

⚠️ **Common Mistake:** Confusing ker(T) with the zero vector itself.
✓ **How to Avoid:** ker(T) is a SET of vectors, not just {0}. It can be multi-dimensional.

J. Similar Practice Question

Practice Problem 103: Prove that Range(T) is a subspace of W.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 104: Rank-Nullity Theorem Statement and Verification

FUNDAMENTAL

A. Full Question:
State the Rank-Nullity Theorem (Dimension Theorem) for a linear transformation T: V → W where V is finite-dimensional, and verify it for T: $\mathbb{R}^3 \to \mathbb{R}^2$, T(x, y, z) = (x - y, 2z).

B. Meaning of the Question

State dim(V) = rank(T) + nullity(T), and compute rank and nullity of the given transformation to verify the theorem.

C. Basic Concept

The total dimension of the domain V is conserved: every dimension either gets mapped into a non-zero output dimension (contributing to rank) or is squashed into zero (contributing to nullity).

D. Formula and Explanation

Formula: `dim(V) = rank(T) + nullity(T), where rank(T) = dim(Range T) and nullity(T) = dim(ker T).`

Explanation & Conditions: dim(Domain) = Rank + Nullity.

E. Given Information

T: $\mathbb{R}^3 \to \mathbb{R}^2$, T(x, y, z) = (x - y, 2z). Domain V = $\mathbb{R}^3$, dim(V) = 3.

F. Complete Step-by-Step Solution

Step 1: State the Rank-Nullity Theorem

Let V and W be vector spaces, with V finite-dimensional. If T: V → W is a linear transformation, then:
dim(V) = rank(T) + nullity(T)
where rank(T) = dim(Range T) and nullity(T) = dim(ker T).

💡 **Why this step is performed:** Execute step 1: state the rank-nullity theorem to advance the step-by-step mathematical solution toward the final result.

Step 2: Find ker(T) for the given transformation

T(x, y, z) = (0, 0) ⇒ (x - y, 2z) = (0, 0).
x - y = 0 ⇒ x = y.
2z = 0 ⇒ z = 0.
y is a free variable.
Vector: (x, y, z) = (y, y, 0) = y (1, 1, 0).
Basis for ker(T) = {(1, 1, 0)}.
Nullity(T) = dim(ker T) = 1.

💡 **Why this step is performed:** Execute step 2: find ker(t) for the given transformation to advance the step-by-step mathematical solution toward the final result.

Step 3: Find Range(T) and Rank(T)

T(x, y, z) = (x - y, 2z) = x (1, 0) + y (-1, 0) + z (0, 2).
Spanning vectors of Range(T) are (1, 0) and (0, 2).

Since these 2 vectors in $\mathbb{R}^2$ are linearly independent (det |1 0; 0 2| = 2 ≠ 0), they form a basis for Range(T).
Rank(T) = dim(Range T) = 2.

> 💡 **Why this step is performed:** Determine the subspace spanned by the images of the domain basis vectors to construct the range.

Step 4: Verify Rank-Nullity Theorem

dim(Domain V) = dim($\mathbb{R}^3$) = 3.
rank(T) + nullity(T) = 2 + 1 = 3.
Since dim(V) = rank(T) + nullity(T) = 3, the theorem is VERIFIED!

> 💡 **Why this step is performed:** Verify the dimension theorem: the sum of the rank and the nullity equals the dimension of the domain space.

✓ G. Final Answer: Rank(T) = 2, Nullity(T) = 1; Rank + Nullity = 2 + 1 = 3 = dim($\mathbb{R}^3$) (Rank-Nullity Theorem Verified)

H. Examination Presentation Strategy

- State Rank-Nullity theorem formula.
- Solve for ker(T) to find nullity = 1.
- Find Range(T) basis to get rank = 2.
- Add rank + nullity = 3 and compare with dim(Domain).

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Using codomain dimension instead of domain.
✓ **How to Avoid:** dim(V) is the DOMAIN $\mathbb{R}^3$ (dim=3), NOT the codomain $\mathbb{R}^2$ (dim=2)!

⚠️ **Common Mistake:** Arithmetic error: saying nullity = 2.
✓ **How to Avoid:** Only y is free, so basis has 1 vector, nullity = 1.

J. Similar Practice Question

Practice Problem 104: Verify Rank-Nullity theorem for T: $\mathbb{R}^3 \to \mathbb{R}^3$, T(x, y, z) = (x + y, y + z, x - z).
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 105: Standard Matrix of a Linear Transformation

FUNDAMENTAL

A. Full Question:
Find the standard matrix [T] for the linear transformation T: $\mathbb{R}^2 \to \mathbb{R}^2$ that first rotates a vector counterclockwise by 90° and then reflects it across the x-axis.

B. Meaning of the Question

Evaluate T on standard basis vectors $e_1 = (1, 0)$ and $e_2 = (0, 1)$, and place the transformed vectors as columns of matrix [T].

C. Basic Concept

Any linear transformation T: $\mathbb{R}^n \to \mathbb{R}^m$ is completely determined by its action on the standard basis vectors: [T] = [$T(e_1)$ | $T(e_2)$ | ... | T(e_n)].

D. Formula and Explanation

Formula: `[T] = [ T(1, 0) | T(0, 1) ].`

Explanation & Conditions: Rotation by 90°: R = [[0, -1], [1, 0]]; Reflection across x-axis: M = [[1, 0], [0, -1]]. [T] = M · R.

E. Given Information

Operations: R_90 followed by reflection across x-axis.

F. Complete Step-by-Step Solution

Step 1: Track $e_1 = (1, 0)$

• Rotate 90° counterclockwise: (1, 0) → (0, 1).
• Reflect across x-axis: (0, 1) → (0, -1).
Thus: $T(e_1) = [0, -1]^T$ (Column 1 of [T]).

💡 **Why this step is performed:** Execute step 1: track $e_1 = (1, 0)$ to advance the step-by-step mathematical solution toward the final result.

Step 2: Track $e_2 = (0, 1)$

• Rotate 90° counterclockwise: (0, 1) → (-1, 0).
• Reflect across x-axis: (-1, 0) → (-1, 0) (points on x-axis do not change under reflection).
Thus: $T(e_2) = [-1, 0]^T$ (Column 2 of [T]).

💡 **Why this step is performed:** Execute step 2: track $e_2 = (0, 1)$ to advance the step-by-step mathematical solution toward the final result.

Step 3: Construct the Standard Matrix [T]

[T] = [$T(e_1)$ | $T(e_2)$] = [[0, -1], [-1, 0]].

💡 **Why this step is performed:** Evaluate the transformation on each standard basis vector and place the resulting coordinate vectors as columns of [T].

Step 4: Verify by matrix multiplication [T] = M · R

R = [[cos 90°, -sin 90°], [sin 90°, cos 90°]] = [[0, -1], [1, 0]].
M = [[1, 0], [0, -1]].
M · R = [[1, 0], [0, -1]] [[0, -1], [1, 0]] = [[0, -1], [-1, 0]]. Matches!

💡 **Why this step is performed:** Execute step 4: verify by matrix multiplication [t] = m · r to advance the step-by-step mathematical solution toward the final result.

✓ **G. Final Answer: Standard Matrix [T] = [[0, -1], [-1, 0]]**

H. Examination Presentation Strategy

- Show transformation of $e_1 = (1, 0)$ and $e_2 = (0, 1)$.
- Form matrix with $T(e_1)$ and $T(e_2)$ as columns.
- Verify using matrix product M · R.

I. Common Mistakes & How to Avoid Them

⚠ **Common Mistake:** Reversing the order of composition: R · M instead of M · R.
✓ **How to Avoid:** Transformation 'A followed by B' corresponds to matrix B · A!

⚠ **Common Mistake:** Placing $T(e_1)$ as a row instead of a column.
✓ **How to Avoid:** Standard matrix puts images of basis vectors as COLUMNS.

J. Similar Practice Question

Practice Problem 105: Find the standard matrix of reflection across the line y = x in $\mathbb{R}^2$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 106: Finding a Linear Transformation from Its Action on a Basis

INTERMEDIATE

A. Full Question:
A linear transformation T: $\mathbb{R}^2 \to \mathbb{R}^2$ satisfies T(1, 1) = (2, 3) and T(1, -1) = (4, 1). Find a general formula for T(x, y) and evaluate T(3, 5).

B. Meaning of the Question

Express arbitrary vector (x, y) in terms of basis {(1, 1), (1, -1)}, use linearity to find T(x, y), and evaluate for (3, 5).

C. Basic Concept

A linear transformation is completely and uniquely determined by its values on a basis of the domain. Any vector (x, y) = $c_1 v_1 + c_2 v_2$ satisfies $T(x, y) = c_1 T(v_1) + c_2 T(v_2)$.

D. Formula and Explanation

Formula: $(x, y) = c_1 (1, 1) + c_2 (1, -1) \Rightarrow c_1 = (x + y)/2, c_2 = (x - y)/2.$
$T(x, y) = c_1 T(1, 1) + c_2 T(1, -1).$
Explanation & Conditions: Linearity allows distributing T over the basis expansion.

E. Given Information

T(1, 1) = (2, 3), T(1, -1) = (4, 1).

F. Complete Step-by-Step Solution

Step 1: Express (x, y) as linear combination of (1, 1) and (1, -1)

$(x, y) = c_1 (1, 1) + c_2 (1, -1) = (c_1 + c_2, c_1 - c_2)$.

$c_1 + c_2 = x$

$c_1 - c_2 = y$.

Adding: $2c_1 = x + y \Rightarrow c_1 = (x + y) / 2$.

Subtracting: $2c_2 = x - y \Rightarrow c_2 = (x - y) / 2$.

> 💡 **Why this step is performed:** Execute step 1: express (x, y) as linear combination of (1, 1) and (1, -1) to advance the step-by-step mathematical solution toward the final result.

Step 2: Apply linear transformation T

$T(x, y) = c_1 T(1, 1) + c_2 T(1, -1)$

$= [(x + y)/2] (2, 3) + [(x - y)/2] (4, 1)$.

> 💡 **Why this step is performed:** Execute step 2: apply linear transformation t to advance the step-by-step mathematical solution toward the final result.

Step 3: Simplify components

• First component: $[(x + y)/2](2) + [(x - y)/2](4) = (x + y) + 2(x - y) = x + y + 2x - 2y = 3x - y$.

• Second component: $[(x + y)/2](3) + [(x - y)/2](1) = [3x + 3y + x - y] / 2 = [4x + 2y] / 2 = 2x + y$.

Therefore: $T(x, y) = (3x - y, 2x + y)$.

> 💡 **Why this step is performed:** Execute step 3: simplify components to advance the step-by-step mathematical solution toward the final result.

Step 4: Verify with given conditions

$T(1, 1) = (3(1) - 1, 2(1) + 1) = (2, 3)$. Matches!

$T(1, -1) = (3(1) - (-1), 2(1) + (-1)) = (4, 1)$. Matches!

> 💡 **Why this step is performed:** Execute step 4: verify with given conditions to advance the step-by-step mathematical solution toward the final result.

Step 5: Evaluate T(3, 5)

$T(3, 5) = (3(3) - 5, 2(3) + 5) = (9 - 5, 6 + 5) = (4, 11)$.

> 💡 **Why this step is performed:** Execute step 5: evaluate t(3, 5) to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: General formula: $T(x, y) = (3x - y, 2x + y)$; $T(3, 5) = (4, 11)$

H. Examination Presentation Strategy

- Solve for c_1 and c_2 in terms of x and y.
- Use $T(x, y) = c_1 T(v_1) + c_2 T(v_2)$.
- Simplify components clearly.

- Substitute (3, 5) to evaluate.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Arithmetic error in 3(3) - 5 = 4.
✓ **How to Avoid:** Be careful with basic arithmetic.

⚠️ **Common Mistake:** Attempting to guess the formula.
✓ **How to Avoid:** Always use the systematic basis expansion $(x, y) = c_1 v_1 + c_2 v_2$.

J. Similar Practice Question

Practice Problem 106: If T(1, 0) = (1, 2) and T(0, 1) = (-1, 3), find T(x, y) and T(2, -4).
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 107: Kernel, Nullity, Range, and Rank of a Transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$

INTERMEDIATE

A. Full Question:
For the linear transformation T: $\mathbb{R}^3 \to \mathbb{R}^3$ defined by T(x, y, z) = (x + 2y - z, y + z, x + y - 2z), find: (i) a basis and dimension for the Kernel, and (ii) a basis and dimension for the Range.

B. Meaning of the Question

Row reduce the matrix of T to compute the nullity, null space basis, rank, and range basis.

C. Basic Concept

The matrix of T is A. Null space of A gives ker(T); Column space of A gives Range(T).

D. Formula and Explanation

Formula: `A = [ [1, 2, -1], [0, 1, 1], [1, 1, -2] ].`
Explanation & Conditions: Columns of A generate Range(T); solutions of AX = 0 form ker(T).

E. Given Information

T(x, y, z) = (x + 2y - z, y + z, x + y - 2z).

F. Complete Step-by-Step Solution

Step 1: Set up coefficient matrix A

A = [[1, 2, -1], [0, 1, 1], [1, 1, -2]].

💡 **Why this step is performed:** Execute step 1: set up coefficient matrix a to advance the step-by-step mathematical solution toward the final result.

Step 2: Reduce A to row-echelon form

$R_3 \to R_3 - R_1$: [0, -1, -1].
$R_3 \to R_3 + R_2$: [0, 0, 0].
Echelon form:
[[1, 2, -1], [0, 1, 1], [0, 0, 0]].
Pivots in columns 1 and 2 (x, y). Free variable is z.

> 💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 3: Solve for Kernel (Null Space)

From Row 2: $y + z = 0 \Rightarrow y = -z$.
From Row 1: $x + 2y - z = 0 \Rightarrow x = -2y + z = -2(-z) + z = 3z$.
Vector: (x, y, z) = (3z, -z, z) = z (3, -1, 1).
Basis for ker(T): B_ker = {(3, -1, 1)}.
Nullity(T) = dim(ker T) = 1.

> 💡 **Why this step is performed:** Find all input vectors that map to the zero vector by setting T(v) = 0 and solving the resulting homogeneous system.

Step 4: Find Range(T) and Rank(T)

Since pivots are in columns 1 and 2, the corresponding original columns of A form a basis for Range(T):
Basis for Range(T): B_range = {(1, 0, 1), (2, 1, 1)}.
Rank(T) = dim(Range T) = 2.

> 💡 **Why this step is performed:** Determine the subspace spanned by the images of the domain basis vectors to construct the range.

Step 5: Verify Rank-Nullity Theorem

Rank + Nullity = 2 + 1 = 3 = dim(Domain $\mathbb{R}^3$). Verified!

> 💡 **Why this step is performed:** Verify the dimension theorem: the sum of the rank and the nullity equals the dimension of the domain space.

✓ G. Final Answer: Kernel: Basis = {(3, -1, 1)}, Nullity = 1; Range: Basis = {(1, 0, 1), (2, 1, 1)}, Rank = 2

H. Examination Presentation Strategy

- Write matrix A.
- Show row reduction to echelon form.
- Solve for ker(T) with free variable z.
- Select pivot columns for Range(T).
- State Rank and Nullity.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Using echelon columns for Range basis.
✓ **How to Avoid:** You must take the PIVOT COLUMNS OF THE ORIGINAL MATRIX A, not the reduced matrix!

⚠️ **Common Mistake:** Arithmetic error in x = -2(-z) + z = 3z.
✓ **How to Avoid:** 2z + z = 3z.

J. Similar Practice Question

Practice Problem 107: Find basis and dimension of ker(T) and Range(T) for T(x, y, z) = (x - y + z, 2x + y - z, 3x).
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 108: Testing Whether a Linear Transformation is One-to-One and Onto

INTERMEDIATE

A. Full Question:
Determine whether the linear transformation T: $\mathbb{R}^3 \to \mathbb{R}^3$ defined by T(x, y, z) = (x + y, y + z, x + z) is: (i) one-to-one (injective), (ii) onto (surjective), and (iii) an isomorphism.

B. Meaning of the Question

Check if ker(T) = {0} (one-to-one) and if rank(T) = dim(codomain) (onto). If both hold, T is an isomorphism.

C. Basic Concept

For a linear operator between spaces of the same finite dimension: T is one-to-one ⇔ T is onto ⇔ det(A) ≠ 0 ⇔ ker(T) = {0} ⇔ T is invertible (isomorphism).

D. Formula and Explanation

Formula: `det(A) ≠ 0 ⇒ T is both one-to-one and onto.`
Explanation & Conditions: Matrix A = [[1, 1, 0], [0, 1, 1], [1, 0, 1]].

E. Given Information

T(x, y, z) = (x + y, y + z, x + z).

F. Complete Step-by-Step Solution

Step 1: Write matrix representation A

A = [[1, 1, 0], [0, 1, 1], [1, 0, 1]].

💡 **Why this step is performed:** Execute step 1: write matrix representation a to advance the step-by-step mathematical solution toward the final result.

Step 2: Evaluate det(A)

det(A) = 1(1·1 - 0·1) - 1(0·1 - 1·1) + 0
= 1(1) - 1(-1) = 1 + 1 = 2.

💡 **Why this step is performed:** Evaluate the Hessian discriminant $D = rt - s^2$; $D > 0$ and $r > 0$ indicates a local minimum, while $D > 0$ and $r < 0$ indicates a local maximum.

Step 3: Test for One-to-One (Injective)

Since $\det(A) = 2 \neq 0$, the homogeneous system $AX = 0$ has ONLY the trivial solution $X = 0$.
Therefore, $\ker(T) = \{(0, 0, 0)\}$.
Since $\ker(T)$ contains only the zero vector, T is ONE-TO-ONE.

💡 **Why this step is performed:** Show that the kernel contains only the zero vector, which proves the transformation is injective (one-to-one).

Step 4: Test for Onto (Surjective)

$\text{Rank}(T) = \text{rank}(A) = 3$ (since $\det(A) \neq 0$).
Dimension of codomain $W = \dim(\mathbb{R}^3) = 3$.
Since $\text{rank}(T) = \dim(\text{codomain})$, $\text{Range}(T) = \mathbb{R}^3$.
Therefore, T is ONTO.

💡 **Why this step is performed:** Show that the rank of the transformation equals the dimension of the codomain, proving surjectivity (onto).

Step 5: Conclude Isomorphism

Since T is both one-to-one and onto (a bijective linear transformation), T IS AN ISOMORPHISM.

💡 **Why this step is performed:** Execute step 5: conclude isomorphism to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $\det(A) = 2 \neq 0$; (i) T is ONE-TO-ONE, (ii) T is ONTO, (iii) T is an ISOMORPHISM.

H. Examination Presentation Strategy

- Construct matrix A.
- Evaluate $\det(A) = 2 \neq 0$.
- State theorem: $\ker(T) = \{0\} \Leftrightarrow$ one-to-one.
- State theorem: $\text{rank}(T) = \dim(\text{codomain}) \Leftrightarrow$ onto.
- Conclude isomorphism.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Computing $\det(A)$ as 0.
✓ **How to Avoid:** Carefully expand: $1(1) - 1(-1) = 1 + 1 = 2$, NOT $1 - 1 = 0$!

⚠️ **Common Mistake:** Thinking one-to-one and onto must always be checked separately.
✓ **How to Avoid:** When domain and codomain have the same finite dimension, $\det \neq 0$ guarantees BOTH simultaneously!

J. Similar Practice Question

Practice Problem 108: Determine if $T(x, y, z) = (x - y, y - z, z - x)$ is one-to-one and onto.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 109: Matrix Representation Relative to Non-Standard Bases

INTERMEDIATE

A. Full Question:

Find the matrix representation $[T]_{B, B'}$ of the linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^3$ defined by $T(x, y) = (2x - y, x + 3y, x)$ relative to the bases $B = \{u_1 = (1, 1), u_2 = (1, 0)\}$ of $\mathbb{R}^2$ and $B' = \{e_1, e_2, e_3\}$ (standard basis) of $\mathbb{R}^3$.

B. Meaning of the Question

Compute T on each basis vector of B, express the outputs in terms of B', and construct matrix $[T]_{B, B'}$.

C. Basic Concept

The j-th column of the matrix representation $[T]_{B, B'}$ consists of the coordinate vector of $T(u_j)$ relative to the basis B': $Col_j = [\ T(u_j)\ ]_{B'}$.

D. Formula and Explanation

Formula: $[T]_{B, B'} = [\ [T(u_1)]_{B'} \mid [T(u_2)]_{B'}\ ]$.

Explanation & Conditions: Size of matrix is dim(W) × dim(V) = 3 × 2.

E. Given Information

$T(x, y) = (2x - y, x + 3y, x)$, $B = \{(1, 1), (1, 0)\}$, B' = standard basis of $\mathbb{R}^3$.

F. Complete Step-by-Step Solution

Step 1: Apply T to first basis vector $u_1 = (1, 1)$

$T(1, 1) = (\ 2(1) - 1, 1 + 3(1), 1\ ) = (1, 4, 1)$.

Relative to standard basis B', $[T(u_1)]_{B'} = [1, 4, 1]^T$ (Column 1).

> 💡 **Why this step is performed:** Execute step 1: apply t to first basis vector $u_1 = (1, 1)$ to advance the step-by-step mathematical solution toward the final result.

Step 2: Apply T to second basis vector $u_2 = (1, 0)$

$T(1, 0) = (\ 2(1) - 0, 1 + 3(0), 1\ ) = (2, 1, 1)$.

Relative to standard basis B', $[T(u_2)]_{B'} = [2, 1, 1]^T$ (Column 2).

> 💡 **Why this step is performed:** Execute step 2: apply t to second basis vector $u_2 = (1, 0)$ to advance the step-by-step mathematical solution toward the final result.

Step 3: Construct the matrix $[T]_{B, B'}$

Assemble the two column vectors into a 3×2 matrix:

$[T]_{B, B'} = [\ [1, 2], [4, 1], [1, 1]\ ]$.

> 💡 **Why this step is performed:** Execute step 3: construct the matrix [t]_{b, b'} to advance the step-by-step

mathematical solution toward the final result.

Step 4: Verify by applying to vector $u_1 = (1, 1)$

In basis B, coordinate of u_1 is $[1, 0]^T$.
$[T]_{B, B'} [1, 0]^T = [1, 4, 1]^T$. Matches T(1, 1)! Verified.

💡 **Why this step is performed:** Execute step 4: verify by applying to vector $u_1 = (1, 1)$ to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Matrix Representation $[T]_{B, B'} = [[1, 2], [4, 1], [1, 1]]$

H. Examination Presentation Strategy

- Evaluate $T(u_1)$ and $T(u_2)$.
- Write outputs as columns of the 3×2 matrix.
- Verify with test coordinate vector.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Writing matrix as 2×3 instead of 3×2.
✓ **How to Avoid:** The matrix size is always dim(Codomain) × dim(Domain) = 3 × 2!

⚠️ **Common Mistake:** Placing vectors as rows.
✓ **How to Avoid:** Coordinate vectors must form COLUMNS of the transformation matrix.

J. Similar Practice Question

Practice Problem 109: Find matrix of T(x, y) = (x + y, x - y) relative to basis B = {(1, 2), (2, 1)} in $\mathbb{R}^2$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 110: Composition of Linear Transformations and Matrix Multiplication INTERMEDIATE

A. Full Question:
Let T: $\mathbb{R}^2 \to \mathbb{R}^2$ be reflection across the y-axis, and S: $\mathbb{R}^2 \to \mathbb{R}^2$ be counterclockwise rotation by 45°. Find the standard matrices [T] and [S], and find the matrix of the composite transformation (S ∘ T).

B. Meaning of the Question

Find matrices [T] and [S], then multiply them in the correct order $[S \circ T] = [S] \cdot [T]$.

C. Basic Concept

The matrix of a composition of linear transformations is the product of their individual standard matrices: $[S \circ T] = [S] [T]$.

D. Formula and Explanation

Formula: $[S \circ T] = [S] \cdot [T]$.

Explanation & Conditions: Reflection across y-axis: (x, y) → (-x, y); Rotation by θ: [[cos θ, -sin θ], [sin θ, cos θ]].

E. Given Information

T = reflection across y-axis, S = rotation by θ = 45°.

F. Complete Step-by-Step Solution

Step 1: Find matrix [T] for reflection across y-axis

T(1, 0) = (-1, 0)
T(0, 1) = (0, 1).
[T] = [[-1, 0], [0, 1]].

💡 **Why this step is performed:** Execute step 1: find matrix [t] for reflection across y-axis to advance the step-by-step mathematical solution toward the final result.

Step 2: Find matrix [S] for rotation by 45°

cos 45° = 1/√2, sin 45° = 1/√2.
[S] = [[cos 45°, -sin 45°], [sin 45°, cos 45°]] = (1/√2) [[1, -1], [1, 1]].

💡 **Why this step is performed:** Execute step 2: find matrix [s] for rotation by 45° to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute matrix product [S ∘ T] = [S] · [T]

[S ∘ T] = (1/√2) [[1, -1], [1, 1]] [[-1, 0], [0, 1]]
= (1/√2) [[1(-1) + (-1)(0), 1(0) + (-1)(1)], [1(-1) + 1(0), 1(0) + 1(1)]]
= (1/√2) [[-1, -1], [-1, 1]].

💡 **Why this step is performed:** Execute step 3: compute matrix product [s ∘ t] = [s] · [t] to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: [T] = [[-1, 0], [0, 1]]; [S] = (1/√2)[[1, -1], [1, 1]]; [S ∘ T] = (1/√2)[[-1, -1], [-1, 1]]

H. Examination Presentation Strategy

- State [T] and [S] clearly.
- State composition formula [S ∘ T] = [S] [T].
- Show matrix multiplication step by step.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Multiplying in reverse order [T] [S].
✓ **How to Avoid:** S ∘ T means T is applied first, then S, so the matrix is [S] · [T]!

⚠️ **Common Mistake:** Sign error in reflection: (-x, y).
✓ **How to Avoid:** Reflecting across the y-axis flips the sign of x, leaving y unchanged.

J. Similar Practice Question

Practice Problem 110: Find the matrix of (T ∘ S) and verify that [S ∘ T] ≠ [T ∘ S] (non-commutativity).

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 111: Linear Transformation on Polynomial Space: Differentiation Operator

INTERMEDIATE

A. Full Question:

Let D: $P_3 \to P_2$ be the differentiation operator defined by D(p(x)) = p'(x). Find the matrix representation of D relative to the standard bases B = {1, x, x^2, x^3} and B' = {1, x, x^2}, and find its rank and nullity.

B. Meaning of the Question

Differentiate each basis vector of P_3, express outputs in basis B', construct matrix [D], and state rank and nullity.

C. Basic Concept

Calculus operations like differentiation and integration are linear transformations on polynomial and function spaces. Differential equations can thus be solved using matrix methods.

D. Formula and Explanation

Formula: $D(x^n) = n\,x^{n-1}$; $\dim(P_3) = \text{rank}(D) + \text{nullity}(D) = 4$.

Explanation & Conditions: B has 4 basis elements; B' has 3 basis elements.

E. Given Information

D: $P_3 \to P_2$, D(p) = p'(x).

F. Complete Step-by-Step Solution

Step 1: Differentiate each basis vector of B = {1, x, x^2, x^3}

- D(1) = 0 = 0(1) + 0(x) + 0(x^2)
- D(x) = 1 = 1(1) + 0(x) + 0(x^2)
- D(x^2) = 2x = 0(1) + 2(x) + 0(x^2)
- D(x^3) = $3x^2$ = 0(1) + 0(x) + 3(x^2).

💡 **Why this step is performed:** Execute step 1: differentiate each basis vector of b = {1, x, x^2, x^3} to advance the step-by-step mathematical solution toward the final result.

Step 2: Construct the 3×4 matrix [D]

Columns are the coordinate vectors:
[D] = [[0, 1, 0, 0], [0, 0, 2, 0], [0, 0, 0, 3]].

💡 **Why this step is performed:** Execute step 2: construct the 3×4 matrix [d] to advance the step-by-step mathematical solution toward the final result.

Step 3: Determine Rank(D)

The matrix [D] is in row-echelon form with 3 non-zero rows (pivots at cols 2, 3, 4).
Therefore, Rank(D) = 3.

💡 **Why this step is performed:** Execute step 3: determine rank(d) to advance the step-by-step mathematical solution toward the final result.

Step 4: Determine Nullity(D) and Kernel

ker(D) consists of polynomials whose derivative is 0: p'(x) = 0 ⇒ p(x) = c (constant polynomials).
Basis for ker(D) = {1}.
Nullity(D) = dim(ker D) = 1.

💡 **Why this step is performed:** Find all input vectors that map to the zero vector by setting T(v) = 0 and solving the resulting homogeneous system.

Step 5: Verify Rank-Nullity Theorem

$\dim(P_3) = 4$.
Rank + Nullity = 3 + 1 = 4 = $\dim(P_3)$. Verified!

💡 **Why this step is performed:** Verify the dimension theorem: the sum of the rank and the nullity equals the dimension of the domain space.

✓ G. Final Answer: Matrix [D] = [[0, 1, 0, 0], [0, 0, 2, 0], [0, 0, 0, 3]]; Rank = 3, Nullity = 1

H. Examination Presentation Strategy

- Differentiate each basis vector.
- Write outputs as columns of 3×4 matrix.
- Identify rank = 3 from non-zero rows.
- Identify nullity = 1 from constants.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Thinking D(1) = 1.
✓ **How to Avoid:** The derivative of constant 1 is ZERO: D(1) = 0, so column 1 is all zeros!

⚠️ **Common Mistake:** Matrix size 4×3 instead of 3×4.
✓ **How to Avoid:** dim(Codomain P_2) = 3 rows, dim(Domain P_3) = 4 columns.

J. Similar Practice Question

Practice Problem 111: Find matrix of integration operator T: $P_2 \to P_3$, T(p) = $\int_0^x p(t)\,dt$.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 112: Inverse of an Invertible Linear Transformation

INTERMEDIATE

A. Full Question:
Show that the linear operator T: $\mathbb{R}^2 \to \mathbb{R}^2$ defined by T(x, y) = (3x + 5y, 2x + 3y) is invertible, and find the formula for $T^{-1}(x, y)$.

B. Meaning of the Question

Show det([T]) ≠ 0 to prove invertibility, invert the matrix $[T]^{-1}$, and multiply by (x, y) to find $T^{-1}(x, y)$.

C. Basic Concept

A linear operator is invertible if and only if its standard matrix [T] is invertible: $[T^{-1}] = [T]^{-1}$.

D. Formula and Explanation

Formula: `[T] = [[3, 5], [2, 3]]; [T⁻¹] = [T]⁻¹ = (1/det) [[d, -b], [-c, a]].`

Explanation & Conditions: det = ad - bc = 9 - 10 = -1.

E. Given Information

T(x, y) = (3x + 5y, 2x + 3y).

F. Complete Step-by-Step Solution

Step 1: Write standard matrix [T]

[T] = [[3, 5], [2, 3]].

💡 **Why this step is performed:** Evaluate the transformation on each standard basis vector and place the resulting coordinate vectors as columns of [T].

Step 2: Check invertibility via determinant

det[T] = (3)(3) - (5)(2) = 9 - 10 = -1 ≠ 0.
Since det[T] ≠ 0, the matrix [T] is invertible, and therefore T is an INVERTIBLE LINEAR OPERATOR.

💡 **Why this step is performed:** Execute step 2: check invertibility via determinant to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute matrix inverse $[T]^{-1}$

$[T]^{-1}$ = (1 / -1) [[3, -5], [-2, 3]] = [[-3, 5], [2, -3]].

💡 **Why this step is performed:** Execute step 3: compute matrix inverse $[t]^{-1}$ to advance the step-by-step mathematical solution toward the final result.

Step 4: Form the formula for $T^{-1}(x, y)$

$T^{-1}(x, y)$ = [[-3, 5], [2, -3]] $[x, y]^T$ = [-3x + 5y, 2x - 3y]^T.
$T^{-1}(x, y)$ = (-3x + 5y, 2x - 3y).

💡 **Why this step is performed:** Execute step 4: form the formula for $t^{-1}(x, y)$ to advance the step-by-step mathematical solution toward the final result.

Step 5: Verify $T \circ T^{-1}$ = Identity

$T(T^{-1}(x, y)) = T(-3x + 5y, 2x - 3y)$
$= (3(-3x + 5y) + 5(2x - 3y), 2(-3x + 5y) + 3(2x - 3y))$
$= (-9x + 15y + 10x - 15y, -6x + 10y + 6x - 9y)$
$= (x, y)$. Exactly identity! Verified.

💡 **Why this step is performed:** Execute step 5: verify $t \circ t^{-1}$ = identity to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: det[T] = -1 ≠ 0 (Invertible); Inverse Formula: $T^{-1}(x, y) = (-3x + 5y, 2x - 3y)$

H. Examination Presentation Strategy

- Find standard matrix [T].
- Show det[T] = -1 ≠ 0 to prove invertibility.
- Compute $[T]^{-1}$.
- Write $T^{-1}(x, y)$ and verify.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Sign error in 2×2 matrix inverse.
✓ **How to Avoid:** Remember formula: swap diagonal entries, negate off-diagonal entries, divide by det.

⚠️ **Common Mistake:** Forgetting to divide by det = -1.
✓ **How to Avoid:** Dividing by -1 flips all signs in the adjugate matrix.

J. Similar Practice Question

Practice Problem 112: Find $T^{-1}(x, y)$ for $T(x, y) = (x + 2y, 2x + 5y)$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 113: Similarity of Matrix Representations under Change of Basis — INTERMEDIATE

A. Full Question:
Let T: $\mathbb{R}^2 \to \mathbb{R}^2$ be defined by T(x, y) = (x + 2y, 3x + 2y). Find the matrix $[T]_B$ relative to standard basis B, and the matrix $[T]_{B'}$ relative to basis B' = {(1, 1), (2, -1)}, and verify that $[T]_{B'} = P^{-1} [T]_B P$.

B. Meaning of the Question

Verify the similarity transformation relating matrix representations of the same linear operator under different bases.

C. Basic Concept

Two matrices represent the same linear operator in different coordinate systems if and only if they are similar: A' = P^{-1} A P, where P is the change of basis matrix.

D. Formula and Explanation

Formula: $[T]_{B'} = P^{-1} [T]_B P$, where $P = [v_1 \mid v_2]$.

Explanation & Conditions: v_1 = (1, 1), v_2 = (2, -1).

E. Given Information

T(x, y) = (x + 2y, 3x + 2y), B = standard, B' = {(1, 1), (2, -1)}.

F. Complete Step-by-Step Solution

Step 1: Standard matrix [T]_B

[T]_B = [[1, 2], [3, 2]].

💡 **Why this step is performed:** Evaluate the transformation on each standard basis vector and place the resulting coordinate vectors as columns of [T].

Step 2: Transition matrix P and P^{-1}

P = [[1, 2], [1, -1]].
det(P) = (1)(-1) - (2)(1) = -1 - 2 = -3.
P^{-1} = (1 / -3) [[-1, -2], [-1, 1]] = (1/3) [[1, 2], [1, -1]].

💡 **Why this step is performed:** Execute step 2: transition matrix p and p^{-1} to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute P^{-1} [T]_B P

[T]_B P = [[1, 2], [3, 2]] [[1, 2], [1, -1]] = [[1+2, 2-2], [3+2, 6-2]] = [[3, 0], [5, 4]].
P^{-1} ([T]_B P) = (1/3) [[1, 2], [1, -1]] [[3, 0], [5, 4]]
= (1/3) [[3+10, 0+8], [3-5, 0-4]] = (1/3) [[13, 8], [-2, -4]] = [[13/3, 8/3], [-2/3, -4/3]].

💡 **Why this step is performed:** Execute step 3: compute p^{-1} [t]_b p to advance the step-by-step mathematical solution toward the final result.

Step 4: Compute [T]_{B'} directly

• T(1, 1) = (3, 5). Write (3, 5) = c_1 (1, 1) + c_2 (2, -1):
$c_1 + 2c_2 = 3$, $c_1 - c_2 = 5 \Rightarrow 3c_2 = -2 \Rightarrow c_2 = -2/3$, $c_1 = 13/3$.
• T(2, -1) = (2 - 2, 6 - 2) = (0, 4) = d_1 (1, 1) + d_2 (2, -1):
$d_1 + 2d_2 = 0$, $d_1 - d_2 = 4 \Rightarrow 3d_2 = -4 \Rightarrow d_2 = -4/3$, $d_1 = 8/3$.
[T]_{B'} = [[13/3, 8/3], [-2/3, -4/3]]. Exactly matches! Verified.

💡 **Why this step is performed:** Execute step 4: compute [t]_{b'} directly to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: $[T]_{B'} = P^{-1} [T]_B P = [[13/3, 8/3], [-2/3, -4/3]]$ (Verified)

H. Examination Presentation Strategy

- Find [T]_B and P.
- Compute P^{-1}.
- Show matrix multiplications to get P^{-1} [T] P.
- Verify with direct basis transformation.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Computing P [T] P^{-1} instead of P^{-1} [T] P.
✓ **How to Avoid:** Remember: [T]_{B'} = P^{-1} [T]_B P!

⚠️ **Common Mistake:** Determinant error in det(P) = -3.
✓ **How to Avoid:** 1(-1) - 2(1) = -3.

J. Similar Practice Question

Practice Problem 113: Find [T]_{B'} for T(x, y) = (2x, 3y) with B' = {(1, 1), (1, -1)}.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 114: Rank and Nullity of a Matrix Transformation

INTERMEDIATE

A. Full Question:
For the matrix transformation T_A: $\mathbb{R}^4 \to \mathbb{R}^3$ defined by T_A(x) = A x, where A = [[1, -2, 1, 0], [2, -4, 3, -1], [-1, 2, 1, -2]], find rank(T_A) and nullity(T_A).

B. Meaning of the Question

Reduce matrix A to echelon form to find rank = number of pivot rows, and use Rank-Nullity theorem to find nullity = 4 - rank.

C. Basic Concept

For any matrix transformation T_A(x) = A x, rank(T_A) = rank(A) and nullity(T_A) = n - rank(A).

D. Formula and Explanation

Formula: `rank(T) = rank(A); nullity(T) = 4 - rank(A).`
Explanation & Conditions: Number of columns n = 4.

E. Given Information

A = [[1, -2, 1, 0], [2, -4, 3, -1], [-1, 2, 1, -2]].

F. Complete Step-by-Step Solution

Step 1: Reduce A to row-echelon form

A = [[1, -2, 1, 0], [2, -4, 3, -1], [-1, 2, 1, -2]].
$R_2 \to R_2 - 2R_1$: [0, 0, 1, -1].
$R_3 \to R_3 + R_1$: [0, 0, 2, -2].

> 💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 2: Eliminate row 3

$R_3 \to R_3 - 2R_2$: [0, 0, 0, 0].
Row-echelon form:
[[1, -2, 1, 0], [0, 0, 1, -1], [0, 0, 0, 0]].

> 💡 **Why this step is performed:** Apply elementary row operations to introduce zeros below the pivot element and reduce the matrix toward echelon form.

Step 3: Determine Rank

Number of non-zero rows = 2.
Therefore: rank(T_A) = rank(A) = 2.

> 💡 **Why this step is performed:** Execute step 3: determine rank to advance the step-by-step mathematical solution toward the final result.

Step 4: Determine Nullity using Rank-Nullity Theorem

Number of columns (dimension of domain $\mathbb{R}^4$) n = 4.
nullity(T_A) = n - rank(T_A) = 4 - 2 = 2.

> 💡 **Why this step is performed:** Execute step 4: determine nullity using rank-nullity theorem to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: rank(T_A) = 2; nullity(T_A) = 2

H. Examination Presentation Strategy

- Show row reduction of A.
- Count non-zero rows for rank = 2.
- Apply nullity = n - rank = 4 - 2 = 2.
- State final values clearly.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Computing nullity as 3 - rank.
✓ **How to Avoid:** n is the number of COLUMNS (4), not rows (3)! Domain is $\mathbb{R}^4$!

⚠️ **Common Mistake:** Counting column 2 as a pivot.
✓ **How to Avoid:** Column 2 has no pivot; it contains a free variable.

J. Similar Practice Question

Practice Problem 114: Find rank and nullity of T_A for A = [[1, 1, 2], [1, 0, 1], [2, 1, 3]].

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 115: Orthogonal Projection Operator in $\mathbb{R}^3$: Matrix and Properties

INTERMEDIATE

A. Full Question:

Find the standard matrix P of the linear operator T: $\mathbb{R}^3 \to \mathbb{R}^3$ that orthogonally projects every vector onto the xy-plane, and show that $P^2 = P$ and $P^T = P$ (orthogonal projection is idempotent and symmetric).

B. Meaning of the Question

Determine matrix P projecting (x, y, z) → (x, y, 0), and verify the algebraic projection properties $P^2 = P$ and $P^T = P$.

C. Basic Concept

An orthogonal projection onto a subspace satisfies two algebraic properties: idempotency $P^2 = P$ (projecting twice gives the same result) and self-adjointness $P^T = P$.

D. Formula and Explanation

Formula: `T(x, y, z) = (x, y, 0); P = diag(1, 1, 0).`

Explanation & Conditions: z-component is zeroed out.

E. Given Information

Projection onto xy-plane in $\mathbb{R}^3$.

F. Complete Step-by-Step Solution

Step 1: Evaluate T on standard basis vectors

T(1, 0, 0) = (1, 0, 0)
T(0, 1, 0) = (0, 1, 0)
T(0, 0, 1) = (0, 0, 0).

💡 **Why this step is performed:** Execute step 1: evaluate t on standard basis vectors to advance the step-by-step mathematical solution toward the final result.

Step 2: Construct standard matrix P

P = [[1, 0, 0], [0, 1, 0], [0, 0, 0]].

💡 **Why this step is performed:** Evaluate the transformation on each standard basis vector and place the resulting coordinate vectors as columns of [T].

Step 3: Verify idempotency $P^2 = P$

P^2 = [[1, 0, 0], [0, 1, 0], [0, 0, 0]] [[1, 0, 0], [0, 1, 0], [0, 0, 0]]
= [[1, 0, 0], [0, 1, 0], [0, 0, 0]] = P. Verified!

💡 **Why this step is performed:** Execute step 3: verify idempotency $p^2 = p$ to advance the step-by-step mathematical solution toward the final result.

Step 4: Verify symmetry $P^T = P$

$P^T = [\,[1, 0, 0], [0, 1, 0], [0, 0, 0]\,]^T = [\,[1, 0, 0], [0, 1, 0], [0, 0, 0]\,] = P$. Verified!

💡 **Why this step is performed:** Exploit the algebraic symmetry between x and y to write the corresponding partial derivative directly.

✓ G. Final Answer: Standard Matrix P = [[1, 0, 0], [0, 1, 0], [0, 0, 0]]; $P^2 = P$ (Idempotent) and $P^T = P$ (Symmetric) (Both verified)

H. Examination Presentation Strategy

- Show images of e_1, e_2, e_3.
- Write diagonal matrix P.
- Show $P^2 = P$ multiplication.
- Show $P^T = P$.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Thinking projection matrix has non-zero entry at (3, 3).
✓ **How to Avoid:** Projection onto xy-plane eliminates z, so entry (3, 3) must be 0!

⚠️ **Common Mistake:** Claiming P is invertible.
✓ **How to Avoid:** det(P) = 0, so projection matrices are NEVER invertible.

J. Similar Practice Question

Practice Problem 115: Find standard matrix for orthogonal projection onto the line $y = 2x$ in $\mathbb{R}^2$.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 116: Complete Proof of the Rank-Nullity Theorem

EXAM-LEVEL

A. Full Question:
State and prove the Rank-Nullity Theorem (Dimension Theorem) for a linear transformation T: V → W where V is a finite-dimensional vector space.

B. Meaning of the Question

Start with a basis of ker(T), extend it to a basis of V using the basis extension theorem, and prove that the images of the added vectors form a basis for Range(T).

C. Basic Concept

Foundational theorem of linear algebra. Shows that every dimension of V is accounted for: either mapped injectively into Range(T) or mapped to 0 in ker(T).

D. Formula and Explanation

Formula: `dim(V) = dim(Range T) + dim(ker T) = rank(T) + nullity(T).`

Explanation & Conditions: Let nullity(T) = k, dim(V) = n. Show rank(T) = n - k.

E. Given Information

Linear transformation T: V → W, dim(V) = n.

F. Complete Step-by-Step Solution

Step 1: Choose a basis for ker(T)

Let nullity(T) = dim(ker T) = k.
Let B_ker = {u_1, u_2, ..., u_k} be a basis for ker(T).
If k = 0, ker(T) = {0} and basis is empty.

> 💡 **Why this step is performed:** Execute step 1: choose a basis for ker(t) to advance the step-by-step mathematical solution toward the final result.

Step 2: Extend B_ker to a basis of V

By the Basis Extension Theorem, any linearly independent set in a finite-dimensional vector space can be extended to a basis of V.
Extend B_ker by adding (n - k) vectors {v_1, v_2, ..., v_{n-k}} so that:
B_V = {u_1, ..., u_k, v_1, ..., v_{n-k}} is a basis for V.
Here dim(V) = k + (n - k) = n.

> 💡 **Why this step is performed:** Execute step 2: extend b_ker to a basis of v to advance the step-by-step mathematical solution toward the final result.

Step 3: Propose a basis for Range(T)

We claim that S = {T(v_1), T(v_2), ..., T(v_{n-k})} is a BASIS for Range(T).
To prove this, we must show two things: (i) S spans Range(T), and (ii) S is linearly independent.

> 💡 **Why this step is performed:** Determine the subspace spanned by the images of the domain basis vectors to construct the range.

Step 4: Prove S spans Range(T)

Let w ∈ Range(T). Then w = T(v) for some v ∈ V.
Since B_V is a basis for V, v can be written as:
v = c_1 u_1 + ... + c_k u_k + d_1 v_1 + ... + d_{n-k} v_{n-k}.
Applying T and using linearity:
w = T(v) = Σ c_i T(u_i) + Σ d_j T(v_j).
Since each u_i ∈ ker(T), T(u_i) = 0.
Therefore: w = 0 + d_1 T(v_1) + ... + d_{n-k} T(v_{n-k}).
Thus, any vector in Range(T) is a linear combination of S. S spans Range(T).

💡 **Why this step is performed:** Determine the subspace spanned by the images of the domain basis vectors to construct the range.

Step 5: Prove S is linearly independent

Set a linear combination of S to zero:
$d_1 T(v_1) + d_2 T(v_2) + \dots + d_{n-k} T(v_{n-k}) = 0$.
By linearity of T:
$T(d_1 v_1 + \dots + d_{n-k} v_{n-k}) = 0$.
This means the vector $x = (d_1 v_1 + \dots + d_{n-k} v_{n-k})$ belongs to ker(T)!
Since $\{u_1, \dots, u_k\}$ is a basis for ker(T), x can be written as:
$d_1 v_1 + \dots + d_{n-k} v_{n-k} = c_1 u_1 + \dots + c_k u_k$
$\Rightarrow c_1 u_1 + \dots + c_k u_k - d_1 v_1 - \dots - d_{n-k} v_{n-k} = 0$.
Since B_V is a basis for V, all vectors are linearly independent. Therefore, all coefficients MUST be zero:
$c_1 = \dots = c_k = 0$ and $d_1 = \dots = d_{n-k} = 0$.
Since d_j = 0, the set S is LINEARLY INDEPENDENT.

💡 **Why this step is performed:** Execute step 5: prove s is linearly independent to advance the step-by-step mathematical solution toward the final result.

Step 6: Conclude the theorem

Since S is a linearly independent spanning set for Range(T), S is a basis for Range(T).
Therefore: rank(T) = dim(Range T) = |S| = n - k.
rank(T) = dim(V) - nullity(T)
$\Rightarrow$ dim(V) = rank(T) + nullity(T). Hence proved!

💡 **Why this step is performed:** Execute step 6: conclude the theorem to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: dim(V) = rank(T) + nullity(T) (Rank-Nullity Theorem Hence proved)

H. Examination Presentation Strategy

- State theorem statement.
- Define basis of ker(T) with k vectors.
- Extend to basis of V with n vectors.
- Prove $\{T(v_1), \dots, T(v_{n-k})\}$ spans Range(T).
- Prove independence using ker(T) property.
- Conclude rank = n - k.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Forgetting to prove independence of S.
✓ **How to Avoid:** A basis proof REQUIRES proving BOTH spanning AND linear independence!

⚠️ **Common Mistake:** Applying T to the whole equation without justifying $T(u_i) = 0$.
✓ **How to Avoid:** Explicitly state that $T(u_i) = 0$ because $u_i \in \ker(T)$.

J. Similar Practice Question

Practice Problem 116: Apply the Rank-Nullity theorem to show that a linear transformation T: $\mathbb{R}^5 \to \mathbb{R}^3$ cannot be one-to-one.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 117: Theorem: T is One-to-One if and only if Kernel(T) = {0}

EXAM-LEVEL

A. Full Question:

Prove that a linear transformation T: V → W is one-to-one (injective) if and only if ker(T) = {0}.

B. Meaning of the Question

Prove both directions: if T is injective, only 0 maps to 0; conversely, if only 0 maps to 0, T(u) = T(v) implies u = v.

C. Basic Concept

In general non-linear functions, f(a) = f(b) cannot be tested by checking f(x) = 0. But for linear transformations, T(u) - T(v) = T(u - v), so injectivity globally reduces to testing the kernel at the origin.

D. Formula and Explanation

Formula: `T(u) = T(v) ⇔ T(u - v) = 0 ⇔ (u - v) ∈ ker(T).`

Explanation & Conditions: If ker(T) = {0}, u - v = 0 ⇒ u = v.

E. Given Information

Linear transformation T: V → W.

F. Complete Step-by-Step Solution

Step 1: Prove Forward Direction (⇒)

Assume T is one-to-one.
We must show ker(T) = {0}.
Let v ∈ ker(T). Then T(v) = 0.
We already know T(0) = 0.
Therefore: T(v) = T(0).
Since T is assumed one-to-one: v = 0.
Therefore, the only vector in ker(T) is 0, so ker(T) = {0}.

💡 **Why this step is performed:** Execute step 1: prove forward direction (⇒) to advance the step-by-step mathematical solution toward the final result.

Step 2: Prove Reverse Direction (⇐)

Assume ker(T) = {0}.
We must show T is one-to-one.
Let u, v ∈ V such that T(u) = T(v).
Subtract T(v) from both sides in W:
T(u) - T(v) = 0.

By linearity of T: T(u - v) = T(u) - T(v) = 0.
This means (u - v) ∈ ker(T).
Since ker(T) = {0}, the only vector in ker(T) is zero:
u - v = 0 ⇒ u = v.
Since T(u) = T(v) implies u = v, T is ONE-TO-ONE.

💡 **Why this step is performed:** Execute step 2: prove reverse direction (⇐) to advance the step-by-step mathematical solution toward the final result.

Step 3: Conclude the proof

Both implications have been proven.
Therefore, T is one-to-one if and only if ker(T) = {0}. Hence proved.

💡 **Why this step is performed:** Execute step 3: conclude the proof to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: T is one-to-one if and only if ker(T) = {0} (Hence proved)

H. Examination Presentation Strategy

- State proof structure: part 1 (⇒), part 2 (⇐).
- Use linearity T(u) - T(v) = T(u - v).
- Conclude u = v.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Proving only one direction.
✓ **How to Avoid:** 'If and only if' demands proving BOTH forward and backward directions!

⚠️ **Common Mistake:** Writing T(u/v).
✓ **How to Avoid:** Division of vectors is completely undefined in linear algebra.

J. Similar Practice Question

Practice Problem 117: Use this theorem to prove that T(x, y) = (2x - y, x + y) is one-to-one.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 118: Equivalence of One-to-One and Onto for Equal Finite Dimensions EXAM-LEVEL

A. Full Question:
Prove that if V and W are vector spaces of the same finite dimension n, then a linear transformation T: V → W is one-to-one if and only if it is onto.

B. Meaning of the Question

Use the Rank-Nullity Theorem dim(V) = rank(T) + nullity(T) with dim(V) = dim(W) = n to establish the equivalence of injectivity and surjectivity.

C. Basic Concept

Pigeonhole principle for vector spaces. If a linear map between spaces of equal dimension loses no information (nullity = 0), it must cover the entire target space (rank = n).

D. Formula and Explanation

Formula: `n = rank(T) + nullity(T).`

Explanation & Conditions: T is 1-to-1 ⇔ nullity = 0 ⇔ rank = n ⇔ T is onto.

E. Given Information

dim(V) = dim(W) = n, T: V → W linear.

F. Complete Step-by-Step Solution

Step 1: State the Rank-Nullity formula

By the Rank-Nullity Theorem:
dim(V) = rank(T) + nullity(T)
n = rank(T) + nullity(T) --- (1).

> 💡 **Why this step is performed:** Execute step 1: state the rank-nullity formula to advance the step-by-step mathematical solution toward the final result.

Step 2: Prove Forward Implication (One-to-One ⇒ Onto)

Assume T is one-to-one.
By Q117, ker(T) = {0}, so nullity(T) = dim(ker T) = 0.
Substitute nullity(T) = 0 into (1):
n = rank(T) + 0 ⇒ rank(T) = n.
Since rank(T) = dim(Range T) = n = dim(W), the subspace Range(T) has the same dimension as the ambient space W.
A subspace of dimension n inside a space of dimension n must be the entire space:
Range(T) = W.
Therefore, T is ONTO.

> 💡 **Why this step is performed:** Show that the kernel contains only the zero vector, which proves the transformation is injective (one-to-one).

Step 3: Prove Reverse Implication (Onto ⇒ One-to-One)

Assume T is onto.
Then Range(T) = W, so rank(T) = dim(Range T) = dim(W) = n.
Substitute rank(T) = n into (1):
n = n + nullity(T) ⇒ nullity(T) = 0.
Since nullity(T) = 0, ker(T) = {0}.
By Q117, this implies T is ONE-TO-ONE.

> 💡 **Why this step is performed:** Show that the kernel contains only the zero vector, which proves the transformation is

injective (one-to-one).

Step 4: Conclude the theorem

Both implications hold. Thus, T is one-to-one if and only if it is onto. Hence proved.

💡 **Why this step is performed:** Execute step 4: conclude the theorem to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: For dim(V) = dim(W) = n: T is one-to-one ⇔ nullity = 0 ⇔ rank = n ⇔ T is onto (Hence proved)

H. Examination Presentation Strategy

- Quote Rank-Nullity Theorem: n = rank + nullity.
- Show 1-to-1 ⇒ nullity = 0 ⇒ rank = n ⇒ onto.
- Show onto ⇒ rank = n ⇒ nullity = 0 ⇒ 1-to-1.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Applying this theorem when dim(V) ≠ dim(W).
✓ **How to Avoid:** This theorem is FALSE if dimensions differ! (e.g., T: $\mathbb{R}^3 \to \mathbb{R}^2$ can be onto but cannot be 1-to-1).

⚠️ **Common Mistake:** Forgetting to state nullity = 0 ⇔ 1-to-1.
✓ **How to Avoid:** Quote the theorem from Q117.

J. Similar Practice Question

Practice Problem 118: Show that T: $\mathbb{R}^3 \to \mathbb{R}^2$, T(x, y, z) = (x, y) is onto but not one-to-one.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 119: Linear Transformation on Matrix Space $M_{22}(\mathbb{R})$: Trace and Transpose

EXAM-LEVEL

A. Full Question:

Let T: $M_{22}(\mathbb{R}) \to M_{22}(\mathbb{R})$ be defined by T(A) = A + A^T. Show that T is a linear transformation, find a basis for its Kernel, and find its Rank and Nullity.

B. Meaning of the Question

Verify linearity for matrix operation A + A^T, find which matrices satisfy A + A^T = 0 (skew-symmetric matrices), and find rank and nullity.

C. Basic Concept

$M_{22}(\mathbb{R})$ is a 4-dimensional vector space. $A + A^T = 0$ means $A^T = -A$ (skew-symmetric). Thus ker(T) is the subspace of skew-symmetric matrices.

D. Formula and Explanation

Formula: $T(A + B) = (A + B) + (A + B)^T$; $T(c\ A) = c\ T(A)$; $\dim(M_{22}) = 4$.

Explanation & Conditions: ker(T) = {A : $A^T = -A$}; Range(T) = {A : $A^T = A$}.

E. Given Information

T: $M_{22}(\mathbb{R}) \to M_{22}(\mathbb{R})$, $T(A) = A + A^T$.

F. Complete Step-by-Step Solution

Step 1: Verify Linearity of T

• $T(A + B) = (A + B) + (A + B)^T = A + B + A^T + B^T = (A + A^T) + (B + B^T) = T(A) + T(B)$.
• $T(c\ A) = (c\ A) + (c\ A)^T = c\ A + c\ A^T = c\ (A + A^T) = c\ T(A)$.
Therefore, T is a LINEAR TRANSFORMATION.

💡 **Why this step is performed:** Verify that the mapping preserves vector addition T(u + v) = T(u) + T(v) and scalar scaling T(cu) = c T(u).

Step 2: Find the Kernel of T

$T(A) = O \Rightarrow A + A^T = O \Rightarrow A^T = -A$.
Thus, ker(T) is the set of all 2×2 SKEW-SYMMETRIC matrices.
For A = [[a, b], [c, d]]: $A^T = -A$ implies a = 0, d = 0, and c = -b.
A = [[0, b], [-b, 0]] = b [[0, 1], [-1, 0]].
Basis for ker(T): B_ker = { [[0, 1], [-1, 0]] }.
Nullity(T) = dim(ker T) = 1.

💡 **Why this step is performed:** Find all input vectors that map to the zero vector by setting T(v) = 0 and solving the resulting homogeneous system.

Step 3: Find Rank(T) using Rank-Nullity Theorem

dim(Domain M_{22}) = 2 · 2 = 4.
rank(T) = dim(M_{22}) - nullity(T) = 4 - 1 = 3.

💡 **Why this step is performed:** Execute step 3: find rank(t) using rank-nullity theorem to advance the step-by-step mathematical solution toward the final result.

Step 4: Identify Range(T)

Since $(A + A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T$, every output of T is a SYMMETRIC matrix!
The space of 2×2 symmetric matrices has dimension 3 (from Q89).
Matches Rank = 3 exactly!

💡 **Why this step is performed:** Determine the subspace spanned by the images of the domain basis vectors to construct the range.

✓ G. Final Answer: T is linear; Kernel Basis: { [[0, 1], [-1, 0]] }, Nullity = 1; Rank = 3

H. Examination Presentation Strategy

- Show T(A+B) = T(A)+T(B) and T(cA) = cT(A).
- Show A + A^T = 0 gives A^T = -A (skew-symmetric).
- Extract basis matrix for ker(T).
- Apply Rank-Nullity: Rank = 4 - 1 = 3.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Thinking dim(M_{22}) = 2.
✓ **How to Avoid:** $M_{22}(\mathbb{R})$ has 4 entries, so its dimension is 2 × 2 = 4!

⚠️ **Common Mistake:** Missing the condition a = 0 and d = 0 for skew-symmetric.
✓ **How to Avoid:** Diagonal entries of a skew-symmetric matrix must be zero: a = -a ⇒ 2a = 0 ⇒ a = 0.

J. Similar Practice Question

Practice Problem 119: Find rank and nullity of T(A) = A - A^T on $M_{22}(\mathbb{R})$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 120: Rank and Nullity of Composite Linear Transformation (Sylvester's Inequality)

EXAM-LEVEL

A. Full Question:
Prove that for two linear transformations T: U → V and S: V → W: (i) rank(S ∘ T) ≤ min(rank S, rank T), and (ii) nullity(S ∘ T) ≤ nullity(T) + nullity(S).

B. Meaning of the Question

Prove the fundamental inequalities governing how dimension is lost when linear transformations are composed.

C. Basic Concept

A composition cannot recover lost information. Applying T first squashes the domain into Range(T) ⊆ V, and applying S can only further reduce dimension.

D. Formula and Explanation

Formula: `Range(S ∘ T) = S(Range T) ⊆ Range(S); ker(T) ⊆ ker(S ∘ T).`
Explanation & Conditions: Subspace inclusions directly imply dimension inequalities.

E. Given Information

T: U → V, S: V → W.

F. Complete Step-by-Step Solution

Step 1: Prove rank(S ∘ T) ≤ rank(S)

By definition, Range(S ∘ T) = { S(T(u)) : u ∈ U } = S(Range T).
Since Range(T) ⊆ V: S(Range T) ⊆ S(V) = Range(S).
Since Range(S ∘ T) is a subspace of Range(S):
dim(Range(S ∘ T)) ≤ dim(Range(S))
⇒ rank(S ∘ T) ≤ rank(S) --- (1).

💡 **Why this step is performed:** Execute step 1: prove rank(s ∘ t) ≤ rank(s) to advance the step-by-step mathematical solution toward the final result.

Step 2: Prove rank(S ∘ T) ≤ rank(T)

Consider the restriction of S to Range(T), denoted S|_{Range T} : Range(T) → W.
The image of this restricted map is Range(S ∘ T).
By the Rank-Nullity theorem on the restricted map:
dim(Range T) = dim(Image S|_{Range T}) + dim(ker S|_{Range T})
rank(T) = rank(S ∘ T) + nullity(S|_{Range T}).
Since nullity is non-negative (≥ 0):
rank(S ∘ T) ≤ rank(T) --- (2).

💡 **Why this step is performed:** Execute step 2: prove rank(s ∘ t) ≤ rank(t) to advance the step-by-step mathematical solution toward the final result.

Step 3: Combine equations (1) and (2)

From (1) and (2):
rank(S ∘ T) ≤ min(rank(S), rank(T)). Proved!

💡 **Why this step is performed:** Execute step 3: combine equations (1) and (2) to advance the step-by-step mathematical solution toward the final result.

Step 4: Prove nullity(S ∘ T) ≤ nullity(T) + nullity(S)

Let u ∈ ker(T) ⇒ T(u) = 0 ⇒ S(T(u)) = S(0) = 0 ⇒ u ∈ ker(S ∘ T).
Thus ker(T) ⊆ ker(S ∘ T).
The dimension of ker(S ∘ T) restricted to V gives:
nullity(S ∘ T) = nullity(T) + dim(Range T ∩ ker S) ≤ nullity(T) + nullity(S). Hence proved!

💡 **Why this step is performed:** Execute step 4: prove nullity(s ∘ t) ≤ nullity(t) + nullity(s) to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: rank(S ∘ T) ≤ min(rank S, rank T) and nullity(S ∘ T) ≤ nullity(T) + nullity(S) (Both proved)

H. Examination Presentation Strategy

- Show Range(ST) ⊆ Range(S).
- Use Rank-Nullity on restricted map S|_{Range T}.
- Combine to state min(rank S, rank T).
- State nullity inequality.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Claiming rank(ST) = rank(S) · rank(T).
✓ **How to Avoid:** Rank is dimension (addition), NOT multiplication!

⚠️ **Common Mistake:** Assuming ker(S ∘ T) = ker(T).
✓ **How to Avoid:** ker(S ∘ T) contains ker(T), but can be strictly larger if Range(T) hits ker(S)!

J. Similar Practice Question

Practice Problem 120: If A is 3×4 and B is 4×2, what is the maximum possible rank of AB?

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 121: Matrix Representation on Polynomial Space with Quadratic Basis EXAM-LEVEL

A. Full Question:
Find the matrix of the linear operator T: $P_2 \to P_2$ defined by T(p(x)) = p(x + 1) relative to the standard basis B = {1, x, x^2}.

B. Meaning of the Question

Evaluate T on 1, x, x^2 (substituting $x \to x + 1$), expand in powers of x, and construct the 3×3 upper triangular matrix.

C. Basic Concept

Shift operator in polynomial spaces. Represents Taylor shift algebraically as an upper triangular Pascal matrix.

D. Formula and Explanation

Formula: $T(1) = 1;\ T(x) = x + 1;\ T(x^2) = (x + 1)^2 = x^2 + 2x + 1.$

Explanation & Conditions: Expansion coefficients become column entries.

E. Given Information

T: $P_2 \to P_2$, T(p(x)) = p(x + 1), B = {1, x, x^2}.

F. Complete Step-by-Step Solution

Step 1: Apply T to basis vector 1

T(1) = 1 = 1(1) + 0(x) + 0(x^2).
Column 1: $[1, 0, 0]^T$.

💡 **Why this step is performed:** Execute step 1: apply t to basis vector 1 to advance the step-by-step mathematical solution toward the final result.

Step 2: Apply T to basis vector x

$T(x) = x + 1 = 1(1) + 1(x) + 0(x^2)$.
Column 2: $[1, 1, 0]^T$.

> 💡 **Why this step is performed:** Execute step 2: apply t to basis vector x to advance the step-by-step mathematical solution toward the final result.

Step 3: Apply T to basis vector x^2

$T(x^2) = (x + 1)^2 = x^2 + 2x + 1 = 1(1) + 2(x) + 1(x^2)$.
Column 3: $[1, 2, 1]^T$.

> 💡 **Why this step is performed:** Execute step 3: apply t to basis vector x^2 to advance the step-by-step mathematical solution toward the final result.

Step 4: Assemble the matrix [T]_B

$[T]_B = [\ [1, 1, 1], [0, 1, 2], [0, 0, 1]\ ]$.

> 💡 **Why this step is performed:** Execute step 4: assemble the matrix [t]_b to advance the step-by-step mathematical solution toward the final result.

Step 5: Properties of the matrix

Notice this is an upper triangular matrix with 1s on the diagonal (Pascal matrix).
$\det([T]) = 1 \cdot 1 \cdot 1 = 1 \neq 0$, so T is an isomorphism (invertible)!
Eigenvalues are all $\lambda = 1$.

> 💡 **Why this step is performed:** Execute step 5: properties of the matrix to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: Matrix [T]_B = [[1, 1, 1], [0, 1, 2], [0, 0, 1]]

H. Examination Presentation Strategy

- Evaluate T on 1, x, x^2.
- Show expansion in basis $\{1, x, x^2\}$.
- Form 3×3 matrix with coordinate columns.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Reversing the order of basis: writing powers as x^2, x, 1.
✓ **How to Avoid:** The basis is given as $\{1, x, x^2\}$, so row 1 is constant, row 2 is x, row 3 is x^2!

⚠️ **Common Mistake:** Expanding $(x + 1)^2$ incorrectly.
✓ **How to Avoid:** $(x + 1)^2 = x^2 + 2x + 1$.

J. Similar Practice Question

Practice Problem 121: Find the matrix of $T(p(x)) = p(2x + 1)$ relative to $\{1, x, x^2\}$.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 122: Finding Linear Transformation from Kernel and Image Specifications

EXAM-LEVEL

A. Full Question:
Find a linear transformation T: $\mathbb{R}^3 \to \mathbb{R}^3$ whose Kernel is spanned by (1, 1, 1) and whose Image is spanned by (1, 0, -1) and (1, 2, 1).

B. Meaning of the Question

Extend the kernel basis to a basis of $\mathbb{R}^3$, assign output values in the image, and solve for the standard matrix.

C. Basic Concept

We can construct a linear transformation by specifying its action on a basis: assign $T(v_1) = 0$ for v_1 spanning the kernel, and assign $T(v_2)$ and $T(v_3)$ to span the image.

D. Formula and Explanation

Formula: $B = \{v_1, v_2, v_3\}$ basis of $\mathbb{R}^3$; $T(v_1) = 0$, $T(v_2) = w_1$, $T(v_3) = w_2$.

Explanation & Conditions: $v_1 = (1, 1, 1)$, $w_1 = (1, 0, -1)$, $w_2 = (1, 2, 1)$.

E. Given Information

ker(T) = span{(1, 1, 1)}; Im(T) = span{(1, 0, -1), (1, 2, 1)}.

F. Complete Step-by-Step Solution

Step 1: Choose a basis for $\mathbb{R}^3$ containing (1, 1, 1)

Let $v_1 = (1, 1, 1)$ (basis for kernel).
Choose $v_2 = (0, 1, 0)$ and $v_3 = (0, 0, 1)$ to extend to a basis of $\mathbb{R}^3$.
Matrix $[v_1 \mid v_2 \mid v_3]$ has det = $1 \neq 0$, so $\{v_1, v_2, v_3\}$ is a valid basis.

💡 **Why this step is performed:** Execute step 1: choose a basis for $\mathbb{R}^3$ containing (1, 1, 1) to advance the step-by-step mathematical solution toward the final result.

Step 2: Assign images to basis vectors

- $T(v_1) = T(1, 1, 1) = (0, 0, 0)$ (must be zero for kernel)
- $T(v_2) = T(0, 1, 0) = (1, 0, -1)$
- $T(v_3) = T(0, 0, 1) = (1, 2, 1)$.

💡 **Why this step is performed:** Determine the subspace spanned by the images of the domain basis vectors to construct the range.

Step 3: Find T(1, 0, 0) = $T(e_1)$

Notice that $(1, 0, 0) = v_1 - v_2 - v_3 = (1, 1, 1) - (0, 1, 0) - (0, 0, 1)$.
Applying T:
$T(1, 0, 0) = T(v_1) - T(v_2) - T(v_3)$

= (0, 0, 0) - (1, 0, -1) - (1, 2, 1)
= (-1 - 1, 0 - 2, 1 - 1) = (-2, -2, 0).

💡 **Why this step is performed:** Execute step 3: find t(1, 0, 0) = t(e_1) to advance the step-by-step mathematical solution toward the final result.

Step 4: Write the general formula for T(x, y, z)

(x, y, z) = x (1, 0, 0) + y (0, 1, 0) + z (0, 0, 1).
T(x, y, z) = x (-2, -2, 0) + y (1, 0, -1) + z (1, 2, 1)
= (-2x + y + z, -2x + 2z, -y + z).

💡 **Why this step is performed:** Execute step 4: write the general formula for t(x, y, z) to advance the step-by-step mathematical solution toward the final result.

Step 5: Verify Kernel condition

T(1, 1, 1) = (-2(1) + 1 + 1, -2(1) + 2(1), -1 + 1) = (0, 0, 0). Verified!

💡 **Why this step is performed:** Find all input vectors that map to the zero vector by setting T(v) = 0 and solving the resulting homogeneous system.

✓ G. Final Answer: T(x, y, z) = (-2x + y + z, -2x + 2z, -y + z) (or standard matrix [[-2, 1, 1], [-2, 0, 2], [0, -1, 1]])

H. Examination Presentation Strategy

- Extend kernel vector to a full basis.
- Assign image vectors to the non-kernel basis vectors.
- Solve for T(e_1), T(e_2), T(e_3).
- Write final formula and verify.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Choosing dependent vectors to extend the basis.
✓ **How to Avoid:** Verify det [v_1 v_2 v_3] ≠ 0 when extending the basis!

⚠️ **Common Mistake:** Assigning non-zero image to (1, 1, 1).
✓ **How to Avoid:** Any vector in the kernel MUST map to (0, 0, 0).

J. Similar Practice Question

Practice Problem 122: Find a linear transformation T: $\mathbb{R}^3 \to \mathbb{R}^3$ whose kernel is spanned by (1, 0, 0) and (0, 1, 0).

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 123: Idempotent Transformation (Projection) Properties

EXAM-LEVEL

A. Full Question:

Prove that if T: V → V is a linear operator satisfying $T^2 = T$ (projection), then: (i) V = Range(T) ⊕ Kernel(T) (direct sum), and (ii) The only possible eigenvalues of T are 0 and 1.

B. Meaning of the Question

Prove that every vector can be uniquely decomposed as v = T(v) + (v - T(v)) and that the operator polynomial $\lambda^2 - \lambda = 0$ restricts eigenvalues to {0, 1}.

C. Basic Concept

Projections decompose the entire vector space into the subspace being projected onto Range(T) and the direction of projection along Kernel(T).

D. Formula and Explanation

Formula: `v = T(v) + (v - T(v)); Range ∩ Kernel = {0}.`

Explanation & Conditions: T(T(v)) = T(v) ⇒ T(v - T(v)) = T(v) - T^2(v) = 0.

E. Given Information

T: V → V linear, $T^2 = T$.

F. Complete Step-by-Step Solution

Step 1: Prove v - T(v) belongs to Kernel(T)

For any vector v ∈ V, consider u = v - T(v).
Applying T:
T(u) = T(v - T(v)) = T(v) - T^2(v).
Since $T^2 = T$: T(v) - T(v) = 0.
Therefore, (v - T(v)) ∈ ker(T) for all v ∈ V.

💡 **Why this step is performed:** Find all input vectors that map to the zero vector by setting T(v) = 0 and solving the resulting homogeneous system.

Step 2: Show V = Range(T) + Kernel(T)

Any vector v can be written as:
v = T(v) + (v - T(v)).
Here T(v) ∈ Range(T) and (v - T(v)) ∈ ker(T).
Thus, every vector in V is the sum of a vector in Range(T) and a vector in ker(T): V = Range(T) + ker(T).

💡 **Why this step is performed:** Find all input vectors that map to the zero vector by setting T(v) = 0 and solving the resulting homogeneous system.

Step 3: Show Range(T) ∩ Kernel(T) = {0}

Let w ∈ Range(T) ∩ ker(T).
• Since w ∈ Range(T): w = T(x) for some x ∈ V.
• Since w ∈ ker(T): T(w) = 0.
Substitute w = T(x) into T(w) = 0:

$T(T(x)) = 0 \Rightarrow T^2(x) = 0$.
Since $T^2 = T$: $T(x) = 0$.
Therefore: $w = T(x) = 0$.
Thus, Range(T) ∩ ker(T) = {0}.

> 💡 **Why this step is performed:** Find all input vectors that map to the zero vector by setting T(v) = 0 and solving the resulting homogeneous system.

Step 4: Prove Eigenvalues of T are strictly 0 or 1

Let λ be an eigenvalue of T with non-zero eigenvector v:
$T(v) = \lambda v$.
Applying T again: $T^2(v) = T(\lambda v) = \lambda T(v) = \lambda (\lambda v) = \lambda^2 v$.
Since $T^2 = T$:
$T^2(v) = T(v) \Rightarrow \lambda^2 v = \lambda v \Rightarrow (\lambda^2 - \lambda) v = 0$.
Since $v \neq 0$ (eigenvector):
$\lambda^2 - \lambda = 0 \Rightarrow \lambda(\lambda - 1) = 0 \Rightarrow \lambda = 0$ or $\lambda = 1$. Proved!

> 💡 **Why this step is performed:** Execute step 4: prove eigenvalues of t are strictly 0 or 1 to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: V = Range(T) ⊕ Kernel(T) (Direct sum); Eigenvalues of T are strictly λ = 0 or λ = 1 (Hence proved)

H. Examination Presentation Strategy

- Show $v = T(v) + (v - T(v))$.
- Show $T(v - T(v)) = 0$.
- Show Range ∩ Kernel = {0} to prove direct sum ⊕.
- Show $(\lambda^2 - \lambda)v = 0$ to prove eigenvalues 0 and 1.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Proving V = Range + Kernel without showing intersection is {0}.
✓ **How to Avoid:** A DIRECT SUM ⊕ strictly requires showing Range ∩ Kernel = {0}!

⚠️ **Common Mistake:** Assuming T is invertible.
✓ **How to Avoid:** $T^2 = T$ with $T \neq I$ has det(T) = 0, so projections are not invertible.

J. Similar Practice Question

Practice Problem 123: Prove that if $T^2 = I$ (involution/reflection), then its only possible eigenvalues are 1 and -1.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 124: Non-Commutativity of Geometric Transformations (Rotation vs Reflection)

EXAM-LEVEL

A. Full Question:
In $\mathbb{R}^2$, let R be counterclockwise rotation by 90° and M be reflection across the line y = x. Find the standard matrices [R] and [M], compute [R ∘ M] and [M ∘ R], and show that R and M do not commute.

B. Meaning of the Question

Show that the order of performing 2D geometric operations matters by proving [R][M] ≠ [M][R].

C. Basic Concept

Matrix multiplication is generally non-commutative (AB ≠ BA), reflecting the physical fact that spatial transformations yield different outcomes when their sequence is reversed.

D. Formula and Explanation

Formula: `[R] = [[0, -1], [1, 0]]; [M] = [[0, 1], [1, 0]].`

Explanation & Conditions: Line y = x swaps x and y coordinates.

E. Given Information

R = 90° rotation, M = reflection across y = x.

F. Complete Step-by-Step Solution

Step 1: Determine Matrix [R] for 90° rotation

R(1, 0) = (0, 1), R(0, 1) = (-1, 0).
[R] = [[0, -1], [1, 0]].

💡 **Why this step is performed:** Execute step 1: determine matrix [r] for 90° rotation to advance the step-by-step mathematical solution toward the final result.

Step 2: Determine Matrix [M] for reflection across y = x

M(1, 0) = (0, 1), M(0, 1) = (1, 0).
[M] = [[0, 1], [1, 0]].

💡 **Why this step is performed:** Execute step 2: determine matrix [m] for reflection across y = x to advance the step-by-step mathematical solution toward the final result.

Step 3: Compute [R ∘ M] = [R] · [M]

[R] [M] = [[0, -1], [1, 0]] [[0, 1], [1, 0]]
= [[0(0) + (-1)(1), 0(1) + (-1)(0)], [1(0) + 0(1), 1(1) + 0(0)]]
= [[-1, 0], [0, 1]].
(This is reflection across the y-axis!).

💡 **Why this step is performed:** Execute step 3: compute [r ∘ m] = [r] · [m] to advance the step-by-step mathematical solution toward the final result.

Step 4: Compute [M ∘ R] = [M] · [R]

[M] [R] = [[0, 1], [1, 0]] [[0, -1], [1, 0]]
= [[0(0) + 1(1), 0(-1) + 1(0)], [1(0) + 0(1), 1(-1) + 0(0)]]
= [[1, 0], [0, -1]].
(This is reflection across the x-axis!).

💡 **Why this step is performed:** Execute step 4: compute [m ∘ r] = [m] · [r] to advance the step-by-step mathematical solution toward the final result.

Step 5: Compare results and conclude non-commutativity

Since [R ∘ M] = [[-1, 0], [0, 1]] ≠ [M ∘ R] = [[1, 0], [0, -1]]:
[R ∘ M] ≠ [M ∘ R].
Therefore, rotation by 90° and reflection across y = x DO NOT COMMUTE. Hence proved.

💡 **Why this step is performed:** Execute step 5: compare results and conclude non-commutativity to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: [R ∘ M] = [[-1, 0], [0, 1]] ≠ [M ∘ R] = [[1, 0], [0, -1]] (Transformations do not commute)

H. Examination Presentation Strategy

- State standard matrices [R] and [M].
- Compute [R][M] step by step.
- Compute [M][R] step by step.
- Show that the matrices differ and conclude non-commutativity.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Wrong matrix for reflection across y = x.
✓ **How to Avoid:** Swapping x and y gives matrix [[0, 1], [1, 0]], NOT [[1, 0], [0, -1]]!

⚠️ **Common Mistake:** Arithmetic error in matrix multiplication.
✓ **How to Avoid:** Check: (-1)(1) = -1.

J. Similar Practice Question

Practice Problem 124: Check whether rotation by 90° and rotation by 180° commute.
(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

QUESTION 125: Application of Rank-Nullity Theorem: Systems of Linear Equations EXAM-LEVEL

A. Full Question:
Use the Rank-Nullity Theorem to prove that a system of m linear equations in n variables AX = B can have a

unique solution only if m ≥ n (number of equations is at least the number of unknowns).

B. Meaning of the Question

Explain why underdetermined systems (m < n) with fewer equations than unknowns can NEVER have a unique solution.

C. Basic Concept

For underdetermined systems (m < n), rank(A) ≤ m < n. By Rank-Nullity, nullity(A) = n - rank(A) ≥ n - m > 0, which guarantees that non-trivial solutions to AX = 0 always exist, meaning solutions are either non-existent or infinite.

D. Formula and Explanation

Formula: `nullity(A) = n - rank(A) ≥ n - m.`

Explanation & Conditions: If m < n, nullity(A) ≥ 1, so there is at least 1 free parameter.

E. Given Information

System AX = B, A is m×n matrix, m < n.

F. Complete Step-by-Step Solution

Step 1: Set up the matrix transformation T_A

Let T_A: $\mathbb{R}^n \to \mathbb{R}^m$ be defined by T_A(X) = A X.
Domain dimension = n (number of variables).
Codomain dimension = m (number of equations).

💡 **Why this step is performed:** Execute step 1: set up the matrix transformation t_a to advance the step-by-step mathematical solution toward the final result.

Step 2: Apply the Rank-Nullity Theorem

By the Rank-Nullity Theorem:
dim(Domain) = rank(T_A) + nullity(T_A)
n = rank(A) + nullity(A)
⇒ nullity(A) = n - rank(A) --- (1).

💡 **Why this step is performed:** Execute step 2: apply the rank-nullity theorem to advance the step-by-step mathematical solution toward the final result.

Step 3: Bound the rank of matrix A

The rank of an m×n matrix cannot exceed the number of rows: rank(A) ≤ m.
Since m < n (fewer equations than variables):
rank(A) ≤ m < n.

💡 **Why this step is performed:** Execute step 3: bound the rank of matrix a to advance the step-by-step mathematical solution toward the final result.

Step 4: Analyze nullity(A)

Substitute rank(A) ≤ m into (1):
nullity(A) = n - rank(A) ≥ n - m.

Since m < n: n - m > 0 ⇒ nullity(A) ≥ 1.
Therefore, the null space Null(A) = ker(T_A) contains NON-ZERO vectors.

💡 **Why this step is performed:** Execute step 4: analyze nullity(a) to advance the step-by-step mathematical solution toward the final result.

Step 5: Conclude solution behavior for AX = B

If AX = B is inconsistent, it has NO solution.
If AX = B is consistent with a particular solution X_p, then any vector of the form X = X_p + X_h (where X_h ∈ Null(A)) is also a solution.
Since nullity(A) ≥ 1, there are infinitely many vectors X_h.
Therefore, AX = B has INFINITELY MANY solutions.
In no case can it have a UNIQUE solution when m < n.
Hence, a unique solution can exist ONLY IF m ≥ n. Hence proved!

💡 **Why this step is performed:** Execute step 5: conclude solution behavior for ax = b to advance the step-by-step mathematical solution toward the final result.

✓ G. Final Answer: When m < n, nullity(A) ≥ n - m > 0, so AX = B can only have infinite solutions or no solution; a unique solution is possible ONLY IF m ≥ n (Hence proved)

H. Examination Presentation Strategy

- Define T_A: $\mathbb{R}^n \to \mathbb{R}^m$.
- Apply Rank-Nullity Theorem: n = rank + nullity.
- Show rank ≤ m implies nullity ≥ n - m > 0.
- Explain why nullity > 0 rules out unique solutions.

I. Common Mistakes & How to Avoid Them

⚠️ **Common Mistake:** Confusing rows m with columns n.
✓ **How to Avoid:** m = number of equations (rows), n = number of variables (columns).

⚠️ **Common Mistake:** Claiming m < n always means infinite solutions.
✓ **How to Avoid:** Remember: the system might be INCONSISTENT (no solution)! State 'either no solution or infinite solutions'.

J. Similar Practice Question

Practice Problem 125: Explain using Rank-Nullity theorem why a homogeneous system of 3 equations in 5 variables always has non-trivial solutions.

(Answer with working provided in the Consolidated Practice Answer Key at the end of this book).

🔑 Appendix A: Consolidated Practice Question Answer Key (Problems 1 to 125)

Complete working, intermediate values, and final verified answers for every similar practice question presented at the end of each solved problem.

Practice Q1 (Module 1): $\partial u/\partial x = 6x^2 - 10xy + 4y^2$; $\partial u/\partial y = -5x^2 + 8xy - 3y^2$.

Practice Q2 (Module 1): $\partial^2 u/\partial x \partial y = \partial^2 u/\partial y \partial x = 8x^3 y - 24x^2 y^3$ (Verified by Clairaut's theorem).

Practice Q3 (Module 1): Degree n = 3 - 1 = 2; Euler verified: x u_x + y u_y = 2u.

Practice Q4 (Module 1): $J = \partial(x, y)/\partial(u, v) = \begin{vmatrix} 1 & -v \\ -u & v \end{vmatrix} = u(1 - v) - (-uv) = u - uv + uv = u$.

Practice Q5 (Module 1): $f_x = 4x - 4 = 0 \Rightarrow x = 1$; $f_y = 2y + 4 = 0 \Rightarrow y = -2$. r = 4, s = 0, t = 2. $D = 8 > 0$, $r > 0 \Rightarrow$ Local Minimum at (1, -2) with $f_{min} = 2(1) + 4 - 4 - 8 + 10 = 4$.

Practice Q6 (Module 1): $z = \cos^2 t + \sin^2 t = 1 \Rightarrow dz/dt = 0$. Chain rule: $(2x)(-\sin t) + (2y)(\cos t) = -2 \cos t \sin t + 2 \sin t \cos t = 0$.

Practice Q7 (Module 1): $z = \cos u = (x+y)/(\sqrt{x}+\sqrt{y})$ has degree n = 1/2. $-\sin u(x u_x + y u_y) = (1/2) \cos u \Rightarrow x u_x + y u_y = -(1/2) \cot u$.

Practice Q8 (Module 1): $J = \partial(u, v)/\partial(x, y) = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2$. Inverse: $x = (u+v)/2$, $y = (u-v)/2 \Rightarrow J' = \begin{vmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{vmatrix} = -1/2$. $J \cdot J' = (-2)(-1/2) = 1$.

Practice Q9 (Module 1): f(0,0)=0; fx=0, fy=1; fxx=0, fxy=1, fyy=0. $f(x, y) = y + xy + \ldots$ up to 2nd degree.

Practice Q10 (Module 1): f(1, 2) = 1 + 2 + 4 = 7; fx(1, 2) = 4, fy(1, 2) = 5; fxx = 2, fxy = 1, fyy = 2. $f(x, y) = 7 + 4(x-1) + 5(y-2) + (x-1)^2 + (x-1)(y-2) + (y-2)^2$.

Practice Q11 (Module 1): $J = \partial(u, v)/\partial(x, y) = 0$ identically; Relationship: v = (u - y)/u or u v = x.

Practice Q12 (Module 1): Stationary points: (0, 0) and (6, 0). At (6, 0): local maximum $f_{max} = 108$.

Practice Q13 (Module 1): $\partial(x, y, z)/\partial(\rho, \varphi, z) = \rho$ (Cylindrical coordinates Jacobian).

Practice Q14 (Module 1): u is homogeneous of degree 0: x u_x + y u_y = 0.

Practice Q15 (Module 1): Use chain rule $\partial z/\partial x = z_u$, $\partial z/\partial y = -z_u + z_v$, $\partial z/\partial z = -z_v$; summing gives 0.

Practice Q16 (Module 1): $z = \tan u = (x^2+y^2)/(x+y)$ has degree n = 1. $\sec^2 u (x u_x + y u_y) = \tan u \Rightarrow x u_x + y u_y = (1/2) \sin 2u$.

Practice Q17 (Module 1): Applying second order Euler formula: $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = (1/4) \tan u (\tan^2 u - 1)$.

Practice Q18 (Module 1): $f''(r) + (2/r) f'(r) = 0 \Rightarrow r^2 f'' + 2r f' = (r^2 f')' = 0 \Rightarrow f(r) = -A/r + B$ (Coulomb potential).

Practice Q19 (Module 1): e^y log(1 + x) = $x + xy - (1/2)x^2 + (1/2)xy^2 - (1/2)x^2 y + (1/3)x^3 + \ldots$

Practice Q20 (Module 1): Dimensions: Length = 6 cm, Breadth = 6 cm, Height = 3 cm; Minimum Surface Area = 108 sq. cm.

Practice Q21 (Module 1): At (π/3, π/3): f_max = 3/2.

Practice Q22 (Module 1): ∂u/∂x = (y - z) / [(u - v)(u - w)].

Practice Q23 (Module 1): Shortest distance d = 2√3 at point (2, 2, 2).

Practice Q24 (Module 1): Differentiating: u_xx = f'' + g''; u_yy = -f'' - g''. Sum u_xx + u_yy = 0 (Laplace equation verified).

Practice Q25 (Module 1): e^u has degree n = 3 - 2 = 1. Therefore: x u_x + y u_y + z u_z = 1.

Practice Q26 (Module 2): ∇φ(2, -2, 1) = -4 î - 4 ĵ + 32 k̂.

Practice Q27 (Module 2): div F⃗ = $3x^2 + 3y^2 + 3z^2$; at (1, 1, 1), div F⃗ = 3(1 + 1 + 1) = 9.

Practice Q28 (Module 2): curl F⃗ = -4 ĵ + 4 k̂ at (1, 2, 3).

Practice Q29 (Module 2): ∇φ = 2(x î + y ĵ + z k̂); at (1, 2, 2): n̂ = (î + 2ĵ + 2k̂) / 3.

Practice Q30 (Module 2): ∇φ(2, -1, 1) = -4 î + 5 ĵ - 2 k̂; Directional derivative = (-8 - 5 + 4)/3 = -3.

Practice Q31 (Module 2): ∇φ(1, -2, 1) = -12 î + 4 ĵ - 4 k̂; Max value = √[144 + 16 + 16] = 4√11.

Practice Q32 (Module 2): div F⃗ = 2 + k - 4 = k - 2 = 0 ⇒ k = 2.

Practice Q33 (Module 2): a = 6, c = 1, b = 0.

Practice Q34 (Module 2): φ(x, y, z) = $x^2y + x z^3$ + C.

Practice Q35 (Module 2): θ = $\cos^{-1}(19 / (3\sqrt{62}))$ ≈ 36.4°.

Practice Q36 (Module 2): curl(grad φ) = ∇ × ∇φ = î(φ_zy - φ_yz) - ĵ(φ_zx - φ_xz) + k̂(φ_yx - φ_xy) = 0⃗.

Practice Q37 (Module 2): div(r⃗ / r^3) = ∇⃗ · (r^{-3} r⃗) = (-3 + 3) r^{-3} = 0 (for r ≠ 0).

Practice Q38 (Module 2): ∇⃗(1/r) = ∇⃗(r^{-1}) = -r^{-3} r⃗ = -r⃗ / r^3.

Practice Q39 (Module 2): $\nabla^2(1/r) = \nabla^2(r^{-1})$ = (-1)(-1 + 1)r^{-3} = 0 (Harmonic for r ≠ 0).

Practice Q40 (Module 2): h_r = 1, h_θ = r, h_φ = r sin θ.

Practice Q41 (Module 2): φ(x, y, z) = $3x^2y + x z^3$ - y z + C.

Practice Q42 (Module 2): ∇⃗ × (r^n r⃗) = n r^{n-2} (r⃗ × r⃗) = 0⃗ for all n.

Practice Q43 (Module 2): ∇⃗ × (f(r) r⃗) = f'(r)/r (r⃗ × r⃗) = 0⃗.

Practice Q44 (Module 2): Directional derivative = 104 / √26.

Practice Q45 (Module 2): □ · (F□ × G□) = G□ · (□ × F□) - F□ · (□ × G□) = G · 0 - F · 0 = 0.

Practice Q46 (Module 2): grad φ = (∂φ/∂r) ê_r + (1/r)(∂φ/∂θ) ê_θ + (1/(r sin θ))(∂φ/∂φ) ê_φ.

Practice Q47 (Module 2): $\psi = 3x^2y - y^3 + C$.

Practice Q48 (Module 2): Tangent Plane: x + 2y + 3z = 14; Normal Line: (x-1)/1 = (y-2)/2 = (z-3)/3.

Practice Q49 (Module 2): Follows by setting A□ = B□ = v□ in Q49.

Practice Q50 (Module 2): $\nabla^2\psi = (1/r^2)\,\partial/\partial r(r^2\,\partial\psi/\partial r) + (1/(r^2 \sin\theta))\,\partial/\partial\theta(\sin\theta\,\partial\psi/\partial\theta) + (1/(r^2 \sin^2\theta))\,\partial^2\psi/\partial\varphi^2$.

Practice Q51 (Module 3): Rank ρ(A) = 1 (Rows 2 and 3 are multiples of Row 1).

Practice Q52 (Module 3): $\lambda^2 - 7\lambda + 6 = 0 \Rightarrow \lambda = 1, 6$.

Practice Q53 (Module 3): Number of arbitrary parameters = n - r = 4 - 2 = 2 free parameters.

Practice Q54 (Module 3): Rank = 1 < 3; Non-trivial solutions: $X = s\,[-1, 1, 0]^T + t\,[-1, 0, 1]^T$.

Practice Q55 (Module 3): Third eigenvalue λ_3 = Trace - (1 + 2) = 6 - 3 = 3.

Practice Q56 (Module 3): Rank ρ(A) = 2.

Practice Q57 (Module 3): Unique solution: x = 1, y = 2, z = 3.

Practice Q58 (Module 3): Solution by Gauss-Jordan: x = 1, y = 1, z = 1.

Practice Q59 (Module 3): λ = 0, 5; Eigenvectors: $v_1 = [-1, 2]^T$ for λ = 0, $v_2 = [2, 1]^T$ for λ = 5.

Practice Q60 (Module 3): Characteristic equation: $\lambda^2 - 7\lambda + 7 = 0$; $A^{-1} = (1/7)(7I - A) = [[5/7, -1/7], [-3/7, 2/7]]$.

Practice Q61 (Module 3): Consistent; Infinitely many solutions: x = 17/3 + (7/3)k, y = -5/3 - (4/3)k, z = k.

Practice Q62 (Module 3): $\rho(A) = 2 \neq \rho([A|B]) = 3 \Rightarrow$ Inconsistent (no solution).

Practice Q63 (Module 3): k = 4 (makes row 2 dependent on rows 1 and 3).

Practice Q64 (Module 3): $|\lambda| = 1$ because $A^T A = I \Rightarrow \|A v\| = \|v\| \Rightarrow |\lambda|\,\|v\| = \|v\| \Rightarrow |\lambda| = 1$.

Practice Q65 (Module 3): P = [[1, 1], [1, -1]]; D = [[4, 0], [0, 2]].

Practice Q66 (Module 3): (i) No solution: a = 3, b ≠ 8; (ii) Unique: a ≠ 3, b ∈ ℝ; (iii) Infinite: a = 3, b = 8.

Practice Q67 (Module 3): $\lambda = 6, 3, -1$; Eigenvectors: $v_1 = [1, 2, 1]^T$, $v_2 = [1, -1, 1]^T$, $v_3 = [1, 0, -1]^T$.

Practice Q68 (Module 3): $P = (1/\sqrt{5})\ [[2, -1], [1, 2]]$; $D = [[6, 0], [0, 1]]$.

Practice Q69 (Module 3): $k = -1$ or $k = 2$.

Practice Q70 (Module 3): Eigenvalues of real skew-symmetric matrix satisfy $\bar{\lambda} = -\lambda \Rightarrow \text{Re}(\lambda) = 0$ (pure imaginary or zero).

Practice Q71 (Module 3): Eigenvalues $\lambda = 3, -1$; $v_1 = [1, 1]^T$, $v_2 = [1, -1]^T$; $v_1 \cdot v_2 = 0$ (Orthogonal verified).

Practice Q72 (Module 3): $w = 1, x = 1, y = 1, z = 1$.

Practice Q73 (Module 3): $A^8 = [[-1/3 + (4/3)4^8, 4/3 - (4/3)4^8], [-1/3 + (1/3)4^8, 4/3 - (1/3)4^8]]$.

Practice Q74 (Module 3): Rank $\rho(A) = 3$.

Practice Q75 (Module 3): Characteristic equation: $\lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$; Verified; $A^{-1} = (1/4)(A^2 - 6A + 9I)$.

Practice Q76 (Module 4): $M_{22}(\mathbb{R})$ satisfies all 10 vector space axioms with zero matrix O as additive identity.

Practice Q77 (Module 4): Yes, W is a subspace: $(0, 0, 0) \in W$ and $(x_1, y_1, 0) + c(x_2, y_2, 0) = (x_1+cx_2, y_1+cy_2, 0) \in W$.

Practice Q78 (Module 4): $v = 3\ v_1 - 1\ v_2 + 1\ v_3$.

Practice Q79 (Module 4): $\det = 1 \neq 0 \Rightarrow$ Linearly Independent.

Practice Q80 (Module 4): Basis $B = \{(-1, 1, 0), (-1, 0, 1)\}$; $\dim(W) = 2$.

Practice Q81 (Module 4): Not a subspace: $y = 3x^2$ fails scalar multiplication closure (e.g. $2(1, 3) = (2, 6)$, but $3(2)^2 = 12 \neq 6$).

Practice Q82 (Module 4): Basis $B = \{(-1, 1, 0, 1), (1, 0, 1, -2)\}$; $\dim(W) = 2$.

Practice Q83 (Module 4): $[v]_B = [1/2, 1/2]^T$.

Practice Q84 (Module 4): Fails positive-definiteness: for $u = (0, 1)$, $\langle u, u \rangle = -1 < 0$. Not an inner product.

Practice Q85 (Module 4): $||u|| = \sqrt{3}$, $||v|| = \sqrt{2}$; $d(u, v) = \sqrt{3}$; $\cos\theta = 0 \Rightarrow \theta = 90°$ (Orthogonal).

Practice Q86 (Module 4): $\langle u, v \rangle = 10$, $||u||^2 = 13$, $||v||^2 = 17$; $100 \leq 221$ (Verified).

Practice Q87 (Module 4): $\text{proj}_u(v) = (2, 2, 2)$; $v_{perp} = (0, 1, -1)$.

Practice Q88 (Module 4): $c = -1/3 \Rightarrow p(x) = x^2 - 1/3$ is orthogonal to 1 on $[-1, 1]$.

Practice Q89 (Module 4): Basis for skew-symmetric matrices: $\{ [[0, 1], [-1, 0]] \}$; $\dim = 1$.

Practice Q90 (Module 4): $||u|| = \sqrt{5}$, $||v|| = \sqrt{10}$, $||u+v|| = \sqrt{17}$; $\sqrt{17} \approx 4.12 \leq \sqrt{5} + \sqrt{10} \approx 5.40$ (Verified).

Practice Q91 (Module 4): $e_1 = (1/\sqrt{3})(1, 1, 1)$, $e_2 = (1/\sqrt{2})(-1, 1, 0)$, $e_3 = (1/\sqrt{6})(1, 1, -2)$.

Practice Q92 (Module 4): x-axis and y-axis in $\mathbb{R}^2$: $(1, 0) + (0, 1) = (1, 1) \notin$ (x-axis $\cup$ y-axis).

Practice Q93 (Module 4): Line $y = 0$ and line $x = 0$: neither contains the other, union is not a subspace.

Practice Q94 (Module 4): Coordinates $[1, 0, 0]^T$, $[1, -1, 0]^T$, $[1, -2, 1]^T$; $\det = -1 \neq 0 \Rightarrow$ Forms a Basis.

Practice Q95 (Module 4): Basis = {(-2, 1, 0), (-2, 0, 1)}; dim = 2.

Practice Q96 (Module 4): Basis for W^⊥ = {(-1, 1, 0), (-1, 0, 1)}; dim = 2.

Practice Q97 (Module 4): $c_1 = 1$, $c_2 = 2$; $c_1^2 + c_2^2 = 5 \leq ||v||^2 = 14$ (Bessel's inequality verified).

Practice Q98 (Module 4): $u_3(x) = x^2 - x + 1/6$ (third shifted Legendre polynomial).

Practice Q99 (Module 4): $P = [[-2, 3/2], [1, -1/2]]$.

Practice Q100 (Module 4): $Q^T Q = I \Rightarrow \det(Q^T Q) = (\det Q)^2 = \det(I) = 1 \Rightarrow \det(Q) = \pm 1$.

Practice Q101 (Module 5): $T(u - v) = T(u + (-1)v) = T(u) + (-1)T(v) = T(u) - T(v)$.

Practice Q102 (Module 5): Not linear: $T(2(1, 1)) = T(2, 2) = (4, 2) \neq 2\,T(1, 1) = 2(1, 1) = (2, 2)$.

Practice Q103 (Module 5): $0 \in$ Range(T) because $0 = T(0)$; $w_1 + c\,w_2 = T(v_1) + c\,T(v_2) = T(v_1 + c\,v_2) \in$ Range(T).

Practice Q104 (Module 5): rank(T) = 2, nullity(T) = 1; $2 + 1 = 3 = \dim(\mathbb{R}^3)$ (Verified).

Practice Q105 (Module 5): $[T] = [[0, 1], [1, 0]]$.

Practice Q106 (Module 5): $T(x, y) = (x - y, 2x + 3y)$; $T(2, -4) = (6, -8)$.

Practice Q107 (Module 5): Kernel: {(1, -1, -2)}, nullity = 1; Range: {(1, 2, 3), (-1, 1, 0)}, rank = 2.

Practice Q108 (Module 5): $\det(A) = 0 \Rightarrow$ Not one-to-one, Not onto.

Practice Q109 (Module 5): $[T]_B = [[1/3, 5/3], [1/3, -1/3]]$.

Practice Q110 (Module 5): $[T \circ S] = (1/\sqrt{2})\,[[-1, 1], [1, 1]] \neq [S \circ T]$.

Practice Q111 (Module 5): $[T] = [[0, 0, 0], [1, 0, 0], [0, 1/2, 0], [0, 0, 1/3]]$; rank = 3, nullity = 0.

Practice Q112 (Module 5): $T^{-1}(x, y) = (5x - 2y, -2x + y)$.

Practice Q113 (Module 5): [T]_{B'} = [[3, 0], [0, 2]].

Practice Q114 (Module 5): rank = 2, nullity = 1.

Practice Q115 (Module 5): [P] = (1/5) [[1, 2], [2, 4]]; $P^2 = P$ and $P^T = P$.

Practice Q116 (Module 5): By Rank-Nullity: 5 = rank + nullity; rank ≤ 3 ⇒ nullity ≥ 2 > 0 ⇒ ker(T) ≠ {0} ⇒ not one-to-one.

Practice Q117 (Module 5): ker(T) = {(0, 0)} ⇒ T is one-to-one.

Practice Q118 (Module 5): rank(T) = 2 = dim($\mathbb{R}^2$) ⇒ onto; nullity = 3 - 2 = 1 > 0 ⇒ not one-to-one.

Practice Q119 (Module 5): rank = 1, nullity = 3; ker(T) = symmetric matrices, Range(T) = skew-symmetric.

Practice Q120 (Module 5): Max rank of AB is min(3, 4, 2) = 2.

Practice Q121 (Module 5): [T] = [[1, 1, 1], [0, 2, 4], [0, 0, 4]].

Practice Q122 (Module 5): T(x, y, z) = (0, 0, z); ker = xy-plane, Range = z-axis.

Practice Q123 (Module 5): $T^2 = I$ ⇒ (T - I)(T + I) = 0 ⇒ (λ - 1)(λ + 1) = 0 ⇒ λ = 1, -1.

Practice Q124 (Module 5): Rotations commute: $R_{90} \circ R_{180} = R_{270} = R_{180} \circ R_{90}$.

Practice Q125 (Module 5): 5 variables, 3 equations: nullity = 5 - rank ≥ 5 - 3 = 2 > 0 ⇒ non-trivial solutions always exist.

✅ Appendix B: Final Examination Revision Checklist & Strategy

Top 10 High-Yield Exam Hall Presentation Rules:

- [ ] **Always State the Master Formula First:** In VTU valuation, the initial formula carries dedicated method marks. Never jump straight to numbers.
- [ ] **Euler's Theorem Degree Proof:** Always show $f(tx, ty) = t^n f(x, y)$ explicitly before applying $x u_x + y u_y = n u$.
- [ ] **Unit Vector Division in Directional Derivatives:** Always divide by magnitude $|a\square| = \sqrt{(a_1^2 + a_2^2 + a_3^2)}$. Forgetting this loses 50% marks.
- [ ] **Divergence vs Curl Distinction:** Divergence produces a scalar (no $\hat{i}, \hat{j}, \hat{k}$); curl produces a vector. Never write curl as a scalar!
- [ ] **Rank Consistency (Rouché-Capelli):** Always write the explicit rank comparison: $\rho(A) = \rho([A|B]) = r$. State whether $r = n$ or $r < n$ clearly.
- [ ] **Eigenvalue Properties Verification:** Always verify $\Sigma \lambda_i = \text{Trace}(A)$ and $\Pi \lambda_i = \det(A)$ in the rough column to ensure your cubic roots are 100% correct.
- [ ] **Repeated Eigenvalues:** If λ is repeated m times, ensure you provide m linearly independent eigenvectors (or orthogonalize them).
- [ ] **Subspace Testing:** Always test $(0, 0, 0) \in W$ first. If zero is not in the set, conclude 'not a subspace' immediately.
- [ ] **Gram-Schmidt Normalization:** Do not stop at orthogonal vectors u_i; always divide by $||u_i||$ to obtain the final ORTHONORMAL basis e_i.
- [ ] **Rank-Nullity Domain Dimension:** In $\dim(\text{Domain}) = \text{rank} + \text{nullity}$, ensure you use the dimension of the DOMAIN space, not the codomain.

***** END OF TEXTBOOK — ALL 125 QUESTIONS FULLY SOLVED & VERIFIED *****